

# Chemistry—A European Journal

Supporting Information

## **Generation and Characterization of a Tetraradical Embedded in a Curved Cyclic Paraphenylene Unit**

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## 1. Experimental Section

### Reagents and solvents

All commercially available reagents were purchased from TCI, Wako, Oakwood Chemical and BLD Pharmatech and were used without further purification. Dry toluene for synthesis was obtained by distillation over  $\text{CaH}_2$ . Dry-DMF was purchased from Wako. 2-methyl tetrahydrofuran for EPR measurement was obtained by distillation over  $\text{CaH}_2$  and Na. Dry solvents for spectroscopy analysis were purchased from commercial suppliers.

### Procedure

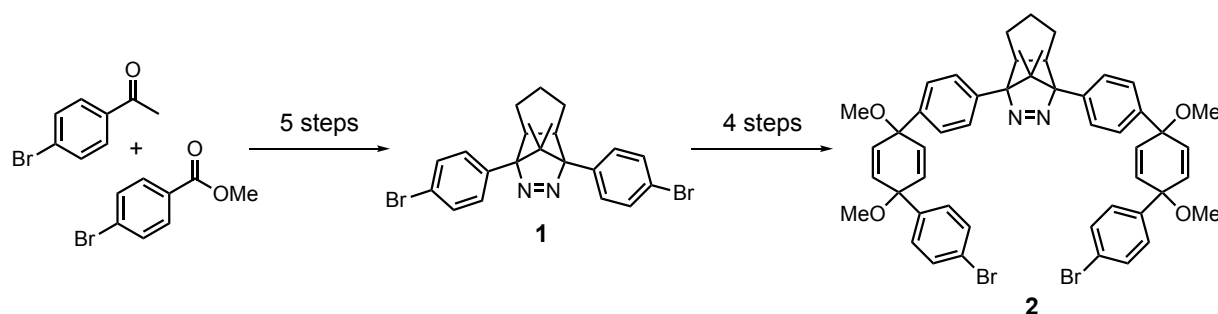
All workup and purification procedures were performed with reagent-grade solvents in the air. Thin-layer chromatography (TLC) analyses were performed using commercial aluminum sheets of Merck silica gel 60  $\text{F}_{254}$  and visualized with an ultraviolet lamp ( $\lambda = 254 \text{ nm}$ ). Column chromatography was performed with silica gel 60N (spherical, neutral, 63-210  $\mu\text{m}$ , Kanto). Recycling chromatographic separation in GPC were performed by LC-9210, Japan Analytical Industry Co., Ltd. with chloroform as mobile phase.

### Measurements

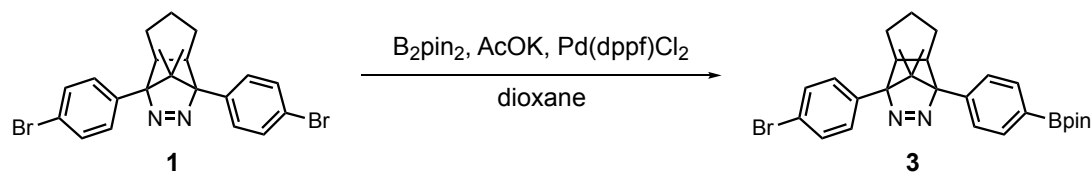
NMR spectra were recorded on a Bruker Ascend 400 ( $^1\text{H}$  NMR : 400 MHz,  $^{13}\text{C}$  NMR : 100 MHz) and AVANCE NEO 700 ( $^1\text{H}$  NMR : 700 MHz,  $^{13}\text{C}$  NMR : 176 MHz) spectrometer at 296 K and referenced to residual solvent peak. Coupling constants ( $J$ ) are denoted in Hz and chemical shifts ( $\delta$ ) in ppm. The abbreviations s,

d, t, dd, dt, td and m stand for the resonance multiplicities singlet, doublet, triplet, doublet of doublets, doublet of triplets, triplet of doublets and multiplet, respectively. The X-ray diffraction data of the single crystals were collected on a Bruker APEX-II Ultra CCD diffractometer. The structures were solved using direct methods and refined by full-matrix least-squares techniques using the SHELX program package. Single crystal X-ray structure was visualized by Mercury (CCDC). A will-V (ARK TECH) or CL-1501 (ASAHI SPECTRA) was used for 365 nm LED light source. UV-vis spectra were recorded on a SHIMADZU UV-3600 Plus spectrometer. The spectra were collected at room temperature using a slit width of 1 nm with middle scan rate. IR spectrum was recorded on a JASCO MCT-6000M. The date of High-resolution Mass spectrometry (HRMS) was measured with Thermo Fisher Scientific LTQ Orbitrap XL using electrospray ionization (ESI) or atmospheric-pressure chemical ionization (APCI) method. Steady-state fluorescence spectra were recorded with a FluoroMax 4 spectrofluorometer Horiba Yobin Yvon using a slit width of 1 nm. Fluorescence lifetime was measured by using FluoroHub-B, TemPro 01, TBX, and 373 nm NanoLED light (HORIBA). The excitation source for sub-microsecond laser flash photolysis (LFP) was a tunable Nd:YAG minilite laser at 355 nm. The monitoring system consisted of a 150 W Xenon arc lamp as light source, a Unisoku MD200 monochromator detection and a photomultiplier. The temperature was controlled by Unisoku CoolSpek USP- 203-B. Sub-nanosecond transient absorption measurements were conducted with Unisoku PicoTAS system in 2 mm cuvette. The excitation source was a passive Q-SWITCH microchip laser at 355 nm. Cw-EPR spectra were recorded on Bruker BioSpin ELEXSYS E500 spectrometer at Hiroshima university and Bruker BioSpin ELEXSYS E600 spectrometer at Osaka Metropolitan University. 2D-pulse EPR spectra were recorded on Bruker ESP380 E at Osaka Metropolitan University. SUPERCURE-351S (SAN-EI) was used for UV light source. Simulation of EPR signal was conducted by Easyspin (open-source MATLAB toolbox).

## 2. Synthesis and spectral data of azoalkane

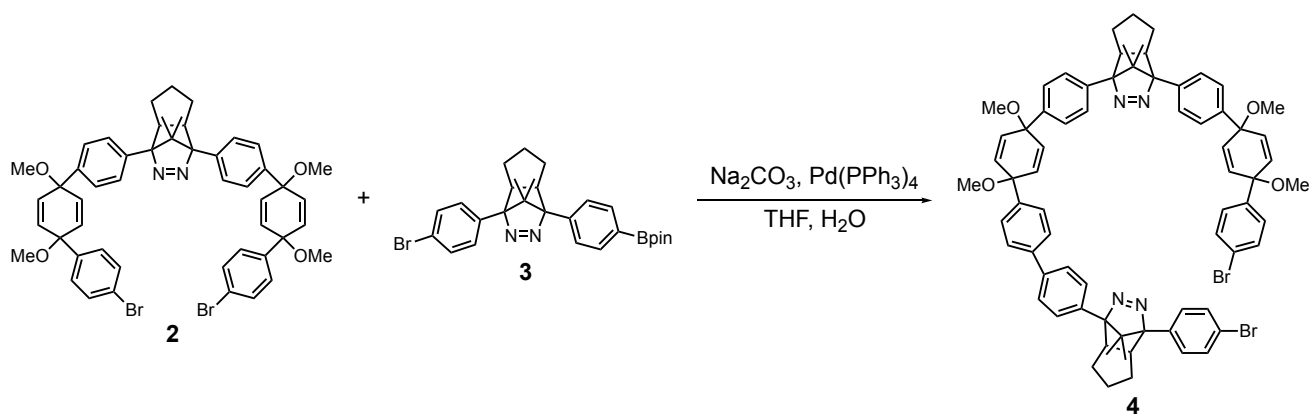


Compounds **1** and **2** were synthesized according to reference 8 in this paper. Compound **1**:  $^1\text{H NMR}$  ( $\text{CDCl}_3$ , 400 MHz)  $\delta$  0.13 (s, 3H), 0.96 (s, 3H), 1.39–1.46 (m, 1H), 1.49–1.56 (m, 5H), 3.44–3.46 (m, 2H), 7.61 (s, 8H).  $^{13}\text{C NMR}$  ( $\text{CDCl}_3$ , 100 MHz)  $\delta$  16.99, 17.93, 25.51, 28.68, 49.21, 66.35, 98.12, 122.15, 129.31, 131.75, 135.22. Compound **2**:  $^1\text{H NMR}$  ( $\text{CDCl}_3$ , 400 MHz)  $\delta$  0.12 (s, 3H), 0.92 (s, 3H), 1.39–1.55 (m, overlap with water), 3.44–3.46 (s, 12H), 3.49 (m, 2H), 6.08–6.20 (d,  $J = 10.2$  Hz and  $J = 10.3$  Hz, 8H), 7.29–7.31 (d,  $J = 8.7$  Hz, 4H), 7.43–7.49 (d,  $J = 8.7$  and  $J = 8.5$  Hz, 8H), 7.67–7.69 (d,  $J = 8.5$  Hz, 4H).  $^{13}\text{C NMR}$  ( $\text{CDCl}_3$ , 100 MHz)  $\delta$  17.16, 17.94, 25.67, 28.75, 49.00, 52.18, 52.23, 66.50, 74.70, 74.82, 98.38, 121.76, 126.05, 127.77, 128.00, 131.61, 133.19, 133.29, 133.87, 133.96, 135.82, 142.73, 142.74.

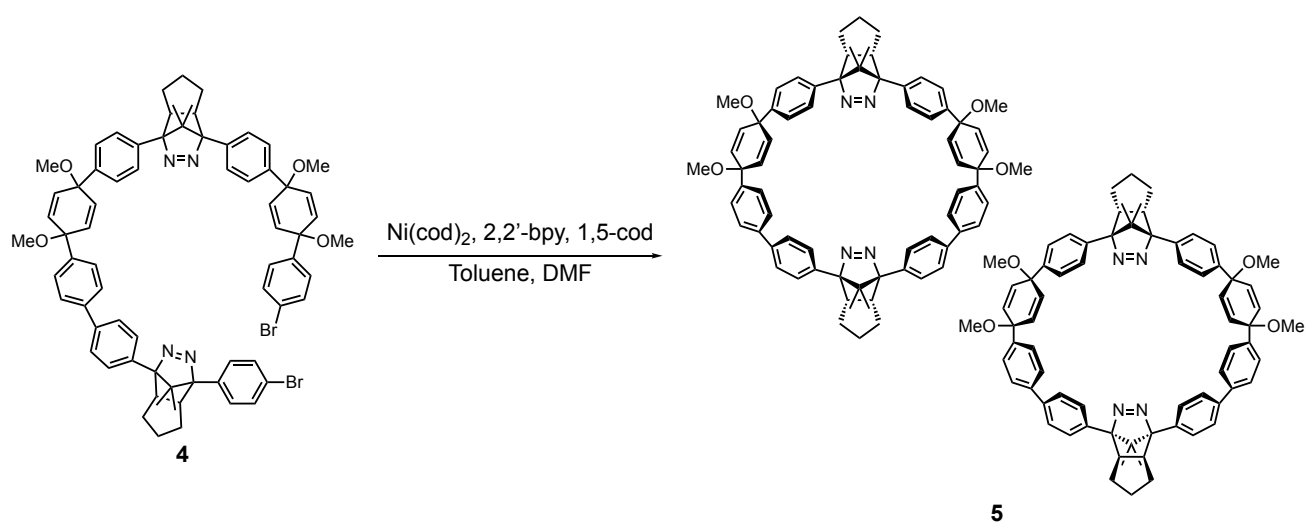


**1** (1.06 g, 2.24 mmol), Bis(pinacolato)diboron (0.53 g, 2.09 mmol),  $\text{Pd(dppf)Cl}_2$  (0.09 g, 0.12 mmol), and potassium acetate (0.69 g, 7.0 mmol) were dissolved in dioxane (20 mL) under nitrogen atmosphere. After freeze-pump thaw cycling (three times), replaced nitrogen atmosphere again. The mixture was stirred at  $65^\circ\text{C}$  for 13.5 h (overnight). Afterwards, the reaction mixture was filtered through a pat of Celite and removed solvent under reduced pressure. The crude mixture was purified by preparative gel permeation chromatography using chloroform as the eluent to give **3** (0.59 g, 1.1 mmol) in 49 % yield.  $^1\text{H NMR}$  ( $\text{CDCl}_3$ , 400 MHz)  $\delta$  0.13 (s, 3H), 0.96 (s, 3H), 1.37 (s, 12H), 1.54–1.57 (m, overlap with water, 6H), 3.42–3.55 (m, 2H), 7.59–7.64 (m, 4H), 7.74–7.76 (d,  $J = 8.2$  Hz, 2H), 7.91–7.93 (d,  $J = 8.2$  Hz, 2H).  $^{13}\text{C NMR}$  ( $\text{CDCl}_3$ , 100 MHz)  $\delta$  17.04, 17.98, 25.04, 25.07, 25.55, 28.73, 49.12, 49.23, 66.51, 84.02, 98.01, 98.73, 122.05, 126.96, 129.37, 131.70, 134.96, 135.45, 139.21. **HRMS** (p-ESI, MeOH sol)  $m/z$ : Calcd for  $\text{C}_{28}\text{H}_{35}\text{N}_2\text{O}_2\text{BBr}$  [ $\text{M} + \text{H}$ ] $^+$ , 521.19695, found 521.19684. **IR** (KBr plate [ $\text{cm}^{-1}$ ]) 524, 660, 676, 706, 744, 805, 823, 860, 962, 1011, 1021, 1076, 1092, 1107, 1145, 1166, 1213, 1271, 1305, 1322, 1362, 1401, 1447, 1469, 1492, 1522, 1613, 2873, 2941, 2976.



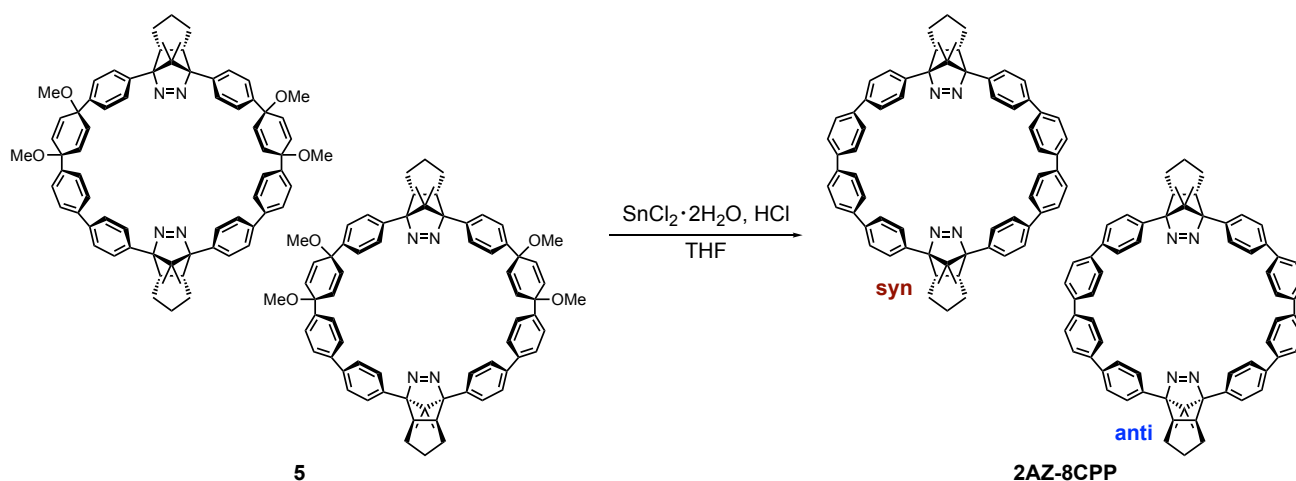


Under nitrogen atmosphere, **2** (555 mg, 0.61 mmol), **3** (126 mg, 0.24 mmol),  $\text{Na}_2\text{CO}_3$  (69 mg, 0.65 mmol), and  $\text{Pd}(\text{PPh}_3)_4$  (19 mg, 0.02 mmol) were dissolved in THF (8 mL) and  $\text{H}_2\text{O}$  (1 mL). The reaction flask was replaced with nitrogen atmosphere after freeze pump thaw cycling (three times). The mixture was stirred at 65 °C for 16.5 h (overnight). The reaction was quenched by the addition of  $\text{NH}_4\text{Cl}$  aq. The organic layer was extracted with EtOAc two times, washed with brine, and dried over  $\text{Na}_2\text{SO}_4$ . After filtration, the solvent was removed under reduced pressure. The crude mixture was purified by silica gel column chromatography to give **4** (113 mg, 0.09 mmol) in 38 % yield. This reaction was repeated several times.  **$^1\text{H}$  NMR** ( $\text{CDCl}_3$ , 400 MHz)  $\delta$  0.12 (s, 3H), 0.19 (s, 3H), 0.92 (s, 3H), 1.00 (s, 3H), 1.41–1.45 (m, 2H), 1.55–1.59 (m, overlap with water, 10H), 3.44–3.57 (s and m, 12H and 4H), 6.07–6.20 (m, 8H), 7.29–7.31 (d,  $J$  = 8.7 Hz, 2H), 7.42–7.48 (d,  $J$  = 8.6 and 8.5 Hz, 4H), 7.53–7.55 (dd,  $J$  = 8.5 Hz,  $J$  = 1.3 and 1.4 Hz, 4H), 7.60–7.70 (m, 12H), 7.78–7.80 (d,  $J$  = 8.4 Hz, 2H).  **$^{13}\text{C}$  NMR** ( $\text{CDCl}_3$ , 100 MHz)  $\delta$  17.14, 17.17, 17.95, 18.02, 25.58, 25.67, 28.74, 49.01, 49.14, 49.25, 52.17, 52.22, 66.49, 74.69, 74.82, 74.91, 74.97, 97.98, 98.36, 98.42, 98.63, 121.75, 122.05, 126.04, 126.13, 126.68, 127.20, 127.27, 127.72, 127.76, 128.01, 128.06, 129.36, 131.69, 131.72, 133.18, 133.29, 133.48, 133.55, 133.75, 133.80, 133.86, 133.95, 135.20, 135.52, 135.67, 135.85, 140.19, 140.31, 142.73, 143.00. **HRMS** (p-ESI, MeOH sol)  $m/z$ : Calcd for  $\text{C}_{72}\text{H}_{72}\text{N}_4\text{O}_4\text{Br}_2\text{Na}$   $[\text{M} + \text{Na}]^+$ , 1237.38125, found 1237.38123. **IR** (KBr plate  $[\text{cm}^{-1}]$ ) 537, 561, 664, 756, 774, 807, 821, 952, 1010, 1027, 1038, 1077, 1174, 1229, 1302, 1372, 1395, 1448, 1469, 1491, 1507, 2822, 2876, 2939, 3033.



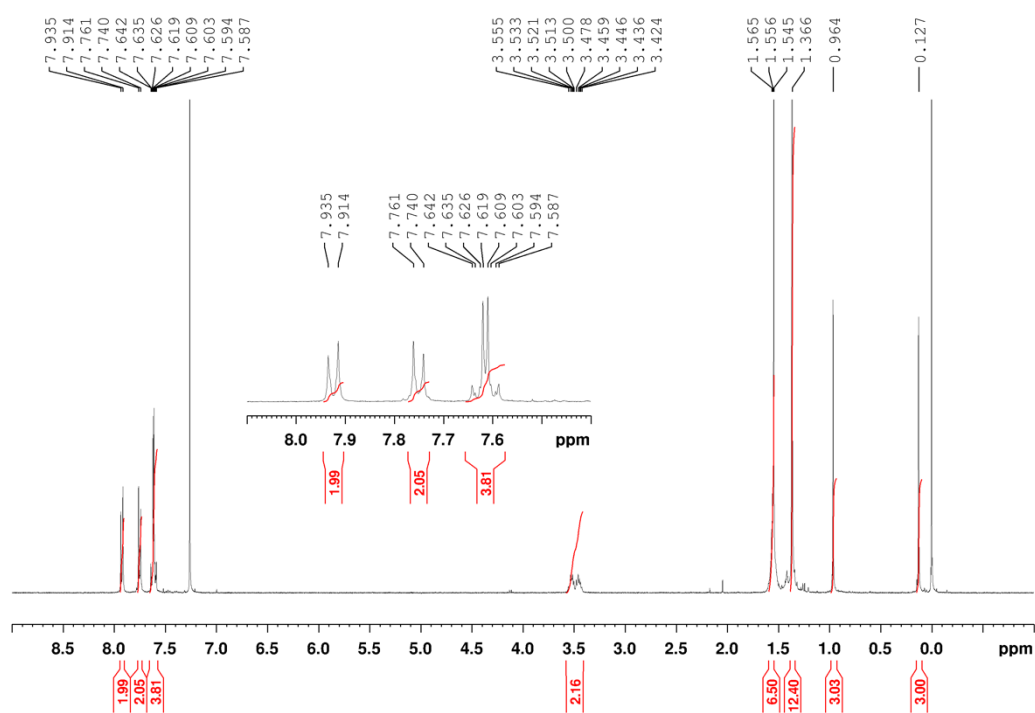
Under nitrogen atmosphere, 1,5-cyclooctadiene (0.1 mL, 0.80 mmol) and 2,2'-bipyridyl (121 mg, 0.77 mmol) were dissolved in dry-toluene (6 mL) and dry-DMF (6 mL). The reaction flask was replaced with argon atmosphere after freeze pump thaw cycling (four times). Ni(cod)<sub>2</sub> (235 mg, 0.85 mmol) was added, and replacement of argon atmosphere was conduct again. The mixture was stirred at 70 °C for 30 min.

On the other hand, under nitrogen atmosphere, **4** (194 mg, 0.16 mmol) was dissolved in dry-toluene (30 mL). The reaction flask was replaced with argon atmosphere after freeze pump thaw cycling (four times). The solution of **4** was added to the Ni-catalyst solution for 1 h at 70 °C. The mixture was stirred for 3 h, then cooled down to room temperature. The reaction mixture was added to 1 M HCl<sub>aq</sub> and stirred for 15 min at room temperature. The organic layer was extracted with EtOAc two times, washed with 1 M HCl<sub>aq</sub> and brine, and dried over Na<sub>2</sub>SO<sub>4</sub>. After filtration, the solvent was removed under reduced pressure. The crude mixture was purified by silica gel column chromatography (Hex : EtOAc = 6 : 1 → 5 : 1) to give **5** (31 mg, 0.03 mmol) in 19 % yield. **<sup>1</sup>H NMR** (CDCl<sub>3</sub>, 400 MHz) δ -0.78 and -0.68 (s, 3H), -0.09 and -0.05 (s, 3H), 0.49, 0.71, 0.76 (s, 6H), 1.48–1.50 (m, 6H), 1.74 (m, 4H), 1.88 (m, 2H), 3.43–3.49 (s and m, 12H and 2H), 3.61–3.65 (m, 2H), 5.79–5.85 (td, 2H), 6.10–6.29 (td, 6H), 7.30–7.36 (d, *J* = 8.4 Hz, 4H), 7.41–7.46 (d, *J* = 8.6 and 8.5 Hz, 4H), 7.54–7.56 (s and d, *J* = 8.1 Hz, 6H), 7.63–7.70 (d and m, *J* = 8.4 Hz, 10H). **<sup>13</sup>C NMR** (CDCl<sub>3</sub>, 100 MHz) δ 17.11, 17.16, 17.51, 17.59, 18.13, 18.29, 25.62, 25.75, 28.61, 28.66, 29.34, 29.85, 47.85, 48.05, 48.40, 48.48, 52.02, 52.34, 52.35, 66.74, 66.83, 67.50, 67.63, 74.78, 75.01, 75.12, 75.17, 97.99, 98.11, 98.70, 98.76, 125.93, 125.95, 126.76, 127.44, 127.78, 127.84, 127.92, 131.27, 131.42, 131.52, 132.07, 135.06, 135.15, 135.22, 135.66, 135.84, 136.11, 140.02, 140.33, 140.61, 141.00, 142.34, 142.36, 142.97, 143.05. **HRMS** (p-ESI, MeOH sol) *m/z*: Calcd for C<sub>72</sub>H<sub>72</sub>N<sub>4</sub>O<sub>4</sub>Na [M + Na]<sup>+</sup>, 1079.54458, found 1079.54358. **IR** (KBr plate [cm<sup>-1</sup>]) 544, 559, 665, 686, 755, 772, 809, 821, 869, 955, 984, 1005, 1016, 1029, 1039, 1082, 1115, 1174, 1183, 1192, 1227, 1261, 1301, 1371, 1389, 1449, 1469, 1494, 1508, 1609, 1735, 2821, 2878, 2937, 3031.

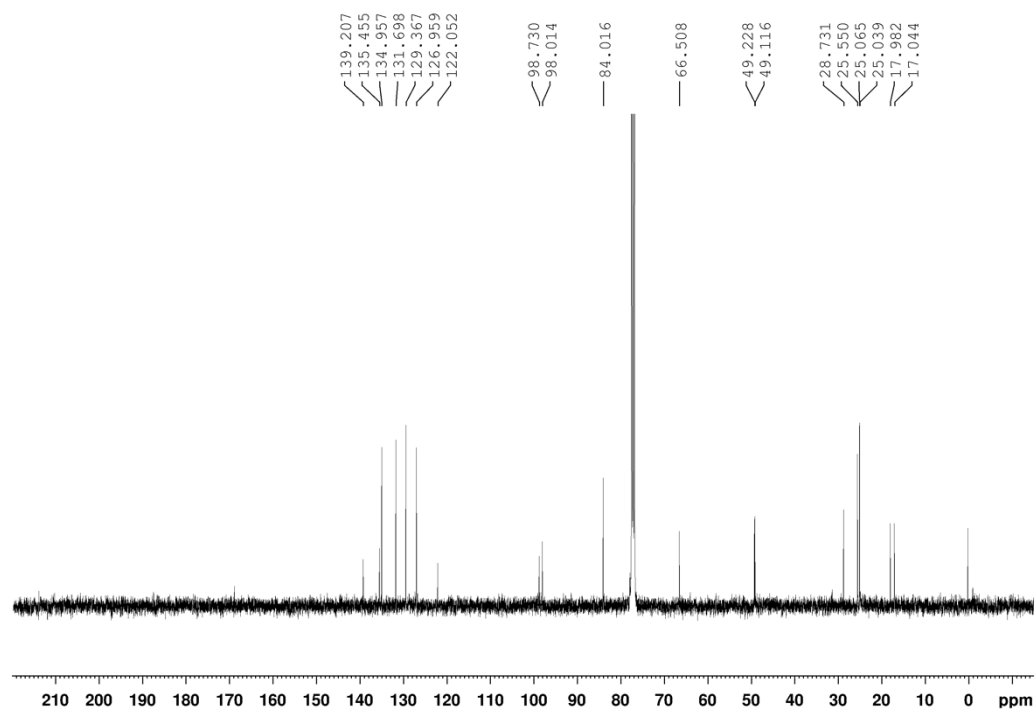


Under nitrogen atmosphere,  $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$  (96 mg, 0.42 mmol) was dissolved in THF (0.5 mL).  $\text{HCl}$ aq (12 M, 0.07 mL, 0.84 mmol) was added to the solution and the mixture was stirred at room temperature for 30 min.

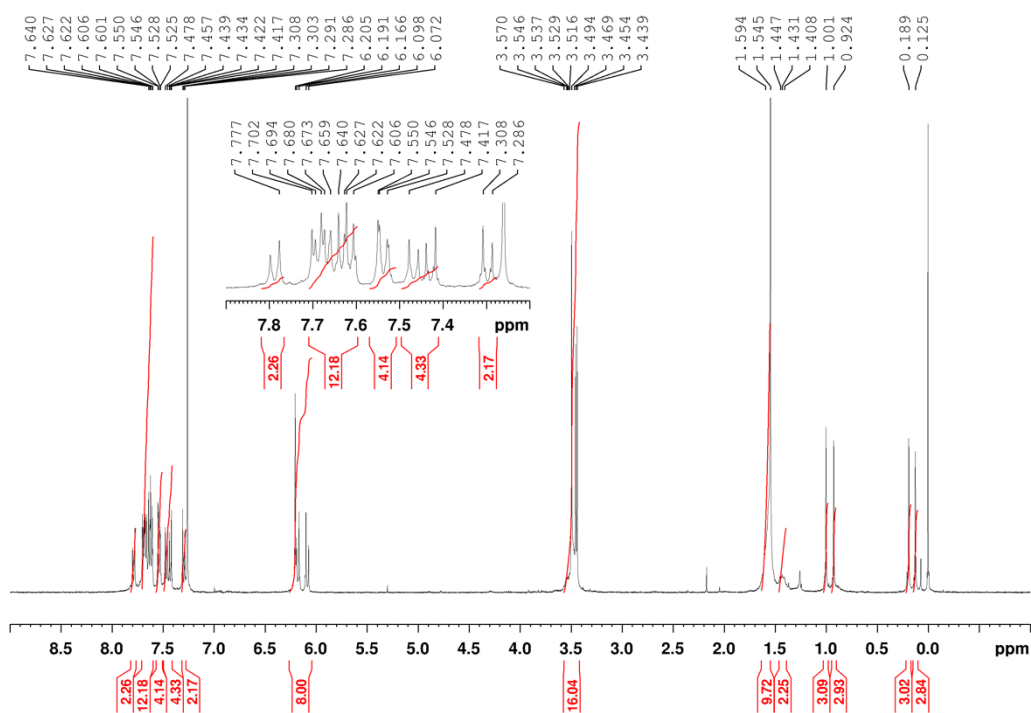
On the other hand, under nitrogen atmosphere, **5** (31 mg, 0.03 mmol) was dissolved in THF (7 mL). The solution of  $\text{H}_2\text{SnCl}_4$ aq (0.2 mL) was added to the solution of **5** and stirred at room temperature to 65 °C and 65 °C for 13 h. After cooled down to room temperature, the solvent was removed under reduced pressure. The crude mixture was purified by silica gel column chromatography ( $\text{CHCl}_3$ ) and preparative gel permeation chromatography using chloroform as the eluent to give **2AZ-8CPP** (6.3 mg, 6.8  $\mu\text{mol}$ ) as white solid in 23 % yield.  $^1\text{H NMR}$  ( $\text{CDCl}_3$ , 400 MHz)  $\delta$  -1.19 (s, 6H), 0.22 (s and s, 6H), 1.67–1.87 (m, 12H), 3.52–3.53 (m, 4H), 7.28–7.29 (m, overlap with water, 2H), 7.51–7.53 (d,  $J = 8.3$  Hz, 2H), 7.57–7.75 (m, 24H), 7.82–7.88 (dd and d,  $J = 8.1$ ,  $J = 1.8$  and 1.2 Hz,  $J = 8.5$  Hz, 4H).  $^{13}\text{C NMR}$  ( $\text{CDCl}_3$ , 100 MHz)  $\delta$  17.08, 17.44, 17.49, 25.59, 29.29, 47.44, 47.53, 67.76, 97.84, 97.92, 124.50, 124.93, 125.20, 126.81, 127.15, 127.36, 127.61, 127.71, 129.36, 134.56, 134.63, 138.00, 138.27, 138.56, 138.70, 138.82, 139.12. **HRMS** (p-ESI, MeOH sol)  $m/z$ : Calcd for  $\text{C}_{68}\text{H}_{61}\text{N}_4$   $[\text{M} + \text{H}]^+$ , 933.48907, found 933.48932. **IR** (on CsI plate /  $\text{cm}^{-1}$ ) 522, 542, 567, 703, 739, 757, 810, 850, 865, 943, 959, 1002, 1018, 1026, 1043, 1068, 1074, 1110, 1154, 1254, 1333, 1346, 1372, 1389, 1447, 1469, 1484, 1497, 1526, 1599, 2860, 2924, 2930, 2946, 2953, 3025, 3035, 3042, 3050, 3068, 3078.



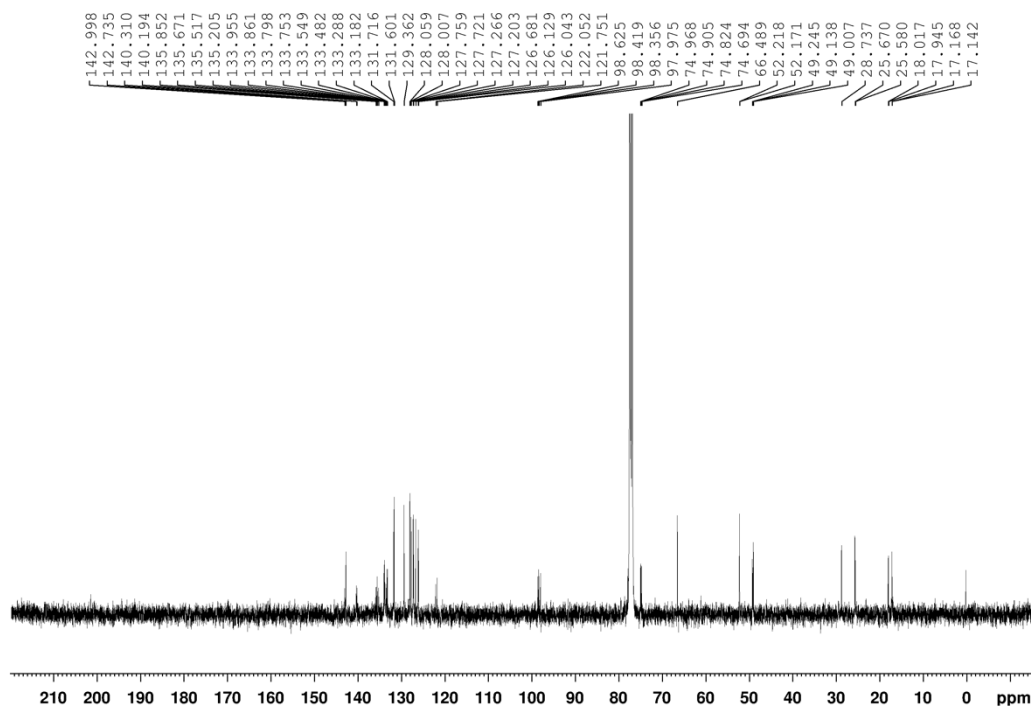
**Figure S1:**  $^1\text{H}$  NMR spectrum of compound **3**.



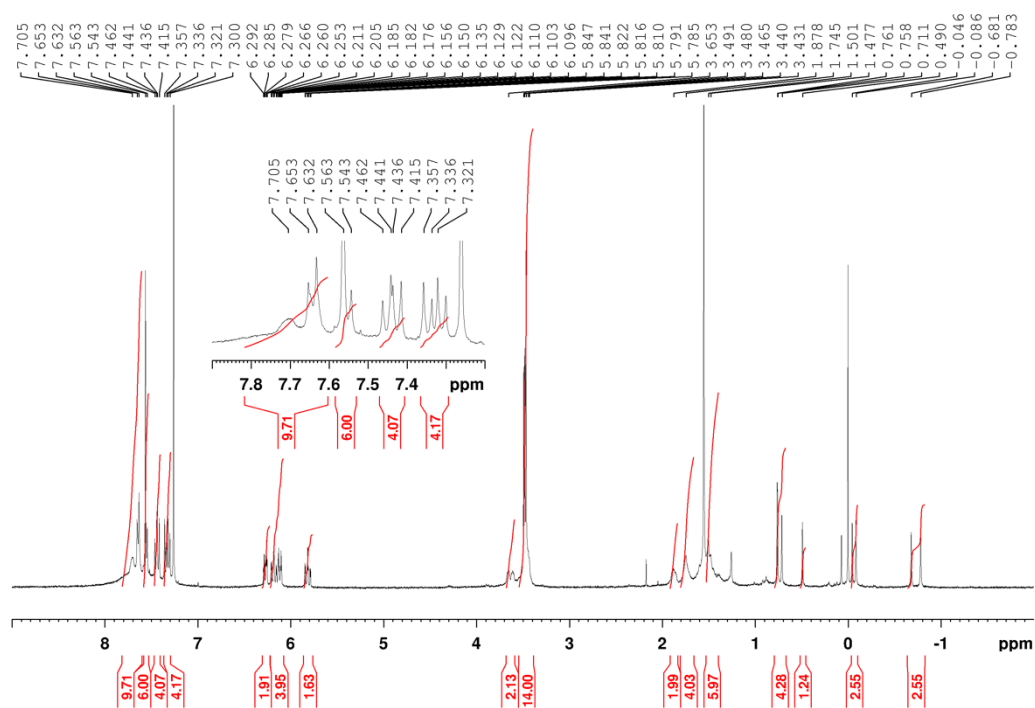
**Figure S2:**  $^{13}\text{C}$  NMR spectrum of compound **3**.



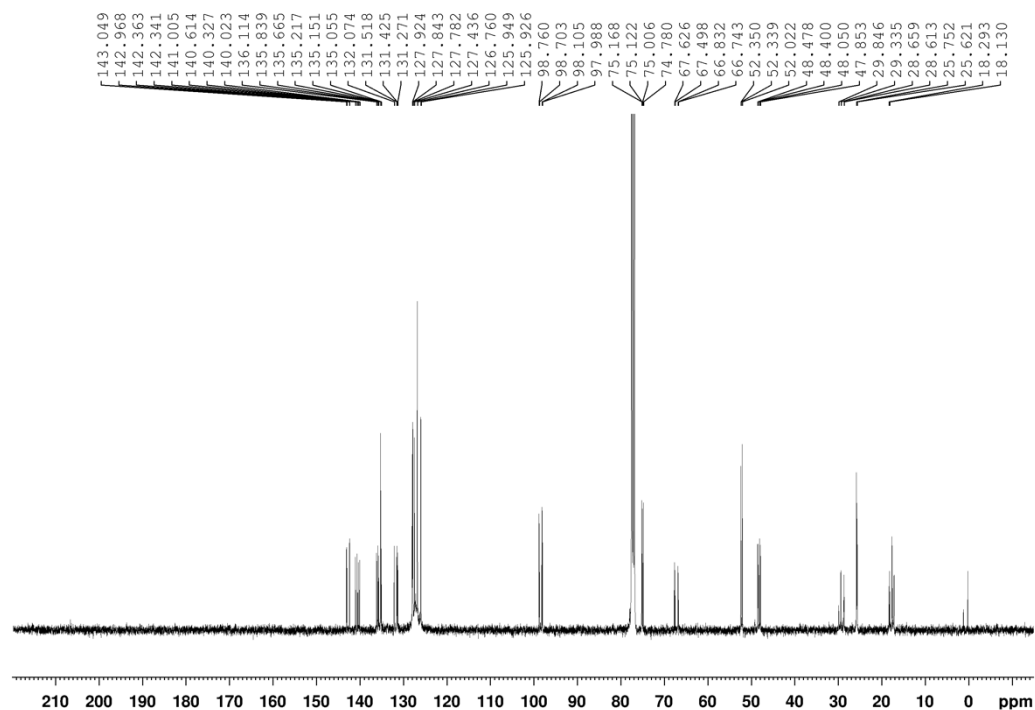
**Figure S3:  $^1\text{H}$  NMR spectrum of compound 4.**



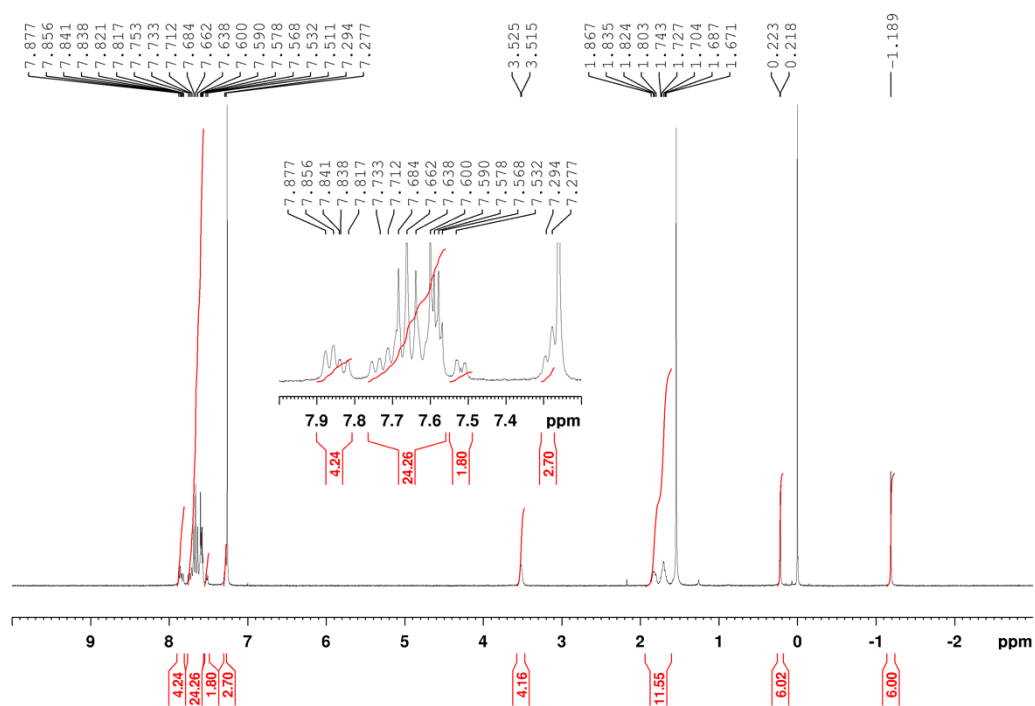
**Figure S4:  $^{13}\text{C}$  NMR spectrum of compound 4.**



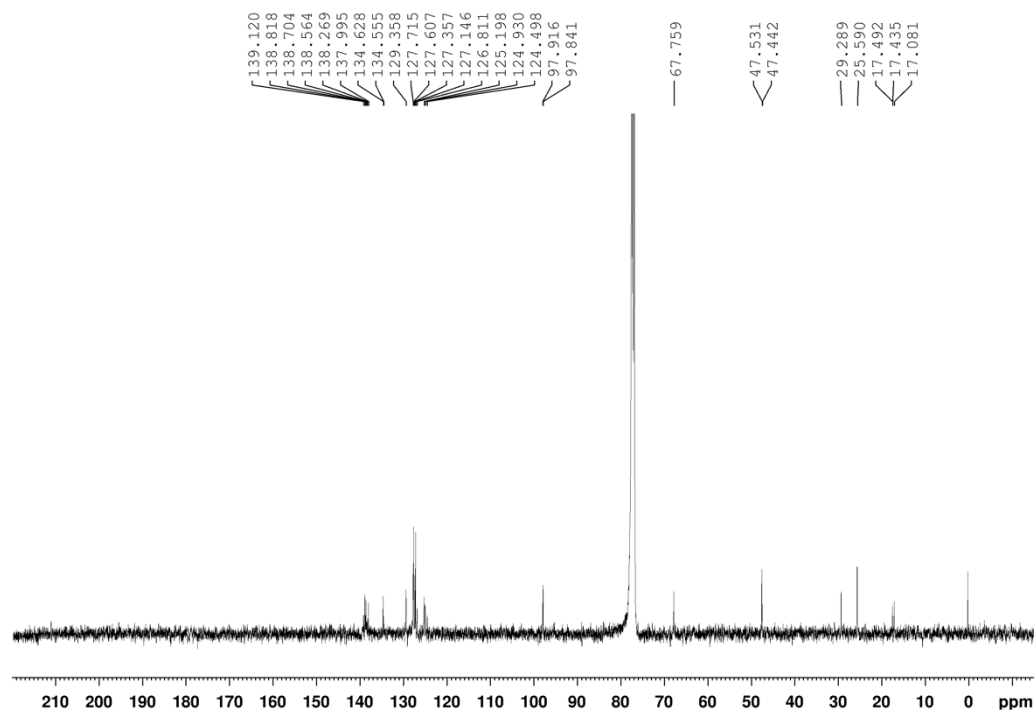
**Figure S5:** <sup>1</sup>H NMR spectrum of compound **5**.



**Figure S6:** <sup>13</sup>C NMR spectrum of compound **5**.



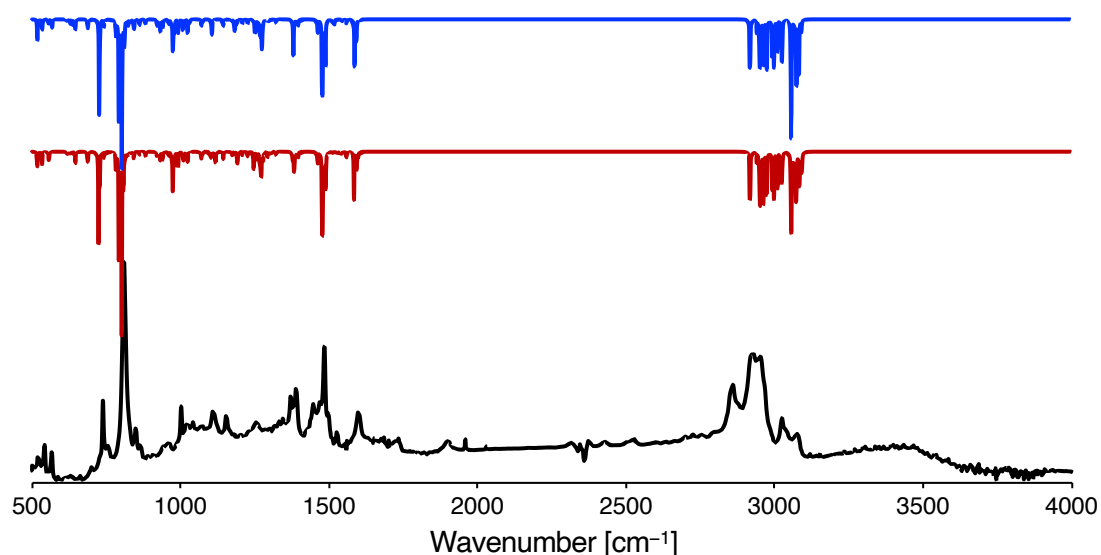
**Figure S7:** <sup>1</sup>H NMR spectrum of compound **2AZ-8CPP**.



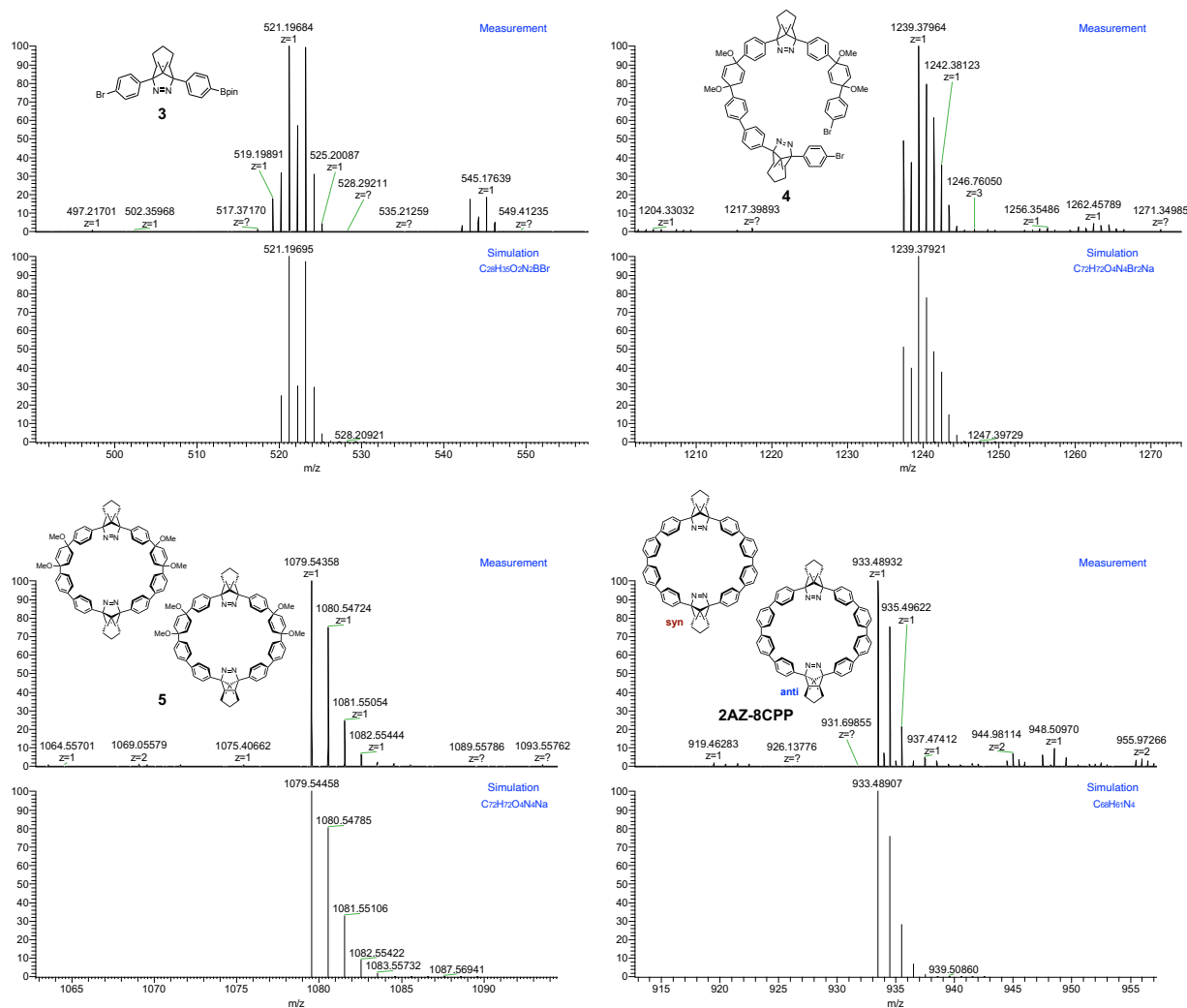
**Figure S8:** <sup>13</sup>C NMR spectrum of compound **2AZ-8CPP**.



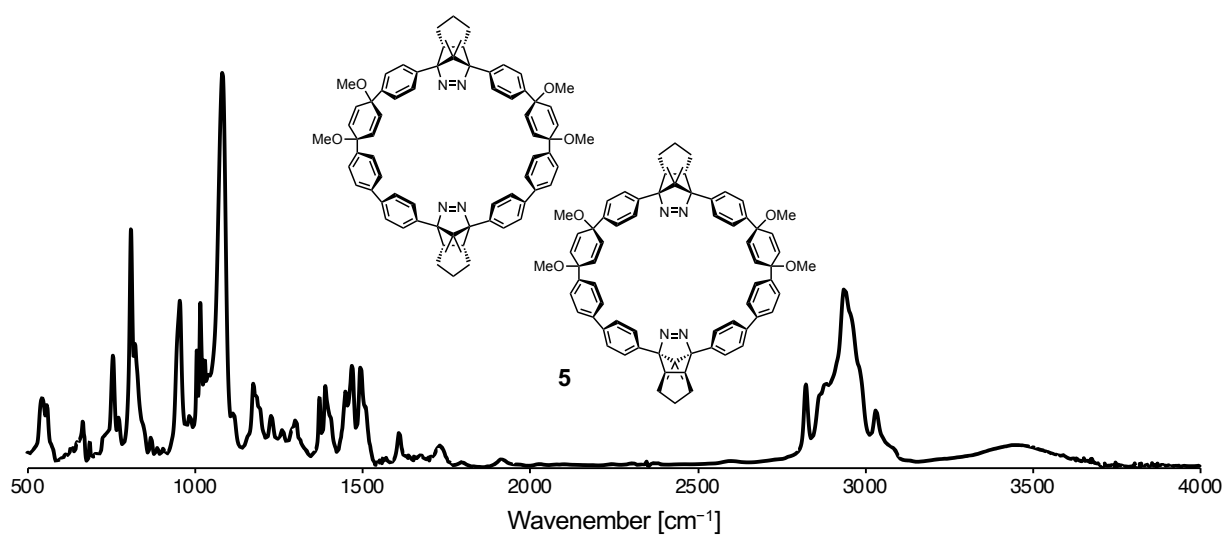
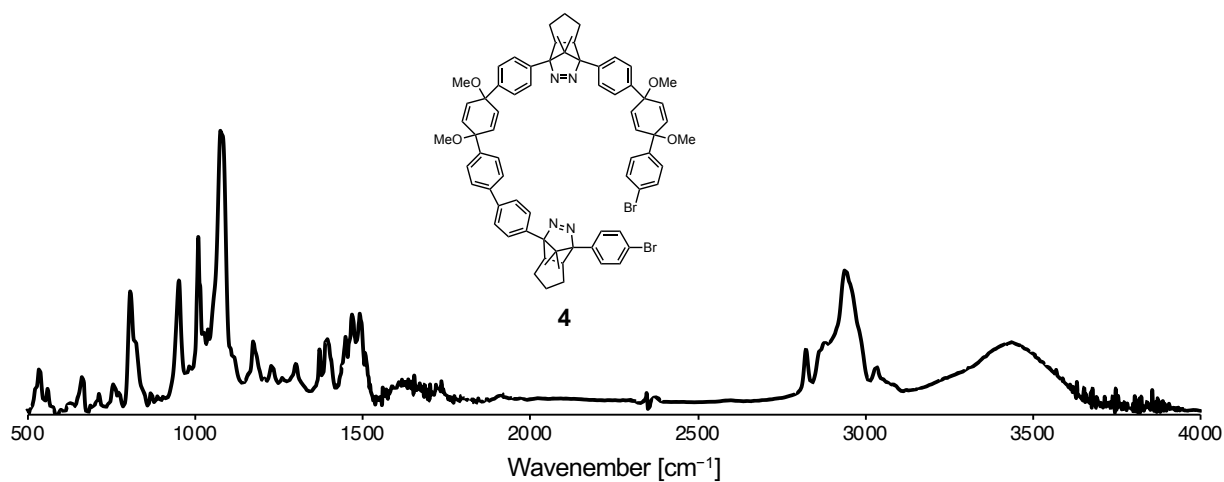
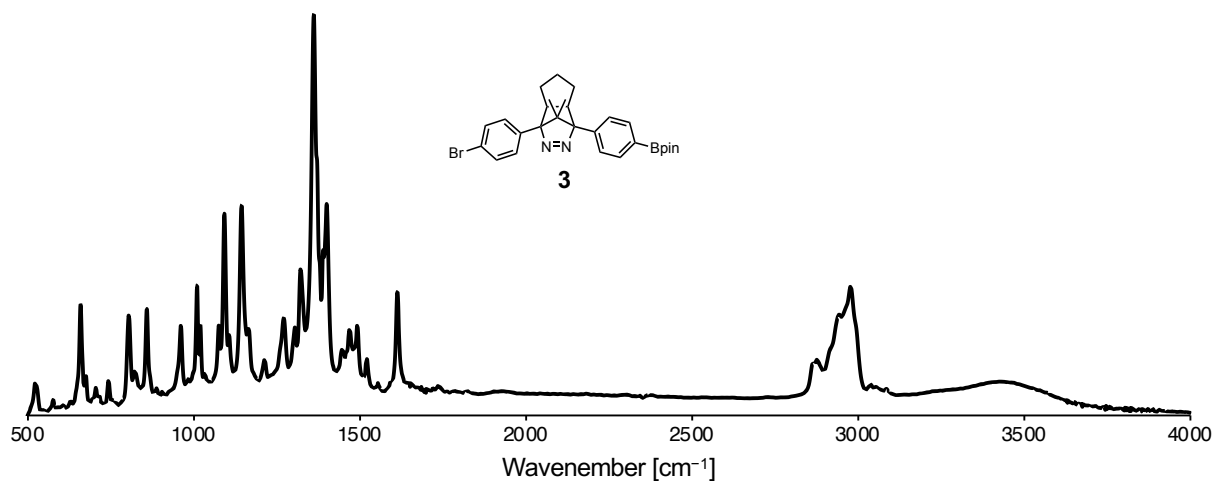




**Figure S11:** Experimental (black) and computed (red: syn, blue: anti) IR spectra of **2AZ-8CPP** on CsI plate.



**Figure S12:** HRMS data and simulation of compound **3**, **4**, **5**, and **2AZ-8CPP**.

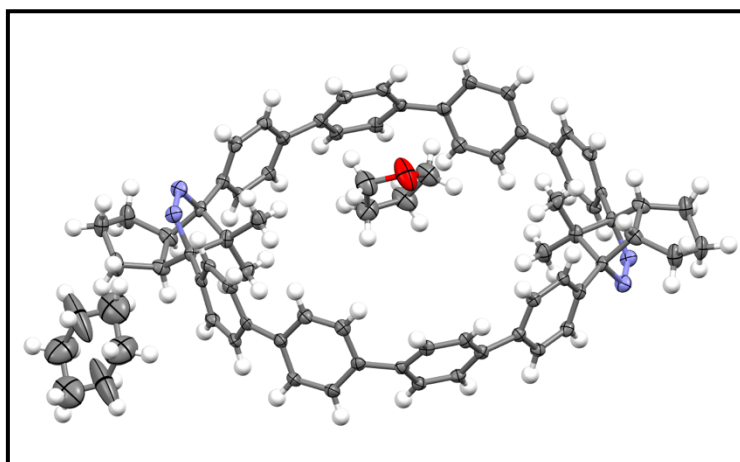


**Figure S13:** IR spectra of compound **3**, **4**, and **5** in KBr plate.

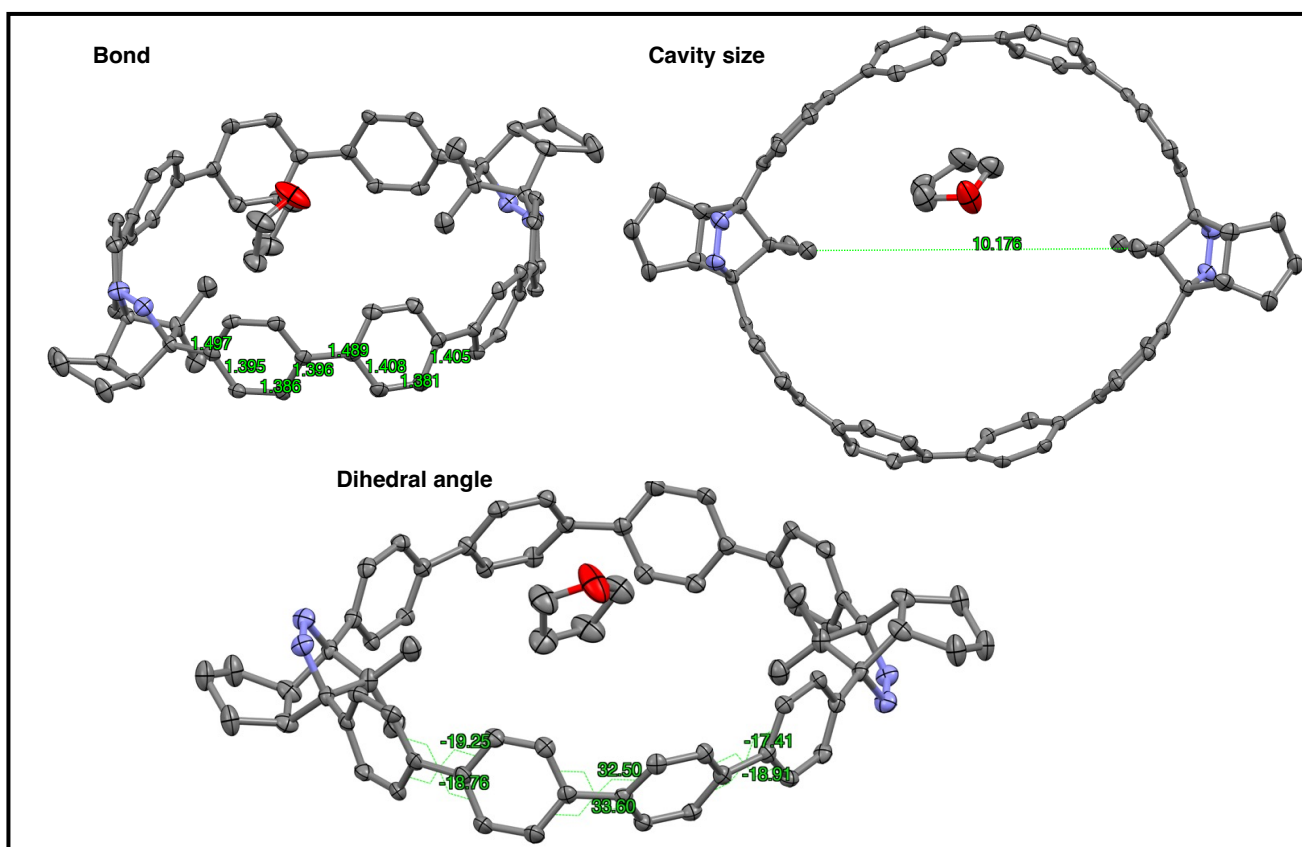
### 3. X-ray crystallographic structure of 2AZ-8CPP

**Table S1:** Crystal data and structure refinement for **2AZ-8CPP** with thermal ellipsoids at 50% probability

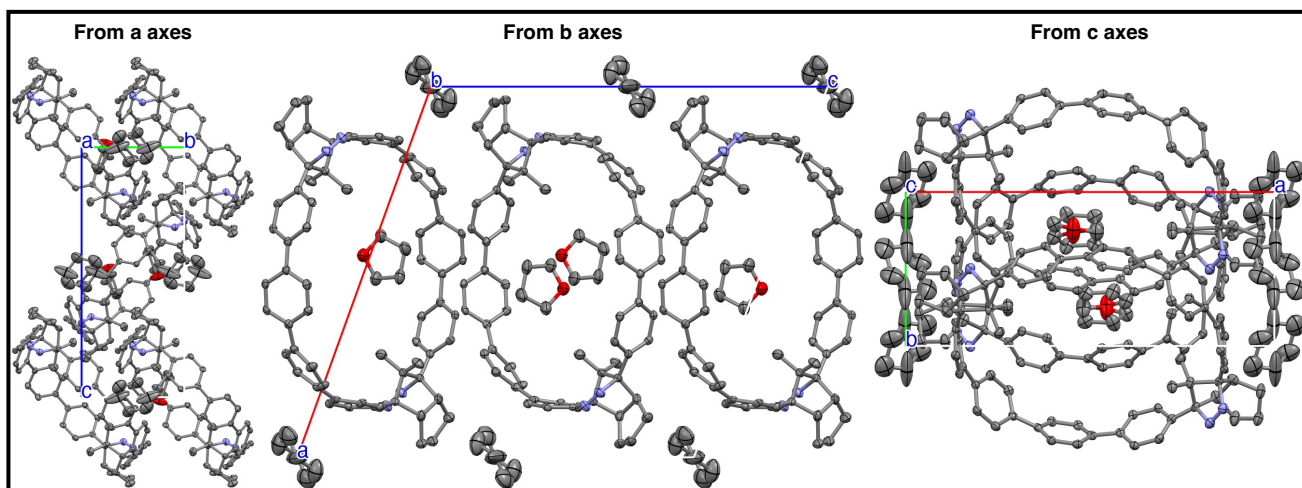
(gray : C, blue : N, red : O, white : H).



|                                                              |                                                                                                                                                    |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of the radiation                                        | MoK $\alpha$                                                                                                                                       |
| Formula                                                      | C <sub>78</sub> H <sub>80</sub> N <sub>4</sub> O                                                                                                   |
| Weight                                                       | 1089.46                                                                                                                                            |
| Temperature <i>T</i> (K)                                     | 90                                                                                                                                                 |
| Wavelength $\lambda$ (Å)                                     | 0.71073                                                                                                                                            |
| Crystal system                                               | Monoclinic                                                                                                                                         |
| Space group                                                  | <i>P</i> 12 <sub>1</sub> / <i>c</i> 1                                                                                                              |
| Unit cell                                                    | <i>a</i> (Å) = 20.514(8) $\alpha$ (deg) = 90<br><i>b</i> (Å) = 8.040(3) $\beta$ (deg) = 109.926(5)<br><i>c</i> (Å) = 20.740(7) $\gamma$ (deg) = 90 |
| Volume                                                       | 3216.(2)                                                                                                                                           |
| <i>z</i>                                                     | 2                                                                                                                                                  |
| Density (calculation)                                        | 1.125                                                                                                                                              |
| $\mu$ (mm <sup>-1</sup> )                                    | 0.066                                                                                                                                              |
| <i>F</i> (000)                                               | 1168                                                                                                                                               |
| Crystal shape                                                | Needle                                                                                                                                             |
| Color                                                        | Colorless                                                                                                                                          |
| Cryst size (mm <sup>3</sup> )                                | 0.01×0.05×0.10                                                                                                                                     |
| Theta range for data collection (deg)                        | 1.99 to 25.39                                                                                                                                      |
| Index ranges                                                 | -13≤ <i>h</i> ≤24, -9≤ <i>k</i> ≤9, -24≤ <i>l</i> ≤24                                                                                              |
| Number of Reflns                                             | 5890                                                                                                                                               |
| Params                                                       | 399                                                                                                                                                |
| Goodness-of-fit on <i>F</i> <sup>2</sup>                     | 1.033                                                                                                                                              |
| <i>R</i> 1, <i>wR</i> 2 [ <i>I</i> >2 $\sigma$ ( <i>I</i> )] | 0.0788, 0.1946                                                                                                                                     |
| <i>R</i> 1, <i>wR</i> 2 (all data)                           | 0.1502, 0.2342                                                                                                                                     |

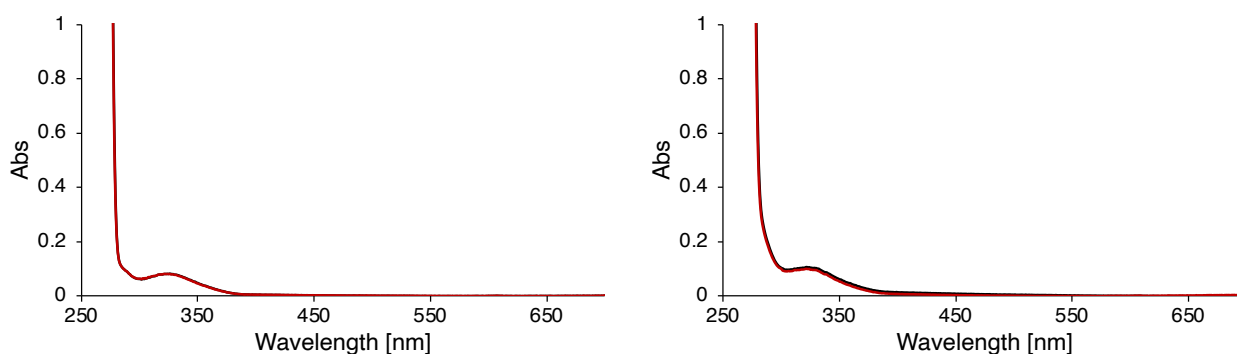


**Figure S14:** Bond length, cavity size, and dihedral angle of **2AZ-8CPP**. Cyclohexane and hydrogen atoms are omitted for clarity.

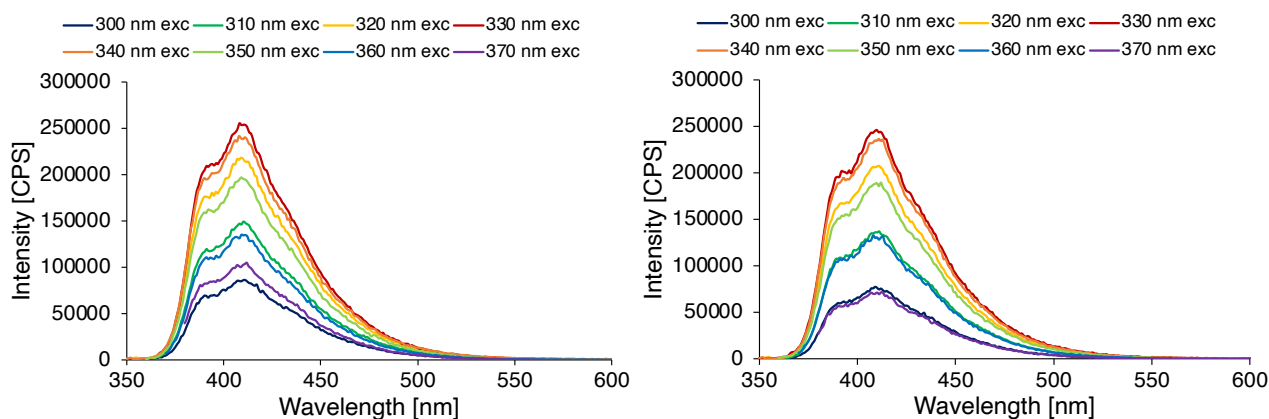


**Figure S15:** Crystal packing structure along a, b, and c axes, respectively. Hydrogen atoms are omitted for clarity.

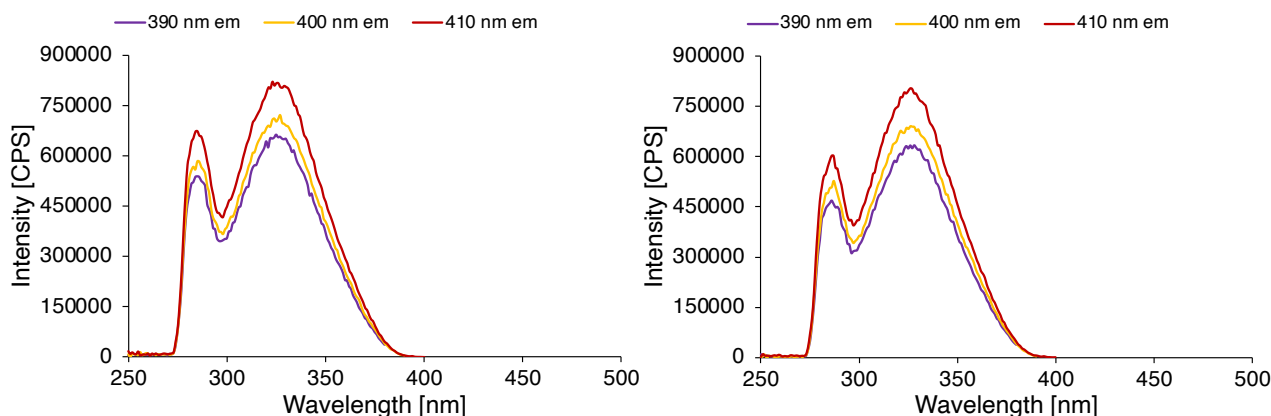
#### 4. Emission spectra of 2AZ-8CPP



**Figure S16:** UV-vis absorption spectra of the sample for emission measurement under nitrogen (left) and air (right) atmosphere (black: before, red: after, benzene solution: 1.9  $\mu$ M).

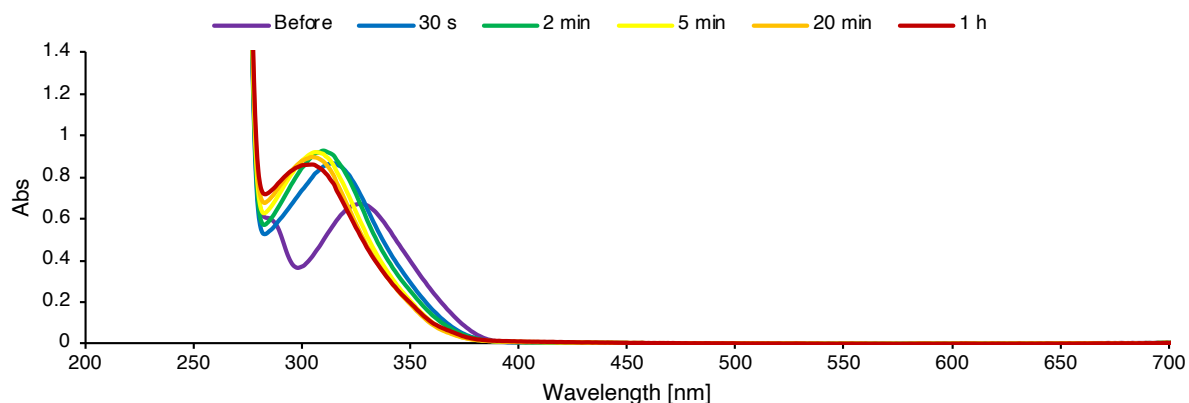


**Figure S17:** Emission spectra of **2AZ-8CPP** under nitrogen (left) and air (right) atmosphere (exc = excitation wavelength). Emission quantum yield  $\Phi_f = 0.39 \pm 0.01$  (under nitrogen) and  $0.34 \pm 0.01$  (under air) calculated by comparing with the emission intensity of 9,10-diphenylanthracene ( $\Phi_f = 0.91$ ). Emission lifetime  $\tau = 1.2 \pm 0.0$  ns (under nitrogen) and  $1.1 \pm 0.0$  ns (under air).

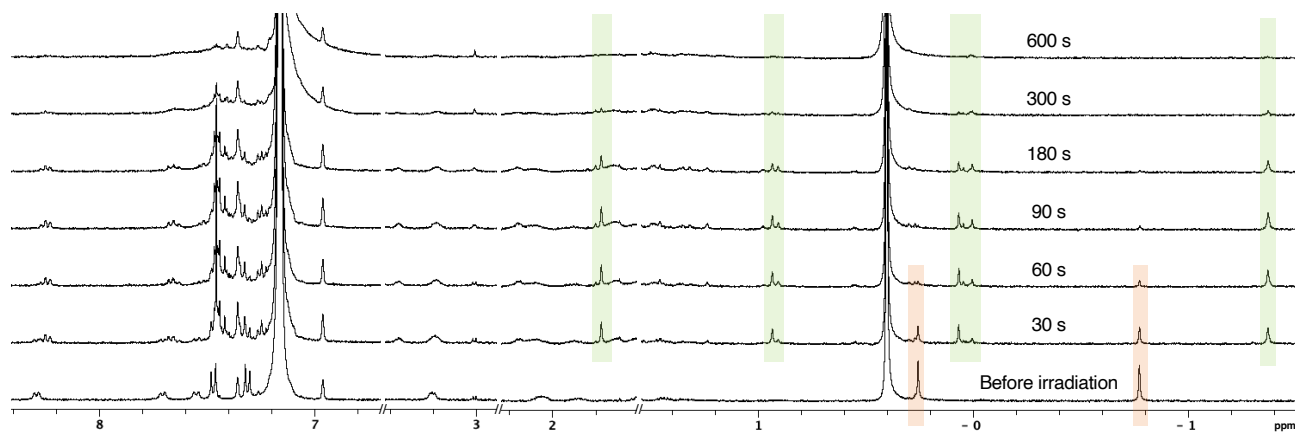


**Figure S18:** Excitation spectra of **2AZ-8CPP** under nitrogen (left) and air (right) atmosphere (em = monitoring emission wavelength).

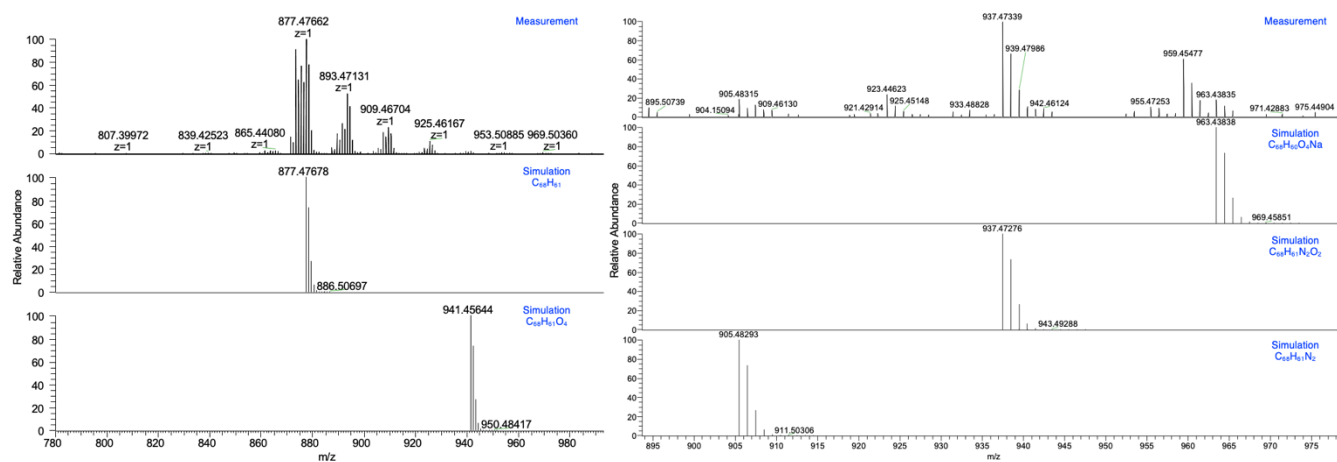
## 5. In-situ photolysis of 2AZ-8CPP



**Figure S19:** Photoreaction of **2AZ-8CPP** with 365 nm LED light monitored by UV-vis absorption spectroscopy in degassed benzene.



**Figure S20:**  $^1\text{H}$  NMR analysis (400 MHz) of the photolysis of **2AZ-8CPP** (0.9 mM) by 365 nm LED light in degassed  $\text{C}_6\text{D}_6$  solution.

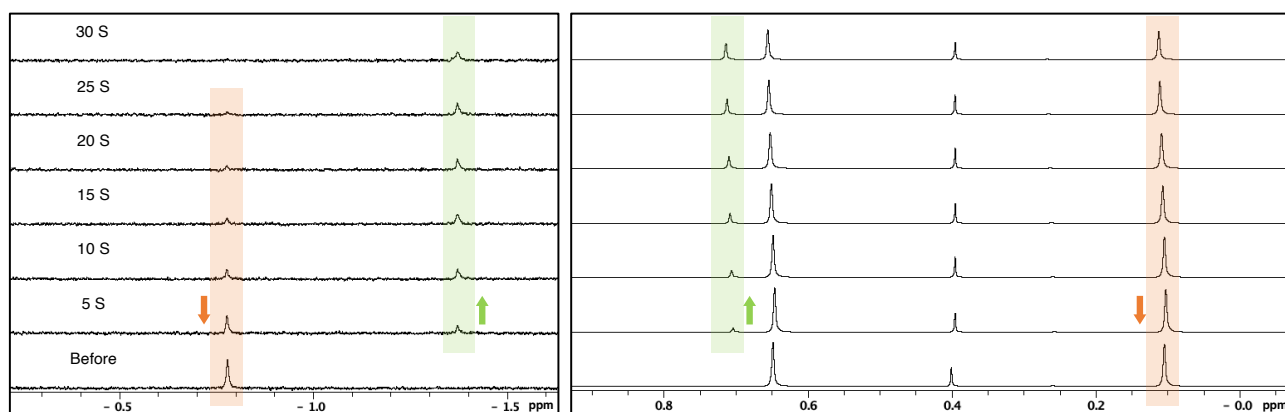


**Figure S21:** HRMS data (APCI and *p*-ESI, MeOH sol) and simulation of products (left: S19, right: S20).

Calcd for  $\text{C}_{68}\text{H}_{61}\text{N}_2$   $[\text{M} + \text{H}]^+$ , 877.47678, found 877.47662 /  $\text{C}_{68}\text{H}_{61}\text{O}_4$   $[\text{M} + \text{H}]^+$ , 941.45644, found 941.45697.

Calcd for  $\text{C}_{68}\text{H}_{60}\text{O}_4\text{Na}$   $[\text{M} + \text{Na}]^+$ , 963.43838, found 963.43835 /  $\text{C}_{68}\text{H}_{61}\text{N}_2\text{O}_2$   $[\text{M} + \text{H}]^+$ , 937.47276, found 937.47339 /  $\text{C}_{68}\text{H}_{61}\text{N}_2$   $[\text{M} + \text{H}]^+$ , 905.48293, found 905.48315.

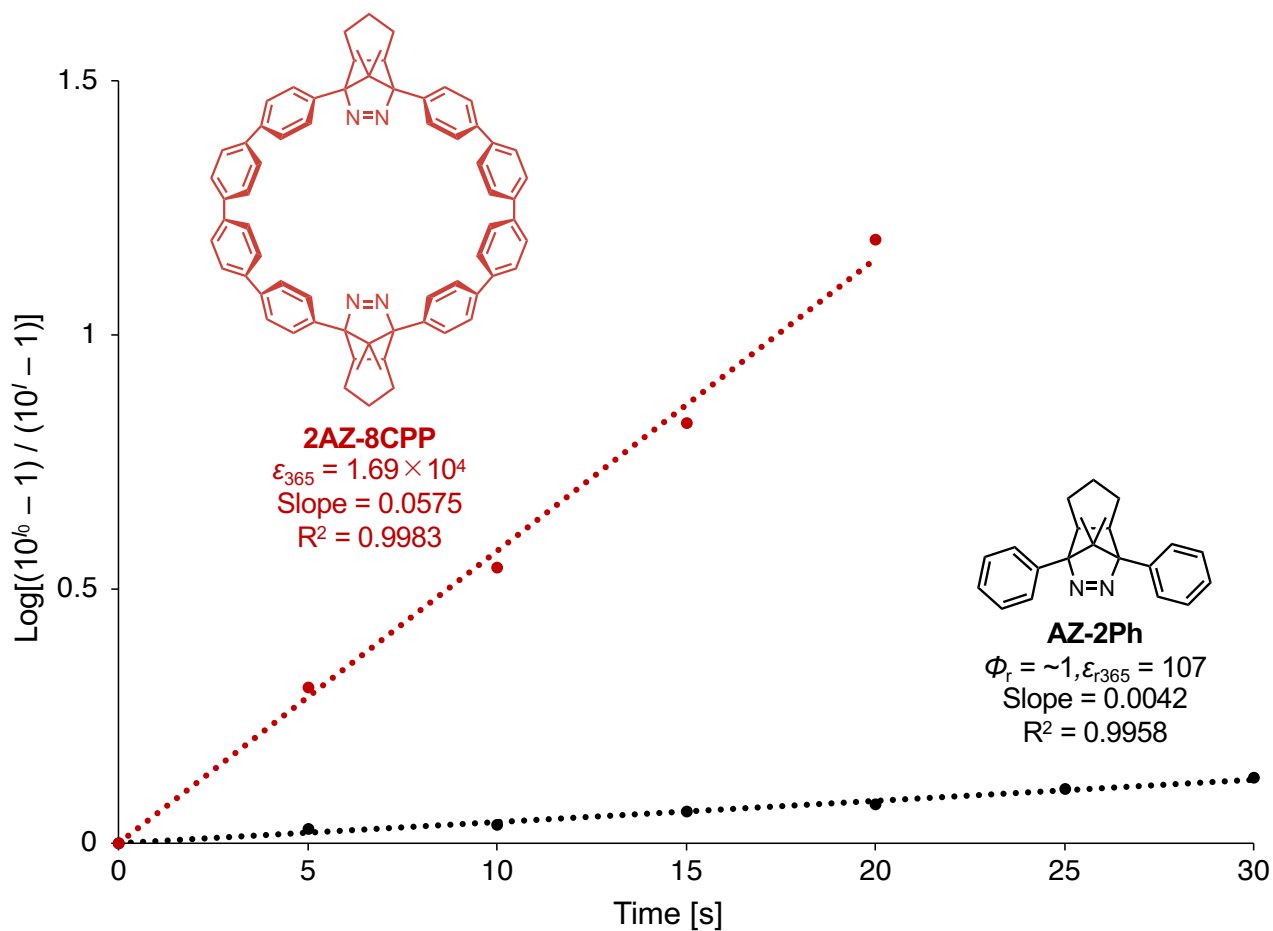
## 6. Denitrogenation quantum yield of 2AZ-8CPP (First step)



**Figure S22:** In-situ  $^1\text{H}$  NMR analysis of determination of first denitrogenation quantum yield of **2AZ-8CPP** (left) and reference of denitrogenation quantum yield of **AZ-2Ph** in degassed  $\text{C}_6\text{D}_6$  (right) with 365 nm LED light.

| Time<br>/ s | Integral ratio                                |                        | <b>2AZ-8CPP</b>           |                            | $\text{Log}[(10^0 - 1)/(10^t - 1)]$ |
|-------------|-----------------------------------------------|------------------------|---------------------------|----------------------------|-------------------------------------|
|             | 0.38 – 0.41 ( $\text{H}_2\text{O}$ )<br>/ ppm | –0.76 – –0.80<br>/ ppm | Ratio<br>/ $\times 100\%$ | Con : I<br>/ $\mu\text{M}$ |                                     |
| 0           | 1                                             | 0.0369                 | 1                         | 113 ( $I_0$ )              | 0                                   |
| 5           | 1                                             | 0.0182                 | 0.493                     | 55.7                       | 0.307                               |
| 10          | 1                                             | 0.0106                 | 0.287                     | 32.5                       | 0.542                               |
| 15          | 1                                             | 0.0055                 | 0.149                     | 16.8                       | 0.827                               |
| 20          | 1                                             | 0.0024                 | 0.065                     | 7.35                       | 1.187                               |
| 25          | 1                                             | 0.0019                 | 0.051                     | 5.82                       | 1.288                               |
| 30          | 1                                             | 0.0012                 | 0.031                     | 3.52                       | 1.506                               |

| Time<br>/ s | Integral ratio                                |                      | <b>AZ-2Ph</b>             |                 | $\text{Log}[(10^0 - 1)/(10^t - 1)]$ |
|-------------|-----------------------------------------------|----------------------|---------------------------|-----------------|-------------------------------------|
|             | 0.38 – 0.41 ( $\text{H}_2\text{O}$ )<br>/ ppm | 0.09 – 0.13<br>/ ppm | Ratio<br>/ $\times 100\%$ | Con : I<br>/ mM |                                     |
| 0           | 1                                             | 3.9396               | 1                         | 19 ( $I_0$ )    | 0                                   |
| 5           | 1                                             | 3.6925               | 0.937                     | 17.8            | 0.029                               |
| 10          | 1                                             | 3.6256               | 0.920                     | 17.5            | 0.037                               |
| 15          | 1                                             | 3.4251               | 0.869                     | 16.5            | 0.062                               |
| 20          | 1                                             | 3.3213               | 0.843                     | 16.0            | 0.076                               |
| 25          | 1                                             | 3.0920               | 0.785                     | 14.9            | 0.107                               |
| 30          | 1                                             | 2.9456               | 0.748                     | 14.2            | 0.129                               |

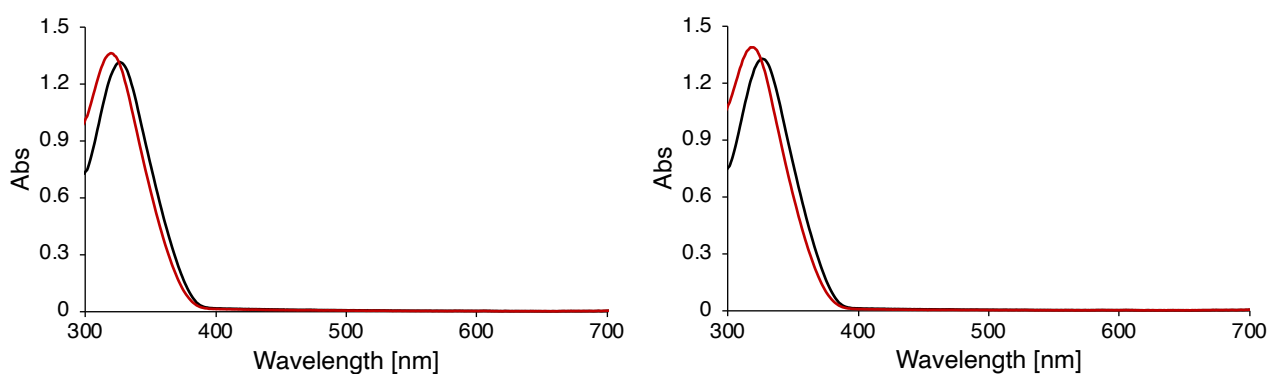


**Figure S23:** Time profiles of decomposition of azoalkane **AZ-2Ph** (black) and **2AZ-8CPP** (red) as monitored by  $^1\text{H}$  NMR spectroscopy using 365 nm LED light.

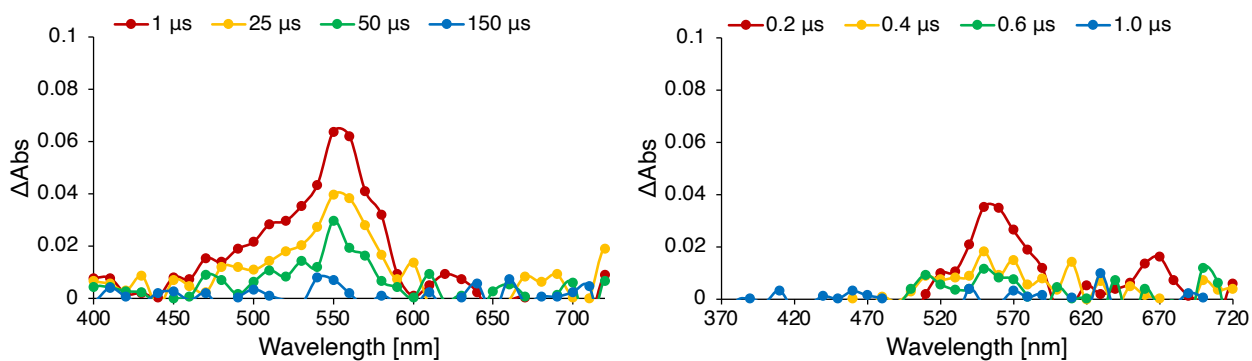
First Quantum yield of denitrogenation  $\Phi_{2\text{AZ-8CPP}} = \text{Slope}_{2\text{AZ-8CPP}} / \text{Slope}_{\text{AZ-2Ph}} \times \epsilon_{r365} / \epsilon_{365} \times \Phi_r = \mathbf{0.086}$



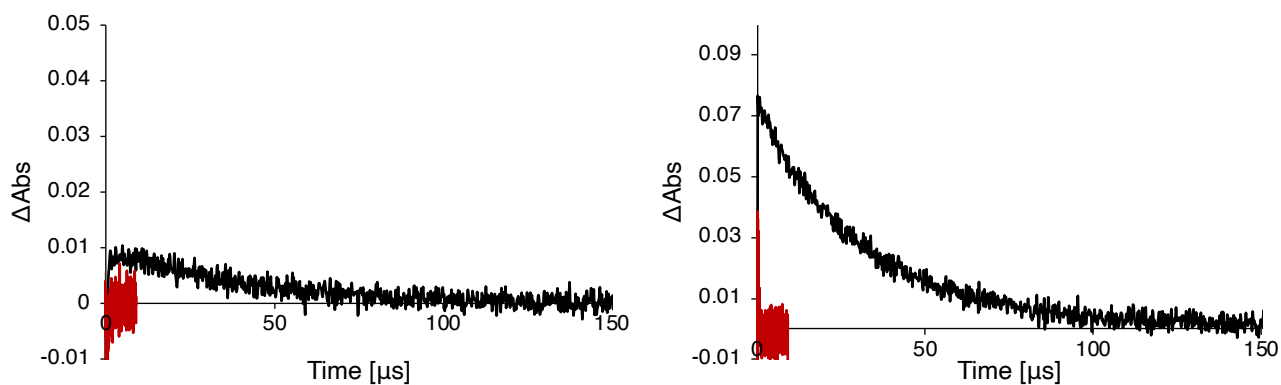
## 7. Sub-microsecond Laser Flash photolysis (LFP) measurement of 2AZ-8CPP



**Figure S24:** UV-vis absorption spectra of sub-microsecond LFP measurement under nitrogen (left) and air (right) atmosphere (black: before, red: after, benzene solution: 38  $\mu\text{M}$ ).



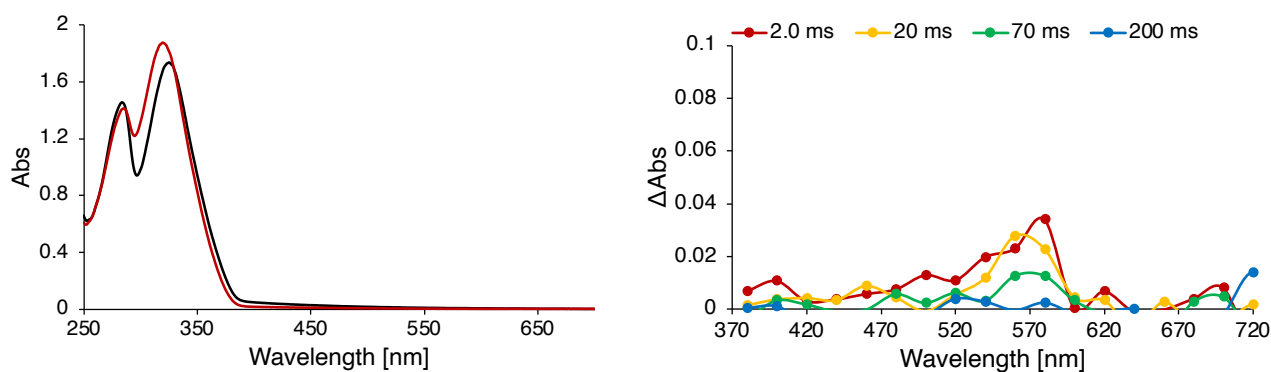
**Figure S25:** Transient absorption spectra of sub-microsecond LFP measurement at 295 K under nitrogen (left) and air (right) atmosphere with 355 nm laser (4.0 mJ) in benzene.



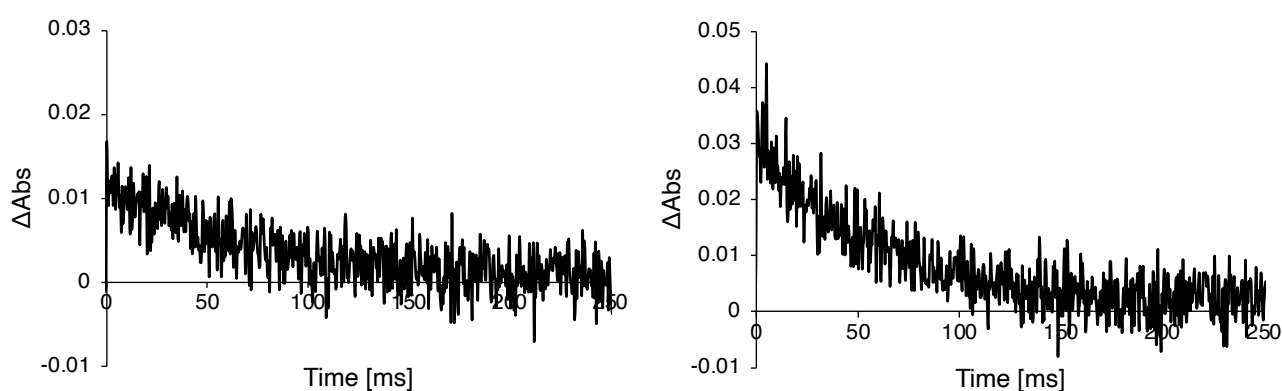
**Figure S26:** Time profile at 400 nm (left) and 560 nm (right) in nitrogen (black) and air (red) atmosphere.

| Lifetime       | At 400 nm                    | At 560 nm                    |
|----------------|------------------------------|------------------------------|
| N <sub>2</sub> | 40.7 $\pm$ 0.2 $\mu\text{s}$ | 30.7 $\pm$ 0.1 $\mu\text{s}$ |
| Air            | n.d                          | 0.3 $\pm$ 0.0 $\mu\text{s}$  |

n.d: Not determined.

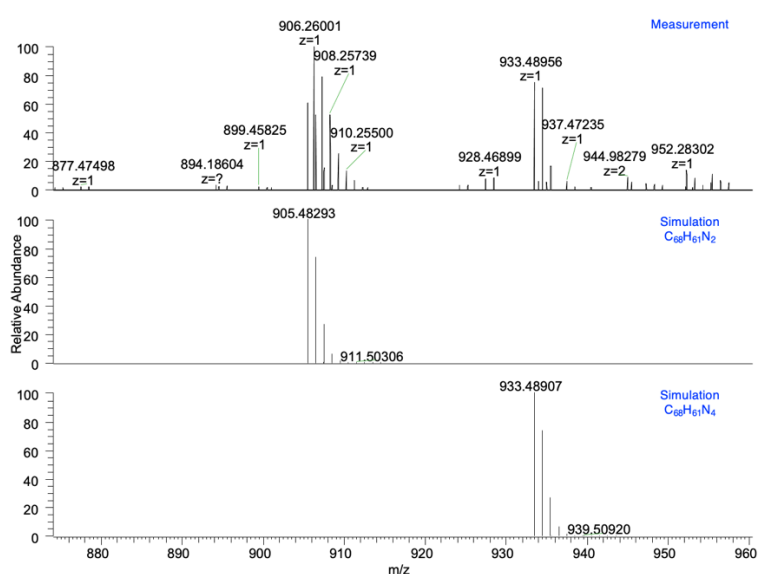


**Figure S27:** UV-vis absorption spectra of sub-microsecond LFP measurement under argon atmosphere (black: before, red: after, 2-MTHF solution: 56  $\mu\text{M}$ ) and transient absorption spectra of sub-microsecond LFP measurement at 124 K under argon atmosphere with 355 nm laser (4.3 mJ) in 2-MTHF.



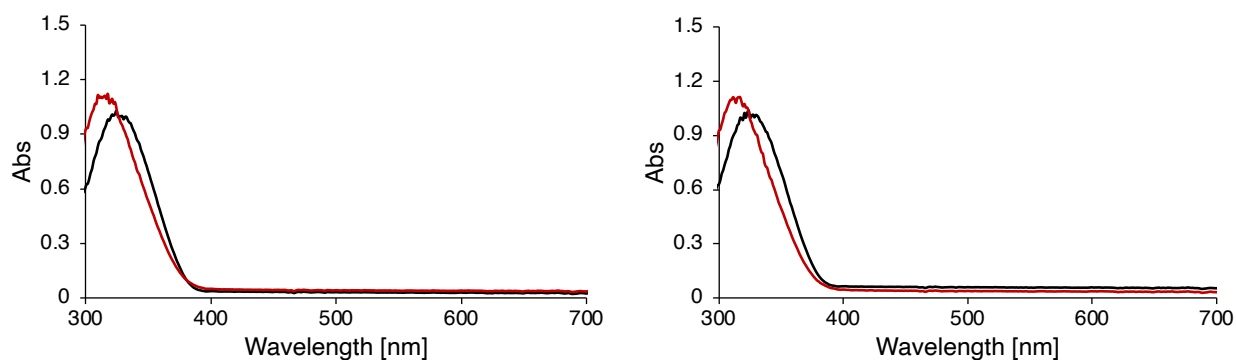
**Figure S28:** Time profile at 400 nm (left) and 560 nm (right) at 124 K in argon atmosphere.

| Lifetime | At 400 nm         | At 560 nm         |
|----------|-------------------|-------------------|
| Ar       | $66.5 \pm 0.5$ ms | $57.7 \pm 0.4$ ms |

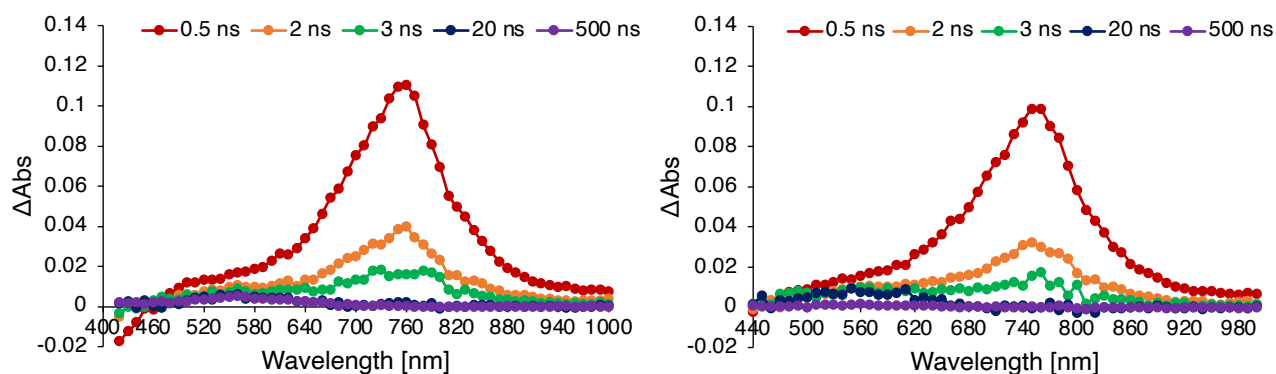


**Figure S29:** HRMS data (*p*-ESI, MeOH sol) and simulation of sample after LFP measurement under  $\text{N}_2$ . Calcd for  $\text{C}_{68}\text{H}_{61}\text{N}_2$   $[\text{M} + \text{H}]^+$ , 905.48293, found 905.48315 /  $\text{C}_{68}\text{H}_{61}\text{N}_4$   $[\text{M} + \text{H}]^+$ , 933.48907, found 933.48956.

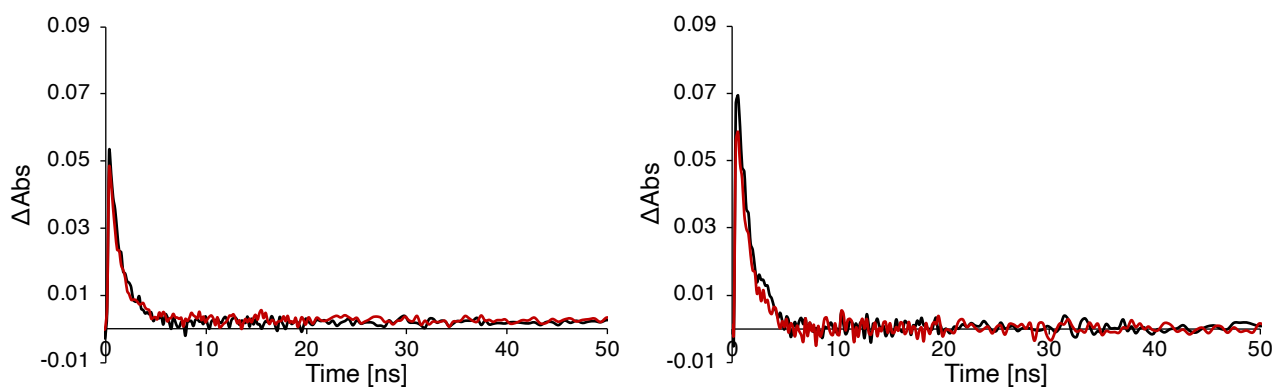
## 8. Sub-nanosecond Laser Flash photolysis (LFP) measurement of 2AZ-8CPP



**Figure S30:** UV-vis absorption spectra of sub-nanosecond LFP measurement under nitrogen (left) and air (right) atmosphere (black: before, red: after, benzene solution: 0.2 mM).



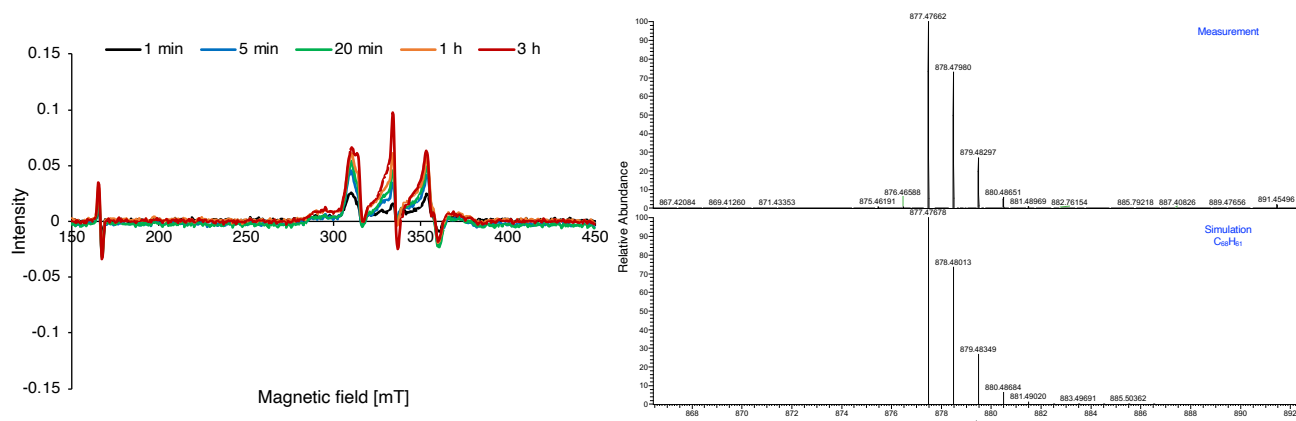
**Figure S31:** Transient absorption spectra of sub-nanosecond LFP measurement at 295 K under argon (left) and air (right) atmosphere with 355 nm laser (10  $\mu$ J) in benzene.



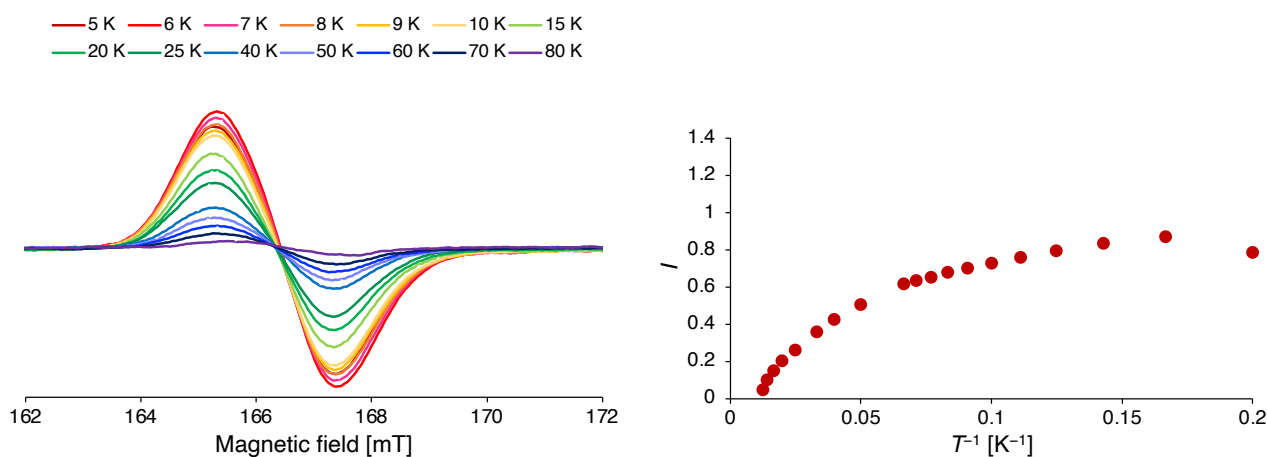
**Figure S32:** Time profile at 660 nm (left) and 800 nm (right) in argon (black) and air (red) atmosphere.

| Lifetime | At 660 nm          | At 800 nm               |
|----------|--------------------|-------------------------|
| Ar       | $1.28 \pm 0.01$ ns | $1.34 \pm 0.02$ ns      |
| Air      | $1.21 \pm 0.01$ ns | $1.24 \pm 0.01$ $\mu$ s |

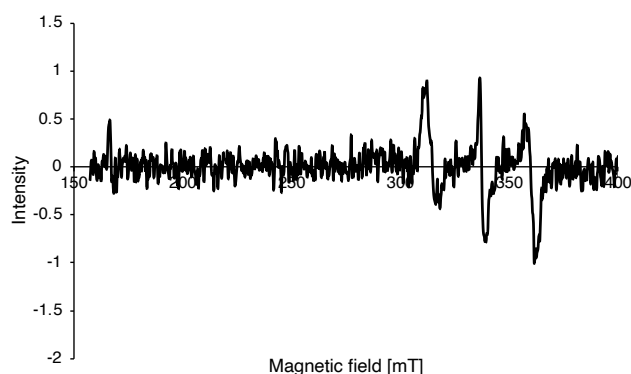
## 9. Electron Paramagnetic Resonance (EPR) measurement of 2AZ-8CPP



**Figure S33:** EPR spectra of the photolysis of **2AZ-8CPP** (0.1 mM) using 365 nm LED light in 2-MTHF matrix in a sealed quartz tube at 5 K under vacuum conditions and HR-mass spectra (p-ESI, MeOH sol) with simulation of sample after EPR measurement. Calcd for  $C_{68}H_{61} [M + H]^+$ , 877.47678, found 877.47662.



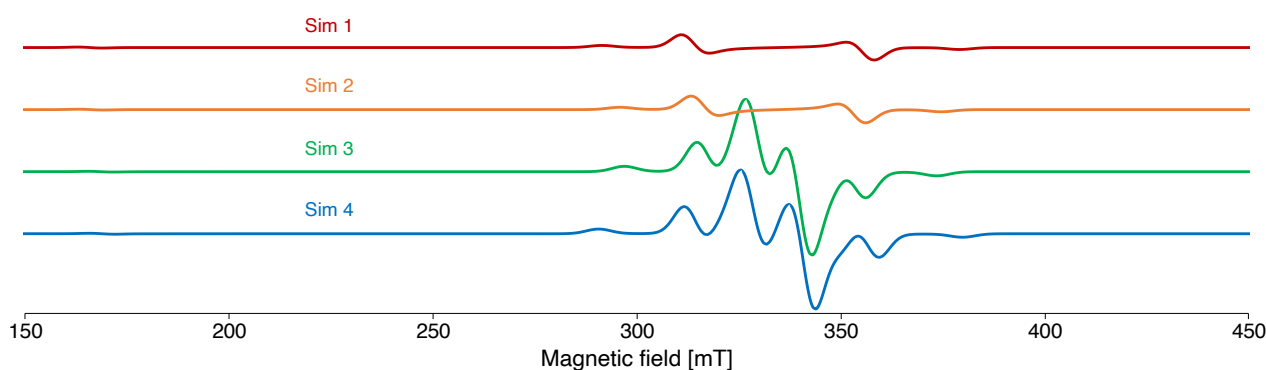
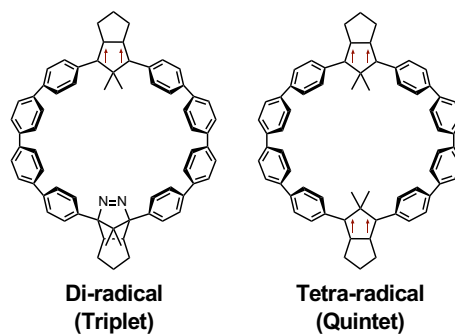
**Figure S34:** Various temperature EPR measurement for monitored around half field regions after photolysis for 2 h at 5 K using high pressure Hg light in 2-MTHF matrix and  $I-T^{-1}$  plot with least-squares fit for Bleaney-Bowers analysis (signal intensity  $I$  and temperature  $T$ ).



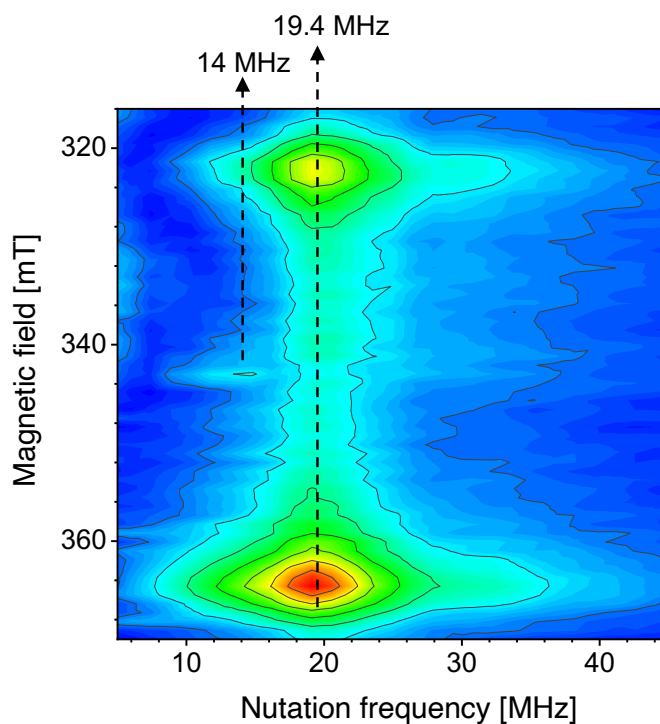
**Figure S35:** EPR spectrum at 20 K after photolysis of **2AZ-8CPP** using UV light in toluene matrix (0.1 mM) in a sealed quartz tube under vacuum condition.

**Table S2:** Computed zfs parameters ( $D$  and  $E$ ) for di-radical and tetra-radical.

| Sim | Spin state                    | $ D/hc $ ( $ E/hc $ )<br>[cm <sup>-1</sup> ] |
|-----|-------------------------------|----------------------------------------------|
| 1   | Di-radical (Triplet, syn)     | 0.041 (0.007)                                |
| 2   | Di-radical (Triplet, anti)    | 0.037 (0.002)                                |
| 3   | Tetra-radical (Quintet, syn)  | 0.012 (< 0.001)                              |
| 4   | Tetra-radical (Quintet, anti) | 0.014 (< 0.001)                              |



**Figure S36:** Simulated EPR spectra using the zfs parameters  $D$  and  $E$  (Table S2) were computed at the (RO)BP/EPR-II. The calculated geometries were optimized at g16/(U)CAM-B3LYP/6-31G(d) level of theory.

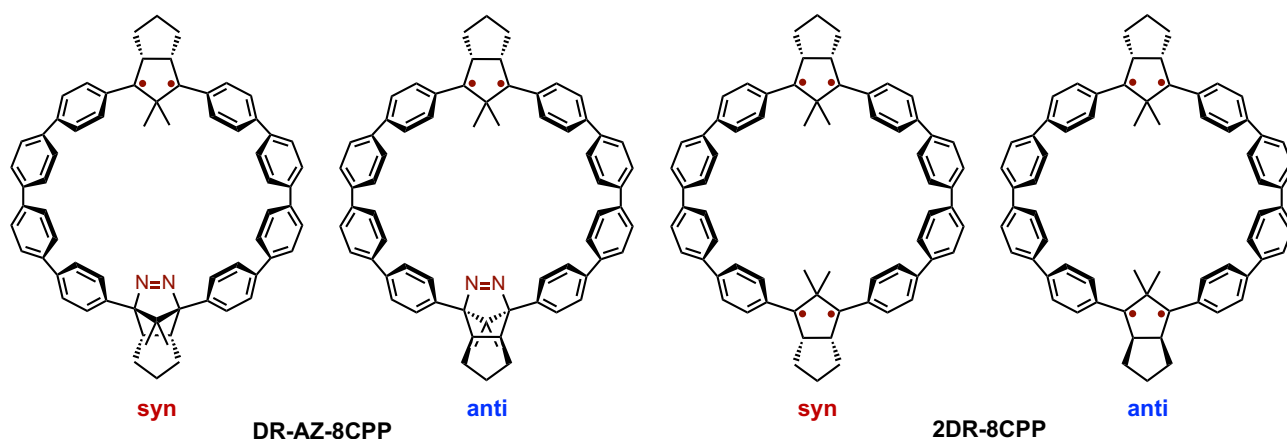


**Figure S37:** 2D-pulse EPR signal at 20 K after photolysis for 10 h using UV light in toluene matrix (0.1 mM) under vacuum condition. Change the width of the microwave pulse.

## 10. Computational details I

Quantum chemical computations in gas phase have been performed with the Gaussian 16 (Revision B.01) suite of programs. Charge, spin multiplicity, number of imaginary frequencies, energy (in Hartree) and Cartesian coordinates (in Å) of computed geometries are listed in following. Zero-field splitting (ZFS) parameters ( $D$  tensor and  $E/D$  ratio) calculation were performed with ORCA 4.2.1 program at restricted open-shell (RO) BP/EPR-II level of theory. The calculated geometries were optimized at (U)CAM-B3LYP/6-31G(d) level of theory with Gaussian 16 (Revision B.01) before ZFS parameters calculations. The automatic auxiliary basis sets (AutoAux) were used in ZFS calculations. The  $D$  tensor calculation included the spin-spin components (SS). The DSS part is an expectation value that involves the spin density of the calculated system directly used the canonical orbitals from the spin density. Anisotropy of the induced current density (ACID) analyses were performed with AICD 3.0.4 software package. The NMR with CSGT method were calculated at UB3LYP/6-31G(d) level of theory. POV-Ray 3.7 on Windows was used for visualization of ACID maps.

**Table S3:** Energy gaps in **DR-AZ-8CPP** and **2DR-8CPP** calculated at the R(U)B3LYP/6-31G(d) or R(U)CAM-B3LYP/6-31G(d) level of theory in gas phase.



g16/DFT/R(U)B3LYP/6-31G(d)

| DR-AZ-8CPP | Spin    | $\langle S^2 \rangle$ | $\Delta E_{S-T}$<br>[kcal mol <sup>-1</sup> ] |
|------------|---------|-----------------------|-----------------------------------------------|
| syn        | Singlet | n.d                   | n.d                                           |
|            | Triplet | 2.0678                | 0                                             |

| DR-AZ-8CPP | Spin    | $\langle S^2 \rangle$ | $\Delta E_{S-T}$<br>[kcal mol <sup>-1</sup> ] |
|------------|---------|-----------------------|-----------------------------------------------|
| anti       | Singlet | 1.0643                | +0.12                                         |
|            | Triplet | 2.0678                | 0                                             |

n.d: Open singlet state couldn't be optimized.

g16/DFT/R(U)CAM-B3LYP/6-31G(d)

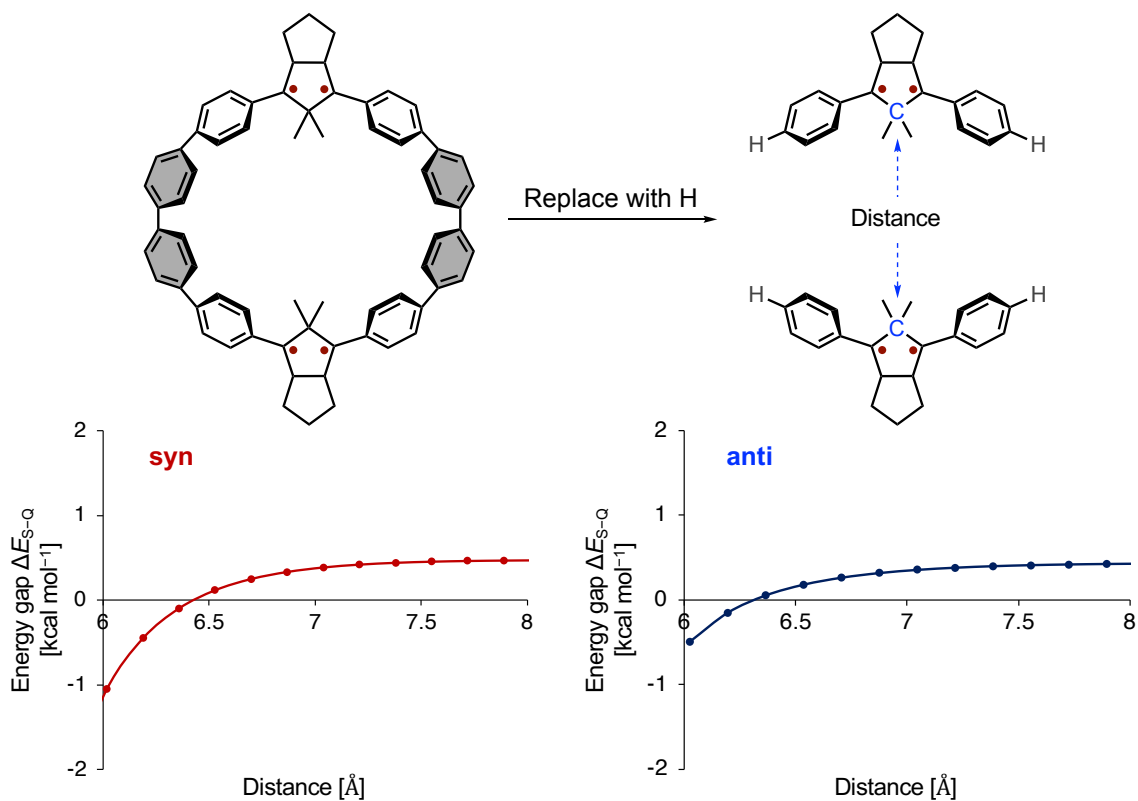
| DR-AZ-8CPP | Spin    | $\langle S^2 \rangle$ | $\Delta E_{S-T}$<br>[kcal mol <sup>-1</sup> ] |
|------------|---------|-----------------------|-----------------------------------------------|
| syn        | Singlet | 1.0144                | -1.54                                         |
|            | Triplet | 2.1301                | 0                                             |

| DR-AZ-8CPP | Spin    | $\langle S^2 \rangle$ | $\Delta E_{S-T}$<br>[kcal mol <sup>-1</sup> ] |
|------------|---------|-----------------------|-----------------------------------------------|
| anti       | Singlet | 1.1309                | +0.19                                         |
|            | Triplet | 2.1345                | 0                                             |

|          |           |                       |                                                 | g16/DFT/R(U)B3LYP/6-31G(d) |           |                       |                                                 |
|----------|-----------|-----------------------|-------------------------------------------------|----------------------------|-----------|-----------------------|-------------------------------------------------|
| 2DR-8CPP | Spin      | $\langle S^2 \rangle$ | $\Delta E_{S,T-Q}$<br>[kcal mol <sup>-1</sup> ] | 2DR-8CPP                   | Spin      | $\langle S^2 \rangle$ | $\Delta E_{S,T-Q}$<br>[kcal mol <sup>-1</sup> ] |
| syn      | Singlet   | 2.1356                | -0.57                                           | anti                       | Singlet   | 2.1213                | +0.13                                           |
|          | Triplet   | 3.1305                | -0.67                                           |                            | Triplet   | 2.0674                | +13.5                                           |
|          | Quintet   | 6.1318                | 0                                               |                            | Quintet   | 6.1235                | 0                                               |
|          | R-singlet | 0                     | +25.0                                           |                            | R-singlet | 0                     | +26.5                                           |

|          |           |                       |                                                 | g16/DFT/R(U)CAM-B3LYP/6-31G(d) |           |                       |                                                 |
|----------|-----------|-----------------------|-------------------------------------------------|--------------------------------|-----------|-----------------------|-------------------------------------------------|
| 2DR-8CPP | Spin      | $\langle S^2 \rangle$ | $\Delta E_{S,T-Q}$<br>[kcal mol <sup>-1</sup> ] | 2DR-8CPP                       | Spin      | $\langle S^2 \rangle$ | $\Delta E_{S,T-Q}$<br>[kcal mol <sup>-1</sup> ] |
| syn      | Singlet   | 2.2866                | -0.05                                           | anti                           | Singlet   | 2.2385                | +0.13                                           |
|          | Triplet   | 3.2724                | -0.04                                           |                                | Triplet   | 3.2350                | +0.06                                           |
|          | Quintet   | 6.2588                | 0                                               |                                | Quintet   | 6.2319                | 0                                               |
|          | R-singlet | 0                     | +57.2                                           |                                | R-singlet | 0                     | +59.2                                           |



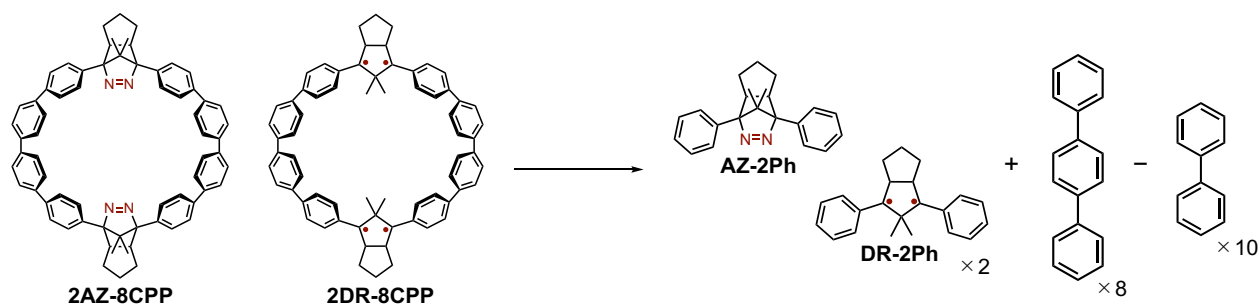
**Figure S38:** Scan calculation between two diradical units using UCAM-B3LYP/6-31G(d) level of theory ( $\Delta E_{S-Q}$  = Singlet energy  $E_S$  – Quintet energy  $E_Q$ ).

**Table S4:** Results of scan calculation between two diradical units (step size 0.17 Å).

| <b>syn</b>   |                                            | <b>anti</b>  |                                            |
|--------------|--------------------------------------------|--------------|--------------------------------------------|
| Distance [Å] | $\Delta E_{S-Q}$ [kcal mol <sup>-1</sup> ] | Distance [Å] | $\Delta E_{S-Q}$ [kcal mol <sup>-1</sup> ] |
| 10.78        | 0.474                                      | 10.96        | 0.432                                      |
| 10.61        | 0.474                                      | 10.79        | 0.432                                      |
| 10.44        | 0.474                                      | 10.62        | 0.432                                      |
| 10.27        | 0.474                                      | 10.45        | 0.432                                      |
| 10.10        | 0.474                                      | 10.28        | 0.432                                      |
| 9.93         | 0.474                                      | 10.11        | 0.432                                      |
| 9.76         | 0.474                                      | 9.94         | 0.432                                      |
| 9.59         | 0.474                                      | 9.77         | 0.432                                      |
| 9.42         | 0.474                                      | 9.60         | 0.432                                      |
| 9.25         | 0.474                                      | 9.43         | 0.432                                      |
| 9.08         | 0.474                                      | 9.26         | 0.432                                      |
| 8.91         | 0.474                                      | 9.09         | 0.432                                      |
| 8.74         | 0.474                                      | 8.92         | 0.432                                      |
| 8.57         | 0.474                                      | 8.75         | 0.432                                      |
| 8.40         | 0.473                                      | 8.58         | 0.432                                      |
| 8.23         | 0.473                                      | 8.41         | 0.431                                      |
| 8.06         | 0.471                                      | 8.24         | 0.430                                      |
| 7.89         | 0.468                                      | 8.07         | 0.427                                      |
| 7.72         | 0.463                                      | 7.90         | 0.424                                      |
| 7.55         | 0.455                                      | 7.73         | 0.418                                      |
| 7.38         | 0.442                                      | 7.56         | 0.410                                      |
| 7.21         | 0.420                                      | 7.39         | 0.398                                      |
| 7.04         | 0.387                                      | 7.22         | 0.380                                      |
| 6.87         | 0.334                                      | 7.05         | 0.354                                      |
| 6.70         | 0.251                                      | 6.88         | 0.317                                      |
| 6.53         | 0.118                                      | 6.71         | 0.262                                      |
| 6.36         | -0.095                                     | 6.54         | 0.180                                      |
| 6.19         | -0.448                                     | 6.37         | 0.053                                      |
| 6.02         | -1.047                                     | 6.20         | -0.150                                     |
| 5.85         | -2.104                                     | 6.03         | -0.491                                     |

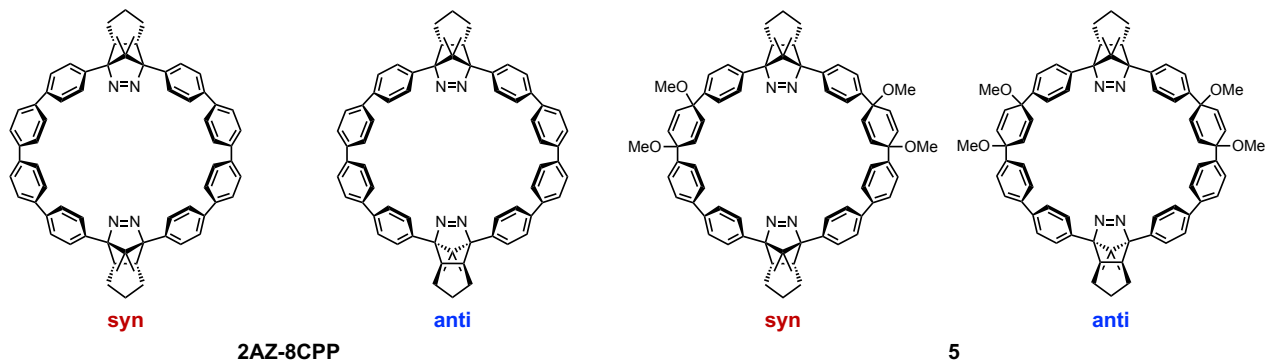


**Table S5:** Strain Energies of **2AZ-8CPP** and **2DR-8CPP** calculated at the R(U)B3LYP/6-31G(d) level of theory.

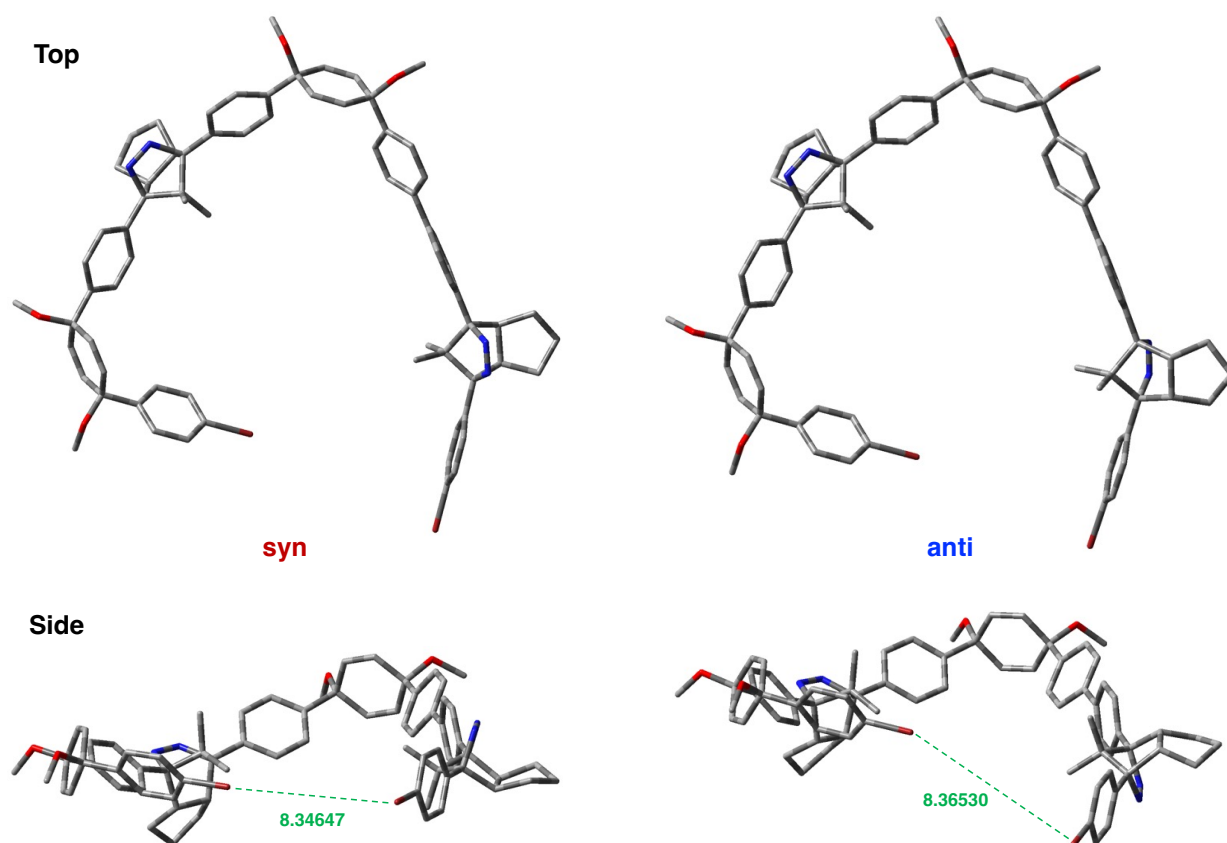


| Compound                        | Strain energy [kcal mol <sup>-1</sup> ] |
|---------------------------------|-----------------------------------------|
| <b>2AZ-8CPP</b> (syn)           | 45.2                                    |
| <b>2AZ-8CPP</b> (anti)          | 42.6                                    |
| <b>2DR-8CPP</b> (syn, Quintet)  | 59.7                                    |
| <b>2DR-8CPP</b> (anti, Quintet) | 58.4                                    |

**Table S6:** Energy difference between syn- and anti-isomers calculated at the RB3LYP/6-31G(d) level of theory.

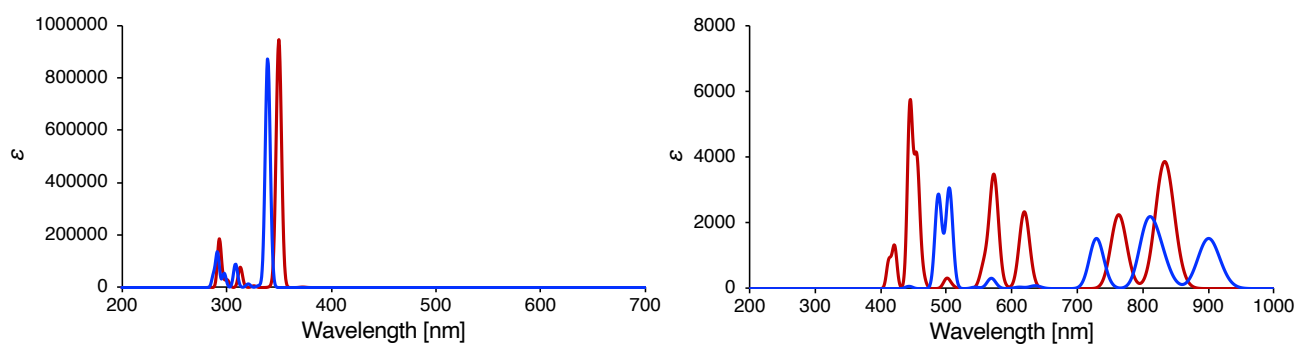


| Compound               | $\Delta E_{\text{syn-anti}}$ [kcal mol <sup>-1</sup> ] |
|------------------------|--------------------------------------------------------|
| <b>2AZ-8CPP</b> (syn)  | +2.60                                                  |
| <b>2AZ-8CPP</b> (anti) | 0                                                      |
| <b>5</b> (syn)         | +0.47                                                  |
| <b>5</b> (anti)        | 0                                                      |

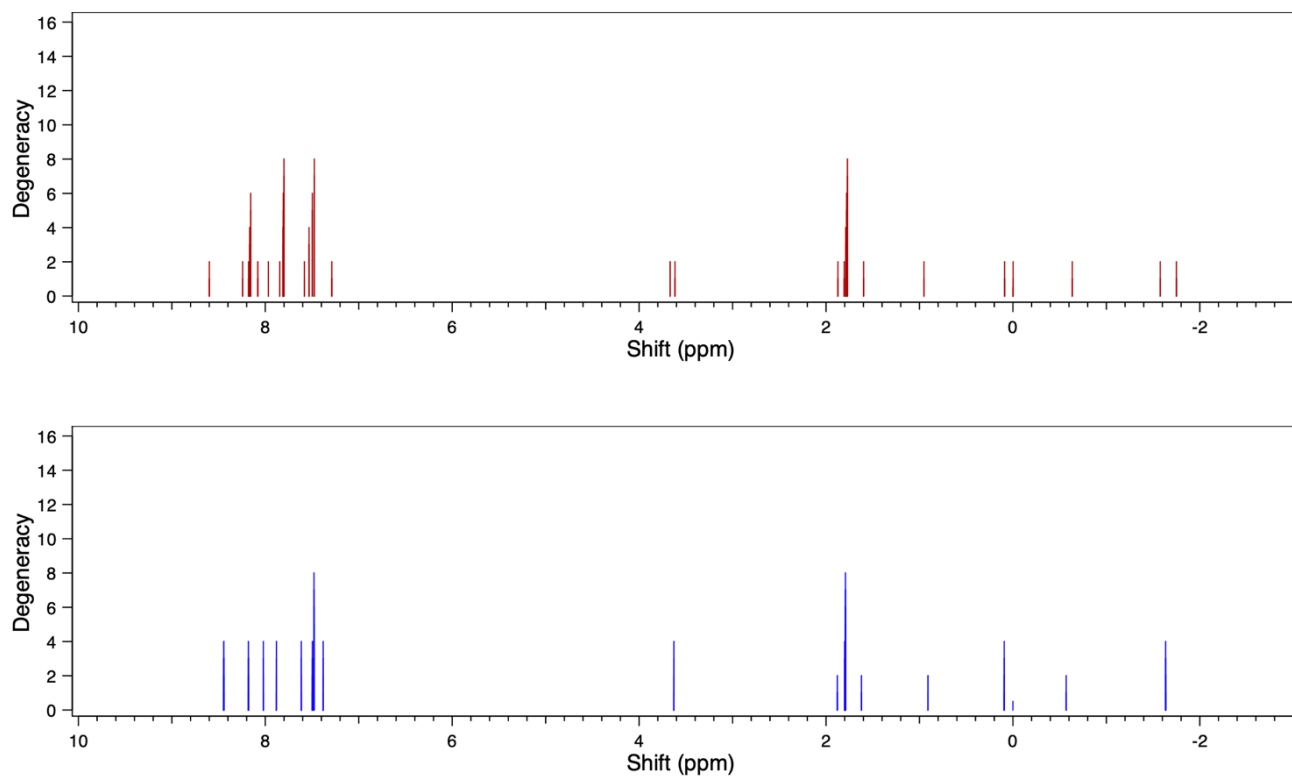


**Figure S39:** Optimized structure of compound **4** (precursor of macrocyclic compound) calculated at the RB3LYP/6-31G(d) level of theory.

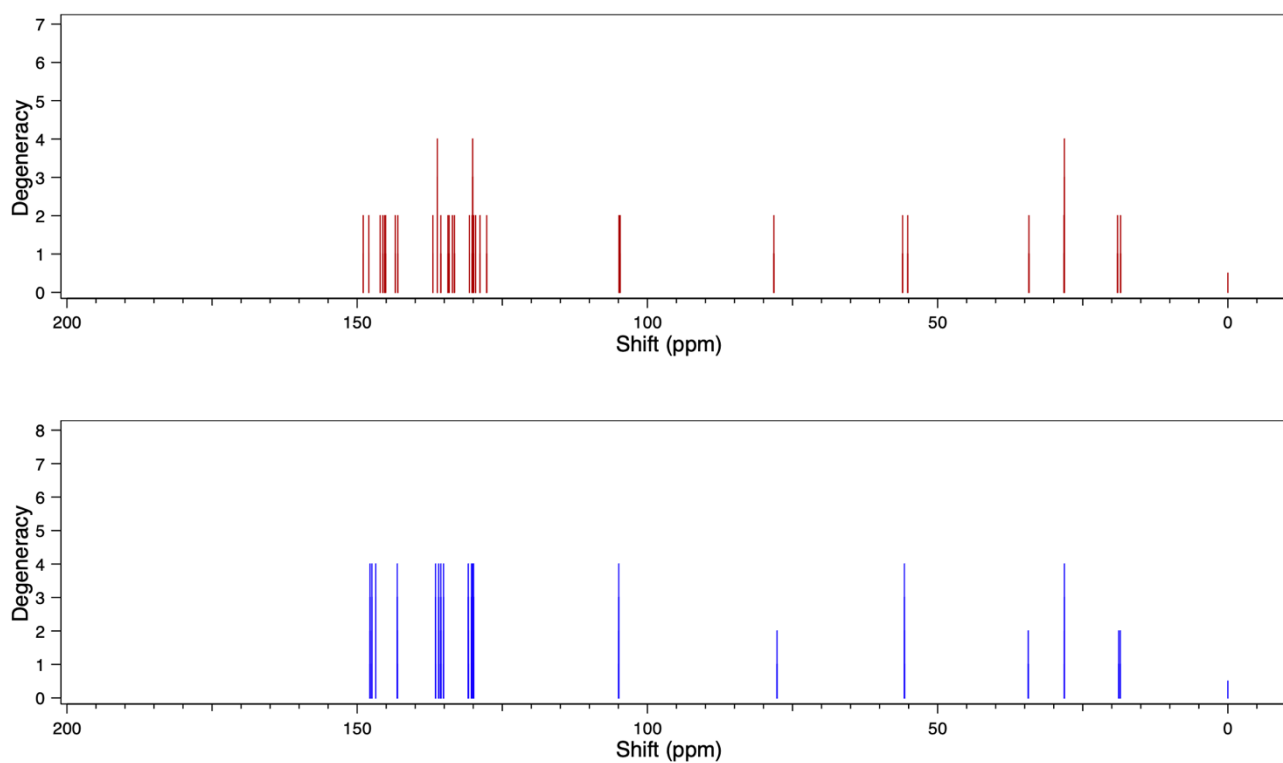
Anti-conformer has more distant and twisted intramolecular reaction points than syn-conformer. Therefore, the activation barrier in intramolecular reactions is expected to be larger than that of the syn-conformer, resulting in differences observed in isomer ratios.



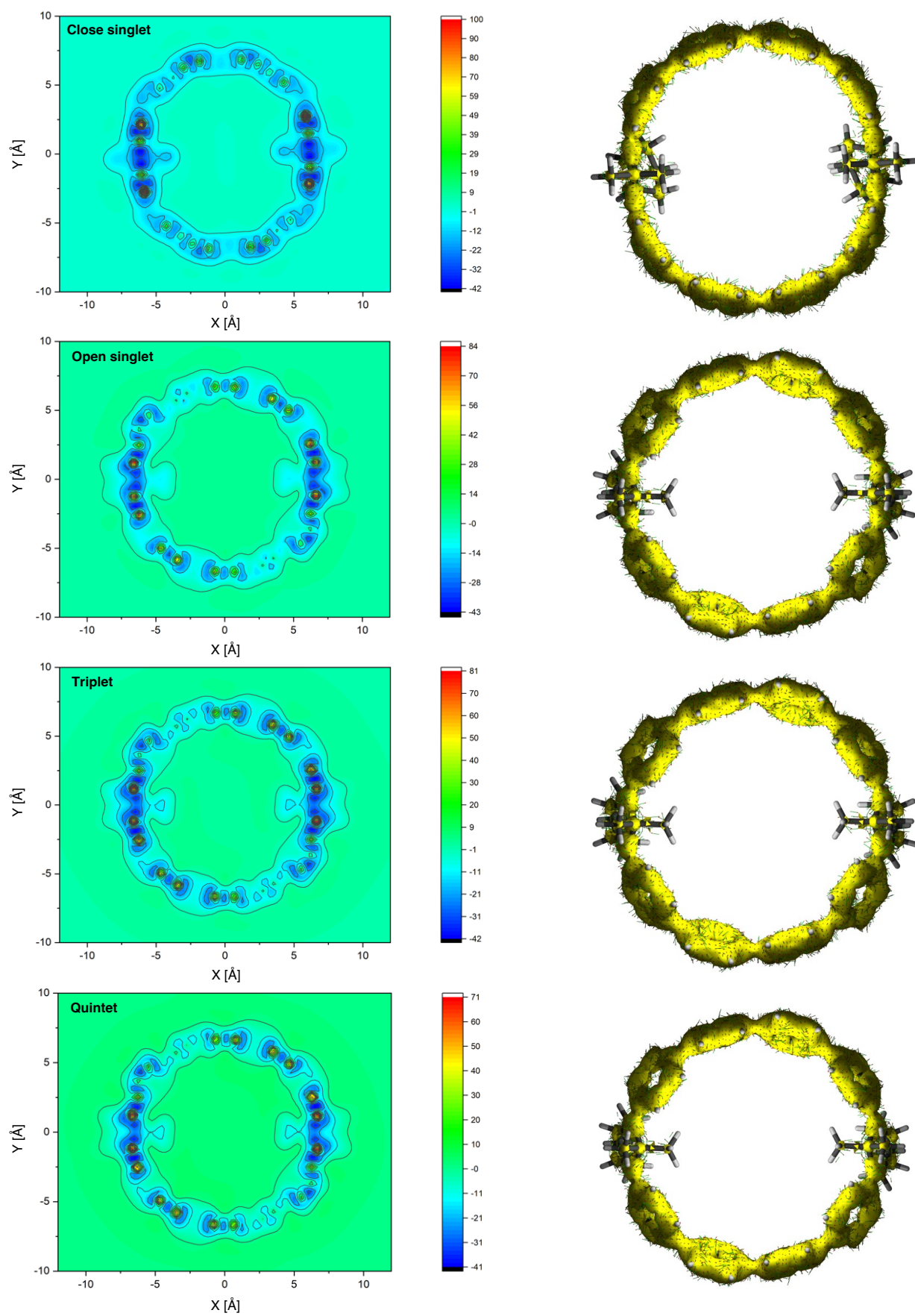
**Figure S40:** Calculated UV-vis absorption spectra of  $S_0$  (left) and  $T_1$  (right) of **2AZ-8CPP** at TD-B3LYP/6-311G+(2d, p) level of theory (red: syn, anti: blue).



**Figure S41:** Calculated  $^1\text{H}$  chemical shifts of **2AZ-8CPP** at the RB3LYP/6-311+G(2d,p) level of theory (red: syn, anti: blue).

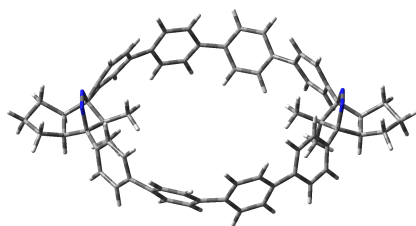


**Figure S42:** Calculated  $^{13}\text{C}$  chemical shifts of **2AZ-8CPP** at the RB3LYP/6-311+G(2d,p) level of theory (red: syn, anti: blue).



**Figure S43:** A grid plot of the NICS(0)<sub>zz</sub> value in the XY plane of 2DR-8CPP (syn) and ACID maps.

### 2AZ-8CPP (syn)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2845.174218 Hartree

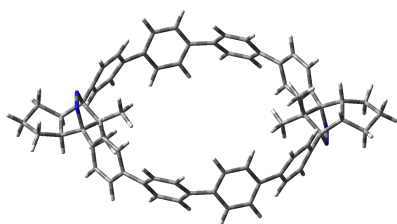
Sum of electronic and thermal Energies = -2845.118057 Hartree

Sum of electronic and enthalpy Energies = -2845.117113 Hartree

Sum of electronic and thermal Free Energies = -2845.117113 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -8.503647             | 0.759708  | 0.546295  | C    | 1.298808              | -5.475381 | 1.436798  |
| C    | -8.495693             | -0.81255  | 0.463364  | C    | 1.595624              | -6.657871 | -0.626625 |
| C    | -7.174396             | -1.104446 | -0.324924 | C    | 2.934116              | -6.271428 | -0.654497 |
| C    | -6.200061             | -0.002503 | 0.247465  | C    | 2.635228              | -5.099330 | 1.415298  |
| C    | -7.185254             | 1.143835  | -0.20605  | H    | 0.669921              | -5.093630 | 2.234308  |
| C    | -4.835552             | 0.053205  | -0.459564 | H    | 1.215771              | -7.285992 | -1.427741 |
| H    | -4.248563             | -0.838023 | -0.217706 | H    | 3.562422              | -6.600315 | -1.478290 |
| H    | -4.274796             | 0.928553  | -0.117292 | H    | 3.005706              | -4.441190 | 2.195598  |
| H    | -4.917782             | 0.115342  | -1.546991 | C    | 4.730842              | -4.657783 | 0.155180  |
| C    | -5.926597             | -0.094903 | 1.754374  | C    | 6.595927              | -2.529809 | -0.121730 |
| H    | -5.356257             | -1.005802 | 1.967387  | C    | 5.055162              | -4.133909 | -1.107912 |
| H    | -6.822266             | -0.111542 | 2.378804  | C    | 5.494986              | -4.214654 | 1.249308  |
| H    | -5.314886             | 0.756     | 2.072339  | C    | 6.410430              | -3.172016 | 1.112942  |
| N    | -7.45253              | -0.530967 | -1.689556 | C    | 5.966994              | -3.090392 | -1.243736 |
| N    | -7.45945              | 0.713861  | -1.62262  | H    | 4.501274              | -4.464932 | -1.981934 |
| C    | -6.595942             | 2.5298    | -0.121694 | H    | 5.329419              | -4.649909 | 2.231385  |
| C    | -4.73084              | 4.657769  | 0.155213  | H    | 6.931431              | -2.813861 | 1.996735  |
| C    | -6.410397             | 3.171975  | 1.11299   | H    | 6.130444              | -2.642945 | -2.218578 |
| C    | -5.967047             | 3.090414  | -1.243704 | C    | 4.721320              | 4.650408  | -0.258217 |
| C    | -5.055206             | 4.133926  | -1.10788  | C    | 6.575384              | 2.486162  | -0.381187 |
| C    | -5.494948             | 4.21461   | 1.249355  | C    | 5.614571              | 4.390287  | 0.797696  |
| H    | -6.931363             | 2.8138    | 1.996796  | C    | 4.910254              | 3.909021  | -1.439513 |
| H    | -6.130526             | 2.643001  | -2.218554 | C    | 5.818831              | 2.858751  | -1.502381 |
| H    | -4.501351             | 4.464972  | -1.981914 | C    | 6.514346              | 3.325668  | 0.741480  |
| H    | -5.329352             | 4.649835  | 2.23144   | H    | 5.572489              | 4.988019  | 1.703666  |
| C    | -6.575367             | -2.486163 | -0.381207 | H    | 4.265116              | 4.080002  | -2.294982 |
| C    | -4.721297             | -4.650401 | -0.258225 | H    | 5.885668              | 2.262225  | -2.406496 |
| C    | -6.514371             | -3.325707 | 0.741436  | H    | 7.130869              | 3.117732  | 1.611822  |
| C    | -5.818742             | -2.858689 | -1.502372 | C    | 7.174394              | 1.104429  | -0.324931 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | -4.910162 | -3.908955 | -1.439495 | C | 7.185225   | -1.143843 | -0.206107 |
| C | -5.614595 | -4.390328 | 0.797656  | C | 6.200031   | 0.002495  | 0.247465  |
| H | -7.130934 | -3.117798 | 1.611756  | C | 4.835540   | -0.053196 | -0.459610 |
| H | -5.885529 | -2.262124 | -2.406465 | H | 4.274772   | -0.928560 | -0.117392 |
| H | -4.264968 | -4.079894 | -2.294931 | H | 4.917817   | -0.115290 | -1.547034 |
| H | -5.572547 | -4.988095 | 1.703606  | H | 4.248537   | 0.838020  | -0.217740 |
| C | -3.46784  | 5.419946  | 0.329539  | C | 5.926531   | 0.094822  | 1.754354  |
| C | -0.72185  | 6.201449  | 0.378864  | H | 5.314800   | -0.756092 | 2.072254  |
| C | -2.63521  | 5.09932   | 1.415305  | H | 5.356197   | 1.005714  | 1.967414  |
| C | -2.934119 | 6.271403  | -0.654496 | H | 6.822184   | 0.111405  | 2.378807  |
| C | -1.595627 | 6.657845  | -0.626643 | C | 8.503639   | -0.759695 | 0.546309  |
| C | -1.298789 | 5.475375  | 1.43679   | H | 8.432508   | -1.093765 | 1.585459  |
| H | -3.005675 | 4.441184  | 2.195613  | C | 8.495673   | 0.812542  | 0.463403  |
| H | -3.562435 | 6.600286  | -1.478284 | H | 8.420814   | 1.250811  | 1.462646  |
| H | -1.21578  | 7.285959  | -1.427767 | H | 1.202215   | 6.113388  | 2.362930  |
| H | -0.669893 | 5.093634  | 2.234297  | N | 7.452593   | 0.530981  | -1.689506 |
| C | -3.489293 | -5.465452 | -0.083712 | N | 7.459426   | -0.713873 | -1.622571 |
| C | -0.757435 | -6.242288 | 0.249594  | H | -8.432512  | 1.093718  | 1.585460  |
| C | -2.925868 | -5.646399 | 1.193364  | H | -8.420925  | -1.250833 | 1.462608  |
| C | -2.703914 | -5.875099 | -1.177279 | C | -9.855244  | 1.214333  | -0.054696 |
| C | -1.370655 | -6.238064 | -1.016724 | H | -10.249819 | 2.107496  | 0.440745  |
| C | -1.602367 | -6.038948 | 1.356081  | H | -9.744625  | 1.452405  | -1.117242 |
| H | -3.496636 | -5.383933 | 2.078629  | C | -10.769184 | -0.016980 | 0.113163  |
| H | -3.111712 | -5.836437 | -2.182819 | H | -11.647611 | 0.013062  | -0.541054 |
| H | -0.774549 | -6.423689 | -1.9052   | H | -11.136666 | -0.073071 | 1.147083  |
| H | -1.202209 | -6.113338 | 2.362954  | C | -9.843382  | -1.215001 | -0.182301 |
| C | 0.75744   | 6.242278  | 0.249568  | H | -9.731131  | -1.339861 | -1.263834 |
| C | 3.489309  | 5.465452  | -0.083717 | H | -10.229192 | -2.158813 | 0.216883  |
| C | 1.602373  | 6.038974  | 1.356058  | C | 9.843345   | 1.215090  | -0.182210 |
| C | 1.370658  | 6.238014  | -1.016750 | H | 9.731084   | 1.340047  | -1.263721 |
| C | 2.703919  | 5.875053  | -1.177295 | H | 10.229163  | 2.158865  | 0.217061  |
| C | 2.925878  | 5.646432  | 1.193353  | C | 9.855233   | -1.214251 | -0.054695 |
| H | 0.774550  | 6.423599  | -1.905232 | H | 10.249838  | -2.107427 | 0.440677  |
| H | 3.111709  | 5.836355  | -2.182836 | H | 9.744565   | -1.452246 | -1.117243 |
| H | 3.496641  | 5.384001  | 2.078630  | C | 10.769177  | 0.017041  | 0.113160  |
| C | 0.721856  | -6.201465 | 0.378884  | H | 11.647561  | -0.012979 | -0.541120 |
| C | 3.467848  | -5.419966 | 0.329526  | H | 11.136716  | 0.073121  | 1.147045  |

**2AZ-8CPP (anti)**

opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2845.178368 Hartree

Sum of electronic and thermal Energies = -2845.122223 Hartree

Sum of electronic and enthalpy Energies = -2845.121279 Hartree

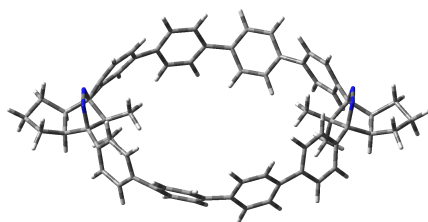
Sum of electronic and thermal Free Energies = -2845.266134 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 7.160543              | 1.125654  | 0.168995  | H    | -1.146627             | 6.997287  | -1.959086 |
| C    | 6.157333              | 0         | -0.296621 | H    | -3.497837             | 6.311070  | -1.968889 |
| C    | 7.160549              | -1.125644 | 0.169001  | H    | -3.086132             | 4.877336  | 2.064491  |
| C    | 5.811306              | -0.000008 | -1.791319 | C    | -4.717836             | 4.673043  | -0.081121 |
| H    | 5.207782              | 0.882631  | -2.029261 | C    | -6.568309             | 2.512068  | -0.184922 |
| H    | 5.20779               | -0.882655 | -2.029253 | C    | -5.028221             | 4.033525  | -1.294342 |
| H    | 6.676005              | -0.000007 | -2.458213 | C    | -5.486725             | 4.317800  | 1.041695  |
| C    | 4.828656              | -0.000005 | 0.477555  | C    | -6.389960             | 3.256279  | 0.991750  |
| H    | 4.241304              | 0.884852  | 0.213547  | C    | -5.934547             | 2.979227  | -1.346233 |
| H    | 4.963685              | -0.000001 | 1.561503  | H    | -4.472474             | 4.287410  | -2.192072 |
| H    | 4.241315              | -0.884872 | 0.213553  | H    | -5.338737             | 4.839641  | 1.983304  |
| C    | 6.568308              | -2.512056 | 0.184913  | H    | -6.911988             | 2.971019  | 1.901117  |
| C    | 4.717847              | -4.673031 | 0.081177  | H    | -6.093113             | 2.453199  | -2.281989 |
| C    | 5.934559              | -2.979185 | 1.34624   | C    | -4.717826             | -4.673051 | -0.081146 |
| C    | 6.389961              | -3.256293 | -0.991741 | C    | -6.568303             | -2.512081 | -0.184954 |
| C    | 5.486727              | -4.317816 | -1.041654 | C    | -5.486732             | -4.317820 | 1.041662  |
| C    | 5.028234              | -4.033484 | 1.294382  | C    | -5.028196             | -4.033524 | -1.294366 |
| H    | 6.093133              | -2.453132 | 2.281983  | C    | -5.934523             | -2.979226 | -1.346260 |
| H    | 6.911981              | -2.97105  | -1.901117 | C    | -6.389971             | -3.256303 | 0.991713  |
| H    | 5.338728              | -4.839678 | -1.983251 | H    | -5.338758             | -4.839671 | 1.983268  |
| H    | 4.47249               | -4.287346 | 2.192119  | H    | -4.472436             | -4.287399 | -2.192090 |
| C    | 6.568299              | 2.512065  | 0.184899  | H    | -6.093076             | -2.453189 | -2.282013 |
| C    | 4.717836              | 4.673038  | 0.081154  | H    | -6.912012             | -2.971053 | 1.901075  |
| C    | 5.934555              | 2.979202  | 1.346225  | C    | -7.160545             | -1.125660 | -0.169056 |
| C    | 6.389948              | 3.256294  | -0.99176  | C    | -7.160550             | 1.125647  | -0.169032 |
| C    | 5.486713              | 4.317816  | -1.041677 | C    | -6.157375             | -0.000010 | 0.296627  |
| C    | 5.028228              | 4.033499  | 1.294363  | C    | -4.828632             | 0.000001  | -0.477431 |
| H    | 6.093134              | 2.453156  | 2.281971  | H    | -4.241318             | 0.884863  | -0.213354 |
| H    | 6.911967              | 2.971046  | -1.901135 | H    | -4.963551             | 0.000011  | -1.561392 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| H | 5.338711  | 4.839672  | -1.983276 | H | -4.241315  | -0.884863 | -0.213370 |
| H | 4.472489  | 4.287368  | 2.192101  | C | -5.811484  | -0.000022 | 1.791357  |
| C | 3.472799  | -5.477286 | -0.026243 | H | -5.207984  | 0.882616  | 2.029363  |
| C | 0.741123  | -6.29089  | -0.042325 | H | -5.207996  | -0.882672 | 2.029352  |
| C | 2.903399  | -6.143998 | 1.074826  | H | -6.676253  | -0.000021 | 2.458164  |
| C | 2.688931  | -5.383445 | -1.189727 | C | -8.441908  | 0.787198  | 0.665069  |
| C | 1.357487  | -5.783478 | -1.199068 | H | -8.320260  | 1.173870  | 1.680912  |
| C | 1.567626  | -6.536966 | 1.06901   | C | -8.441923  | -0.787225 | 0.665025  |
| H | 3.497847  | -6.310986 | 1.969005  | H | -8.320309  | -1.173953 | 1.680850  |
| H | 3.086152  | -4.877395 | -2.064428 | H | -0.754679  | -5.580382 | 2.079563  |
| H | 0.754696  | -5.580448 | -2.079479 | N | -7.503725  | -0.623279 | -1.546592 |
| H | 1.146639  | -6.997213 | 1.959218  | N | -7.503777  | 0.623287  | -1.546567 |
| C | 3.472788  | 5.477292  | -0.026265 | N | 7.503720   | -0.623272 | 1.546547  |
| C | 0.74111   | 6.290893  | -0.042338 | N | 7.503714   | 0.623294  | 1.546545  |
| C | 2.903393  | 6.144009  | 1.074803  | C | 8.441909   | -0.787202 | -0.665086 |
| C | 2.688914  | 5.383444  | -1.189745 | H | 8.320277   | -1.173853 | -1.680946 |
| C | 1.35747   | 5.783475  | -1.199081 | C | 8.441910   | 0.787213  | -0.665084 |
| C | 1.567619  | 6.536975  | 1.068992  | H | 8.320284   | 1.173867  | -1.680943 |
| H | 3.497845  | 6.311002  | 1.968978  | H | 3.086131   | 4.877390  | -2.064445 |
| H | 0.754674  | 5.580439  | -2.079488 | C | 10.727532  | 0.000006  | -0.382888 |
| H | 1.146636  | 6.997226  | 1.959199  | H | 11.636557  | 0.000004  | 0.228817  |
| C | -0.741094 | -6.290895 | 0.042433  | H | 11.044584  | 0.000010  | -1.434869 |
| C | -3.472775 | -5.4773   | 0.026311  | C | 9.819941   | -1.216371 | -0.106715 |
| C | -1.357466 | -5.783446 | 1.199156  | H | 9.760182   | -1.398113 | 0.970927  |
| C | -1.567592 | -6.537012 | -1.068896 | H | 10.188185  | -2.136103 | -0.572981 |
| C | -2.903367 | -6.14405  | -1.074731 | C | 9.819939   | 1.216379  | -0.106704 |
| C | -2.688912 | -5.383421 | 1.189796  | H | 10.188182  | 2.136116  | -0.572960 |
| H | -1.146599 | -6.997288 | -1.959086 | H | 9.760178   | 1.398110  | 0.970940  |
| H | -3.497810 | -6.311075 | -1.968906 | C | -9.819896  | -1.216424 | 0.106504  |
| H | -3.086137 | -4.877343 | 2.064479  | H | -9.759947  | -1.398408 | -0.971076 |
| C | -0.741106 | 6.290894  | 0.042429  | H | -10.188234 | -2.136050 | 0.572916  |
| C | -3.472786 | 5.477293  | 0.026326  | C | -9.819885  | 1.216453  | 0.106603  |
| C | -1.357468 | 5.783443  | 1.199157  | H | -10.188209 | 2.136045  | 0.573098  |
| C | -1.567612 | 6.537010  | -1.068894 | H | -9.759946  | 1.398529  | -0.970961 |
| C | -2.903386 | 6.144046  | -1.074719 | C | -10.727549 | 0.000007  | 0.382354  |
| C | -2.688914 | 5.383415  | 1.189806  | H | -11.636323 | 0.000036  | -0.229737 |
| H | -0.754675 | 5.580381  | 2.079559  | H | -11.045037 | -0.000036 | 1.434188  |



### 2AZ-8CPP (syn)



opt freq RM06-2x/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2844.020511 Hartree

Sum of electronic and thermal Energies = -2843.965282 Hartree

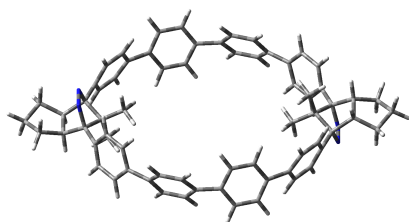
Sum of electronic and enthalpy Energies = -2843.964338 Hartree

Sum of electronic and thermal Free Energies = -2844.107376 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.620734              | -0.715584 | 0.514062  | C    | -1.317074             | 5.165517  | 1.436946  |
| C    | 8.587791              | 0.85031   | 0.420802  | C    | -1.630763             | 6.501088  | -0.525757 |
| C    | 7.271065              | 1.109055  | -0.36727  | C    | -2.973223             | 6.138394  | -0.557222 |
| C    | 6.32397               | 0.008431  | 0.211561  | C    | -2.657688             | 4.814553  | 1.41403   |
| C    | 7.315201              | -1.116276 | -0.231245 | H    | -0.674203             | 4.713859  | 2.185216  |
| C    | 4.96517               | -0.079042 | -0.485685 | H    | -1.247284             | 7.176131  | -1.285709 |
| H    | 4.362906              | 0.802117  | -0.243712 | H    | -3.613979             | 6.526927  | -1.344391 |
| H    | 4.429923              | -0.967311 | -0.134279 | H    | -3.030691             | 4.103312  | 2.145051  |
| H    | 5.048906              | -0.145848 | -1.573422 | C    | -4.787094             | 4.525121  | 0.192047  |
| C    | 6.062028              | 0.116367  | 1.712041  | C    | -6.716789             | 2.489769  | -0.128208 |
| H    | 5.506304              | 1.03885   | 1.913731  | C    | -5.130857             | 4.054245  | -1.080259 |
| H    | 6.96211               | 0.124067  | 2.329837  | C    | -5.547827             | 4.079679  | 1.28076   |
| H    | 5.437073              | -0.722076 | 2.035907  | C    | -6.501291             | 3.079532  | 1.122172  |
| N    | 7.555249              | 0.53604   | -1.721937 | C    | -6.078816             | 3.050023  | -1.238961 |
| N    | 7.580639              | -0.700087 | -1.645193 | H    | -4.564444             | 4.391344  | -1.943929 |
| C    | 6.716789              | -2.489769 | -0.128201 | H    | -5.34642              | 4.476758  | 2.272123  |
| C    | 4.787095              | -4.525121 | 0.192061  | H    | -7.022786             | 2.700871  | 1.997711  |
| C    | 6.501292              | -3.079528 | 1.122181  | H    | -6.258853             | 2.625055  | -2.221327 |
| C    | 6.078816              | -3.050028 | -1.238952 | C    | -4.699861             | -4.530582 | -0.262382 |
| C    | 5.130857              | -4.054249 | -1.080247 | C    | -6.631138             | -2.464847 | -0.423877 |
| C    | 5.547827              | -4.079675 | 1.280772  | C    | -5.625533             | -4.323546 | 0.769138  |
| H    | 7.022787              | -2.700865 | 1.997718  | C    | -4.872317             | -3.786304 | -1.437731 |
| H    | 6.258852              | -2.625064 | -2.221319 | C    | -5.824997             | -2.781293 | -1.5217   |
| H    | 4.564444              | -4.391352 | -1.943915 | C    | -6.568004             | -3.301421 | 0.694346  |
| H    | 5.346421              | -4.476751 | 2.272137  | H    | -5.576879             | -4.929547 | 1.669075  |
| C    | 6.631139              | 2.464846  | -0.423889 | H    | -4.18365              | -3.917426 | -2.265886 |
| C    | 4.699861              | 4.53058   | -0.2624   | H    | -5.886185             | -2.168435 | -2.416065 |
| C    | 6.568005              | 3.301423  | 0.694332  | H    | -7.216136             | -3.118378 | 1.547515  |
| C    | 5.824996              | 2.781287  | -1.521712 | C    | -7.271065             | -1.109056 | -0.367264 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | 4.872316  | 3.786298  | -1.437746 | C | -7.315201  | 1.116275  | -0.231247 |
| C | 5.625533  | 4.323548  | 0.76912   | C | -6.32397   | -0.00843  | 0.211562  |
| H | 7.216137  | 3.118383  | 1.547501  | C | -4.965171  | 0.079044  | -0.485686 |
| H | 5.886184  | 2.168425  | -2.416074 | H | -4.429926  | 0.967316  | -0.134284 |
| H | 4.183648  | 3.917416  | -2.265902 | H | -5.048908  | 0.145844  | -1.573422 |
| H | 5.57688   | 4.929553  | 1.669055  | H | -4.362905  | -0.802113 | -0.243708 |
| C | 3.498353  | -5.237903 | 0.378342  | C | -6.062025  | -0.116356 | 1.712043  |
| C | 0.758082  | -5.96615  | 0.432628  | H | -5.437071  | 0.72209   | 2.035903  |
| C | 2.657689  | -4.814548 | 1.414046  | H | -5.5063    | -1.038837 | 1.913738  |
| C | 2.973222  | -6.138396 | -0.557201 | H | -6.962106  | -0.124053 | 2.32984   |
| C | 1.630763  | -6.50109  | -0.525735 | C | -8.620734  | 0.715585  | 0.514063  |
| C | 1.317075  | -5.165512 | 1.436964  | H | -8.560695  | 1.048002  | 1.554578  |
| H | 3.030691  | -4.103304 | 2.145064  | C | -8.587792  | -0.850309 | 0.420805  |
| H | 3.613978  | -6.526933 | -1.344369 | H | -8.508481  | -1.296795 | 1.416081  |
| H | 1.247283  | -7.176137 | -1.285683 | H | -1.153804  | -5.815622 | 2.401349  |
| H | 0.674204  | -4.713851 | 2.185232  | N | -7.55525   | -0.536047 | -1.721933 |
| C | 3.444932  | 5.30246   | -0.063426 | N | -7.580637  | 0.70008   | -1.645194 |
| C | 0.720026  | 6.015893  | 0.297964  | H | 8.560696   | -1.048    | 1.554577  |
| C | 2.886865  | 5.419058  | 1.216869  | H | 8.508476   | 1.296796  | 1.416078  |
| C | 2.655982  | 5.72253   | -1.142599 | C | 9.967241   | -1.147125 | -0.096869 |
| C | 1.318849  | 6.054963  | -0.967889 | H | 10.373057  | -2.0412   | 0.383544  |
| C | 1.560206  | 5.784443  | 1.394849  | H | 9.851634   | -1.365939 | -1.162774 |
| H | 3.466635  | 5.127926  | 2.087117  | C | 10.858698  | 0.091748  | 0.083249  |
| H | 3.067086  | 5.717536  | -2.147445 | H | 11.748996  | 0.073239  | -0.551624 |
| H | 0.710336  | 6.253447  | -1.845096 | H | 11.19616   | 0.159141  | 1.125188  |
| H | 1.153805  | 5.81563   | 2.401328  | C | 9.917781   | 1.264014  | -0.236731 |
| C | -0.720026 | -6.015893 | 0.297986  | H | 9.79699    | 1.354929  | -1.320531 |
| C | -3.444933 | -5.302461 | -0.063406 | H | 10.284846  | 2.223356  | 0.137577  |
| C | -1.560206 | -5.784438 | 1.39487   | C | -9.917781  | -1.264012 | -0.236731 |
| C | -1.31885  | -6.054967 | -0.967867 | H | -9.796988  | -1.354925 | -1.320531 |
| C | -2.655982 | -5.722535 | -1.142577 | H | -10.284846 | -2.223355 | 0.137574  |
| C | -2.886864 | -5.419054 | 1.21689   | C | -9.967241  | 1.147127  | -0.096867 |
| H | -0.710337 | -6.253454 | -1.845073 | H | -10.373058 | 2.041201  | 0.383547  |
| H | -3.067087 | -5.717545 | -2.147424 | H | -9.851635  | 1.365941  | -1.162771 |
| H | -3.466634 | -5.127919 | 2.087137  | C | -10.858698 | -0.091747 | 0.083249  |
| C | -0.758082 | 5.966151  | 0.432608  | H | -11.748996 | -0.073237 | -0.551624 |
| C | -3.498353 | 5.237904  | 0.378325  | H | -11.196161 | -0.159142 | 1.125189  |

### 2AZ-8CPP (anti)



opt freq RM06-2x/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2844.025329 Hartree

Sum of electronic and thermal Energies = -2843.970043 Hartree

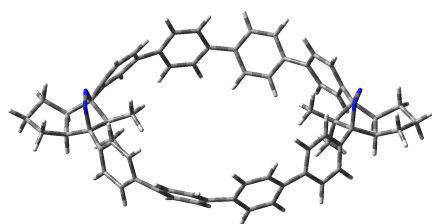
Sum of electronic and enthalpy Energies = -2843.969098 Hartree

Sum of electronic and thermal Free Energies = -2844.112081 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 7.267649              | 1.114863  | 0.176182  | H    | -1.130763             | 6.834759  | -1.942399 |
| C    | 6.269628              | 0         | -0.274945 | H    | -3.500524             | 6.182891  | -1.948803 |
| C    | 7.267649              | -1.114863 | 0.176182  | H    | -3.081439             | 4.660268  | 2.041905  |
| C    | 5.917563              | 0         | -1.760636 | C    | -4.731279             | 4.556667  | -0.079452 |
| H    | 5.31171               | 0.883246  | -1.989409 | C    | -6.65035              | 2.482906  | -0.19337  |
| H    | 5.311711              | -0.883247 | -1.989409 | C    | -5.05334              | 3.935144  | -1.292588 |
| H    | 6.779325              | 0         | -2.430857 | C    | -5.503443             | 4.227436  | 1.042144  |
| C    | 4.955118              | 0         | 0.507273  | C    | -6.444695             | 3.203752  | 0.987289  |
| H    | 4.369322              | 0.887143  | 0.246238  | C    | -5.998598             | 2.919773  | -1.350735 |
| H    | 5.1047                | 0         | 1.59006   | H    | -4.479246             | 4.170325  | -2.183751 |
| H    | 4.369323              | -0.887145 | 0.246239  | H    | -5.32794              | 4.738771  | 1.984539  |
| C    | 6.65035               | -2.482906 | 0.193371  | H    | -6.97263              | 2.922015  | 1.894863  |
| C    | 4.731279              | -4.556666 | 0.07945   | H    | -6.171197             | 2.393715  | -2.284595 |
| C    | 5.998597              | -2.919773 | 1.350735  | C    | -4.731279             | -4.556667 | -0.079452 |
| C    | 6.444695              | -3.203751 | -0.987289 | C    | -6.65035              | -2.482906 | -0.19337  |
| C    | 5.503443              | -4.227434 | -1.042145 | C    | -5.503443             | -4.227436 | 1.042145  |
| C    | 5.053339              | -3.935144 | 1.292586  | C    | -5.05334              | -3.935144 | -1.292587 |
| H    | 6.171196              | -2.393716 | 2.284595  | C    | -5.998598             | -2.919772 | -1.350735 |
| H    | 6.972631              | -2.922012 | -1.894862 | C    | -6.444694             | -3.203752 | 0.98729   |
| H    | 5.32794               | -4.738769 | -1.984541 | H    | -5.327939             | -4.738771 | 1.984539  |
| H    | 4.479244              | -4.170325 | 2.183749  | H    | -4.479246             | -4.170325 | -2.183751 |
| C    | 6.650349              | 2.482906  | 0.193372  | H    | -6.171197             | -2.393715 | -2.284594 |
| C    | 4.731278              | 4.556666  | 0.079451  | H    | -6.97263              | -2.922015 | 1.894863  |
| C    | 5.998596              | 2.919773  | 1.350736  | C    | -7.267648             | -1.114862 | -0.176179 |
| C    | 6.444695              | 3.203751  | -0.987289 | C    | -7.267648             | 1.114863  | -0.17618  |
| C    | 5.503443              | 4.227434  | -1.042145 | C    | -6.269626             | 0         | 0.274945  |
| C    | 5.053339              | 3.935144  | 1.292587  | C    | -4.955118             | -0.000001 | -0.507276 |
| H    | 6.171195              | 2.393716  | 2.284595  | H    | -4.369323             | 0.887144  | -0.246244 |
| H    | 6.972631              | 2.922013  | -1.894862 | H    | -5.104703             | -0.000002 | -1.590063 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| H | 5.32794   | 4.738769  | -1.98454  | H | -4.369322  | -0.887145 | -0.246242 |
| H | 4.479244  | 4.170325  | 2.18375   | C | -5.917556  | 0         | 1.760635  |
| C | 3.464711  | -5.324905 | -0.029457 | H | -5.311705  | 0.883247  | 1.989407  |
| C | 0.740614  | -6.095918 | -0.043481 | H | -5.311701  | -0.883246 | 1.989406  |
| C | 2.898866  | -5.996113 | 1.063671  | H | -6.779316  | -0.000003 | 2.430859  |
| C | 2.679869  | -5.184902 | -1.180307 | C | -8.530363  | 0.784525  | 0.671083  |
| C | 1.345609  | -5.567258 | -1.189252 | H | -8.399624  | 1.174782  | 1.684455  |
| C | 1.560288  | -6.368783 | 1.059491  | C | -8.530362  | -0.784524 | 0.671085  |
| H | 3.500524  | -6.182892 | 1.948799  | H | -8.39962   | -1.174777 | 1.684458  |
| H | 3.081438  | -4.660264 | -2.041907 | H | -0.733433  | -5.337356 | 2.056526  |
| H | 0.733433  | -5.337353 | -2.056529 | N | -7.627079  | -0.619304 | -1.543182 |
| H | 1.130763  | -6.834761 | 1.942394  | N | -7.62708   | 0.619304  | -1.543182 |
| C | 3.464711  | 5.324905  | -0.029457 | N | 7.627077   | -0.619304 | 1.543185  |
| C | 0.740614  | 6.095919  | -0.043481 | N | 7.627077   | 0.619304  | 1.543185  |
| C | 2.898866  | 5.996113  | 1.063671  | C | 8.530364   | -0.784525 | -0.67108  |
| C | 2.679868  | 5.184903  | -1.180306 | H | 8.399625   | -1.174778 | -1.684453 |
| C | 1.345609  | 5.567259  | -1.189251 | C | 8.530365   | 0.784525  | -0.671079 |
| C | 1.560288  | 6.368783  | 1.059492  | H | 8.399625   | 1.17478   | -1.684452 |
| H | 3.500523  | 6.182892  | 1.9488    | H | 3.081438   | 4.660264  | -2.041907 |
| H | 0.733433  | 5.337353  | -2.056529 | C | 10.803992  | 0         | -0.425367 |
| H | 1.130762  | 6.834761  | 1.942394  | H | 11.731315  | 0         | 0.154346  |
| C | -0.740614 | -6.095919 | 0.043477  | H | 11.077464  | 0.000001  | -1.488034 |
| C | -3.464711 | -5.324906 | 0.029454  | C | 9.904513   | -1.207874 | -0.118901 |
| C | -1.345609 | -5.56726  | 1.189248  | H | 9.851171   | -1.362882 | 0.963035  |
| C | -1.560288 | -6.368782 | -1.059496 | H | 10.264444  | -2.136347 | -0.569848 |
| C | -2.898867 | -5.996112 | -1.063674 | C | 9.904513   | 1.207874  | -0.1189   |
| C | -2.679869 | -5.184905 | 1.180304  | H | 10.264444  | 2.136347  | -0.569845 |
| H | -1.130763 | -6.834758 | -1.942399 | H | 9.851171   | 1.36288   | 0.963037  |
| H | -3.500524 | -6.18289  | -1.948803 | C | -9.904512  | -1.207875 | 0.118911  |
| H | -3.081438 | -4.660268 | 2.041905  | H | -9.851172  | -1.362884 | -0.963025 |
| C | -0.740615 | 6.095919  | 0.043477  | H | -10.264442 | -2.136348 | 0.569859  |
| C | -3.464712 | 5.324906  | 0.029454  | C | -9.904512  | 1.207872  | 0.118904  |
| C | -1.34561  | 5.567261  | 1.189248  | H | -10.264443 | 2.136347  | 0.569846  |
| C | -1.560288 | 6.368782  | -1.059496 | H | -9.851171  | 1.362874  | -0.963033 |
| C | -2.898867 | 5.996113  | -1.063674 | C | -10.803991 | -0.000001 | 0.425376  |
| C | -2.679869 | 5.184905  | 1.180304  | H | -11.731316 | -0.000003 | -0.154333 |
| H | -0.733434 | 5.337356  | 2.056526  | H | -11.077457 | 0.000002  | 1.488045  |

### 2AZ-8CPP (syn)



opt freq UM06-2x/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.0068$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2843.927201 Hartree

Sum of electronic and thermal Energies = -2843.871476 Hartree

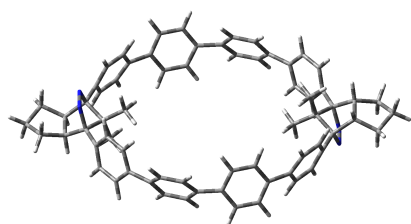
Sum of electronic and enthalpy Energies = -2843.870532 Hartree

Sum of electronic and thermal Free Energies = -2844.015566 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.585616              | -0.673589 | 0.589099  | C    | -1.326203             | 5.152278  | 1.45667   |
| C    | 8.584705              | 0.883669  | 0.364211  | C    | -1.649461             | 6.506344  | -0.491744 |
| C    | 7.284182              | 1.145984  | -0.434089 | C    | -2.99126              | 6.14115   | -0.522194 |
| C    | 6.307512              | 0.02335   | 0.144446  | C    | -2.666236             | 4.798996  | 1.435084  |
| C    | 7.338089              | -1.116638 | -0.216522 | H    | -0.679863             | 4.694573  | 2.198254  |
| C    | 4.984503              | -0.111333 | -0.604638 | H    | -1.270147             | 7.189602  | -1.246418 |
| H    | 4.382378              | 0.792662  | -0.469982 | H    | -3.635461             | 6.53583   | -1.303471 |
| H    | 4.422091              | -0.957455 | -0.195182 | H    | -3.035407             | 4.080004  | 2.160471  |
| H    | 5.125011              | -0.278363 | -1.674    | C    | -4.798596             | 4.516061  | 0.216695  |
| C    | 5.987413              | 0.189506  | 1.628016  | C    | -6.723288             | 2.47811   | -0.116313 |
| H    | 5.314224              | 1.044015  | 1.754393  | C    | -5.142597             | 4.053664  | -1.058617 |
| H    | 6.860125              | 0.357006  | 2.26307   | C    | -5.556929             | 4.060787  | 1.303025  |
| H    | 5.462827              | -0.699071 | 1.995348  | C    | -6.508188             | 3.059585  | 1.138089  |
| N    | 7.426825              | 0.509402  | -1.774055 | C    | -6.087965             | 3.04799   | -1.223631 |
| N    | 7.739304              | -0.711156 | -1.61114  | H    | -4.577876             | 4.398456  | -1.920366 |
| C    | 6.740174              | -2.484453 | -0.097581 | H    | -5.355225             | 4.451007  | 2.29706   |
| C    | 4.804886              | -4.51216  | 0.220626  | H    | -7.027935             | 2.67334   | 2.011352  |
| C    | 6.503701              | -3.05291  | 1.159248  | H    | -6.267624             | 2.629487  | -2.208828 |
| C    | 6.127417              | -3.066053 | -1.210722 | C    | -4.68645              | -4.535193 | -0.279466 |
| C    | 5.174902              | -4.065696 | -1.053369 | C    | -6.623286             | -2.474656 | -0.438421 |
| C    | 5.546886              | -4.049941 | 1.315448  | C    | -5.618229             | -4.337606 | 0.748336  |
| H    | 7.012035              | -2.66229  | 2.036959  | C    | -4.85522              | -3.784585 | -1.45133  |
| H    | 6.331742              | -2.653861 | -2.194312 | C    | -5.81066              | -2.78208  | -1.534106 |
| H    | 4.626977              | -4.419    | -1.922408 | C    | -6.563525             | -3.317991 | 0.674872  |
| H    | 5.329077              | -4.432137 | 2.30908   | H    | -5.572178             | -4.948886 | 1.644848  |
| C    | 6.626483              | 2.492634  | -0.479129 | H    | -4.161612             | -3.908561 | -2.276494 |
| C    | 4.685037              | 4.546537  | -0.279399 | H    | -5.869174             | -2.163885 | -2.424996 |
| C    | 6.566603              | 3.316143  | 0.649535  | H    | -7.216538             | -3.142192 | 1.525827  |
| C    | 5.810045              | 2.816821  | -1.567189 | C    | -7.266981             | -1.120901 | -0.376519 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | 4.853814  | 3.81621   | -1.464046 | C | -7.317754  | 1.103511  | -0.228089 |
| C | 5.618514  | 4.331508  | 0.742876  | C | -6.324405  | -0.020867 | 0.210751  |
| H | 7.218654  | 3.12777   | 1.498003  | C | -4.964118  | 0.074961  | -0.482553 |
| H | 5.865648  | 2.208687  | -2.46412  | H | -4.432743  | 0.962894  | -0.124445 |
| H | 4.160164  | 3.954346  | -2.286898 | H | -5.04522   | 0.148096  | -1.570082 |
| H | 5.573698  | 4.926003  | 1.650621  | H | -4.359528  | -0.805581 | -0.244211 |
| C | 3.513138  | -5.221417 | 0.398272  | C | -6.065679  | -0.137494 | 1.711195  |
| C | 0.7731    | -5.952163 | 0.435707  | H | -5.441782  | 0.699172  | 2.041543  |
| C | 2.662899  | -4.789708 | 1.422812  | H | -5.509944  | -1.060925 | 1.908532  |
| C | 2.996276  | -6.129098 | -0.535003 | H | -6.967073  | -0.149359 | 2.327015  |
| C | 1.653678  | -6.492397 | -0.512256 | C | -8.623938  | 0.695114  | 0.511806  |
| C | 1.322826  | -5.142357 | 1.437795  | H | -8.567365  | 1.022104  | 1.554232  |
| H | 3.028404  | -4.071726 | 2.150927  | C | -8.586272  | -0.870143 | 0.409946  |
| H | 3.643807  | -6.524142 | -1.31332  | H | -8.507957  | -1.321883 | 1.402902  |
| H | 1.276736  | -7.173247 | -1.270357 | H | -1.146617  | -5.812024 | 2.395501  |
| H | 0.673011  | -4.685918 | 2.177141  | N | -7.549513  | -0.541158 | -1.728658 |
| C | 3.429521  | 5.31283   | -0.06512  | N | -7.578737  | 0.694392  | -1.644989 |
| C | 0.705164  | 6.016857  | 0.31745   | H | 8.426743   | -0.914244 | 1.646321  |
| C | 2.877534  | 5.413607  | 1.219142  | H | 8.540643   | 1.40727   | 1.324605  |
| C | 2.634561  | 5.743409  | -1.1357   | C | 9.961128   | -1.178985 | 0.126168  |
| C | 1.297792  | 6.071367  | -0.950757 | H | 10.343581  | -1.982111 | 0.761713  |
| C | 1.551028  | 5.774349  | 1.40761   | H | 9.886448   | -1.55806  | -0.897082 |
| H | 3.462208  | 5.113628  | 2.083118  | C | 10.848513  | 0.07721   | 0.151312  |
| H | 3.040833  | 5.750553  | -2.142486 | H | 11.749275  | -0.025237 | -0.460366 |
| H | 0.684748  | 6.27857   | -1.822753 | H | 11.169467  | 0.287402  | 1.179564  |
| H | 1.149432  | 5.793283  | 2.416319  | C | 9.91448    | 1.19288   | -0.338247 |
| C | -0.70399  | -6.007022 | 0.2937    | H | 9.77904    | 1.114458  | -1.422331 |
| C | -3.429508 | -5.30305  | -0.077857 | H | 10.285252  | 2.196872  | -0.114225 |
| C | -1.549435 | -5.780674 | 1.387555  | C | -9.913478  | -1.284027 | -0.253058 |
| C | -1.296831 | -6.045787 | -0.974911 | H | -9.789823  | -1.368697 | -1.33702  |
| C | -2.634441 | -5.717537 | -1.154701 | H | -10.278747 | -2.246418 | 0.115157  |
| C | -2.876678 | -5.420218 | 1.20467   | C | -9.970147  | 1.126168  | -0.100142 |
| H | -0.683282 | -6.240323 | -1.849568 | H | -10.379744 | 2.016426  | 0.384146  |
| H | -3.041381 | -5.711303 | -2.161264 | H | -9.85254   | 1.35113   | -1.164527 |
| H | -3.461464 | -5.132954 | 2.07285   | C | -10.858485 | -0.116201 | 0.07107   |
| C | -0.772277 | 5.963561  | 0.458054  | H | -11.747344 | -0.096747 | -0.565793 |
| C | -3.511225 | 5.230416  | 0.406212  | H | -11.19815  | -0.190229 | 1.111819  |

### 2AZ-8CPP (anti)



opt freq UM06-2x/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.0074$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2843.931323 Hartree

Sum of electronic and thermal Energies = -2843.876148 Hartree

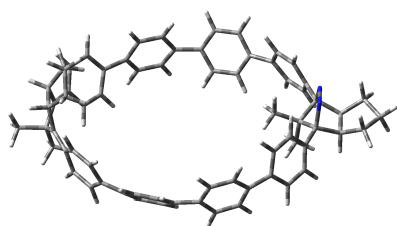
Sum of electronic and enthalpy Energies = -2843.875204 Hartree

Sum of electronic and thermal Free Energies = -2844.018871 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 7.265697              | 1.114901  | 0.171744  | H    | -1.136159             | 6.859335  | -1.916767 |
| C    | 6.266414              | -0.000001 | -0.276375 | H    | -3.506602             | 6.211372  | -1.923228 |
| C    | 7.265696              | -1.114905 | 0.171743  | H    | -3.07597              | 4.639137  | 2.047379  |
| C    | 5.909971              | -0.000001 | -1.760945 | C    | -4.734559             | 4.564772  | -0.06869  |
| H    | 5.303468              | 0.883266  | -1.987915 | C    | -6.65913              | 2.498352  | -0.206176 |
| H    | 5.303468              | -0.883267 | -1.987916 | C    | -5.073683             | 3.971557  | -1.291462 |
| H    | 6.769833              | -0.000001 | -2.433613 | C    | -5.495733             | 4.213273  | 1.053556  |
| C    | 4.954277              | -0.000001 | 0.509868  | C    | -6.440029             | 3.19331   | 0.987859  |
| H    | 4.367646              | 0.887134  | 0.250694  | C    | -6.02208              | 2.960232  | -1.361373 |
| H    | 5.107268              | -0.000002 | 1.592173  | H    | -4.511982             | 4.226566  | -2.185017 |
| H    | 4.367646              | -0.887137 | 0.250694  | H    | -5.310518             | 4.704233  | 2.004763  |
| C    | 6.648577              | -2.482956 | 0.190926  | H    | -6.957721             | 2.894044  | 1.89534   |
| C    | 4.729935              | -4.557654 | 0.0829    | H    | -6.208363             | 2.449641  | -2.300864 |
| C    | 5.999393              | -2.91906  | 1.350016  | C    | -4.734563             | -4.564769 | -0.068685 |
| C    | 6.440487              | -3.2048   | -0.988654 | C    | -6.659134             | -2.498348 | -0.206167 |
| C    | 5.499602              | -4.229008 | -1.04058  | C    | -5.495734             | -4.213268 | 1.053563  |
| C    | 5.054321              | -3.934681 | 1.294716  | C    | -5.073692             | -3.971555 | -1.291457 |
| H    | 6.173811              | -2.392195 | 2.283068  | C    | -6.022089             | -2.96023  | -1.361366 |
| H    | 6.966279              | -2.923592 | -1.897633 | C    | -6.440029             | -3.193304 | 0.987868  |
| H    | 5.322603              | -4.741334 | -1.982143 | H    | -5.310516             | -4.704227 | 2.004769  |
| H    | 4.482303              | -4.169137 | 2.187373  | H    | -4.511994             | -4.226566 | -2.185014 |
| C    | 6.648579              | 2.482953  | 0.190928  | H    | -6.208375             | -2.44964  | -2.300857 |
| C    | 4.729938              | 4.557652  | 0.082904  | H    | -6.957718             | -2.894037 | 1.89535   |
| C    | 5.999395              | 2.919056  | 1.350018  | C    | -7.283274             | -1.135902 | -0.204538 |
| C    | 6.44049               | 3.204798  | -0.988651 | C    | -7.283273             | 1.135907  | -0.204546 |
| C    | 5.499606              | 4.229007  | -1.040576 | C    | -6.251063             | 0.000002  | 0.194584  |
| C    | 5.054324              | 3.934678  | 1.294719  | C    | -4.986907             | -0.000003 | -0.659063 |
| H    | 6.173813              | 2.39219   | 2.28307   | H    | -4.386237             | 0.886132  | -0.42951  |
| H    | 6.966282              | 2.923591  | -1.897631 | H    | -5.208561             | -0.000014 | -1.727885 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| H | 5.322608  | 4.741334  | -1.982139 | H | -4.386234  | -0.886132 | -0.429493 |
| H | 4.482305  | 4.169133  | 2.187377  | C | -5.829275  | 0.000009  | 1.661386  |
| C | 3.463702  | -5.327056 | -0.022186 | H | -5.213504  | 0.883009  | 1.86331   |
| C | 0.739986  | -6.098932 | -0.029794 | H | -5.213524  | -0.883003 | 1.863324  |
| C | 2.899369  | -5.994058 | 1.074316  | H | -6.660491  | 0.000024  | 2.370128  |
| C | 2.677548  | -5.19243  | -1.172834 | C | -8.50443   | 0.786046  | 0.681398  |
| C | 1.343449  | -5.575546 | -1.178753 | H | -8.337471  | 1.169888  | 1.693741  |
| C | 1.560829  | -6.366711 | 1.073468  | C | -8.504425  | -0.786037 | 0.681413  |
| H | 3.50213   | -6.177207 | 1.959424  | H | -8.337451  | -1.169859 | 1.69376   |
| H | 3.077896  | -4.671348 | -2.037132 | H | -0.728417  | -5.31589  | 2.063361  |
| H | 0.730039  | -5.350103 | -2.046355 | N | -7.638133  | -0.628973 | -1.565206 |
| H | 1.132166  | -6.828563 | 1.958957  | N | -7.638102  | 0.628971  | -1.565218 |
| C | 3.463706  | 5.327055  | -0.022181 | N | 7.628954   | -0.619343 | 1.537714  |
| C | 0.739991  | 6.098933  | -0.02979  | N | 7.628956   | 0.619338  | 1.537714  |
| C | 2.899372  | 5.994056  | 1.07432   | C | 8.525977   | -0.784522 | -0.679286 |
| C | 2.677553  | 5.192431  | -1.17283  | H | 8.392054   | -1.1746   | -1.692328 |
| C | 1.343454  | 5.575548  | -1.17875  | C | 8.525978   | 0.784518  | -0.679286 |
| C | 1.560833  | 6.36671   | 1.073473  | H | 8.392055   | 1.174597  | -1.692327 |
| H | 3.502133  | 6.177204  | 1.95943   | H | 3.077902   | 4.67135   | -2.037129 |
| H | 0.730044  | 5.350107  | -2.046353 | C | 10.800448  | -0.000003 | -0.440473 |
| H | 1.13217   | 6.828561  | 1.958962  | H | 11.72938   | -0.000003 | 0.13667   |
| C | -0.741086 | -6.097871 | 0.059436  | H | 11.071042  | -0.000003 | -1.503871 |
| C | -3.465966 | -5.328919 | 0.044399  | C | 9.901873   | -1.207909 | -0.131489 |
| C | -1.343054 | -5.555882 | 1.200581  | H | 9.851949   | -1.363184 | 0.950564  |
| C | -1.563574 | -6.383638 | -1.038029 | H | 10.260437  | -2.136256 | -0.58378  |
| C | -2.902655 | -6.012592 | -1.042328 | C | 9.901874   | 1.207904  | -0.131488 |
| C | -2.67732  | -5.174066 | 1.190857  | H | 10.260439  | 2.136251  | -0.583779 |
| H | -1.136166 | -6.859332 | -1.916769 | H | 9.851949   | 1.363178  | 0.950565  |
| H | -3.506609 | -6.211368 | -1.923226 | C | -9.893306  | -1.206871 | 0.179525  |
| H | -3.075971 | -4.639135 | 2.04738   | H | -9.856121  | -1.351555 | -0.905339 |
| C | -0.741082 | 6.097873  | 0.059438  | H | -10.24044  | -2.137557 | 0.636185  |
| C | -3.465962 | 5.328922  | 0.044397  | C | -9.893308  | 1.206859  | 0.179487  |
| C | -1.343052 | 5.555884  | 1.200582  | H | -10.240448 | 2.13756   | 0.636112  |
| C | -1.563568 | 6.383641  | -1.038028 | H | -9.856117  | 1.351505  | -0.905382 |
| C | -2.902649 | 6.012595  | -1.042329 | C | -10.784148 | -0.000001 | 0.511773  |
| C | -2.677318 | 5.174069  | 1.190856  | H | -11.727317 | -0.000011 | -0.041867 |
| H | -0.728415 | 5.315891  | 2.063363  | H | -11.028369 | 0.000015  | 1.581784  |



**DR-AZ-8CPP (syn)**

opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.0678$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2735.646099 Hartree

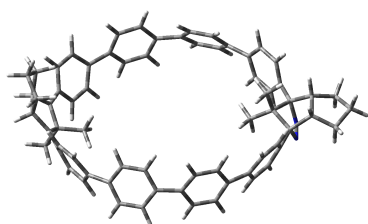
Sum of electronic and thermal Energies = -2735.590860 Hartree

Sum of electronic and enthalpy Energies = -2735.589916 Hartree

Sum of electronic and thermal Free Energies = -2735.735014 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.368916              | 0.71636   | -0.296252 | C    | -3.778971             | -5.090009 | -0.742832 |
| C    | 8.348707              | -0.857835 | -0.240863 | C    | -1.535044             | -5.001807 | -1.698164 |
| C    | 6.982312              | -1.154848 | 0.464203  | C    | -1.960654             | -6.451695 | 0.162768  |
| C    | 6.049076              | -0.037665 | -0.146148 | C    | -3.298809             | -6.067592 | 0.153744  |
| C    | 7.010918              | 1.095156  | 0.384673  | C    | -2.867944             | -4.620721 | -1.709877 |
| C    | 4.644807              | 0.011263  | 0.478508  | H    | -0.854638             | -4.521468 | -2.39389  |
| H    | 4.071282              | -0.873452 | 0.186083  | H    | -1.630228             | -7.185694 | 0.893151  |
| H    | 4.10772               | 0.894335  | 0.118962  | H    | -3.975636             | -6.513816 | 0.877243  |
| H    | 4.661845              | 0.053532  | 1.569865  | H    | -3.180745             | -3.85763  | -2.415021 |
| C    | 5.865824              | -0.099615 | -1.668317 | H    | -1.245776             | 6.202041  | -2.660888 |
| H    | 5.300132              | -0.999782 | -1.932562 | H    | 8.360054              | 1.067815  | -1.332032 |
| H    | 6.797168              | -0.11442  | -2.238256 | H    | 8.330063              | -1.279044 | -1.250024 |
| H    | 5.282757              | 0.763862  | -2.00566  | C    | 9.685558              | 1.152612  | 0.3899    |
| N    | 7.183861              | -0.606159 | 1.851895  | H    | 10.112596             | 2.051828  | -0.066166 |
| N    | 7.200469              | 0.639553  | 1.807229  | H    | 9.51458               | 1.373068  | 1.448248  |
| C    | 6.425515              | 2.481984  | 0.290651  | C    | 10.601772             | -0.0809   | 0.253776  |
| C    | 4.545425              | 4.58498   | -0.068817 | H    | 11.441232             | -0.066955 | 0.957834  |
| C    | 6.316592              | 3.151466  | -0.938614 | H    | 11.027889             | -0.121611 | -0.758137 |
| C    | 5.713297              | 3.008858  | 1.378695  | C    | 9.654797              | -1.2784   | 0.474805  |
| C    | 4.793457              | 4.038517  | 1.202275  | H    | 9.47953               | -1.420557 | 1.545775  |
| C    | 5.395188              | 4.183172  | -1.114964 | H    | 10.058706             | -2.217595 | 0.08266   |
| H    | 6.902713              | 2.821743  | -1.792247 | C    | -5.08136              | -4.415653 | -0.594972 |
| H    | 5.816193              | 2.541631  | 2.352562  | C    | -5.723069             | -4.291127 | 0.658098  |
| H    | 4.174103              | 4.337168  | 2.043172  | C    | -5.671016             | -3.687937 | -1.655476 |
| H    | 5.292731              | 4.639156  | -2.096193 | C    | -6.70577              | -3.345974 | 0.880073  |
| C    | 6.364711              | -2.52965  | 0.461582  | H    | -5.389463             | -4.89549  | 1.497216  |
| C    | 4.451465              | -4.623624 | 0.183725  | C    | -6.646974             | -2.736477 | -1.442161 |
| C    | 6.33314               | -3.333667 | -0.68818  | H    | -5.332788             | -3.859299 | -2.673826 |
| C    | 5.55054               | -2.915443 | 1.537162  | C    | -7.100495             | -2.390577 | -0.121286 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | 4.61309   | -3.933001 | 1.399015  | H | -7.143791  | -3.292477 | 1.865841  |
| C | 5.402761  | -4.364557 | -0.820509 | H | -7.034447  | -2.222738 | -2.30624  |
| H | 6.995055  | -3.117787 | -1.522582 | C | -7.762605  | -1.186042 | 0.225919  |
| H | 5.592523  | -2.347856 | 2.461194  | C | -8.061017  | -0.738069 | 1.657401  |
| H | 3.922036  | -4.110246 | 2.21685   | C | -8.195832  | -0.007074 | -0.675605 |
| H | 5.380491  | -4.931561 | -1.746882 | H | -9.057157  | -1.104385 | 1.950832  |
| C | 3.279177  | 5.322618  | -0.312631 | C | -8.093195  | 0.851859  | 1.608311  |
| C | 0.522267  | 6.020981  | -0.539144 | C | -7.080991  | -1.073799 | 2.817639  |
| C | 2.548006  | 5.045429  | -1.48087  | C | -7.727169  | 1.19116   | 0.173128  |
| C | 2.639985  | 6.097695  | 0.672204  | C | -9.770497  | 0.022616  | -0.700895 |
| C | 1.294545  | 6.440466  | 0.560401  | C | -7.800239  | -0.003667 | -2.161115 |
| C | 1.205172  | 5.384143  | -1.590591 | C | -7.165899  | 1.322473  | 2.760788  |
| H | 3.001198  | 4.446679  | -2.265299 | H | -9.114843  | 1.193183  | 1.835506  |
| H | 3.189404  | 6.396596  | 1.561135  | C | -7.155829  | 0.146238  | 3.74038   |
| H | 0.825832  | 7.0031    | 1.363394  | H | -7.343169  | -1.995546 | 3.345166  |
| H | 0.651106  | 5.038785  | -2.45795  | H | -6.060455  | -1.190231 | 2.435666  |
| C | 3.19191   | -5.368581 | -0.079888 | C | -6.943156  | 2.331334  | -0.189292 |
| C | 0.441945  | -5.985677 | -0.54787  | H | -10.145892 | -0.873812 | -1.207163 |
| C | 2.661556  | -5.419875 | -1.382594 | H | -10.214444 | 0.055491  | 0.296056  |
| C | 2.355603  | -5.820889 | 0.957741  | H | -10.116141 | 0.903033  | -1.254834 |
| C | 1.013061  | -6.109604 | 0.732075  | H | -8.070579  | 0.958716  | -2.606544 |
| C | 1.327303  | -5.73289  | -1.611752 | H | -6.737326  | -0.178524 | -2.331669 |
| H | 3.272246  | -5.113917 | -2.226381 | H | -8.361607  | -0.770423 | -2.704076 |
| H | 2.73509   | -5.875253 | 1.97392   | H | -7.506847  | 2.250662  | 3.228888  |
| H | 0.378652  | -6.337527 | 1.58384   | H | -6.152019  | 1.499985  | 2.386011  |
| H | 0.951568  | -5.706987 | -2.630502 | H | -8.08978   | 0.124454  | 4.319324  |
| C | -0.96202  | 6.00701   | -0.525881 | H | -6.328171  | 0.191047  | 4.457328  |
| C | -3.693515 | 5.188687  | -0.508678 | C | -6.860206  | 3.466543  | 0.682884  |
| C | -1.718662 | 5.962014  | -1.712455 | C | -6.084691  | 2.402217  | -1.329488 |
| C | -1.65365  | 5.793751  | 0.680298  | C | -5.890091  | 4.444845  | 0.544795  |
| C | -2.988183 | 5.40111   | 0.689762  | H | -7.568341  | 3.553244  | 1.498545  |
| C | -3.04858  | 5.556835  | -1.705278 | C | -5.12671   | 3.390574  | -1.466882 |
| H | -1.110808 | 5.819615  | 1.620594  | H | -6.086137  | 1.616046  | -2.066022 |
| H | -3.45248  | 5.135573  | 1.635412  | C | -4.943049  | 4.399107  | -0.499841 |
| H | -3.586404 | 5.484097  | -2.646869 | H | -5.851227  | 5.256022  | 1.268054  |
| C | -1.026984 | -5.880803 | -0.723529 | H | -4.428596  | 3.319824  | -2.296293 |

**DR-AZ-8CPP (anti)**

opt freq BS-UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 1.0643$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2735.646777 Hartree

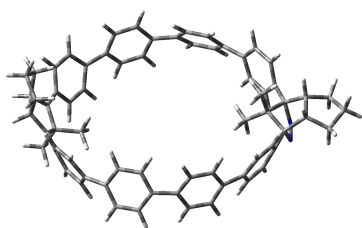
Sum of electronic and thermal Energies = -2735.590780 Hartree

Sum of electronic and enthalpy Energies = -2735.589836 Hartree

Sum of electronic and thermal Free Energies = -2735.736742 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.086933              | -0.810534 | 0.730737  | C    | -3.764556             | 5.615206  | -0.465728 |
| C    | 8.093483              | 0.76158   | 0.812692  | C    | -1.953584             | 6.310686  | 1.019085  |
| C    | 6.825885              | 1.148252  | -0.021197 | C    | -1.586716             | 6.278387  | -1.347464 |
| C    | 5.811285              | 0.004764  | 0.370912  | C    | -2.91102              | 5.909522  | -1.548687 |
| C    | 6.816695              | -1.100276 | -0.138236 | C    | -3.273303             | 5.923168  | 0.819731  |
| C    | 4.493851              | 0.062702  | -0.419966 | H    | -1.604855             | 6.473934  | 2.03498   |
| H    | 3.914315              | 0.939142  | -0.113655 | H    | -0.936422             | 6.372088  | -2.212267 |
| H    | 3.891403              | -0.827506 | -0.21498  | H    | -3.26108              | 5.771452  | -2.567221 |
| H    | 4.644013              | 0.124352  | -1.500157 | H    | -3.900504             | 5.754628  | 1.689804  |
| C    | 5.444407              | -0.084067 | 1.858063  | H    | -1.127953             | -5.591437 | 2.087009  |
| H    | 4.819051              | 0.770841  | 2.136601  | H    | 7.949                 | -1.247637 | 1.723697  |
| H    | 6.29972               | -0.104552 | 2.53666   | H    | 7.957875              | 1.096678  | 1.845061  |
| H    | 4.856757              | -0.991423 | 2.036543  | C    | 9.4713                | -1.216893 | 0.170855  |
| N    | 7.187377              | 0.715881  | -1.417337 | H    | 9.828981              | -2.161395 | 0.593909  |
| N    | 7.181543              | -0.5289   | -1.483345 | H    | 9.426504              | -1.342252 | -0.915505 |
| C    | 6.224407              | -2.482676 | -0.233151 | C    | 10.379833             | -0.020959 | 0.52293   |
| C    | 4.395476              | -4.674205 | -0.240028 | H    | 11.297614             | 0.006388  | -0.074991 |
| C    | 6.103044              | -3.329951 | 0.878543  | H    | 10.681684             | -0.077188 | 1.577884  |
| C    | 5.540014              | -2.854763 | -1.399944 | C    | 9.4813                | 1.212525  | 0.297526  |
| C    | 4.64409               | -3.917299 | -1.40051  | H    | 9.438188              | 1.450362  | -0.769946 |
| C    | 5.216776              | -4.407357 | 0.871393  | H    | 9.846296              | 2.104958  | 0.816419  |
| H    | 6.661834              | -3.12123  | 1.786878  | C    | -4.979869             | 4.791795  | -0.626714 |
| H    | 5.655188              | -2.251577 | -2.294692 | C    | -5.103278             | 3.914075  | -1.72743  |
| H    | 4.057258              | -4.089835 | -2.296458 | C    | -5.959468             | 4.643923  | 0.382315  |
| H    | 5.128823              | -5.01198  | 1.769307  | C    | -5.955106             | 2.829045  | -1.700565 |
| C    | 6.240541              | 2.537693  | 0.022777  | H    | -4.452412             | 4.030253  | -2.588248 |
| C    | 4.396867              | 4.698823  | 0.174763  | C    | -6.825626             | 3.564134  | 0.409029  |
| C    | 5.993377              | 3.193192  | 1.239422  | H    | -6.014822             | 5.370502  | 1.188801  |
| C    | 5.682046              | 3.095208  | -1.137344 | C    | -6.742045             | 2.499849  | -0.549041 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| C | 4.780674  | 4.15335   | -1.062489 | H | -5.979967 | 2.188795  | -2.571398 |
| C | 5.09001   | 4.252698  | 1.314184  | H | -7.524094 | 3.49651   | 1.230832  |
| H | 6.457909  | 2.835965  | 2.154566  | C | -7.295364 | -1.226865 | -0.18118  |
| H | 5.893831  | 2.6367    | -2.09762  | C | -8.284587 | -0.732342 | 0.851078  |
| H | 4.282265  | 4.482918  | -1.969672 | C | -6.829288 | -0.066169 | -1.088177 |
| H | 4.879153  | 4.701399  | 2.281358  | H | -9.285432 | -1.118245 | 0.597536  |
| C | 3.176981  | -5.522754 | -0.151954 | C | -8.292701 | 0.837243  | 0.743573  |
| C | 0.452304  | -6.390534 | 0.020744  | C | -8.050214 | -1.03741  | 2.363308  |
| C | 2.560381  | -5.776308 | 1.087612  | C | -7.275898 | 1.197155  | -0.319072 |
| C | 2.452683  | -5.906808 | -1.295928 | C | -7.553961 | -0.122794 | -2.469541 |
| C | 1.125439  | -6.312336 | -1.213202 | C | -5.27972  | -0.103449 | -1.236538 |
| C | 1.243683  | -6.214038 | 1.171626  | C | -8.097874 | 1.35254   | 2.204025  |
| H | 3.080935  | -5.539949 | 2.010342  | H | -9.288651 | 1.171     | 0.409564  |
| H | 2.902889  | -5.817106 | -2.279687 | C | -8.595593 | 0.199973  | 3.080902  |
| H | 0.581863  | -6.476819 | -2.138191 | H | -8.54415  | -1.957253 | 2.691004  |
| H | 0.806704  | -6.345258 | 2.156727  | H | -6.978073 | -1.140441 | 2.568298  |
| C | 3.146992  | 5.496015  | 0.267407  | C | -6.776251 | -2.56064  | -0.269746 |
| C | 0.422598  | 6.340964  | 0.153824  | H | -7.28072  | -1.014937 | -3.040825 |
| C | 2.257058  | 5.240104  | 1.324839  | H | -8.640379 | -0.139653 | -2.329787 |
| C | 2.681813  | 6.321659  | -0.772436 | H | -7.315198 | 0.749287  | -3.085938 |
| C | 1.353367  | 6.737284  | -0.825729 | H | -4.903049 | 0.673794  | -1.899219 |
| C | 0.930971  | 5.649172  | 1.268502  | H | -4.81233  | 0.043879  | -0.257464 |
| H | 2.57289   | 4.604852  | 2.14704   | H | -4.932092 | -1.059736 | -1.623998 |
| H | 3.35667   | 6.604819  | -1.576095 | H | -8.638819 | 2.28336   | 2.400149  |
| H | 1.024999  | 7.339205  | -1.668816 | H | -7.035605 | 1.53617   | 2.402549  |
| H | 0.253624  | 5.316658  | 2.048577  | H | -9.694068 | 0.177729  | 3.090337  |
| C | -1.032178 | -6.400976 | 0.082218  | H | -8.260553 | 0.276343  | 4.121616  |
| C | -3.786219 | -5.64081  | -0.030957 | C | -6.106732 | -3.058366 | -1.432701 |
| C | -1.692501 | -5.853409 | 1.19852   | C | -6.746668 | -3.456944 | 0.845255  |
| C | -1.837977 | -6.71877  | -1.029868 | C | -5.262924 | -4.153176 | -1.389124 |
| C | -3.175703 | -6.340396 | -1.08992  | H | -6.209664 | -2.538379 | -2.375978 |
| C | -3.032254 | -5.492303 | 1.147738  | C | -5.877025 | -4.535008 | 0.893136  |
| H | -1.402572 | -7.222343 | -1.888141 | H | -7.339771 | -3.24732  | 1.723429  |
| H | -3.743143 | -6.549517 | -1.993167 | C | -5.025586 | -4.850933 | -0.186628 |
| H | -3.464172 | -4.964844 | 1.993094  | H | -4.69441  | -4.410382 | -2.278751 |
| C | -1.045857 | 6.397635  | -0.053152 | H | -5.819197 | -5.120441 | 1.807608  |

**DR-AZ-8CPP (anti)**

opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.0678$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2735.646970 Hartree

Sum of electronic and thermal Energies = -2735.590996 Hartree

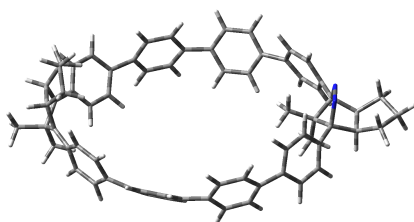
Sum of electronic and enthalpy Energies = -2735.590052 Hartree

Sum of electronic and thermal Free Energies = -2735.737671 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.071471              | -0.814841 | 0.737159  | C    | -3.764769             | 5.631123  | -0.476772 |
| C    | 8.07932               | 0.757257  | 0.818239  | C    | -1.95527              | 6.331108  | 1.007664  |
| C    | 6.813594              | 1.144659  | -0.018183 | C    | -1.585863             | 6.290759  | -1.358401 |
| C    | 5.797201              | 0.002335  | 0.372688  | C    | -2.910184             | 5.921981  | -1.559835 |
| C    | 6.80249               | -1.103911 | -0.133935 | C    | -3.274982             | 5.943672  | 0.80817   |
| C    | 4.481327              | 0.060953  | -0.420734 | H    | -1.60743              | 6.497544  | 2.023351  |
| H    | 3.901921              | 0.938039  | -0.116019 | H    | -0.934547             | 6.38117   | -2.222788 |
| H    | 3.877776              | -0.828667 | -0.216414 | H    | -3.259228             | 5.780616  | -2.578276 |
| H    | 4.633624              | 0.121875  | -1.500669 | H    | -3.903282             | 5.778239  | 1.678064  |
| C    | 5.427455              | -0.085116 | 1.85921   | H    | -1.146348             | -5.622571 | 2.081848  |
| H    | 4.80283               | 0.770873  | 2.136086  | H    | 7.931343              | -1.251274 | 1.7301    |
| H    | 6.281503              | -0.106425 | 2.539373  | H    | 7.94213               | 1.093066  | 1.850162  |
| H    | 4.838177              | -0.991519 | 2.037187  | C    | 9.456448              | -1.222832 | 0.179977  |
| N    | 7.177267              | 0.711148  | -1.413426 | H    | 9.812494              | -2.167402 | 0.604259  |
| N    | 7.170256              | -0.53366  | -1.478723 | H    | 9.413478              | -1.348822 | -0.90638  |
| C    | 6.20961               | -2.48604  | -0.229254 | C    | 10.365465             | -0.027533 | 0.532973  |
| C    | 4.382087              | -4.67889  | -0.239381 | H    | 11.284303             | -0.001366 | -0.063376 |
| C    | 6.08628               | -3.33301  | 0.882494  | H    | 10.665433             | -0.083452 | 1.588478  |
| C    | 5.527536              | -2.858626 | -1.397197 | C    | 9.468474              | 1.206662  | 0.30532   |
| C    | 4.632302              | -3.921767 | -1.399347 | H    | 9.427481              | 1.44401   | -0.762341 |
| C    | 5.20074               | -4.410973 | 0.873778  | H    | 9.833368              | 2.099012  | 0.824428  |
| H    | 6.643185              | -3.123806 | 1.791873  | C    | -4.978501             | 4.805081  | -0.635827 |
| H    | 5.644152              | -2.255643 | -2.291894 | C    | -5.097769             | 3.921498  | -1.732382 |
| H    | 4.04749               | -4.094804 | -2.296511 | C    | -5.958982             | 4.658964  | 0.372674  |
| H    | 5.111412              | -5.015456 | 1.771646  | C    | -5.944822             | 2.832873  | -1.70081  |
| C    | 6.230021              | 2.534887  | 0.023935  | H    | -4.446334             | 4.035742  | -2.593028 |
| C    | 4.391273              | 4.700608  | 0.17161   | C    | -6.820429             | 3.575594  | 0.404066  |
| C    | 5.983091              | 3.192309  | 1.239568  | H    | -6.01854              | 5.389832  | 1.174995  |
| C    | 5.673661              | 3.092162  | -1.13735  | C    | -6.730455             | 2.505894  | -0.547515 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| C | 4.774675  | 4.152446  | -1.064605 | H | -5.964441 | 2.187689  | -2.568107 |
| C | 5.082214  | 4.254105  | 1.312198  | H | -7.520712 | 3.509528  | 1.224568  |
| H | 6.446001  | 2.835195  | 2.155577  | C | -7.28059  | -1.224046 | -0.172799 |
| H | 5.885395  | 2.632096  | -2.096893 | C | -8.250248 | -0.727541 | 0.877872  |
| H | 4.277838  | 4.482041  | -1.972636 | C | -6.812628 | -0.061569 | -1.073165 |
| H | 4.871704  | 4.704567  | 2.278628  | H | -9.256811 | -1.110093 | 0.643737  |
| C | 3.164828  | -5.529444 | -0.153935 | C | -8.25538  | 0.842904  | 0.772949  |
| C | 0.441322  | -6.402913 | 0.012603  | C | -7.984473 | -1.036142 | 2.384584  |
| C | 2.546109  | -5.785017 | 1.08414   | C | -7.256454 | 1.202108  | -0.307405 |
| C | 2.443631  | -5.914438 | -1.299556 | C | -7.536385 | -0.114756 | -2.45671  |
| C | 1.117078  | -6.322684 | -1.219827 | C | -5.261878 | -0.101256 | -1.2172   |
| C | 1.230131  | -6.22552  | 1.165154  | C | -8.026859 | 1.355799  | 2.229709  |
| H | 3.064172  | -5.548238 | 2.008168  | H | -9.256359 | 1.180944  | 0.459712  |
| H | 2.895672  | -5.823573 | -2.282359 | C | -8.50564  | 0.203091  | 3.116625  |
| H | 0.576127  | -6.487921 | -2.146188 | H | -8.477892 | -1.952453 | 2.722576  |
| H | 0.791662  | -6.358077 | 2.14939   | H | -6.908862 | -1.148433 | 2.565213  |
| C | 3.143682  | 5.501601  | 0.262054  | C | -6.771444 | -2.560118 | -0.273228 |
| C | 0.421884  | 6.354151  | 0.144693  | H | -7.266859 | -1.008176 | -3.027294 |
| C | 2.252297  | 5.250419  | 1.319338  | H | -8.622799 | -0.126839 | -2.317023 |
| C | 2.681733  | 6.326752  | -0.779685 | H | -7.293542 | 0.756161  | -3.072693 |
| C | 1.354556  | 6.746075  | -0.834862 | H | -4.880527 | 0.690643  | -1.859421 |
| C | 0.927427  | 5.663245  | 1.261182  | H | -4.798479 | 0.022149  | -0.232904 |
| H | 2.565596  | 4.615804  | 2.142999  | H | -4.916014 | -1.049944 | -1.624608 |
| H | 3.358129  | 6.606501  | -1.583246 | H | -8.562822 | 2.286337  | 2.439984  |
| H | 1.028458  | 7.347315  | -1.679319 | H | -6.960277 | 1.539678  | 2.403551  |
| H | 0.248483  | 5.334167  | 2.041325  | H | -9.603596 | 0.185144  | 3.154299  |
| C | -1.043339 | -6.416008 | 0.070832  | H | -8.143838 | 0.2766    | 4.148562  |
| C | -3.796929 | -5.654675 | -0.045828 | C | -6.09916  | -3.051891 | -1.437473 |
| C | -1.707613 | -5.87755  | 1.189267  | C | -6.75428  | -3.468006 | 0.833067  |
| C | -1.845808 | -6.726233 | -1.045868 | C | -5.261449 | -4.151369 | -1.398822 |
| C | -3.183201 | -6.34714  | -1.107627 | H | -6.19334  | -2.523437 | -2.376801 |
| C | -3.047235 | -5.516051 | 1.136798  | C | -5.890166 | -4.550591 | 0.876478  |
| H | -1.407873 | -7.223571 | -1.906453 | H | -7.353716 | -3.26455  | 1.708448  |
| H | -3.74761  | -6.54931  | -2.014357 | C | -5.033533 | -4.860533 | -0.201054 |
| H | -3.482068 | -4.99515  | 1.984748  | H | -4.689389 | -4.40361  | -2.287592 |
| C | -1.046255 | 6.413392  | -0.063889 | H | -5.841108 | -5.145026 | 1.785665  |

### DR-AZ-8CPP (syn)



opt freq BS-UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 1.0144$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2734.054305 Hartree

Sum of electronic and thermal Energies = -2733.999966 Hartree

Sum of electronic and enthalpy Energies = -2733.999021 Hartree

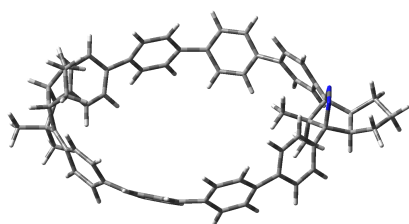
Sum of electronic and thermal Free Energies = -2734.141605 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.460299              | 0.691084  | -0.271669 | C    | -3.763646             | -4.893198 | -0.70222  |
| C    | 8.427428              | -0.87632  | -0.221003 | C    | -1.528264             | -4.743864 | -1.633497 |
| C    | 7.063848              | -1.159359 | 0.474803  | C    | -1.956352             | -6.28402  | 0.14085   |
| C    | 6.150833              | -0.042124 | -0.132742 | C    | -3.291422             | -5.902134 | 0.147643  |
| C    | 7.110303              | 1.07325   | 0.402028  | C    | -2.858844             | -4.367324 | -1.631512 |
| C    | 4.749037              | 0.015808  | 0.480013  | H    | -0.843873             | -4.228917 | -2.298502 |
| H    | 4.171588              | -0.862662 | 0.181058  | H    | -1.623506             | -7.049405 | 0.835663  |
| H    | 4.222568              | 0.903741  | 0.120364  | H    | -3.972238             | -6.377315 | 0.847514  |
| H    | 4.759383              | 0.054223  | 1.570643  | H    | -3.178228             | -3.571809 | -2.296006 |
| C    | 5.976946              | -0.098877 | -1.649884 | H    | -1.173371             | 5.953382  | -2.66759  |
| H    | 5.408378              | -0.994377 | -1.917897 | H    | 8.460766              | 1.046982  | -1.304645 |
| H    | 6.910302              | -0.116924 | -2.213991 | H    | 8.411537              | -1.296134 | -1.229438 |
| H    | 5.400494              | 0.767417  | -1.986614 | C    | 9.769089              | 1.111836  | 0.423349  |
| N    | 7.262179              | -0.621735 | 1.855439  | H    | 10.20454              | 2.01126   | -0.020696 |
| N    | 7.28871               | 0.617541  | 1.814493  | H    | 9.591764              | 1.322991  | 1.481384  |
| C    | 6.522998              | 2.454981  | 0.308193  | C    | 10.67289              | -0.122075 | 0.283608  |
| C    | 4.61859               | 4.509929  | -0.061791 | H    | 11.51198              | -0.117066 | 0.985756  |
| C    | 6.409325              | 3.114192  | -0.9184   | H    | 11.096035             | -0.163254 | -0.727935 |
| C    | 5.804641              | 2.969606  | 1.389021  | C    | 9.718789              | -1.30592  | 0.500785  |
| C    | 4.869469              | 3.97782   | 1.206399  | H    | 9.534492              | -1.441819 | 1.569901  |
| C    | 5.473146              | 4.12473   | -1.100323 | H    | 10.115952             | -2.249125 | 0.115492  |
| H    | 6.999876              | 2.786741  | -1.76878  | C    | -5.077549             | -4.238826 | -0.541753 |
| H    | 5.911182              | 2.503781  | 2.36228   | C    | -5.667059             | -4.078326 | 0.721272  |
| H    | 4.240201              | 4.268078  | 2.041766  | C    | -5.715486             | -3.582045 | -1.60758  |
| H    | 5.358932              | 4.57152   | -2.083499 | C    | -6.678812             | -3.166044 | 0.937034  |
| C    | 6.424337              | -2.519857 | 0.465011  | H    | -5.26957              | -4.623154 | 1.572228  |
| C    | 4.468764              | -4.54787  | 0.170985  | C    | -6.726115             | -2.669183 | -1.400179 |
| C    | 6.374029              | -3.307516 | -0.686981 | H    | -5.393131             | -3.779645 | -2.625672 |
| C    | 5.604803              | -2.888819 | 1.533205  | C    | -7.156152             | -2.29603  | -0.089781 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | 4.643207  | -3.875497 | 1.385834  | H | -7.0723    | -3.080074 | 1.936665  |
| C | 5.419757  | -4.308739 | -0.827649 | H | -7.172953  | -2.215533 | -2.268511 |
| H | 7.038545  | -3.097338 | -1.519672 | C | -7.914707  | -1.129453 | 0.205901  |
| H | 5.659877  | -2.32658  | 2.458861  | C | -8.321941  | -0.674086 | 1.606581  |
| H | 3.944726  | -4.040284 | 2.198735  | C | -8.368282  | -0.005019 | -0.741995 |
| H | 5.377631  | -4.864385 | -1.759167 | H | -9.326794  | -1.062529 | 1.830354  |
| C | 3.333752  | 5.210731  | -0.314025 | C | -8.364725  | 0.89899   | 1.515965  |
| C | 0.577676  | 5.836449  | -0.546581 | C | -7.433816  | -0.958741 | 2.8411    |
| C | 2.607896  | 4.874719  | -1.46061  | C | -7.866701  | 1.166067  | 0.111637  |
| C | 2.687234  | 5.995217  | 0.647311  | C | -9.921299  | 0.028866  | -0.811209 |
| C | 1.338006  | 6.303641  | 0.531849  | C | -7.925211  | -0.003593 | -2.208827 |
| C | 1.261825  | 5.178404  | -1.573223 | C | -7.557652  | 1.412047  | 2.726935  |
| H | 3.071964  | 4.260187  | -2.224938 | H | -9.401149  | 1.247596  | 1.626703  |
| H | 3.233539  | 6.327562  | 1.525161  | C | -7.6312    | 0.265317  | 3.729218  |
| H | 0.857874  | 6.875836  | 1.320024  | H | -7.702782  | -1.885182 | 3.355024  |
| H | 0.709788  | 4.79267   | -2.423478 | H | -6.381231  | -1.030461 | 2.549717  |
| C | 3.188275  | -5.254325 | -0.097902 | C | -6.950515  | 2.223005  | -0.215765 |
| C | 0.439183  | -5.79926  | -0.552213 | H | -10.284697 | -0.867373 | -1.324397 |
| C | 2.648637  | -5.251655 | -1.389277 | H | -10.398753 | 0.064672  | 0.168598  |
| C | 2.358949  | -5.711168 | 0.932524  | H | -10.251156 | 0.90625   | -1.377976 |
| C | 1.012605  | -5.96444  | 0.713098  | H | -8.128728  | 0.980203  | -2.64125  |
| C | 1.310472  | -5.53128  | -1.613133 | H | -6.872422  | -0.229751 | -2.358984 |
| H | 3.257603  | -4.930056 | -2.227284 | H | -8.50715   | -0.732116 | -2.780394 |
| H | 2.747623  | -5.796313 | 1.942129  | H | -7.948192  | 2.350289  | 3.129576  |
| H | 0.379681  | -6.200538 | 1.562628  | H | -6.514367  | 1.586227  | 2.447578  |
| H | 0.922427  | -5.464372 | -2.624431 | H | -8.621095  | 0.233656  | 4.202094  |
| C | -0.906571 | 5.799686  | -0.534872 | H | -6.88739   | 0.340508  | 4.528272  |
| C | -3.61347  | 4.967412  | -0.514447 | C | -6.891702  | 3.39946   | 0.580958  |
| C | -1.650725 | 5.729163  | -1.718639 | C | -5.981989  | 2.151688  | -1.24754  |
| C | -1.591378 | 5.593034  | 0.666925  | C | -5.880113  | 4.334226  | 0.452174  |
| C | -2.919924 | 5.192792  | 0.677922  | H | -7.656872  | 3.565719  | 1.329111  |
| C | -2.972875 | 5.313164  | -1.709714 | C | -4.983728  | 3.095352  | -1.379738 |
| H | -1.049261 | 5.636068  | 1.606044  | H | -5.942282  | 1.296576  | -1.900423 |
| H | -3.390272 | 4.938953  | 1.623026  | C | -4.867898  | 4.181484  | -0.503944 |
| H | -3.506898 | 5.210617  | -2.649757 | H | -5.864974  | 5.194214  | 1.116071  |
| C | -1.030562 | -5.672382 | -0.71273  | H | -4.213955  | 2.937241  | -2.128716 |



### DR-AZ-8CPP (syn)



opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.1301$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2734.051848 Hartree

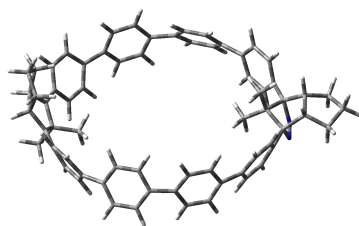
Sum of electronic and thermal Energies = -2733.997549 Hartree

Sum of electronic and enthalpy Energies = -2733.996605 Hartree

Sum of electronic and thermal Free Energies = -2734.139917 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 8.372373              | 0.708327  | -0.286805 | C    | -3.767276             | -5.033106 | -0.731665 |
| C    | 8.349536              | -0.85872  | -0.231181 | C    | -1.527601             | -4.883427 | -1.655122 |
| C    | 6.988345              | -1.148524 | 0.466586  | C    | -1.957197             | -6.412222 | 0.127921  |
| C    | 6.067373              | -0.038897 | -0.143237 | C    | -3.294157             | -6.03658  | 0.125233  |
| C    | 7.020465              | 1.084083  | 0.386668  | C    | -2.858976             | -4.511648 | -1.661022 |
| C    | 4.666297              | 0.012954  | 0.471656  | H    | -0.842617             | -4.36971  | -2.320434 |
| H    | 4.093216              | -0.869716 | 0.176619  | H    | -1.624326             | -7.170461 | 0.830617  |
| H    | 4.134331              | 0.896817  | 0.109954  | H    | -3.974935             | -6.510717 | 0.825778  |
| H    | 4.678301              | 0.055167  | 1.562128  | H    | -3.175189             | -3.720217 | -2.331412 |
| C    | 5.891878              | -0.102005 | -1.659889 | H    | -1.237059             | 6.157324  | -2.646848 |
| H    | 5.329637              | -1.002652 | -1.924196 | H    | 8.36978               | 1.061047  | -1.320863 |
| H    | 6.824705              | -0.115192 | -2.224994 | H    | 8.335541              | -1.281665 | -1.23831  |
| H    | 5.308741              | 0.758955  | -1.99894  | C    | 9.678876              | 1.139802  | 0.405954  |
| N    | 7.184226              | -0.605403 | 1.845425  | H    | 10.108231             | 2.040558  | -0.041307 |
| N    | 7.202935              | 0.633874  | 1.800502  | H    | 9.500871              | 1.353253  | 1.463411  |
| C    | 6.429321              | 2.463983  | 0.28999   | C    | 10.590493             | -0.088739 | 0.269595  |
| C    | 4.542547              | 4.537345  | -0.074667 | H    | 11.42988              | -0.076161 | 0.971294  |
| C    | 6.316221              | 3.122948  | -0.936762 | H    | 11.013391             | -0.130365 | -0.742032 |
| C    | 5.715102              | 2.982124  | 1.371823  | C    | 9.644106              | -1.278004 | 0.490958  |
| C    | 4.78885               | 3.99882   | 1.191776  | H    | 9.461385              | -1.411955 | 1.560589  |
| C    | 5.38911               | 4.142387  | -1.115997 | H    | 10.047004             | -2.219792 | 0.108163  |
| H    | 6.90234               | 2.791163  | -1.788524 | C    | -5.078838             | -4.374436 | -0.57658  |
| H    | 5.820378              | 2.51552   | 2.344818  | C    | -5.694979             | -4.239754 | 0.678838  |
| H    | 4.164419              | 4.294418  | 2.02884   | C    | -5.684172             | -3.673729 | -1.635904 |
| H    | 5.277776              | 4.591446  | -2.098458 | C    | -6.685457             | -3.307579 | 0.900472  |
| C    | 6.361889              | -2.515033 | 0.461674  | H    | -5.335109             | -4.824238 | 1.520151  |
| C    | 4.441537              | -4.578565 | 0.177286  | C    | -6.666634             | -2.734337 | -1.422993 |
| C    | 6.328886              | -3.312874 | -0.68382  | H    | -5.35051              | -3.853134 | -2.653553 |
| C    | 5.542167              | -2.886903 | 1.528738  | C    | -7.103062             | -2.379019 | -0.10619  |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | 4.597768  | -3.89068  | 1.385974  | H | -7.106437  | -3.241242 | 1.891436  |
| C | 5.392489  | -4.331505 | -0.819446 | H | -7.073303  | -2.236827 | -2.286657 |
| H | 6.993912  | -3.09957  | -1.515285 | C | -7.773421  | -1.179448 | 0.228875  |
| H | 5.584223  | -2.316236 | 2.449924  | C | -8.081709  | -0.720575 | 1.650096  |
| H | 3.898359  | -4.059667 | 2.197189  | C | -8.199123  | -0.012731 | -0.680865 |
| H | 5.364708  | -4.896429 | -1.745875 | H | -9.076518  | -1.089627 | 1.937846  |
| C | 3.270903  | 5.264173  | -0.319978 | C | -8.119203  | 0.860098  | 1.587889  |
| C | 0.528723  | 5.95029   | -0.540367 | C | -7.111576  | -1.040505 | 2.813251  |
| C | 2.536563  | 4.956304  | -1.469058 | C | -7.740186  | 1.186337  | 0.158716  |
| C | 2.642947  | 6.05272   | 0.650463  | C | -9.762638  | 0.014966  | -0.715628 |
| C | 1.300338  | 6.390064  | 0.541498  | C | -7.791467  | -0.013978 | -2.156592 |
| C | 1.196941  | 5.290294  | -1.576187 | C | -7.205399  | 1.342392  | 2.736102  |
| H | 2.986282  | 4.340475  | -2.24092  | H | -9.141868  | 1.201604  | 1.800856  |
| H | 3.197983  | 6.364916  | 1.530186  | C | -7.2033    | 0.180423  | 3.722888  |
| H | 0.832991  | 6.962774  | 1.33699   | H | -7.368968  | -1.959514 | 3.344942  |
| H | 0.634562  | 4.926473  | -2.429474 | H | -6.088346  | -1.14957  | 2.439623  |
| C | 3.176571  | -5.313186 | -0.089974 | C | -6.933059  | 2.310384  | -0.200399 |
| C | 0.440375  | -5.9117   | -0.553895 | H | -10.134457 | -0.884752 | -1.215648 |
| C | 2.645993  | -5.339803 | -1.384852 | H | -10.212138 | 0.055504  | 0.277313  |
| C | 2.348648  | -5.770841 | 0.941325  | H | -10.106346 | 0.888934  | -1.278197 |
| C | 1.008238  | -6.050471 | 0.71705   | H | -8.039     | 0.954194  | -2.599705 |
| C | 1.314442  | -5.645698 | -1.613165 | H | -6.732466  | -0.209219 | -2.319524 |
| H | 3.254734  | -5.020151 | -2.223741 | H | -8.362506  | -0.767218 | -2.705733 |
| H | 2.731869  | -5.834968 | 1.954552  | H | -7.548401  | 2.274889  | 3.190719  |
| H | 0.373524  | -6.284908 | 1.565759  | H | -6.189629  | 1.514609  | 2.367413  |
| H | 0.931867  | -5.599708 | -2.627717 | H | -8.14461   | 0.159039  | 4.286892  |
| C | -0.95562  | 5.938088  | -0.521942 | H | -6.38718   | 0.234415  | 4.449905  |
| C | -3.671434 | 5.132649  | -0.503783 | C | -6.864229  | 3.452035  | 0.650357  |
| C | -1.710184 | 5.911651  | -1.700966 | C | -6.046314  | 2.344638  | -1.309643 |
| C | -1.633555 | 5.705392  | 0.678908  | C | -5.887263  | 4.41965   | 0.517501  |
| C | -2.965114 | 5.317063  | 0.688833  | H | -7.586853  | 3.552809  | 1.450558  |
| C | -3.037415 | 5.511609  | -1.693179 | C | -5.081395  | 3.321676  | -1.442446 |
| H | -1.083322 | 5.714937  | 1.614242  | H | -6.0328    | 1.540771  | -2.025296 |
| H | -3.427708 | 5.036714  | 1.630085  | C | -4.924646  | 4.34653   | -0.500052 |
| H | -3.579519 | 5.445811  | -2.631843 | H | -5.854013  | 5.240446  | 1.228676  |
| C | -1.029439 | -5.801962 | -0.724708 | H | -4.360592  | 3.231229  | -2.249079 |

**DR-AZ-8CPP (anti)**

opt freq BS-UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 1.1309$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2734.052199 Hartree

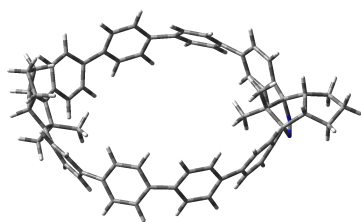
Sum of electronic and thermal Energies = -2733.997376 Hartree

Sum of electronic and enthalpy Energies = -2733.996432 Hartree

Sum of electronic and thermal Free Energies = -2734.140296 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -8.026802             | 0.826657  | 0.731038  | C    | 3.743722              | -5.623775 | -0.505523 |
| C    | -8.041981             | -0.73804  | 0.813694  | C    | 1.954128              | -6.3183   | 0.98486   |
| C    | -6.783503             | -1.126763 | -0.015616 | C    | 1.568225              | -6.27073  | -1.368676 |
| C    | -5.771804             | 0.003002  | 0.373014  | C    | 2.888458              | -5.906745 | -1.579021 |
| C    | -6.762425             | 1.104228  | -0.133973 | C    | 3.269066              | -5.932885 | 0.776509  |
| C    | -4.459513             | -0.063742 | -0.412559 | H    | 1.609919              | -6.479817 | 2.001632  |
| H    | -3.88751              | -0.943238 | -0.104976 | H    | 0.90747               | -6.354298 | -2.225489 |
| H    | -3.85247              | 0.821938  | -0.207892 | H    | 3.231826              | -5.757582 | -2.597461 |
| H    | -4.608989             | -0.125637 | -1.491856 | H    | 3.904918              | -5.763771 | 1.639227  |
| C    | -5.408148             | 0.092907  | 1.854299  | H    | 1.174856              | 5.580109  | 2.058714  |
| H    | -4.781986             | -0.759626 | 2.132785  | H    | -7.888699             | 1.264886  | 1.722055  |
| H    | -6.263338             | 0.112299  | 2.531018  | H    | -7.911245             | -1.074763 | 1.844818  |
| H    | -4.823103             | 1.000703  | 2.031063  | C    | -9.400627             | 1.23726   | 0.167699  |
| N    | -7.139393             | -0.698649 | -1.403004 | H    | -9.753391             | 2.185524  | 0.582639  |
| N    | -7.126816             | 0.539673  | -1.469809 | H    | -9.35115              | 1.354543  | -0.91823  |
| C    | -6.159096             | 2.477601  | -0.229672 | C    | -10.31219             | 0.053086  | 0.522481  |
| C    | -4.331158             | 4.648729  | -0.239512 | H    | -11.231442            | 0.030778  | -0.07046  |
| C    | -6.039567             | 3.323494  | 0.874354  | H    | -10.607586            | 0.11132   | 1.57762   |
| C    | -5.467792             | 2.83346   | -1.388996 | C    | -9.423762             | -1.178934 | 0.294642  |
| C    | -4.568843             | 3.88696   | -1.389872 | H    | -9.377547             | -1.409849 | -0.773062 |
| C    | -5.151679             | 4.393665  | 0.865335  | H    | -9.794091             | -2.071662 | 0.806111  |
| H    | -6.601737             | 3.119252  | 1.780573  | C    | 4.951909              | -4.787439 | -0.668734 |
| H    | -5.580155             | 2.223454  | -2.278503 | C    | 5.007937              | -3.851159 | -1.71613  |
| H    | -3.973553             | 4.05032   | -2.280719 | C    | 5.961859              | -4.685199 | 0.305377  |
| H    | -5.062336             | 5.0017    | 1.759672  | C    | 5.847603              | -2.762902 | -1.669595 |
| C    | -6.20003              | -2.51267  | 0.029284  | H    | 4.305982              | -3.921481 | -2.539765 |
| C    | -4.369086             | -4.662764 | 0.180394  | C    | 6.813925              | -3.599963 | 0.35517   |
| C    | -5.950381             | -3.158869 | 1.242513  | H    | 6.050139              | -5.451668 | 1.069856  |
| C    | -5.64745              | -3.0674   | -1.126507 | C    | 6.681107              | -2.498511 | -0.543039 |

|   |           |           |           |   |          |           |           |
|---|-----------|-----------|-----------|---|----------|-----------|-----------|
| C | -4.749232 | -4.121947 | -1.051397 | H | 5.800041 | -2.052722 | -2.482509 |
| C | -5.050897 | -4.215265 | 1.316886  | H | 7.546114 | -3.560155 | 1.149216  |
| H | -6.408503 | -2.794552 | 2.157089  | C | 7.226863 | 1.200912  | -0.13718  |
| H | -5.860832 | -2.608444 | -2.085284 | C | 8.144622 | 0.704698  | 0.952104  |
| H | -4.251903 | -4.454906 | -1.9569   | C | 6.797154 | 0.05102   | -1.058224 |
| H | -4.831766 | -4.658803 | 2.283599  | H | 9.159359 | 1.091147  | 0.777425  |
| C | -3.113947 | 5.500111  | -0.154487 | C | 8.153858 | -0.857189 | 0.841169  |
| C | -0.409738 | 6.378634  | 0.009205  | C | 7.778042 | 1.007286  | 2.430279  |
| C | -2.499379 | 5.751528  | 1.077519  | C | 7.199436 | -1.205639 | -0.274273 |
| C | -2.400486 | 5.88276   | -1.295934 | C | 7.623498 | 0.099218  | -2.373358 |
| C | -1.078966 | 6.293097  | -1.217227 | C | 5.260054 | 0.093138  | -1.335665 |
| C | -1.189406 | 6.197102  | 1.157474  | C | 7.83542  | -1.377213 | 2.269572  |
| H | -3.015282 | 5.508394  | 1.999986  | H | 9.164973 | -1.198732 | 0.575212  |
| H | -2.852569 | 5.786515  | -2.27712  | C | 8.242344 | -0.230865 | 3.189874  |
| H | -0.537444 | 6.456556  | -2.142418 | H | 8.248558 | 1.918353  | 2.807499  |
| H | -0.74849  | 6.327098  | 2.139892  | H | 6.693966 | 1.126541  | 2.533086  |
| C | -3.121534 | -5.463094 | 0.270305  | C | 6.740068 | 2.533481  | -0.269918 |
| C | -0.421531 | -6.324909 | 0.143318  | H | 7.420336 | 1.006629  | -2.947932 |
| C | -2.221326 | -5.181388 | 1.301723  | H | 8.695013 | 0.081209  | -2.151836 |
| C | -2.680477 | -6.311623 | -0.751173 | H | 7.402855 | -0.761403 | -3.011002 |
| C | -1.359605 | -6.736658 | -0.811134 | H | 5.033592 | 0.040091  | -2.405597 |
| C | -0.903163 | -5.599636 | 1.238315  | H | 4.748206 | -0.737    | -0.85088  |
| H | -2.523434 | -4.521836 | 2.108614  | H | 4.80942  | 1.007722  | -0.956986 |
| H | -3.367427 | -6.6041   | -1.539916 | H | 8.361735 | -2.303753 | 2.511567  |
| H | -1.043082 | -7.356447 | -1.644677 | H | 6.762384 | -1.572539 | 2.367649  |
| H | -0.211016 | -5.251361 | 1.996971  | H | 9.33328  | -0.210961 | 3.30606   |
| C | 1.076292  | 6.396774  | 0.065155  | H | 7.808264 | -0.309212 | 4.191436  |
| C | 3.81253   | 5.644285  | -0.064195 | C | 6.021088 | 2.968099  | -1.42093  |
| C | 1.735711  | 5.847111  | 1.170616  | C | 6.780549 | 3.481542  | 0.792436  |
| C | 1.869013  | 6.718521  | -1.045129 | C | 5.200588 | 4.07485   | -1.395525 |
| C | 3.202591  | 6.341247  | -1.114013 | H | 6.042169 | 2.373922  | -2.324227 |
| C | 3.07196   | 5.489295  | 1.112091  | C | 5.932973 | 4.572583  | 0.822766  |
| H | 1.426257  | 7.218661  | -1.900397 | H | 7.416551 | 3.30459   | 1.647522  |
| H | 3.764038  | 6.54274   | -2.021727 | C | 5.042885 | 4.839904  | -0.229417 |
| H | 3.511588  | 4.958859  | 1.950637  | H | 4.580915 | 4.288822  | -2.260997 |
| C | 1.045135  | -6.395493 | -0.076312 | H | 5.920262 | 5.20526   | 1.70597   |

**DR-AZ-8CPP (anti)**

opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.1345$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2734.052512 Hartree

Sum of electronic and thermal Energies = -2733.997699 Hartree

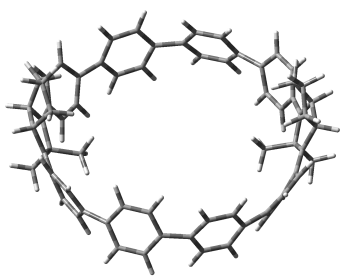
Sum of electronic and enthalpy Energies = -2733.996755 Hartree

Sum of electronic and thermal Free Energies = -2734.141608 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -8.021735             | 0.827968  | 0.73265   | C    | 3.743817              | -5.629491 | -0.508061 |
| C    | -8.037392             | -0.736741 | 0.814797  | C    | 1.95438               | -6.324979 | 0.982052  |
| C    | -6.779334             | -1.125607 | -0.015079 | C    | 1.568212              | -6.275834 | -1.371382 |
| C    | -5.767161             | 0.003735  | 0.37358   | C    | 2.888495              | -5.911928 | -1.581643 |
| C    | -6.757601             | 1.105415  | -0.132751 | C    | 3.26934               | -5.939565 | 0.773798  |
| C    | -4.455115             | -0.063129 | -0.412395 | H    | 1.610275              | -6.487076 | 1.998767  |
| H    | -3.883241             | -0.942852 | -0.10522  | H    | 0.907354              | -6.358713 | -2.228185 |
| H    | -3.847794             | 0.822347  | -0.207661 | H    | 3.231757              | -5.762119 | -2.600028 |
| H    | -4.604916             | -0.124689 | -1.491665 | H    | 3.905273              | -5.770996 | 1.636565  |
| C    | -5.40308              | 0.093056  | 1.854791  | H    | 1.181209              | 5.590815  | 2.057685  |
| H    | -4.777216             | -0.759832 | 2.132869  | H    | -7.883105             | 1.26582   | 1.723761  |
| H    | -6.258102             | 0.11261   | 2.531722  | H    | -7.9064               | -1.07384  | 1.845765  |
| H    | -4.817612             | 1.000556  | 2.031679  | C    | -9.395637             | 1.239213  | 0.169971  |
| N    | -7.135598             | -0.696965 | -1.402203 | H    | -9.747945             | 2.18744   | 0.585386  |
| N    | -7.122654             | 0.541373  | -1.468628 | H    | -9.346525             | 1.356878  | -0.915933 |
| C    | -6.154084             | 2.478728  | -0.228281 | C    | -10.307448            | 0.055203  | 0.524663  |
| C    | -4.326828             | 4.650448  | -0.23863  | H    | -11.226917            | 0.033391  | -0.067962 |
| C    | -6.034188             | 3.3244    | 0.875874  | H    | -10.602457            | 0.113166  | 1.579924  |
| C    | -5.463307             | 2.834894  | -1.387825 | C    | -9.419495             | -1.177021 | 0.296081  |
| C    | -4.56469              | 3.888677  | -1.388951 | H    | -9.373729             | -1.407583 | -0.771719 |
| C    | -5.146634             | 4.394852  | 0.866618  | H    | -9.789924             | -2.069811 | 0.807373  |
| H    | -6.595888             | 3.119844  | 1.782313  | C    | 4.95145               | -4.792141 | -0.670713 |
| H    | -5.575918             | 2.225013  | -2.277385 | C    | 5.006083              | -3.854481 | -1.716892 |
| H    | -3.969836             | 4.052376  | -2.280028 | C    | 5.961246              | -4.689632 | 0.303502  |
| H    | -5.057042             | 5.002786  | 1.761002  | C    | 5.843983              | -2.76483  | -1.668993 |
| C    | -6.196476             | -2.511797 | 0.029216  | H    | 4.304013              | -3.924649 | -2.540444 |
| C    | -4.367277             | -4.663489 | 0.179122  | C    | 6.811644              | -3.603099 | 0.354587  |
| C    | -5.947119             | -3.158686 | 1.242133  | H    | 6.050603              | -5.456828 | 1.067132  |
| C    | -5.644441             | -3.066425 | -1.126887 | C    | 6.677004              | -2.500789 | -0.542101 |

|   |           |           |           |   |          |           |           |
|---|-----------|-----------|-----------|---|----------|-----------|-----------|
| C | -4.747078 | -4.121741 | -1.052366 | H | 5.794534 | -2.053412 | -2.48075  |
| C | -5.048515 | -4.215874 | 1.31591   | H | 7.543764 | -3.562783 | 1.148763  |
| H | -6.404826 | -2.794408 | 2.156932  | C | 7.2219   | 1.200064  | -0.135204 |
| H | -5.857627 | -2.60691  | -2.08544  | C | 8.135236 | 0.70398   | 0.958474  |
| H | -4.250156 | -4.454718 | -1.958086 | C | 6.789748 | 0.048869  | -1.049627 |
| H | -4.829647 | -4.660037 | 2.282395  | H | 9.151058 | 1.089259  | 0.788832  |
| C | -3.110145 | 5.502639  | -0.154281 | C | 8.143207 | -0.85878  | 0.849218  |
| C | -0.406384 | 6.383186  | 0.00739   | C | 7.759743 | 1.008656  | 2.434062  |
| C | -2.494686 | 5.754066  | 1.077263  | C | 7.193067 | -1.207477 | -0.270346 |
| C | -2.397979 | 5.886328  | -1.296179 | C | 7.617266 | 0.095828  | -2.366851 |
| C | -1.076717 | 6.297654  | -1.218459 | C | 5.25258  | 0.091706  | -1.32904  |
| C | -1.184985 | 6.200637  | 1.156238  | C | 7.81449  | -1.376765 | 2.27613   |
| H | -3.009576 | 5.510136  | 2.000091  | H | 9.155111 | -1.202322 | 0.589682  |
| H | -2.850877 | 5.790217  | -2.277004 | C | 8.216332 | -0.229791 | 3.197883  |
| H | -0.536302 | 6.461993  | -2.144125 | H | 8.229722 | 1.918993  | 2.813675  |
| H | -0.74343  | 6.330505  | 2.138372  | H | 6.675298 | 1.130654  | 2.529728  |
| C | -3.120533 | -5.465149 | 0.268507  | C | 6.737639 | 2.53306   | -0.272176 |
| C | -0.42139  | -6.329605 | 0.140833  | H | 7.414929 | 1.004145  | -2.939997 |
| C | -2.21999  | -5.185    | 1.300049  | H | 8.688666 | 0.075785  | -2.145404 |
| C | -2.68039  | -6.313537 | -0.753487 | H | 7.394219 | -0.764326 | -3.004008 |
| C | -1.359947 | -6.73987  | -0.813784 | H | 5.027805 | 0.046432  | -2.399673 |
| C | -0.902252 | -5.604538 | 1.236306  | H | 4.74116  | -0.742696 | -0.851309 |
| H | -2.521367 | -4.525623 | 2.107357  | H | 4.801364 | 1.003377  | -0.943943 |
| H | -3.367676 | -6.60485  | -1.542369 | H | 8.338096 | -2.303567 | 2.522899  |
| H | -1.044096 | -7.359447 | -1.647741 | H | 6.740612 | -1.571066 | 2.367062  |
| H | -0.209718 | -5.257377 | 1.995117  | H | 9.30641  | -0.211535 | 3.322124  |
| C | 1.07975   | 6.402196  | 0.062043  | H | 7.774669 | -0.306414 | 4.196259  |
| C | 3.815717  | 5.648933  | -0.069207 | C | 6.01601  | 2.964733  | -1.422593 |
| C | 1.740719  | 5.855461  | 1.168043  | C | 6.783266 | 3.484967  | 0.786528  |
| C | 1.871151  | 6.721415  | -1.04993  | C | 5.197706 | 4.073151  | -1.398618 |
| C | 3.204554  | 6.343726  | -1.119733 | H | 6.032548 | 2.36713   | -2.323694 |
| C | 3.076859  | 5.49729   | 1.108577  | C | 5.93758  | 4.577538  | 0.815668  |
| H | 1.427443  | 7.219539  | -1.905864 | H | 7.422048 | 3.30993   | 1.64002   |
| H | 3.764757  | 6.542791  | -2.028752 | C | 5.044833 | 4.842678  | -0.234831 |
| H | 3.517585  | 4.968895  | 1.947847  | H | 4.575631 | 4.285016  | -2.262883 |
| C | 1.045209  | -6.401154 | -0.079042 | H | 5.92862  | 5.213366  | 1.696664  |

## 2DR-8CPP (syn)



opt freq BS-UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 2.1356$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.121064 Hartree

Sum of electronic and thermal Energies = -2626.065274 Hartree

Sum of electronic and enthalpy Energies = -2626.064330 Hartree

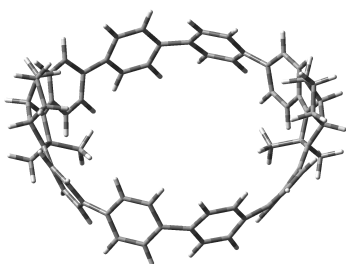
Sum of electronic and thermal Free Energies = -2626.211777 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.064009             | -2.731872 | -0.091296 | H    | 5.226035              | 5.499608  | 1.765974  |
| C    | -4.487553             | -5.139817 | -0.2391   | C    | 5.499554              | 3.202164  | -1.322746 |
| C    | -5.499563             | -3.202186 | -1.322873 | H    | 4.226168              | 4.586167  | -2.315163 |
| C    | -6.058847             | -3.707744 | 0.961056  | C    | 6.063985              | 2.731861  | -0.091157 |
| C    | -5.272237             | -4.843818 | 0.899761  | H    | 6.622386              | 3.518347  | 1.866371  |
| C    | -4.735808             | -4.351148 | -1.385063 | H    | 5.597806              | 2.614458  | -2.225231 |
| H    | -5.597811             | -2.614492 | -2.225366 | C    | 6.456147              | 1.385064  | 0.142765  |
| H    | -6.622436             | -3.518339 | 1.866233  | C    | 7.020377              | 0.902249  | 1.457914  |
| H    | -5.226076             | -5.499597 | 1.765875  | C    | 6.163794              | 0.202016  | -0.798659 |
| H    | -4.226159             | -4.586194 | -2.315259 | H    | 7.979478              | 1.397323  | 1.673215  |
| C    | -6.206083             | 2.376824  | -0.328358 | C    | 7.232964              | -0.645403 | 1.319067  |
| C    | -4.728834             | 4.841965  | -0.655491 | C    | 6.121651              | 1.060913  | 2.724038  |
| C    | -5.555229             | 2.727881  | -1.558758 | C    | 6.547205              | -1.045521 | 0.024906  |
| C    | -6.351798             | 3.486456  | 0.575     | C    | 7.055202              | 0.28203   | -2.074957 |
| C    | -5.620292             | 4.650599  | 0.42902   | C    | 4.638288              | 0.187918  | -1.133508 |
| C    | -4.837215             | 3.895247  | -1.702155 | C    | 6.698013              | -1.267337 | 2.654305  |
| H    | -5.561824             | 2.039575  | -2.3916   | H    | 8.30979               | -0.863425 | 1.240274  |
| H    | -6.997987             | 3.393322  | 1.438074  | C    | 6.574259              | -0.087691 | 3.626297  |
| H    | -5.709554             | 5.417331  | 1.194247  | H    | 6.21711               | 2.037779  | 3.203765  |
| H    | -4.26871              | 4.031576  | -2.61633  | H    | 5.066635              | 0.935032  | 2.448563  |
| C    | -3.289272             | -5.999107 | -0.195735 | C    | 6.206094              | -2.376828 | -0.328214 |
| C    | -0.553785             | -6.827047 | -0.274302 | H    | 6.82617               | 1.159748  | -2.685307 |
| C    | -2.7418               | -6.614346 | -1.340435 | H    | 8.111602              | 0.343673  | -1.792505 |
| C    | -2.489071             | -5.998301 | 0.963199  | H    | 6.935093              | -0.601546 | -2.70804  |
| C    | -1.158406             | -6.392939 | 0.921926  | H    | 4.375005              | -0.573317 | -1.867383 |
| C    | -1.414262             | -7.024861 | -1.374843 | H    | 4.063678              | -0.020525 | -0.225072 |
| H    | -3.35023              | -6.723246 | -2.23452  | H    | 4.302058              | 1.150044  | -1.518878 |
| H    | -2.871734             | -5.548312 | 1.874559  | H    | 7.351014              | -2.056274 | 3.041152  |
| H    | -0.551048             | -6.234245 | 1.806945  | H    | 5.71118               | -1.712476 | 2.491133  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.025716 | -7.450583 | -2.295609 | H | 7.551456  | 0.149534  | 4.069135  |
| C | -3.594628 | 5.782587  | -0.621735 | H | 5.881754  | -0.288728 | 4.451728  |
| C | -0.92223  | 6.792391  | -0.411687 | C | 6.351791  | -3.486468 | 0.575136  |
| C | -2.814589 | 6.051898  | -1.766575 | C | 5.555257  | -2.727868 | -1.558626 |
| C | -3.0743   | 6.246952  | 0.605313  | C | 5.620279  | -4.650604 | 0.429134  |
| C | -1.781137 | 6.745233  | 0.704971  | H | 6.997968  | -3.393346 | 1.438221  |
| C | -1.515754 | 6.529114  | -1.66296  | C | 4.83724   | -3.895228 | -1.702047 |
| H | -3.19552  | 5.805527  | -2.75284  | H | 5.561872  | -2.03955  | -2.391457 |
| H | -3.647818 | 6.115118  | 1.517784  | C | 4.728838  | -4.841955 | -0.655393 |
| H | -1.409167 | 7.024497  | 1.686378  | H | 5.709524  | -5.417344 | 1.194355  |
| H | -0.919761 | 6.593586  | -2.567865 | H | 4.268745  | -4.031543 | -2.616231 |
| C | 0.922225  | -6.792378 | -0.411676 | C | -6.547197 | 1.045516  | 0.024758  |
| C | 3.594629  | -5.782573 | -0.621664 | C | -6.163791 | -0.202026 | -0.7988   |
| C | 1.515773  | -6.52909  | -1.662935 | C | -7.232958 | 0.645403  | 1.318921  |
| C | 1.78111   | -6.745227 | 0.704999  | C | -6.456175 | -1.385074 | 0.142613  |
| C | 3.074275  | -6.246946 | 0.60537   | C | -7.055158 | -0.282029 | -2.075126 |
| C | 2.81461   | -6.051873 | -1.766522 | C | -4.638273 | -0.187956 | -1.133576 |
| H | 1.40912   | -7.024496 | 1.686398  | C | -7.020379 | -0.902249 | 1.457768  |
| H | 3.647775  | -6.115117 | 1.517853  | C | -6.698002 | 1.267335  | 2.654158  |
| H | 3.195559  | -5.805492 | -2.752777 | H | -8.309783 | 0.863433  | 1.240125  |
| C | 0.553778  | 6.827057  | -0.274284 | H | -6.826144 | -1.159772 | -2.685446 |
| C | 3.289261  | 5.999107  | -0.195666 | H | -8.111568 | -0.34362  | -1.792702 |
| C | 1.414275  | 7.024863  | -1.374811 | H | -6.934997 | 0.601522  | -2.708232 |
| C | 1.158376  | 6.39295   | 0.921956  | H | -4.374859 | 0.573626  | -1.867039 |
| C | 2.489039  | 5.998308  | 0.963254  | H | -4.063684 | 0.019948  | -0.225003 |
| C | 2.741811  | 6.614343  | -1.340378 | H | -4.302149 | -1.149932 | -1.519417 |
| H | 1.025747  | 7.450583  | -2.295586 | H | -7.979478 | -1.397317 | 1.673095  |
| H | 0.551002  | 6.23426   | 1.806964  | C | -6.12163  | -1.060912 | 2.723877  |
| H | 2.871683  | 5.54832   | 1.874621  | C | -6.574234 | 0.087685  | 3.626145  |
| H | 3.350257  | 6.723236  | -2.234454 | H | -7.351004 | 2.056269  | 3.04101   |
| H | 0.919797  | -6.593552 | -2.567852 | H | -5.711172 | 1.712478  | 2.490982  |
| C | 4.487538  | 5.139811  | -0.239007 | H | -6.21707  | -2.037783 | 3.203598  |
| C | 5.272205  | 4.84382   | 0.899867  | H | -5.06662  | -0.935018 | 2.448386  |
| C | 4.735805  | 4.351129  | -1.384959 | H | -7.551426 | -0.149546 | 4.068993  |
| C | 6.058811  | 3.707744  | 0.961184  | H | -5.881722 | 0.288723  | 4.451569  |



### 2DR-8CPP (syn)



opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 3.1305$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.121227 Hartree

Sum of electronic and thermal Energies = -2626.065949 Hartree

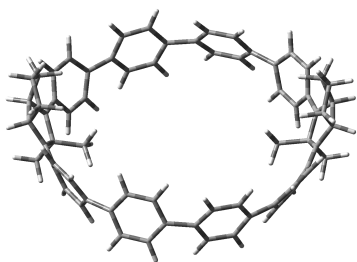
Sum of electronic and enthalpy Energies = -2626.065004 Hartree

Sum of electronic and thermal Free Energies = -2626.212122 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 6.193687              | 2.600237  | -0.066347 | H    | -5.391966             | -5.403309 | 1.741658  |
| C    | 4.619116              | 5.007163  | -0.234447 | C    | -5.584389             | -3.032646 | -1.298211 |
| C    | 5.595723              | 3.038942  | -1.293092 | H    | -4.282921             | -4.395305 | -2.288231 |
| C    | 6.202014              | 3.590437  | 0.970297  | C    | -6.192533             | -2.599959 | -0.075366 |
| C    | 5.416526              | 4.72773   | 0.898978  | H    | -6.784089             | -3.418745 | 1.8573    |
| C    | 4.836857              | 4.190356  | -1.366539 | H    | -5.639881             | -2.408287 | -2.180324 |
| H    | 5.658963              | 2.418273  | -2.177291 | C    | -6.602473             | -1.255212 | 0.152756  |
| H    | 6.773395              | 3.414143  | 1.872813  | C    | -7.2773               | -0.766751 | 1.413668  |
| H    | 5.382108              | 5.398104  | 1.754441  | C    | -6.292243             | -0.087745 | -0.797346 |
| H    | 4.30538               | 4.405703  | -2.289275 | H    | -8.281346             | -1.208963 | 1.505151  |
| C    | 6.202682              | -2.476421 | -0.314027 | C    | -7.392302             | 0.795257  | 1.281864  |
| C    | 4.623649              | -4.872836 | -0.654551 | C    | -6.540527             | -1.000173 | 2.770012  |
| C    | 5.488839              | -2.760759 | -1.524234 | C    | -6.623364             | 1.166656  | 0.030122  |
| C    | 6.321138              | -3.603619 | 0.567161  | C    | -7.261698             | -0.142681 | -2.024739 |
| C    | 5.543757              | -4.738194 | 0.412613  | C    | -4.785058             | -0.1115   | -1.247967 |
| C    | 4.72366               | -3.897891 | -1.674196 | C    | -6.887873             | 1.373874  | 2.64493   |
| H    | 5.473111              | -2.034451 | -2.32477  | H    | -8.450349             | 1.07603   | 1.159319  |
| H    | 6.985848              | -3.550126 | 1.419747  | C    | -6.961876             | 0.197392  | 3.624426  |
| H    | 5.620295              | -5.524603 | 1.159123  | H    | -6.794368             | -1.953996 | 3.239608  |
| H    | 4.110439              | -3.983776 | -2.565356 | H    | -5.454845             | -0.991107 | 2.611278  |
| C    | 3.431315              | 5.882323  | -0.195388 | C    | -6.201287             | 2.476833  | -0.314593 |
| C    | 0.708519              | 6.749185  | -0.27764  | H    | -7.110203             | -1.048085 | -2.62024  |
| C    | 2.891836              | 6.496067  | -1.344244 | H    | -8.30399              | -0.137758 | -1.688175 |
| C    | 2.632578              | 5.900405  | 0.964038  | H    | -7.11905              | 0.72315   | -2.679175 |
| C    | 1.307364              | 6.313717  | 0.920876  | H    | -4.678941             | -0.028705 | -2.335836 |
| C    | 1.570495              | 6.9263    | -1.380307 | H    | -4.227886             | 0.711874  | -0.798907 |
| H    | 3.501232              | 6.58847   | -2.239561 | H    | -4.293752             | -1.035459 | -0.944916 |
| H    | 3.01043               | 5.451761  | 1.878161  | H    | -7.478375             | 2.231199  | 2.983101  |
| H    | 0.698416              | 6.170042  | 1.807394  | H    | -5.849712             | 1.709818  | 2.547857  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 1.187504  | 7.351447  | -2.303631 | H | -7.993424 | 0.059353  | 3.976861  |
| C | 3.464481  | -5.787203 | -0.623442 | H | -6.330871 | 0.339107  | 4.509343  |
| C | 0.773159  | -6.739511 | -0.41472  | C | -6.316863 | 3.602169  | 0.570104  |
| C | 2.677608  | -6.029788 | -1.768141 | C | -5.488048 | 2.763737  | -1.525288 |
| C | 2.937035  | -6.245091 | 0.601622  | C | -5.537527 | 4.735064  | 0.418369  |
| C | 1.633187  | -6.717673 | 0.700501  | H | -6.98058  | 3.547115  | 1.423356  |
| C | 1.367588  | -6.478147 | -1.664829 | C | -4.723149 | 3.900624  | -1.673952 |
| H | 3.063535  | -5.784884 | -2.752966 | H | -5.473998 | 2.039256  | -2.327651 |
| H | 3.514959  | -6.128629 | 1.513525  | C | -4.618902 | 4.872375  | -0.650685 |
| H | 1.256     | -6.994805 | 1.680596  | H | -5.611619 | 5.519346  | 1.16734   |
| H | 0.767007  | -6.521572 | -2.56804  | H | -4.112497 | 3.989154  | -2.566603 |
| C | -0.768385 | 6.738942  | -0.414098 | C | 6.623065  | -1.16532  | 0.033582  |
| C | -3.460906 | 5.784992  | -0.619618 | C | 6.295789  | 0.090356  | -0.796942 |
| C | -1.366668 | 6.480873  | -1.66404  | C | 7.38639   | -0.795458 | 1.288739  |
| C | -1.627451 | 6.713791  | 0.702934  | C | 6.59916   | 1.256076  | 0.160687  |
| C | -2.93106  | 6.242107  | 0.60553   | C | 7.268505  | 0.14655   | -2.019172 |
| C | -2.675447 | 6.030714  | -1.765526 | C | 4.790165  | 0.115083  | -1.251372 |
| H | -1.249402 | 6.989182  | 1.683105  | C | 7.261251  | 0.764427  | 1.42669   |
| H | -3.507637 | 6.124996  | 1.518201  | C | 6.88261   | -1.383461 | 2.648253  |
| H | -3.063287 | 5.788399  | -2.75023  | H | 8.446724  | -1.068504 | 1.166949  |
| C | -0.70545  | -6.746926 | -0.27505  | H | 7.117953  | 1.050258  | -2.617629 |
| C | -3.42658  | -5.880594 | -0.195505 | H | 8.309616  | 0.144662  | -1.678714 |
| C | -1.570557 | -6.942625 | -1.370816 | H | 7.131105  | -0.720496 | -2.673336 |
| C | -1.296989 | -6.288249 | 0.917408  | H | 4.685901  | 0.018977  | -2.338315 |
| C | -2.62267  | -5.874685 | 0.959378  | H | 4.227979  | -0.699859 | -0.792923 |
| C | -2.893051 | -6.51213  | -1.336137 | H | 4.301908  | 1.044904  | -0.961713 |
| H | -1.190023 | -7.381757 | -2.288723 | H | 8.261481  | 1.212098  | 1.532979  |
| H | -0.682667 | -6.127149 | 1.797252  | C | 6.50735   | 0.985613  | 2.775433  |
| H | -2.997119 | -5.408931 | 1.866361  | C | 6.935817  | -0.208416 | 3.631099  |
| H | -3.506318 | -6.617504 | -2.227458 | H | 7.482401  | -2.234015 | 2.987312  |
| H | -0.769517 | 6.528885  | -2.569176 | H | 5.849434  | -1.732441 | 2.545291  |
| C | -4.617276 | -5.005478 | -0.240338 | H | 6.743053  | 1.941937  | 3.249255  |
| C | -5.421947 | -4.730797 | 0.887649  | H | 5.423655  | 0.962179  | 2.604829  |
| C | -4.823547 | -4.183847 | -1.369815 | H | 7.962927  | -0.059005 | 3.991757  |
| C | -6.207842 | -3.592609 | 0.957503  | H | 6.299792  | -0.359415 | 4.510869  |

### 2DR-8CPP (syn)



opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 5,  $\langle S^2 \rangle = 6.1318$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.120161 Hartree

Sum of electronic and thermal Energies = -2626.064266 Hartree

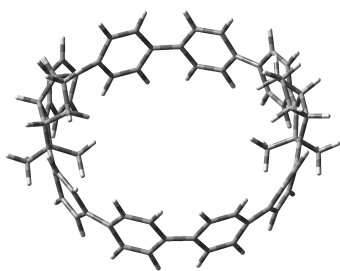
Sum of electronic and enthalpy Energies = -2626.063322 Hartree

Sum of electronic and thermal Free Energies = -2626.213796 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.306495             | -2.563789 | -0.098981 | H    | 5.420628              | 5.276403  | 1.81362   |
| C    | -4.685206             | -4.948443 | -0.195009 | C    | 5.715013              | 3.042575  | -1.312639 |
| C    | -5.714985             | -3.042612 | -1.312586 | H    | 4.412771              | 4.423714  | -2.276022 |
| C    | -6.286047             | -3.517582 | 0.969574  | C    | 6.306555              | 2.563798  | -0.099027 |
| C    | -5.4772               | -4.641951 | 0.932364  | H    | 6.843901              | 3.323114  | 1.874547  |
| C    | -4.931818             | -4.180793 | -1.352836 | H    | 5.821609              | 2.474996  | -2.22684  |
| H    | -5.821631             | -2.475081 | -2.226812 | C    | 6.743228              | 1.213943  | 0.082169  |
| H    | -6.843714             | -3.322982 | 1.874672  | C    | 7.533278              | 0.710353  | 1.271983  |
| H    | -5.420407             | -5.276249 | 1.813787  | C    | 6.345134              | 0.049385  | -0.846254 |
| H    | -4.412792             | -4.423805 | -2.275965 | H    | 8.565938              | 1.088536  | 1.209917  |
| C    | -6.259357             | 2.526484  | -0.365634 | C    | 7.54456               | -0.859548 | 1.169383  |
| C    | -4.612665             | 4.890912  | -0.632693 | C    | 7.011497              | 1.024061  | 2.711159  |
| C    | -5.560729             | 2.847045  | -1.575854 | C    | 6.718068              | -1.215898 | -0.049193 |
| C    | -6.333916             | 3.623233  | 0.557157  | C    | 7.174359              | 0.09518   | -2.168884 |
| C    | -5.523788             | 4.739286  | 0.43897   | C    | 4.806696              | 0.094677  | -1.10236  |
| C    | -4.762516             | 3.965848  | -1.691325 | C    | 7.068973              | -1.372869 | 2.566899  |
| H    | -5.601888             | 2.173613  | -2.420116 | H    | 8.579208              | -1.207747 | 1.021799  |
| H    | -6.984864             | 3.55648   | 1.418034  | C    | 7.377468              | -0.219409 | 3.524624  |
| H    | -5.567661             | 5.494129  | 1.219729  | H    | 7.452233              | 1.931052  | 3.134815  |
| H    | -4.170805             | 4.075583  | -2.594348 | H    | 5.923419              | 1.159603  | 2.697283  |
| C    | -3.48148              | -5.80429  | -0.133077 | C    | 6.259395              | -2.526501 | -0.365569 |
| C    | -0.757412             | -6.662937 | -0.19986  | H    | 6.954824              | 0.985792  | -2.763941 |
| C    | -2.951632             | -6.47026  | -1.255603 | H    | 8.245763              | 0.106363  | -1.941362 |
| C    | -2.66866              | -5.750958 | 1.014276  | H    | 6.981704              | -0.77823  | -2.798241 |
| C    | -1.341932             | -6.161209 | 0.978449  | H    | 4.468698              | -0.699566 | -1.766116 |
| C    | -1.627413             | -6.896243 | -1.284074 | H    | 4.274446              | -0.023284 | -0.152762 |
| H    | -3.570215             | -6.609923 | -2.138476 | H    | 4.493841              | 1.04356   | -1.53576  |
| H    | -3.03896              | -5.254586 | 1.906526  | H    | 7.566007              | -2.298902 | 2.870402  |
| H    | -0.721856             | -5.967068 | 1.847551  | H    | 5.990525              | -1.568715 | 2.548959  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.25024  | -7.365645 | -2.188353 | H | 8.449126  | -0.206942 | 3.767093  |
| C | -3.428494 | 5.771681  | -0.577753 | H | 6.829609  | -0.289812 | 4.471319  |
| C | -0.720456 | 6.670767  | -0.342496 | C | 6.333939  | -3.623246 | 0.557235  |
| C | -2.633125 | 6.022386  | -1.714787 | C | 5.560853  | -2.847098 | -1.575836 |
| C | -2.894028 | 6.189423  | 0.658633  | C | 5.523842  | -4.739313 | 0.439021  |
| C | -1.582336 | 6.636012  | 0.77072   | H | 6.984852  | -3.556464 | 1.418137  |
| C | -1.315461 | 6.444949  | -1.598985 | C | 4.76268   | -3.965923 | -1.691338 |
| H | -3.0196   | 5.805659  | -2.705874 | H | 5.602073  | -2.173688 | -2.420115 |
| H | -3.475669 | 6.064157  | 1.566849  | C | 4.61278   | -4.890967 | -0.632693 |
| H | -1.202206 | 6.884958  | 1.757248  | H | 5.5677    | -5.49416  | 1.219777  |
| H | -0.711184 | 6.497116  | -2.499352 | H | 4.171027  | -4.075697 | -2.594396 |
| C | 0.720551  | -6.670755 | -0.342586 | C | -6.718114 | 1.215895  | -0.049296 |
| C | 3.428612  | -5.771734 | -0.577816 | C | -6.345141 | -0.049404 | -0.846324 |
| C | 1.315524  | -6.444799 | -1.599066 | C | -7.544738 | 0.859562  | 1.169194  |
| C | 1.582467  | -6.636169 | 0.770606  | C | -6.743228 | -1.213933 | 0.082137  |
| C | 2.89417   | -6.189605 | 0.658536  | C | -7.17437  | -0.095251 | -2.16895  |
| C | 2.633196  | -6.02226  | -1.714855 | C | -4.806707 | -0.094678 | -1.102466 |
| H | 1.20236   | -6.885234 | 1.757113  | C | -7.533491 | -0.710338 | 1.271802  |
| H | 3.475842  | -6.064471 | 1.56675   | C | -7.069314 | 1.372865  | 2.566769  |
| H | 3.019642  | -5.805401 | -2.705926 | H | -8.579366 | 1.207777  | 1.021504  |
| C | 0.757506  | 6.662969  | -0.199813 | H | -6.954663 | -0.985763 | -2.764095 |
| C | 3.481582  | 5.804326  | -0.133138 | H | -8.245774 | -0.106656 | -1.941437 |
| C | 1.627473  | 6.896309  | -1.284047 | H | -6.981881 | 0.778255  | -2.798227 |
| C | 1.34207   | 6.161221  | 0.978466  | H | -4.468845 | 0.699086  | -1.766868 |
| C | 2.6688    | 5.750976  | 1.014242  | H | -4.274414 | 0.024078  | -0.152995 |
| C | 2.951697  | 6.470334  | -1.255627 | H | -4.49376  | -1.04385  | -1.535164 |
| H | 1.250273  | 7.365732  | -2.188305 | H | -8.566142 | -1.08851  | 1.209512  |
| H | 0.722028  | 5.967062  | 1.847589  | C | -7.012016 | -1.024067 | 2.711083  |
| H | 3.039131  | 5.254587  | 1.906471  | C | -7.378064 | 0.219444  | 3.524457  |
| H | 3.570255  | 6.610043  | -2.13851  | H | -7.56631  | 2.298947  | 2.870181  |
| H | 0.711215  | -6.496833 | -2.49942  | H | -5.990843 | 1.568605  | 2.54899   |
| C | 4.685305  | 4.948479  | -0.195112 | H | -7.452918 | -1.931021 | 3.134655  |
| C | 5.477364  | 4.642046  | 0.932233  | H | -5.923948 | -1.159689 | 2.697438  |
| C | 4.931854  | 4.18076   | -1.352911 | H | -8.449768 | 0.207043  | 3.766726  |
| C | 6.286187  | 3.517665  | 0.96947   | H | -6.830376 | 0.289824  | 4.471253  |

### 2DR-8CPP (syn)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.080273 Hartree

Sum of electronic and thermal Energies = -2626.024570 Hartree

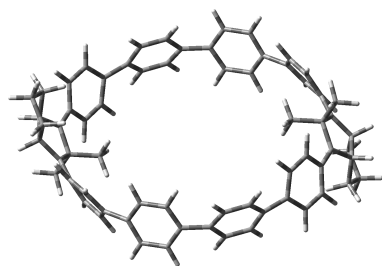
Sum of electronic and enthalpy Energies = -2626.023626 Hartree

Sum of electronic and thermal Free Energies = -2626.169656 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -5.898183             | -2.815475 | 0.004705  | H    | 4.860094              | 5.645899  | 1.660011  |
| C    | -4.327284             | -5.208899 | -0.417272 | C    | 5.551174              | 3.290457  | -1.324856 |
| C    | -5.551095             | -3.290052 | -1.325506 | H    | 4.503645              | 4.664908  | -2.527744 |
| C    | -5.758764             | -3.84228  | 1.027158  | C    | 5.898199              | 2.815512  | 0.005238  |
| C    | -4.990556             | -4.953056 | 0.832966  | H    | 6.225161              | 3.689399  | 1.995101  |
| C    | -4.804834             | -4.416194 | -1.516235 | H    | 5.83894               | 2.715492  | -2.19543  |
| H    | -5.838807             | -2.714849 | -2.195938 | C    | 6.194067              | 1.504681  | 0.335609  |
| H    | -6.225123             | -3.689894 | 1.994333  | C    | 6.432429              | 1.059129  | 1.759496  |
| H    | -4.860028             | -5.646288 | 1.658745  | C    | 6.146822              | 0.291081  | -0.610847 |
| H    | -4.503465             | -4.664135 | -2.528727 | H    | 7.290371              | 1.589109  | 2.198582  |
| C    | -6.17476              | 2.251638  | -0.055218 | C    | 6.711694              | -0.477641 | 1.720666  |
| C    | -4.829085             | 4.771433  | -0.533099 | C    | 5.231935              | 1.214884  | 2.741087  |
| C    | -5.812619             | 2.700525  | -1.392177 | C    | 6.357086              | -0.929757 | 0.311424  |
| C    | -6.201485             | 3.342447  | 0.910893  | C    | 7.298636              | 0.370847  | -1.656389 |
| C    | -5.532687             | 4.512314  | 0.694062  | C    | 4.742791              | 0.227372  | -1.287256 |
| C    | -5.175196             | 3.885885  | -1.610221 | C    | 5.895015              | -1.085741 | 2.913061  |
| H    | -6.013271             | 2.066349  | -2.24526  | H    | 7.782648              | -0.662878 | 1.896304  |
| H    | -6.712608             | 3.201899  | 1.855248  | C    | 5.478595              | 0.115232  | 3.774306  |
| H    | -5.519133             | 5.257165  | 1.48462   | H    | 5.161911              | 2.214597  | 3.176632  |
| H    | -4.860593             | 4.119049  | -2.622182 | H    | 4.29207               | 1.027874  | 2.205966  |
| C    | -3.158994             | -6.022633 | -0.529005 | C    | 6.174691              | -2.25162  | -0.056067 |
| C    | -0.384978             | -6.817588 | -0.666391 | H    | 7.201677              | 1.24756   | -2.302874 |
| C    | -2.49444              | -6.24791  | -1.776768 | H    | 8.266698              | 0.43943   | -1.14917  |
| C    | -2.442945             | -6.477594 | 0.623111  | H    | 7.323061              | -0.514856 | -2.297612 |
| C    | -1.122598             | -6.850299 | 0.556945  | H    | 4.651417              | -0.61551  | -1.972365 |
| C    | -1.178549             | -6.629029 | -1.840663 | H    | 3.967908              | 0.114268  | -0.52229  |
| H    | -3.020896             | -6.06409  | -2.707627 | H    | 4.524367              | 1.139642  | -1.843222 |
| H    | -2.911505             | -6.425688 | 1.60036   | H    | 6.470358              | -1.823089 | 3.481887  |
| H    | -0.615074             | -7.086083 | 1.486209  | H    | 5.003469              | -1.594422 | 2.532712  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.723779 | -6.724594 | -2.820848 | H | 6.297549  | 0.409694  | 4.445045  |
| C | -3.740007 | 5.688919  | -0.634595 | H | 4.603881  | -0.096004 | 4.399936  |
| C | -1.052366 | 6.746611  | -0.693843 | C | 6.20161   | -3.342722 | 0.909702  |
| C | -3.039666 | 5.915798  | -1.862084 | C | 5.812388  | -2.700121 | -1.393103 |
| C | -3.128181 | 6.268192  | 0.522097  | C | 5.532847  | -4.512561 | 0.692608  |
| C | -1.84758  | 6.764016  | 0.493603  | H | 6.712856  | -3.202447 | 1.854023  |
| C | -1.765903 | 6.423307  | -1.889454 | C | 5.175018  | -3.885463 | -1.611428 |
| H | -3.496653 | 5.628096  | -2.803135 | H | 6.012843  | -2.065672 | -2.246027 |
| H | -3.639872 | 6.22337   | 1.478005  | C | 4.829101  | -4.77135  | -0.534537 |
| H | -1.414231 | 7.107046  | 1.427213  | H | 5.519442  | -5.257655 | 1.482937  |
| H | -1.271199 | 6.500321  | -2.851549 | H | 4.860267  | -4.118322 | -2.623415 |
| C | 1.052406  | -6.746527 | -0.695511 | C | -6.35717  | 0.929696  | 0.311894  |
| C | 3.740047  | -5.688855 | -0.636179 | C | -6.146946 | -0.290888 | -0.610719 |
| C | 1.765878  | -6.422971 | -1.89109  | C | -6.711757 | 0.477209  | 1.721016  |
| C | 1.847684  | -6.764196 | 0.491887  | C | -6.194129 | -1.504745 | 0.335419  |
| C | 3.128291  | -6.268389 | 0.520416  | C | -7.298803 | -0.370332 | -1.656219 |
| C | 3.039642  | -5.91546  | -1.863678 | C | -4.742944 | -0.226953 | -1.287179 |
| H | 1.414388  | -7.107425 | 1.425447  | C | -6.43246  | -1.059574 | 1.759434  |
| H | 3.640027  | -6.223767 | 1.476309  | C | -5.895034 | 1.085024  | 2.913512  |
| H | 3.496581  | -5.627547 | -2.804689 | H | -7.782706 | 0.662375  | 1.896749  |
| C | 0.385013  | 6.817696  | -0.664773 | H | -7.201902 | -1.246862 | -2.302959 |
| C | 3.159061  | 6.022796  | -0.52764  | H | -8.266843 | -0.439033 | -1.148975 |
| C | 1.178555  | 6.629405  | -1.839111 | H | -7.323233 | 0.515557  | -2.297193 |
| C | 1.122671  | 6.850169  | 0.558547  | H | -4.65172  | 0.615973  | -1.97226  |
| C | 2.443031  | 6.477496  | 0.624594  | H | -3.96805  | -0.113764 | -0.522238 |
| C | 2.494458  | 6.24831   | -1.775335 | H | -4.524402 | -1.139162 | -1.843199 |
| H | 0.723754  | 6.725154  | -2.819263 | H | -7.290393 | -1.589672 | 2.198394  |
| H | 0.615164  | 7.085741  | 1.487876  | C | -5.231951 | -1.215562 | 2.740963  |
| H | 2.911619  | 6.425405  | 1.601818  | C | -5.478574 | -0.116153 | 3.774457  |
| H | 3.020879  | 6.064671  | -2.70625  | H | -6.470321 | 1.822253  | 3.482541  |
| H | 1.271119  | -6.499778 | -2.853174 | H | -5.003503 | 1.593791  | 2.533223  |
| C | 4.327365  | 5.209066  | -0.416129 | H | -5.161906 | -2.21538  | 3.17626   |
| C | 4.990635  | 4.952892  | 0.834045  | H | -4.292097 | -1.028418 | 2.205867  |
| C | 4.804949  | 4.416668  | -1.515305 | H | -6.297517 | -0.410782 | 4.445138  |
| C | 5.758826  | 3.842056  | 1.027957  | H | -4.603853 | 0.094937  | 4.400125  |

### 2DR-8CPP (anti)



opt freq BS-UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 2.1213$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.122498 Hartree

Sum of electronic and thermal Energies = -2626.066811 Hartree

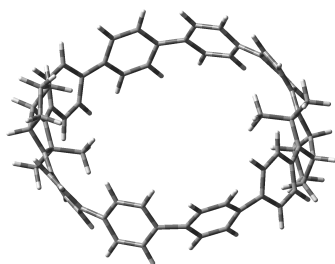
Sum of electronic and enthalpy Energies = -2626.065867 Hartree

Sum of electronic and thermal Free Energies = -2626.213517 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 6.402318              | -2.535434 | -0.51195  | H    | -5.615138             | 5.257661  | -1.42778  |
| C    | 4.671176              | -4.846056 | -0.475416 | C    | -5.652901             | 2.954028  | 1.658624  |
| C    | 6.455529              | -3.514401 | 0.532352  | H    | -4.19102              | 4.250705  | 2.500533  |
| C    | 5.653213              | -2.954295 | -1.658533 | C    | -6.402289             | 2.535344  | 0.512175  |
| C    | 4.818643              | -4.055589 | -1.635185 | H    | -7.114226             | 3.368806  | -1.375739 |
| C    | 5.598243              | -4.601998 | 0.560899  | H    | -5.69125              | 2.371136  | 2.569032  |
| H    | 7.114375              | -3.368907 | 1.375918  | C    | -6.931052             | 1.213662  | 0.365479  |
| H    | 5.691622              | -2.37145  | -2.568964 | C    | -7.963265             | 0.786333  | -0.655788 |
| H    | 4.191652              | -4.251213 | -2.50061  | C    | -6.455037             | -0.000373 | 1.193797  |
| H    | 5.61553               | -5.257985 | 1.427835  | H    | -8.952995             | 1.147365  | -0.331525 |
| C    | 6.401956              | 2.536096  | -0.511885 | C    | -7.962937             | -0.786959 | -0.656114 |
| C    | 4.670494              | 4.846481  | -0.475599 | C    | -7.794577             | 1.197477  | -2.151829 |
| C    | 6.455043              | 3.51519   | 0.532305  | C    | -6.930582             | -1.214299 | 0.365026  |
| C    | 5.652795              | 2.954732  | -1.65851  | C    | -7.147634             | -0.000807 | 2.59258   |
| C    | 4.818057              | 4.055902  | -1.635278 | C    | -4.902673             | 0.000026  | 1.306404  |
| C    | 5.597612              | 4.602676  | 0.560728  | C    | -7.794085             | -1.197401 | -2.152333 |
| H    | 7.113926              | 3.369907  | 1.375872  | H    | -8.952533             | -1.148502 | -0.332012 |
| H    | 5.691277              | 2.371777  | -2.568872 | C    | -8.341877             | 0.00009   | -2.933242 |
| H    | 4.191024              | 4.251332  | -2.500718 | H    | -8.319305             | 2.125835  | -2.397395 |
| H    | 5.614819              | 5.258769  | 1.427586  | H    | -6.733796             | 1.339307  | -2.389417 |
| C    | 3.460715              | -5.673284 | -0.298115 | C    | -6.401723             | -2.535962 | 0.511621  |
| C    | 0.741296              | -6.504507 | -0.035564 | H    | -6.878614             | 0.881715  | 3.180724  |
| C    | 2.817589              | -5.691968 | 0.95426   | H    | -8.237161             | -0.001014 | 2.478109  |
| C    | 2.764262              | -6.252168 | -1.378131 | H    | -6.87824              | -0.883474 | 3.180341  |
| C    | 1.441755              | -6.661599 | -1.248251 | H    | -4.526652             | -0.878309 | 1.828551  |
| C    | 1.492771              | -6.090998 | 1.080171  | H    | -4.461668             | 0.000219  | 0.30425   |
| H    | 3.318021              | -5.262128 | 1.816913  | H    | -4.527146             | 0.878513  | 1.828667  |
| H    | 3.250381              | -6.338427 | -2.346381 | H    | -8.318439             | -2.125866 | -2.398294 |
| H    | 0.928329              | -7.061957 | -2.118245 | H    | -6.733248             | -1.338694 | -2.389988 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 1.000707  | -5.957763 | 2.038738  | H | -9.440282 | -0.00014  | -2.905583 |
| C | 3.459915  | 5.67355   | -0.298353 | H | -8.040609 | 0.000374  | -3.987006 |
| C | 0.74041   | 6.504468  | -0.035737 | C | -6.454507 | -3.514743 | -0.532863 |
| C | 2.816859  | 5.692297  | 0.954055  | C | -5.652659 | -2.954798 | 1.658226  |
| C | 2.763325  | 6.252216  | -1.378399 | C | -5.596995 | -4.602167 | -0.561421 |
| C | 1.440774  | 6.661502  | -1.248487 | H | -7.113223 | -3.369274 | -1.376528 |
| C | 1.492001  | 6.091176  | 1.079998  | C | -4.817819 | -4.055895 | 1.634842  |
| H | 3.317401  | 5.262631  | 1.816732  | H | -5.691348 | -2.372083 | 2.568734  |
| H | 3.249379  | 6.33843   | -2.346685 | C | -4.670054 | -4.846198 | 0.475004  |
| H | 0.927249  | 7.061706  | -2.118494 | H | -5.614009 | -5.258029 | -1.428458 |
| H | 1.000012  | 5.958005  | 2.038613  | H | -4.190908 | -4.251502 | 2.50033   |
| C | -0.740041 | -6.504477 | 0.03532   | C | 6.930733  | -1.213656 | -0.365063 |
| C | -3.459481 | -5.673284 | 0.297779  | C | 6.454684  | 0.00037   | -1.19339  |
| C | -1.44054  | -6.661581 | 1.247982  | C | 7.962562  | -0.786251 | 0.656571  |
| C | -1.491471 | -6.090929 | -1.08043  | C | 6.930516  | 1.21438   | -0.364924 |
| C | -2.816292 | -5.691905 | -0.954562 | C | 7.146926  | 0.000545  | -2.592344 |
| C | -2.763062 | -6.252172 | 1.377815  | C | 4.902269  | 0.000275  | -1.305541 |
| H | -0.999371 | -5.95768  | -2.038977 | C | 7.962408  | 0.787037  | 0.656688  |
| H | -3.316698 | -5.262064 | -1.81723  | C | 7.793113  | -1.197208 | 2.152586  |
| H | -3.249228 | -6.338476 | 2.346037  | H | 8.9524    | -1.147421 | 0.332803  |
| C | -0.740921 | 6.504308  | 0.035272  | H | 6.877517  | 0.883181  | -3.180142 |
| C | -3.460259 | 5.672872  | 0.298038  | H | 8.236479  | 0.000613  | -2.478132 |
| C | -1.441307 | 6.661375  | 1.248003  | H | 6.877647  | -0.882011 | -3.180318 |
| C | -1.492431 | 6.090665  | -1.08039  | H | 4.526369  | -0.878193 | -1.827549 |
| C | -2.817209 | 5.69154   | -0.954375 | H | 4.461579  | 0.000331  | -0.303248 |
| C | -2.76377  | 6.251832  | 1.37799   | H | 4.526277  | 0.878625  | -1.827689 |
| H | -0.927857 | 7.061815  | 2.117946  | H | 8.952185  | 1.148432  | 0.332986  |
| H | -1.00042  | 5.957436  | -2.038985 | C | 7.792881  | 1.197703  | 2.15278   |
| H | -3.317669 | 5.261637  | -1.816982 | C | 8.340064  | 0.000236  | 2.934146  |
| H | -3.249835 | 6.338103  | 2.346266  | H | 8.317711  | -2.125537 | 2.398537  |
| H | -0.927149 | -7.061963 | 2.117986  | H | 6.73221   | -1.339025 | 2.389641  |
| C | -4.670731 | 4.845671  | 0.475424  | H | 8.317304  | 2.126086  | 2.398894  |
| C | -5.597889 | 4.6017    | -0.560824 | H | 6.731953  | 1.33927   | 2.389867  |
| C | -4.818154 | 4.055193  | 1.635186  | H | 9.438486  | 0.000343  | 2.907127  |
| C | -6.455343 | 3.514233  | -0.532193 | H | 8.03819   | 0.00012   | 3.987736  |



**2DR-8CPP (anti)**

opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 2.0674$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.100709 Hartree

Sum of electronic and thermal Energies = -2626.045151 Hartree

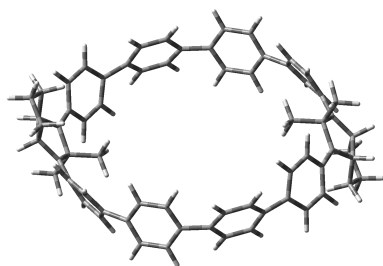
Sum of electronic and enthalpy Energies = -2626.044207 Hartree

Sum of electronic and thermal Free Energies = -2626.190671 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.108875             | -2.553095 | 0.596874  | H    | 5.595445              | 5.406712  | 1.433863  |
| C    | -4.545191             | -4.982703 | 0.75688   | C    | 5.393971              | 2.996082  | -1.569785 |
| C    | -6.268551             | -3.641683 | -0.356049 | H    | 4.0491                | 4.391313  | -2.445737 |
| C    | -5.444448             | -2.969287 | 1.821748  | C    | 6.091255              | 2.542222  | -0.401425 |
| C    | -4.705925             | -4.114931 | 1.889109  | H    | 6.92732               | 3.384331  | 1.432144  |
| C    | -5.49896              | -4.768112 | -0.296785 | H    | 5.368973              | 2.383665  | -2.46013  |
| H    | -6.968512             | -3.52775  | -1.17612  | C    | 6.465921              | 1.187573  | -0.178281 |
| H    | -5.483338             | -2.335967 | 2.698306  | C    | 7.200856              | 0.698186  | 1.056995  |
| H    | -4.161177             | -4.318478 | 2.805152  | C    | 6.074896              | 0.00666   | -1.089361 |
| H    | -5.601353             | -5.510954 | -1.08278  | H    | 8.266118              | 0.966243  | 0.975273  |
| C    | -6.091028             | 2.542578  | 0.401492  | C    | 7.048795              | -0.860496 | 1.05611   |
| C    | -4.587594             | 5.002926  | 0.440493  | C    | 6.681089              | 1.169513  | 2.459713  |
| C    | -6.254479             | 3.556387  | -0.602382 | C    | 6.435748              | -1.247591 | -0.271142 |
| C    | -5.393671             | 2.996448  | 1.5698    | C    | 6.903688              | 0.054308  | -2.408243 |
| C    | -4.662485             | 4.168418  | 1.577456  | C    | 4.536777              | 0.008077  | -1.356082 |
| C    | -5.504101             | 4.718649  | -0.59704  | C    | 6.20828               | -1.178093 | 2.330617  |
| H    | -6.927051             | 3.384627  | -1.432124 | H    | 8.033179              | -1.335259 | 1.181904  |
| H    | -5.36872              | 2.384097  | 2.460189  | C    | 6.643263              | -0.098774 | 3.321712  |
| H    | -4.048675             | 4.391631  | 2.445637  | H    | 7.310246              | 1.949244  | 2.900551  |
| H    | -5.595004             | 5.406904  | -1.434003 | H    | 5.670536              | 1.580752  | 2.366167  |
| C    | -3.404591             | -5.835689 | 0.622596  | C    | 6.10865               | -2.553465 | -0.596687 |
| C    | -0.706507             | -6.742212 | 0.14424   | H    | 6.726873              | 0.979339  | -2.964176 |
| C    | -2.991557             | -6.342321 | -0.649303 | H    | 7.974242              | 0.003841  | -2.183196 |
| C    | -2.476172             | -6.051223 | 1.688637  | H    | 6.666013              | -0.782072 | -3.071363 |
| C    | -1.195636             | -6.491963 | 1.461472  | H    | 4.218791              | -0.90026  | -1.868052 |
| C    | -1.702734             | -6.760965 | -0.87812  | H    | 3.99721               | 0.054513  | -0.404525 |
| H    | -3.675911             | -6.301147 | -1.490536 | H    | 4.216507              | 0.860168  | -1.955231 |
| H    | -2.760396             | -5.805652 | 2.706617  | H    | 6.362142              | -2.193092 | 2.704403  |
| H    | -0.51801              | -6.54781  | 2.306815  | H    | 5.139662              | -1.075279 | 2.102627  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.432367 | -7.062687 | -1.885139 | H | 7.645157  | -0.328889 | 3.709783  |
| C | -3.431738 | 5.907335  | 0.274574  | H | 5.972416  | -0.005787 | 4.183463  |
| C | -0.741468 | 6.827999  | 0.029038  | C | 6.268315  | -3.642173 | 0.356103  |
| C | -2.791484 | 5.971185  | -0.977574 | C | 5.444264  | -2.969519 | -1.821625 |
| C | -2.760938 | 6.503612  | 1.361227  | C | 5.498686  | -4.76857  | 0.296713  |
| C | -1.452204 | 6.958351  | 1.238997  | H | 6.968309  | -3.528375 | 1.176161  |
| C | -1.479304 | 6.412867  | -1.094988 | C | 4.705684  | -4.115118 | -1.88911  |
| H | -3.275895 | 5.535553  | -1.846501 | H | 5.483183  | -2.336095 | -2.698109 |
| H | -3.250785 | 6.55963   | 2.329902  | C | 4.544903  | -4.982996 | -0.75697  |
| H | -0.954127 | 7.363521  | 2.115652  | H | 5.601052  | -5.511507 | 1.082621  |
| H | -0.982044 | 6.30455   | -2.053886 | H | 4.160912  | -4.318531 | -2.805167 |
| C | 0.706092  | -6.742265 | -0.144501 | C | -6.435954 | -1.247198 | 0.271423  |
| C | 3.404249  | -5.835926 | -0.62277  | C | -6.074994 | 0.007052  | 1.08961   |
| C | 1.195251  | -6.491965 | -1.461712 | C | -7.049031 | -0.860099 | -1.055806 |
| C | 1.70231   | -6.761176 | 0.877865  | C | -6.465863 | 1.18796   | 0.178454  |
| C | 2.991166  | -6.342616 | 0.649088  | C | -6.903847 | 0.054899  | 2.408445  |
| C | 2.475824  | -6.051311 | -1.688836 | C | -4.536882 | 0.008315  | 1.356408  |
| H | 1.431906  | -7.062943 | 1.884861  | C | -7.200867 | 0.698599  | -1.056787 |
| H | 3.675514  | -6.301543 | 1.49033   | C | -6.208615 | -1.177929 | -2.330314 |
| H | 2.760077  | -5.805687 | -2.706794 | H | -8.033484 | -1.334737 | -1.181515 |
| C | 0.741994  | 6.827966  | -0.0293   | H | -6.727092 | 0.980049  | 2.964203  |
| C | 3.432227  | 5.907178  | -0.274765 | H | -7.97439  | 0.004347  | 2.183359  |
| C | 1.45274   | 6.95821   | -1.239264 | H | -6.666166 | -0.781342 | 3.071738  |
| C | 1.479808  | 6.412882  | 1.094757  | H | -4.21896  | -0.900221 | 1.868059  |
| C | 2.791968  | 5.971135  | 0.977376  | H | -3.997266 | 0.055085  | 0.404895  |
| C | 2.761456  | 6.503409  | -1.36146  | H | -4.216605 | 0.860164  | 1.9559    |
| H | 0.954682  | 7.363338  | -2.115949 | H | -8.266089 | 0.966815  | -0.975079 |
| H | 0.982537  | 6.304645  | 2.053658  | C | -6.681028 | 1.16976   | -2.459538 |
| H | 3.276351  | 5.535529  | 1.846332  | C | -6.643425 | -0.098589 | -3.321461 |
| H | 3.251304  | 6.559332  | -2.33014  | H | -6.36268  | -2.192931 | -2.704022 |
| H | 0.517626  | -6.547695 | -2.307063 | H | -5.139975 | -1.0753   | -2.102351 |
| C | 4.588031  | 5.002692  | -0.440621 | H | -7.31007  | 1.949569  | -2.900409 |
| C | 5.504503  | 4.718414  | 0.596942  | H | -5.670412 | 1.580845  | -2.366025 |
| C | 4.662883  | 4.168112  | -1.577534 | H | -7.64536  | -0.328555 | -3.709513 |
| C | 6.254786  | 3.556091  | 0.602374  | H | -5.972567 | -0.005771 | -4.183223 |

### 2DR-8CPP (anti)



opt freq UB3LYP/6-31G(d)

Charge = 0, Multiplicity = 5,  $\langle S^2 \rangle = 6.1235$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.122295 Hartree

Sum of electronic and thermal Energies = -2626.066649 Hartree

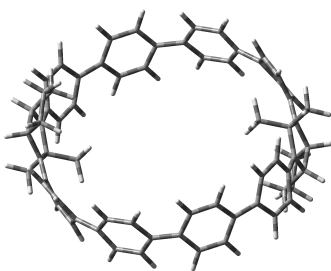
Sum of electronic and enthalpy Energies = -2626.065704 Hartree

Sum of electronic and thermal Free Energies = -2626.214253 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 6.397351              | -2.537151 | -0.514593 | H    | -5.620362             | 5.263632  | -1.423763 |
| C    | 4.669879              | -4.849846 | -0.47571  | C    | -5.64321              | 2.953637  | 1.658171  |
| C    | 6.455719              | -3.517404 | 0.527496  | H    | -4.177479             | 4.250019  | 2.49535   |
| C    | 5.64336               | -2.953462 | -1.65824  | C    | -6.397182             | 2.537133  | 0.514604  |
| C    | 4.809676              | -4.055927 | -1.633394 | H    | -7.116528             | 3.371492  | -1.36952  |
| C    | 5.600158              | -4.607007 | 0.557176  | H    | -5.675338             | 2.368228  | 2.567167  |
| H    | 7.117129              | -3.372073 | 1.369125  | C    | -6.925992             | 1.214216  | 0.367504  |
| H    | 5.675355              | -2.367833 | -2.567094 | C    | -7.960643             | 0.786912  | -0.65171  |
| H    | 4.177693              | -4.249758 | -2.495629 | C    | -6.444154             | -0.000078 | 1.188289  |
| H    | 5.62105               | -5.26433  | 1.423064  | H    | -8.949742             | 1.148328  | -0.326456 |
| C    | 6.397325              | 2.537387  | -0.514501 | C    | -7.960612             | -0.787172 | -0.651682 |
| C    | 4.66972               | 4.849961  | -0.475616 | C    | -7.792751             | 1.197616  | -2.148007 |
| C    | 6.455446              | 3.517475  | 0.527733  | C    | -6.925949             | -1.214401 | 0.367538  |
| C    | 5.643529              | 2.953789  | -1.658221 | C    | -7.127535             | -0.000089 | 2.593521  |
| C    | 4.809757              | 4.056199  | -1.633374 | C    | -4.89053              | -0.000065 | 1.289928  |
| C    | 5.599844              | 4.607045  | 0.557391  | C    | -7.792701             | -1.197931 | -2.147976 |
| H    | 7.116673              | 3.37203   | 1.369487  | H    | -8.949692             | -1.148622 | -0.326409 |
| H    | 5.675707              | 2.368252  | -2.567133 | C    | -8.340069             | -0.000183 | -2.928857 |
| H    | 4.177895              | 4.250077  | -2.495688 | H    | -8.317927             | 2.125688  | -2.393722 |
| H    | 5.620563              | 5.264245  | 1.423375  | H    | -6.732148             | 1.339871  | -2.386221 |
| C    | 3.45949               | -5.679031 | -0.295793 | C    | -6.397066             | -2.537293 | 0.51466   |
| C    | 0.741657              | -6.511357 | -0.034261 | H    | -6.854113             | 0.88257   | 3.179097  |
| C    | 2.81306               | -5.686524 | 0.954439  | H    | -8.217712             | -0.000121 | 2.486141  |
| C    | 2.767761              | -6.26746  | -1.372943 | H    | -6.854057             | -0.882726 | 3.179109  |
| C    | 1.444767              | -6.6775   | -1.2435   | H    | -4.511485             | -0.878844 | 1.809177  |
| C    | 1.488108              | -6.086565 | 1.079962  | H    | -4.457009             | -0.000098 | 0.284514  |
| H    | 3.310741              | -5.248    | 1.814356  | H    | -4.511486             | 0.87875   | 1.809117  |
| H    | 3.256849              | -6.359727 | -2.339183 | H    | -8.317841             | -2.126033 | -2.393648 |
| H    | 0.933236              | -7.084343 | -2.111682 | H    | -6.732093             | -1.340159 | -2.386179 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 0.992068  | -5.945683 | 2.035453  | H | -9.438472 | -0.000206 | -2.90132  |
| C | 3.459257  | 5.679052  | -0.295708 | H | -8.038626 | -0.000191 | -3.982569 |
| C | 0.741404  | 6.511347  | -0.034258 | C | -6.455124 | -3.517197 | -0.527745 |
| C | 2.812752  | 5.686424  | 0.954484  | C | -5.643096 | -2.95375  | 1.658233  |
| C | 2.767577  | 6.267565  | -1.372847 | C | -5.599426 | -4.606684 | -0.557617 |
| C | 1.444571  | 6.677578  | -1.243453 | H | -7.11637  | -3.371665 | -1.369469 |
| C | 1.487796  | 6.086461  | 1.07997   | C | -4.809237 | -4.056094 | 1.633178  |
| H | 3.31037   | 5.247828  | 1.814399  | H | -5.675263 | -2.368338 | 2.567229  |
| H | 3.256716  | 6.359921  | -2.339053 | C | -4.669253 | -4.849706 | 0.475318  |
| H | 0.933059  | 7.084469  | -2.111625 | H | -5.620114 | -5.263744 | -1.423707 |
| H | 0.991694  | 5.945512  | 2.035418  | H | -4.177288 | -4.250039 | 2.495414  |
| C | -0.74102  | -6.51126  | 0.033632  | C | 6.925884  | -1.214207 | -0.367069 |
| C | -3.458808 | -5.678795 | 0.295267  | C | 6.444254  | 0.000126  | -1.187886 |
| C | -1.444206 | -6.677699 | 1.242785  | C | 7.959905  | -0.786947 | 0.652822  |
| C | -1.487375 | -6.086053 | -1.080497 | C | 6.925928  | 1.214434  | -0.36705  |
| C | -2.812294 | -5.685939 | -0.954924 | C | 7.127822  | 0.000083  | -2.593021 |
| C | -2.76718  | -6.267595 | 1.372277  | C | 4.890631  | 0.000205  | -1.289685 |
| H | -0.991278 | -5.944927 | -2.035922 | C | 7.959843  | 0.787141  | 0.652902  |
| H | -3.309876 | -5.247119 | -1.814745 | C | 7.790891  | -1.197802 | 2.148963  |
| H | -3.256328 | -6.36013  | 2.338462  | H | 8.949218  | -1.148336 | 0.328198  |
| C | -0.741284 | 6.511279  | 0.033616  | H | 6.854418  | 0.882679  | -3.178699 |
| C | -3.459058 | 5.678783  | 0.295216  | H | 8.217982  | 0.000138  | -2.485466 |
| C | -1.444452 | 6.677604  | 1.242802  | H | 6.854507  | -0.88261  | -3.17859  |
| C | -1.487647 | 6.086178  | -1.080542 | H | 4.511544  | -0.878737 | -1.808625 |
| C | -2.812572 | 5.686038  | -0.954983 | H | 4.457012  | 0.000584  | -0.28431  |
| C | -2.767413 | 6.26749   | 1.372273  | H | 4.511716  | 0.878868  | -1.809226 |
| H | -0.932961 | 7.084661  | 2.110909  | H | 8.949181  | 1.148662  | 0.328511  |
| H | -0.991552 | 5.945146  | -2.035982 | C | 7.790604  | 1.197798  | 2.149094  |
| H | -3.310165 | 5.247288  | -1.814834 | C | 8.337284  | 0.00002   | 2.930404  |
| H | -3.256559 | 6.359918  | 2.338469  | H | 8.316112  | -2.125747 | 2.395054  |
| H | -0.93276  | -7.084848 | 2.110874  | H | 6.730131  | -1.340404 | 2.386265  |
| C | -4.669473 | 4.849641  | 0.475252  | H | 8.315578  | 2.125845  | 2.395314  |
| C | -5.599641 | 4.60657   | -0.557675 | H | 6.729789  | 1.3401    | 2.386333  |
| C | -4.809415 | 4.056025  | 1.633116  | H | 9.435715  | 0.000163  | 2.903947  |
| C | -6.455291 | 3.517047  | -0.527794 | H | 8.03482   | -0.00008  | 3.983824  |

### 2DR-8CPP (anti)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2626.080039 Hartree

Sum of electronic and thermal Energies = -2626.024462 Hartree

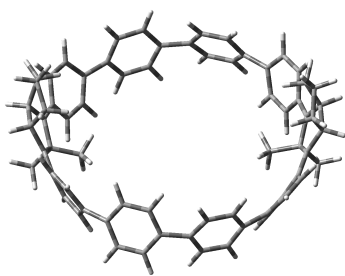
Sum of electronic and enthalpy Energies = -2626.023518 Hartree

Sum of electronic and thermal Free Energies = -2626.168556 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 6.109892              | -2.436508 | -0.453175 | H    | -5.628976             | 5.392418  | -1.239251 |
| C    | 4.619615              | -4.913405 | -0.639457 | C    | -5.50278              | 2.889661  | 1.696295  |
| C    | 6.266739              | -3.5127   | 0.516166  | H    | -4.302012             | 4.289076  | 2.712563  |
| C    | 5.502883              | -2.889712 | -1.696122 | C    | -6.109887             | 2.436506  | 0.453385  |
| C    | 4.799827              | -4.055769 | -1.776879 | H    | -6.938496             | 3.377673  | -1.354516 |
| C    | 5.530397              | -4.660078 | 0.443111  | H    | -5.567712             | 2.273106  | 2.582766  |
| H    | 6.938153              | -3.377547 | 1.35495   | C    | -6.401093             | 1.116865  | 0.151129  |
| H    | 5.567953              | -2.273213 | -2.582625 | C    | -7.017871             | 0.659964  | -1.162771 |
| H    | 4.302299              | -4.289214 | -2.712484 | C    | -6.068671             | -0.098259 | 1.044246  |
| H    | 5.628637              | -5.392245 | 1.239632  | H    | -8.094689             | 0.889012  | -1.150887 |
| C    | 6.022287              | 2.638317  | -0.444057 | C    | -6.811836             | -0.888069 | -1.217559 |
| C    | 4.473361              | 5.076462  | -0.607571 | C    | -6.402961             | 1.215385  | -2.494032 |
| C    | 6.113496              | 3.683984  | 0.564356  | C    | -6.330238             | -1.322031 | 0.147368  |
| C    | 5.445388              | 3.106964  | -1.69407  | C    | -7.01162              | -0.12313  | 2.283734  |
| C    | 4.710004              | 4.254499  | -1.761714 | C    | -4.563076             | -0.076289 | 1.451492  |
| C    | 5.356405              | 4.817315  | 0.497254  | C    | -5.82186              | -1.11163  | -2.400508 |
| H    | 6.751796              | 3.529968  | 1.427504  | H    | -7.757175             | -1.386691 | -1.47747  |
| H    | 5.546264              | 2.511425  | -2.592143 | C    | -6.205038             | -0.013398 | -3.392785 |
| H    | 4.231382              | 4.499571  | -2.704184 | H    | -7.037132             | 1.973699  | -2.964067 |
| H    | 5.407164              | 5.525806  | 1.318829  | H    | -5.434894             | 1.684084  | -2.289918 |
| C    | 3.501239              | -5.797257 | -0.534575 | C    | -6.022295             | -2.638359 | 0.443996  |
| C    | 0.814858              | -6.771685 | -0.133083 | H    | -6.886249             | 0.768444  | 2.904665  |
| C    | 3.063675              | -6.309894 | 0.727197  | H    | -8.057975             | -0.158869 | 1.962839  |
| C    | 2.612454              | -6.044394 | -1.627655 | H    | -6.829883             | -0.997293 | 2.915367  |
| C    | 1.338136              | -6.517517 | -1.436964 | H    | -4.278914             | -0.982958 | 1.986532  |
| C    | 1.78032               | -6.761204 | 0.919198  | H    | -3.937968             | -0.013403 | 0.555     |
| H    | 3.721278              | -6.24743  | 1.588211  | H    | -4.317828             | 0.778584  | 2.081743  |
| H    | 2.92127               | -5.796682 | -2.637879 | H    | -5.878434             | -2.118228 | -2.821801 |
| H    | 0.689111              | -6.597077 | -2.302505 | H    | -4.791864             | -0.963918 | -2.051359 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 1.487341  | -7.064704 | 1.919307  | H | -7.145395 | -0.276344 | -3.896654 |
| C | 3.326076  | 5.923223  | -0.506926 | H | -5.451427 | 0.149152  | -4.171775 |
| C | 0.604222  | 6.799191  | -0.11275  | C | -6.113598 | -3.684022 | -0.564417 |
| C | 2.86602   | 6.42091   | 0.753231  | C | -5.445447 | -3.107083 | 1.694006  |
| C | 2.432332  | 6.140702  | -1.60268  | C | -5.356541 | -4.81738  | -0.497351 |
| C | 1.140191  | 6.565174  | -1.415185 | H | -6.751923 | -3.529974 | -1.427538 |
| C | 1.567437  | 6.826529  | 0.941313  | C | -4.710105 | -4.254652 | 1.761625  |
| H | 3.523091  | 6.38785   | 1.616043  | H | -5.546311 | -2.511561 | 2.592089  |
| H | 2.750802  | 5.90232   | -2.612073 | C | -4.473484 | -5.076575 | 0.607453  |
| H | 0.490437  | 6.619106  | -2.282247 | H | -5.407327 | -5.525868 | -1.318928 |
| H | 1.262324  | 7.123329  | 1.939749  | H | -4.231466 | -4.499749 | 2.704078  |
| C | -0.604337 | -6.799257 | 0.112547  | C | 6.401137  | -1.116864 | -0.150992 |
| C | -3.326204 | -5.92334  | 0.506739  | C | 6.06882   | 0.098211  | -1.044225 |
| C | -1.140399 | -6.565554 | 1.415001  | C | 7.017968  | -0.659889 | 1.162846  |
| C | -1.567479 | -6.826328 | -0.941586 | C | 6.330364  | 1.322033  | -0.147411 |
| C | -2.866071 | -6.42074  | -0.753499 | C | 7.011836  | 0.123041  | -2.283637 |
| C | -2.432543 | -6.14109  | 1.602505  | C | 4.56324   | 0.076232  | -1.45149  |
| H | -1.262304 | -7.122849 | -1.940085 | C | 6.812005  | 0.888171  | 1.217519  |
| H | -3.523069 | -6.387406 | -1.616357 | C | 6.403134  | -1.215158 | 2.494182  |
| H | -2.751092 | -5.902961 | 2.611933  | H | 8.094778  | -0.88897  | 1.150902  |
| C | -0.814954 | 6.77167   | 0.13296   | H | 6.830367  | 0.997381  | -2.915106 |
| C | -3.501302 | 5.797283  | 0.534673  | H | 8.058185  | 0.158456  | -1.962678 |
| C | -1.338143 | 6.517574  | 1.436892  | H | 6.886286  | -0.768375 | -2.904765 |
| C | -1.78048  | 6.761157  | -0.919262 | H | 4.317768  | -0.779303 | -2.080749 |
| C | -3.063824 | 6.30986   | -0.727155 | H | 3.938116  | 0.014561  | -0.554928 |
| C | -2.612453 | 6.044474  | 1.627691  | H | 4.279348  | 0.982368  | -1.987581 |
| H | -0.689043 | 6.597166  | 2.302375  | H | 7.75739   | 1.386753  | 1.477344  |
| H | -1.48755  | 7.064625  | -1.919396 | C | 5.822129  | 1.111893  | 2.400498  |
| H | -3.721488 | 6.247326  | -1.588118 | C | 6.205335  | 0.013725  | 3.392845  |
| H | -2.921222 | 5.79682   | 2.637944  | H | 7.037289  | -1.973479 | 2.964237  |
| H | -0.490725 | -6.619746 | 2.282106  | H | 5.43502   | -1.68381  | 2.290178  |
| C | -4.619661 | 4.913426  | 0.639641  | H | 5.878765  | 2.118537  | 2.821684  |
| C | -5.530605 | 4.660172  | -0.442817 | H | 4.792104  | 0.96418   | 2.051437  |
| C | -4.799715 | 4.055716  | 1.77703   | H | 7.145743  | 0.276676  | 3.896616  |
| C | -6.266928 | 3.512783  | -0.515845 | H | 5.451778  | -0.148721 | 4.171908  |

### 2DR-8CPP (syn)



opt freq BS-UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 2.2866$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.560582 Hartree

Sum of electronic and thermal Energies = -2624.506160 Hartree

Sum of electronic and enthalpy Energies = -2624.505216 Hartree

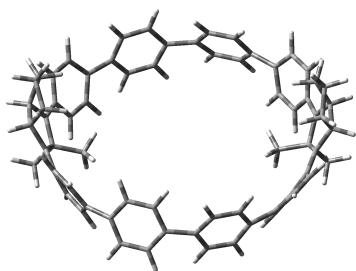
Sum of electronic and thermal Free Energies = -2624.649784 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.198925             | -2.573024 | -0.087982 | H    | 5.416833              | 5.358404  | 1.745494  |
| C    | -4.63028              | -4.964793 | -0.22535  | C    | 5.572976              | 3.000526  | -1.29403  |
| C    | -5.573629             | -3.001128 | -1.295593 | H    | 4.254159              | 4.359498  | -2.258631 |
| C    | -6.228829             | -3.556575 | 0.943771  | C    | 6.198516              | 2.572743  | -0.086442 |
| C    | -5.443638             | -4.691131 | 0.887486  | H    | 6.813538              | 3.377406  | 1.836859  |
| C    | -4.813495             | -4.148182 | -1.353504 | H    | 5.610441              | 2.372807  | -2.173548 |
| H    | -5.610803             | -2.373425 | -2.175143 | C    | 6.61013               | 1.230958  | 0.130651  |
| H    | -6.814771             | -3.377424 | 1.835174  | C    | 7.291667              | 0.737766  | 1.380535  |
| H    | -5.418595             | -5.358968 | 1.744112  | C    | 6.276205              | 0.07265   | -0.814967 |
| H    | -4.25509              | -4.360548 | -2.259917 | H    | 8.294028              | 1.180284  | 1.468177  |
| C    | -6.179036             | 2.482731  | -0.33542  | C    | 7.403807              | -0.815651 | 1.240685  |
| C    | -4.587276             | 4.85486   | -0.651674 | C    | 6.561124              | 0.961816  | 2.731632  |
| C    | -5.44096              | 2.752961  | -1.526157 | C    | 6.614832              | -1.178771 | 0.004854  |
| C    | -6.305577             | 3.60577   | 0.538751  | C    | 7.213951              | 0.121414  | -2.052315 |
| C    | -5.520382             | 4.731686  | 0.39558   | C    | 4.766747              | 0.106779  | -1.223997 |
| C    | -4.666411             | 3.880245  | -1.663199 | C    | 6.918352              | -1.398199 | 2.599472  |
| H    | -5.410774             | 2.02096   | -2.319901 | H    | 8.457068              | -1.097248 | 1.096785  |
| H    | -6.982414             | 3.555599  | 1.380789  | C    | 6.996043              | -0.228896 | 3.577887  |
| H    | -5.600272             | 5.518828  | 1.139588  | H    | 6.805068              | 1.915802  | 3.202604  |
| H    | -4.030261             | 3.95765   | -2.537683 | H    | 5.476286              | 0.941093  | 2.5758    |
| C    | -3.4387               | -5.837672 | -0.165175 | C    | 6.178556              | -2.482766 | -0.334169 |
| C    | -0.733735             | -6.69907  | -0.23282  | H    | 7.060075              | 1.027014  | -2.644321 |
| C    | -2.914822             | -6.49418  | -1.286473 | H    | 8.261281              | 0.107944  | -1.735903 |
| C    | -2.634323             | -5.792653 | 0.978712  | H    | 7.05394               | -0.741349 | -2.704626 |
| C    | -1.312655             | -6.204715 | 0.942093  | H    | 4.633283              | 0.048069  | -2.30932  |
| C    | -1.596396             | -6.923752 | -1.315052 | H    | 4.218793              | -0.724075 | -0.780602 |
| H    | -3.5312               | -6.617382 | -2.172258 | H    | 4.283562              | 1.022716  | -0.88946  |
| H    | -3.006086             | -5.299468 | 1.871129  | H    | 7.513958              | -2.253479 | 2.929321  |
| H    | -0.692378             | -6.015837 | 1.810943  | H    | 5.88159               | -1.737105 | 2.51157   |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.217008 | -7.383048 | -2.222411 | H | 8.029539  | -0.08515  | 3.917398  |
| C | -3.423475 | 5.763836  | -0.60563  | H | 6.376868  | -0.375646 | 4.468276  |
| C | -0.74563  | 6.697382  | -0.376868 | C | 6.304423  | -3.605585 | 0.540324  |
| C | -2.637771 | 6.01305   | -1.739665 | C | 5.440581  | -2.752959 | -1.524964 |
| C | -2.902362 | 6.196156  | 0.622077  | C | 5.518996  | -4.731348 | 0.397149  |
| C | -1.601547 | 6.662394  | 0.731221  | H | 6.980873  | -3.555353 | 1.382655  |
| C | -1.330036 | 6.452743  | -1.626312 | C | 4.665777  | -3.880077 | -1.662001 |
| H | -3.021192 | 5.777351  | -2.726698 | H | 5.410736  | -2.021063 | -2.318828 |
| H | -3.483358 | 6.064766  | 1.528911  | C | 4.586219  | -4.854546 | -0.650373 |
| H | -1.223875 | 6.920591  | 1.715229  | H | 5.598434  | -5.518315 | 1.141387  |
| H | -0.724046 | 6.502046  | -2.524467 | H | 4.029807  | -3.957468 | -2.536619 |
| C | 0.744574  | -6.697384 | -0.376545 | C | -6.61469  | 1.178638  | 0.003864  |
| C | 3.422372  | -5.763494 | -0.604593 | C | -6.276192 | -0.072724 | -0.816154 |
| C | 1.329295  | -6.452694 | -1.625829 | C | -7.402801 | 0.815239  | 1.240234  |
| C | 1.600175  | -6.662216 | 0.731785  | C | -6.609607 | -1.231103 | 0.129497  |
| C | 2.900955  | -6.195808 | 0.622987  | C | -7.21421  | -0.121515 | -2.053254 |
| C | 2.637005  | -6.012825 | -1.738832 | C | -4.766834 | -0.106807 | -1.225627 |
| H | 1.222277  | -6.920443 | 1.715696  | C | -7.288146 | -0.737834 | 1.380891  |
| H | 3.481678  | -6.064323 | 1.52998   | C | -6.917583 | 1.399375  | 2.598509  |
| H | 3.020675  | -5.77713  | -2.725769 | H | -8.456516 | 1.095127  | 1.096328  |
| C | 0.73264   | 6.698771  | -0.232727 | H | -7.060374 | -1.027059 | -2.645355 |
| C | 3.437438  | 5.836951  | -0.164277 | H | -8.261468 | -0.108179 | -1.736609 |
| C | 1.595659  | 6.923345  | -1.314692 | H | -7.054417 | 0.741306  | -2.705542 |
| C | 1.311105  | 6.204251  | 0.942339  | H | -4.633657 | -0.047199 | -2.310934 |
| C | 2.632697  | 5.791974  | 0.97935   | H | -4.218607 | 0.723572  | -0.781648 |
| C | 2.914008  | 6.493555  | -1.285722 | H | -4.283756 | -1.02315  | -0.892035 |
| H | 1.216625  | 7.382751  | -2.222143 | H | -8.289507 | -1.181801 | 1.472423  |
| H | 0.690512  | 6.015448  | 1.810982  | C | -6.552637 | -0.959414 | 2.729649  |
| H | 3.004127  | 5.298712  | 1.871865  | C | -6.989465 | 0.229745  | 3.577062  |
| H | 3.530691  | 6.616688  | -2.171304 | H | -7.515811 | 2.252533  | 2.929161  |
| H | 0.723566  | -6.502119 | -2.524154 | H | -5.882195 | 1.74201   | 2.509259  |
| C | 4.629026  | 4.964044  | -0.224101 | H | -6.791317 | -1.91443  | 3.20119   |
| C | 5.442242  | 4.690594  | 0.888857  | H | -5.468372 | -0.934468 | 2.570369  |
| C | 4.81251   | 4.147354  | -1.352132 | H | -8.021715 | 0.082593  | 3.918891  |
| C | 6.227832  | 3.556296  | 0.945289  | H | -6.368753 | 0.378457  | 4.466055  |



### 2DR-8CPP (syn)



opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 3.2724$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.560559 Hartree

Sum of electronic and thermal Energies = -2624.506081 Hartree

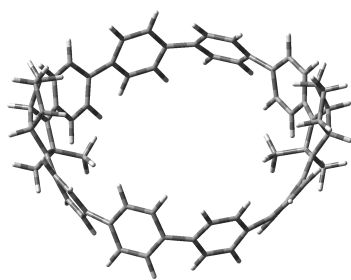
Sum of electronic and enthalpy Energies = -2624.505137 Hartree

Sum of electronic and thermal Free Energies = -2624.652110 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 6.234042              | 2.552044  | -0.085798 | H    | -5.442552             | -5.327533 | 1.747699  |
| C    | 4.661127              | 4.941988  | -0.215556 | C    | -5.592754             | -2.97428  | -1.293931 |
| C    | 5.605821              | 2.982191  | -1.291113 | H    | -4.260854             | -4.328227 | -2.249577 |
| C    | 6.262976              | 3.532852  | 0.94835   | C    | -6.230758             | -2.549651 | -0.092604 |
| C    | 5.475536              | 4.666243  | 0.895761  | H    | -6.846932             | -3.353631 | 1.830104  |
| C    | 4.844161              | 4.128386  | -1.34576  | H    | -5.627495             | -2.347678 | -2.174291 |
| H    | 5.643153              | 2.356925  | -2.172353 | C    | -6.656319             | -1.207691 | 0.113696  |
| H    | 6.848646              | 3.352508  | 1.839398  | C    | -7.397294             | -0.712865 | 1.330294  |
| H    | 5.44978               | 5.331305  | 1.754529  | C    | -6.299421             | -0.05202  | -0.824436 |
| H    | 4.284824              | 4.342315  | -2.251238 | H    | -8.416615             | -1.124246 | 1.34235   |
| C    | 6.191429              | -2.501515 | -0.340928 | C    | -7.456472             | 0.846607  | 1.206933  |
| C    | 4.58045               | -4.861107 | -0.649932 | C    | -6.765514             | -0.983391 | 2.722898  |
| C    | 5.4462                | -2.765829 | -1.527825 | C    | -6.640813             | 1.20138   | -0.013227 |
| C    | 6.311764              | -3.624069 | 0.533621  | C    | -7.224503             | -0.094905 | -2.074595 |
| C    | 5.517685              | -4.744731 | 0.39353   | C    | -4.785931             | -0.094604 | -1.217935 |
| C    | 4.661678              | -3.887135 | -1.661096 | C    | -6.972819             | 1.395605  | 2.57896   |
| H    | 5.41821               | -2.033752 | -2.32159  | H    | -8.498303             | 1.164507  | 1.056552  |
| H    | 6.990451              | -3.578006 | 1.374307  | C    | -7.163742             | 0.234009  | 3.549739  |
| H    | 5.593722              | -5.531672 | 1.138168  | H    | -7.107572             | -1.91586  | 3.176466  |
| H    | 4.020415              | -3.959327 | -2.532314 | H    | -5.674797             | -1.041718 | 2.633895  |
| C    | 3.469121              | 5.81422   | -0.152824 | C    | -6.184237             | 2.501626  | -0.343894 |
| C    | 0.764756              | 6.67743   | -0.218647 | H    | -7.069696             | -1.002703 | -2.662844 |
| C    | 2.946266              | 6.475249  | -1.271889 | H    | -8.275314             | -0.073118 | -1.770525 |
| C    | 2.663891              | 5.76446   | 0.990215  | H    | -7.04985              | 0.765962  | -2.725467 |
| C    | 1.342445              | 6.177401  | 0.954435  | H    | -4.641726             | -0.054666 | -2.302723 |
| C    | 1.628111              | 6.905748  | -1.299526 | H    | -4.241668             | 0.743189  | -0.783725 |
| H    | 3.563343              | 6.60164   | -2.156741 | H    | -4.307216             | -1.004875 | -0.862464 |
| H    | 3.03494               | 5.267413  | 1.880793  | H    | -7.518296             | 2.288703  | 2.894057  |
| H    | 0.72141               | 5.985318  | 1.822048  | H    | -5.912742             | 1.662935  | 2.522118  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 1.24953   | 7.369124  | -2.205145 | H | -8.218389 | 0.15576   | 3.842452  |
| C | 3.409334  | -5.762637 | -0.599642 | H | -6.577176 | 0.342381  | 4.467302  |
| C | 0.727038  | -6.678427 | -0.364157 | C | -6.298427 | 3.620352  | 0.536252  |
| C | 2.620782  | -6.009291 | -1.731597 | C | -5.438042 | 2.767779  | -1.530009 |
| C | 2.887526  | -6.186804 | 0.630025  | C | -5.500976 | 4.738763  | 0.399953  |
| C | 1.583919  | -6.64545  | 0.742482  | H | -6.974667 | 3.57208   | 1.378694  |
| C | 1.309814  | -6.439783 | -1.614869 | C | -4.651296 | 3.887626  | -1.660438 |
| H | 3.004197  | -5.778147 | -2.719737 | H | -5.412604 | 2.03831   | -2.326417 |
| H | 3.470555  | -6.055398 | 1.53556   | C | -4.565783 | 4.857831  | -0.645473 |
| H | 1.205455  | -6.898357 | 1.727618  | H | -5.573014 | 5.522311  | 1.14852   |
| H | 0.701     | -6.486573 | -2.51134  | H | -4.011636 | 3.961746  | -2.532657 |
| C | -0.713526 | 6.679573  | -0.36276  | C | 6.640268  | -1.199828 | -0.004651 |
| C | -3.395436 | 5.758485  | -0.595235 | C | 6.302757  | 0.054156  | -0.820646 |
| C | -1.298619 | 6.441328  | -1.613066 | C | 7.442521  | -0.84324  | 1.223986  |
| C | -1.56946  | 6.643315  | 0.745222  | C | 6.648211  | 1.209471  | 0.124982  |
| C | -2.872256 | 6.183052  | 0.634217  | C | 7.233804  | 0.099541  | -2.063362 |
| C | -2.608285 | 6.007728  | -1.728155 | C | 4.791232  | 0.095707  | -1.220847 |
| H | -1.190753 | 6.896243  | 1.730199  | C | 7.351228  | 0.712236  | 1.361836  |
| H | -3.453905 | 6.050773  | 1.540496  | C | 6.961384  | -1.416704 | 2.587733  |
| H | -2.992729 | 5.777439  | -2.716074 | H | 8.491257  | -1.138257 | 1.073804  |
| C | -0.752131 | -6.671455 | -0.217567 | H | 7.080696  | 1.006026  | -2.654215 |
| C | -3.454401 | -5.805385 | -0.152551 | H | 8.282848  | 0.081226  | -1.752869 |
| C | -1.618514 | -6.911692 | -1.292568 | H | 7.066352  | -0.762222 | -2.715159 |
| C | -1.323611 | -6.154081 | 0.950412  | H | 4.651148  | 0.0401    | -2.305491 |
| C | -2.644748 | -5.739389 | 0.98586   | H | 4.24227   | -0.733748 | -0.776192 |
| C | -2.936846 | -6.47953  | -1.265363 | H | 4.313749  | 1.012753  | -0.881417 |
| H | -1.242714 | -7.385616 | -2.19397  | H | 8.360928  | 1.142553  | 1.422632  |
| H | -0.698217 | -5.950843 | 1.812381  | C | 6.657532  | 0.950692  | 2.730057  |
| H | -3.01252  | -5.230191 | 1.870949  | C | 7.081834  | -0.251267 | 3.565953  |
| H | -3.557304 | -6.614885 | -2.146566 | H | 7.540613  | -2.286389 | 2.908685  |
| H | -0.692267 | 6.491373  | -2.510941 | H | 5.915517  | -1.730344 | 2.513153  |
| C | -4.64867  | -4.933936 | -0.218568 | H | 6.937257  | 1.896872  | 3.19711   |
| C | -5.467915 | -4.662306 | 0.889013  | H | 5.569237  | 0.957989  | 2.599886  |
| C | -4.826299 | -4.117966 | -1.346964 | H | 8.124362  | -0.131093 | 3.886512  |
| C | -6.259577 | -3.530913 | 0.939938  | H | 6.476164  | -0.385895 | 4.467484  |

### 2DR-8CPP (syn)



opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 5,  $\langle S^2 \rangle = 6.2588$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.560503 Hartree

Sum of electronic and thermal Energies = -2624.506021 Hartree

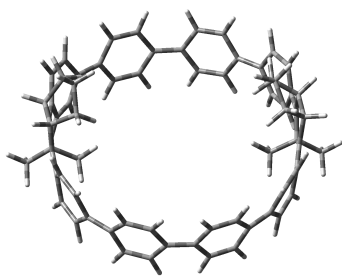
Sum of electronic and enthalpy Energies = -2624.505077 Hartree

Sum of electronic and thermal Free Energies = -2624.651848 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.267135             | -2.528406 | -0.091871 | H    | 5.473598              | 5.297066  | 1.761009  |
| C    | -4.680958             | -4.910962 | -0.207921 | C    | 5.627327              | 2.956616  | -1.289568 |
| C    | -5.627014             | -2.956652 | -1.290702 | H    | 4.293732              | 4.31245   | -2.240309 |
| C    | -6.294355             | -3.505407 | 0.944553  | C    | 6.267174              | 2.528418  | -0.090575 |
| C    | -5.500584             | -4.635728 | 0.898287  | H    | 6.880177              | 3.325657  | 1.836136  |
| C    | -4.859089             | -4.099544 | -1.339382 | H    | 5.66292               | 2.333461  | -2.172329 |
| H    | -5.662395             | -2.333533 | -2.173497 | C    | 6.695646              | 1.185517  | 0.107711  |
| H    | -6.880583             | -3.325573 | 1.834728  | C    | 7.459793              | 0.685749  | 1.308269  |
| H    | -5.473999             | -5.296995 | 1.759995  | C    | 6.326154              | 0.033163  | -0.830094 |
| H    | -4.293204             | -4.312527 | -2.241084 | H    | 8.484685              | 1.082994  | 1.289924  |
| C    | -6.196154             | 2.520578  | -0.35163  | C    | 7.495894              | -0.875139 | 1.189248  |
| C    | -4.55846              | 4.86399   | -0.644552 | C    | 6.87064               | 0.971402  | 2.716625  |
| C    | -5.442108             | 2.78082   | -1.533397 | C    | 6.66605               | -1.222998 | -0.022795 |
| C    | -6.304877             | 3.638678  | 0.528946  | C    | 7.244284              | 0.072111  | -2.085758 |
| C    | -5.49848              | 4.7517    | 0.396443  | C    | 4.810827              | 0.083986  | -1.21465  |
| C    | -4.645392             | 3.894575  | -1.659367 | C    | 7.017272              | -1.412094 | 2.56732   |
| H    | -5.418333             | 2.051355  | -2.329869 | H    | 8.532127              | -1.208117 | 1.032733  |
| H    | -6.983711             | 3.594435  | 1.369457  | C    | 7.255532              | -0.256518 | 3.534482  |
| H    | -5.567222             | 5.53497   | 1.145628  | H    | 7.25064               | 1.893606  | 3.16134   |
| H    | -4.000072             | 3.963545  | -2.527861 | H    | 5.780149              | 1.059988  | 2.655812  |
| C    | -3.48621              | -5.781561 | -0.138684 | C    | 6.196212              | -2.520589 | -0.350067 |
| C    | -0.784414             | -6.649217 | -0.201115 | H    | 7.09032               | 0.98067   | -2.673067 |
| C    | -2.969538             | -6.460943 | -1.248677 | H    | 8.296765              | 0.04528   | -1.787866 |
| C    | -2.675738             | -5.710044 | 0.998794  | H    | 7.061796              | -0.787838 | -2.735722 |
| C    | -1.354809             | -6.125516 | 0.964563  | H    | 4.660034              | 0.046882  | -2.298628 |
| C    | -1.651425             | -6.893874 | -1.274574 | H    | 4.264932              | -0.751878 | -0.778888 |
| H    | -3.590616             | -6.600215 | -2.128833 | H    | 4.338522              | 0.995754  | -0.854684 |
| H    | -3.042849             | -5.196443 | 1.881616  | H    | 7.542676              | -2.319682 | 2.874691  |
| H    | -0.728739             | -5.918497 | 1.825148  | H    | 5.949396              | -1.650196 | 2.525749  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.276362 | -7.372413 | -2.173843 | H | 8.317503  | -0.204976 | 3.805867  |
| C | -3.380855 | 5.757     | -0.589173 | H | 6.685114  | -0.351061 | 4.463682  |
| C | -0.694605 | 6.659918  | -0.348418 | C | 6.304739  | -3.63864  | 0.530594  |
| C | -2.590347 | 6.004135  | -1.719569 | C | 5.442438  | -2.780899 | -1.531991 |
| C | -2.857545 | 6.172783  | 0.642647  | C | 5.498383  | -4.751678 | 0.397965  |
| C | -1.551991 | 6.625211  | 0.757734  | H | 6.983377  | -3.594346 | 1.371261  |
| C | -1.277525 | 6.42836   | -1.600358 | C | 4.645755  | -3.894665 | -1.658081 |
| H | -2.974363 | 5.778904  | -2.708849 | H | 5.418844  | -2.051484 | -2.328514 |
| H | -3.441661 | 6.040123  | 1.547294  | C | 4.558604  | -4.864029 | -0.643237 |
| H | -1.172912 | 6.872297  | 1.744111  | H | 5.566958  | -5.534907 | 1.147209  |
| H | -0.667964 | 6.476296  | -2.496266 | H | 4.000626  | -3.963687 | -2.526714 |
| C | 0.694688  | -6.659954 | -0.347927 | C | -6.666068 | 1.223011  | -0.024393 |
| C | 3.380993  | -5.757046 | -0.58809  | C | -6.325981 | -0.033198 | -0.83154  |
| C | 1.277917  | -6.428492 | -1.599742 | C | -7.496151 | 0.875216  | 1.187504  |
| C | 1.551803  | -6.625162 | 0.758433  | C | -6.695647 | -1.185496 | 0.106263  |
| C | 2.857383  | -6.172735 | 0.643633  | C | -7.243863 | -0.072267 | -2.087381 |
| C | 2.590768  | -6.004272 | -1.718662 | C | -4.810576 | -0.084008 | -1.215787 |
| H | 1.172487  | -6.872177 | 1.744736  | C | -7.460088 | -0.685669 | 1.30661   |
| H | 3.441277  | -6.040001 | 1.548412  | C | -7.017782 | 1.41223   | 2.565638  |
| H | 2.975031  | -5.779116 | -2.707864 | H | -8.532352 | 1.208197  | 1.030782  |
| C | 0.78446   | 6.649191  | -0.201251 | H | -7.089763 | -0.980872 | -2.674583 |
| C | 3.486242  | 5.781553  | -0.138149 | H | -8.296402 | -0.045436 | -1.789697 |
| C | 1.651727  | 6.893833  | -1.274505 | H | -7.061268 | 0.787633  | -2.737381 |
| C | 1.354573  | 6.125513  | 0.964575  | H | -4.659562 | -0.04687  | -2.299734 |
| C | 2.675495  | 5.71005   | 0.999135  | H | -4.264774 | 0.751847  | -0.779888 |
| C | 2.969837  | 6.46091   | -1.248281 | H | -4.338338 | -0.995782 | -0.85575  |
| H | 1.276874  | 7.37235   | -2.173874 | H | -8.484979 | -1.082905 | 1.288018  |
| H | 0.72829   | 5.918505  | 1.825009  | C | -6.871295 | -0.971265 | 2.715129  |
| H | 3.042397  | 5.196464  | 1.882052  | C | -7.256308 | 0.256729  | 3.53282   |
| H | 3.591127  | 6.600166  | -2.12829  | H | -7.543197 | 2.319867  | 2.872849  |
| H | 0.66858   | -6.476501 | -2.495798 | H | -5.949885 | 1.650271  | 2.524269  |
| C | 4.681013  | 4.910961  | -0.207077 | H | -7.251467 | -1.893416 | 3.159811  |
| C | 5.500385  | 4.635772  | 0.899329  | H | -5.780796 | -1.059929 | 2.65459   |
| C | 4.85941   | 4.099502  | -1.338468 | H | -8.318341 | 0.205254  | 3.803975  |
| C | 6.294154  | 3.505456  | 0.945818  | H | -6.68609  | 0.351304  | 4.462141  |

### 2DR-8CPP (syn)



opt freq RCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.469363 Hartree

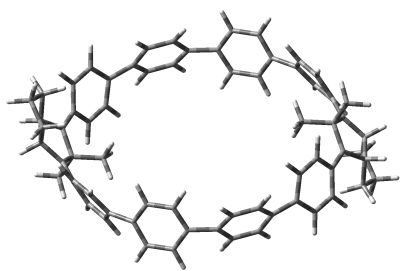
Sum of electronic and thermal Energies = -2624.414351 Hartree

Sum of electronic and enthalpy Energies = -2624.413406 Hartree

Sum of electronic and thermal Free Energies = -2624.557018 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -5.821298             | -2.788372 | 0.071071  | H    | 4.749297              | 5.626617  | 1.679954  |
| C    | -4.2934               | -5.178762 | -0.421791 | C    | 5.583006              | 3.287457  | -1.28113  |
| C    | -5.583049             | -3.287139 | -1.281759 | H    | 4.664947              | 4.697724  | -2.530243 |
| C    | -5.640834             | -3.817511 | 1.094149  | C    | 5.821302              | 2.788357  | 0.071569  |
| C    | -4.906144             | -4.925093 | 0.866942  | H    | 6.06667               | 3.655825  | 2.078894  |
| C    | -4.862364             | -4.404708 | -1.506181 | H    | 5.956804              | 2.733522  | -2.131887 |
| H    | -5.956878             | -2.732997 | -2.132367 | C    | 6.05276               | 1.489694  | 0.41026   |
| H    | -6.066591             | -3.656331 | 2.078193  | C    | 6.150328              | 1.035724  | 1.845715  |
| H    | -4.749233             | -5.627027 | 1.678722  | C    | 6.083421              | 0.283216  | -0.53801  |
| H    | -4.665036             | -4.6971   | -2.531251 | H    | 6.955132              | 1.564692  | 2.37196   |
| C    | -6.078338             | 2.239854  | 0.01672   | C    | 6.431934              | -0.491212 | 1.828195  |
| C    | -4.772037             | 4.750027  | -0.522598 | C    | 4.855015              | 1.184555  | 2.678888  |
| C    | -5.822642             | 2.704554  | -1.346967 | C    | 6.206153              | -0.937005 | 0.392315  |
| C    | -6.070074             | 3.334318  | 0.988792  | C    | 7.319272              | 0.360139  | -1.465049 |
| C    | -5.430263             | 4.495824  | 0.74345   | C    | 4.756204              | 0.229234  | -1.333522 |
| C    | -5.209687             | 3.879393  | -1.594223 | C    | 5.505437              | -1.101985 | 2.923069  |
| H    | -6.108574             | 2.085517  | -2.186344 | H    | 7.48042               | -0.679326 | 2.095905  |
| H    | -6.54853              | 3.187379  | 1.948956  | C    | 4.991356              | 0.091619  | 3.731217  |
| H    | -5.39607              | 5.251271  | 1.521311  | H    | 4.72782               | 2.185361  | 3.096088  |
| H    | -4.99795              | 4.148313  | -2.622869 | H    | 3.984605              | 0.987291  | 2.041785  |
| C    | -3.153865             | -5.962577 | -0.582208 | C    | 6.07834               | -2.239855 | 0.015941  |
| C    | -0.38206              | -6.74259  | -0.773771 | H    | 7.288521              | 1.240168  | -2.111162 |
| C    | -2.483134             | -6.103626 | -1.851494 | H    | 8.236185              | 0.418377  | -0.87146  |
| C    | -2.438937             | -6.529495 | 0.535161  | H    | 7.396678              | -0.520874 | -2.106079 |
| C    | -1.136275             | -6.892859 | 0.444629  | H    | 4.717496              | -0.620551 | -2.013565 |
| C    | -1.183388             | -6.471979 | -1.940349 | H    | 3.912607              | 0.135793  | -0.643902 |
| H    | -3.012252             | -5.856402 | -2.764335 | H    | 4.603176              | 1.136594  | -1.917311 |
| H    | -2.927511             | -6.595932 | 1.500392  | H    | 6.021861              | -1.833938 | 3.549641  |
| H    | -0.637125             | -7.241304 | 1.341224  | H    | 4.66519               | -1.619255 | 2.45146   |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.726393 | -6.505897 | -2.92211  | H | 5.729977  | 0.390772  | 4.485499  |
| C | -3.705756 | 5.632861  | -0.667545 | H | 4.056307  | -0.124686 | 4.256874  |
| C | -1.017245 | 6.673903  | -0.792639 | C | 6.070103  | -3.334556 | 0.987746  |
| C | -3.007707 | 5.806268  | -1.917916 | C | 5.822598  | -2.704223 | -1.347851 |
| C | -3.087398 | 6.291463  | 0.457052  | C | 5.430284  | -4.496002 | 0.742141  |
| C | -1.821607 | 6.77279   | 0.398848  | H | 6.548588  | -3.18785  | 1.947933  |
| C | -1.749173 | 6.301644  | -1.976476 | C | 5.209634  | -3.879001 | -1.595373 |
| H | -3.478945 | 5.485225  | -2.839472 | H | 6.108506  | -2.084983 | -2.187087 |
| H | -3.612931 | 6.330576  | 1.404323  | C | 4.772018  | -4.749895 | -0.523947 |
| H | -1.388484 | 7.189002  | 1.30095   | H | 5.396114  | -5.251637 | 1.51982   |
| H | -1.26388  | 6.35363   | -2.94368  | H | 4.997864  | -4.147669 | -2.624079 |
| C | 1.017218  | -6.673708 | -0.794332 | C | -6.206137 | 0.936911  | 0.392774  |
| C | 3.705733  | -5.632695 | -0.669075 | C | -6.083437 | -0.283072 | -0.537865 |
| C | 1.749105  | -6.301158 | -1.978102 | C | -6.43187  | 0.490752  | 1.828549  |
| C | 1.82162   | -6.772885 | 0.397103  | C | -6.052745 | -1.489793 | 0.410095  |
| C | 3.087413  | -6.291572 | 0.455382  | C | -7.31932  | -0.359751 | -1.464882 |
| C | 3.007641  | -5.805795 | -1.919465 | C | -4.756247 | -0.228878 | -1.333409 |
| H | 1.388529  | -7.189319 | 1.299118  | C | -6.150269 | -1.03619  | 1.845669  |
| H | 3.612979  | -6.330917 | 1.402625  | C | -5.505333 | 1.10124   | 2.923549  |
| H | 3.478847  | -5.484527 | -2.840959 | H | -7.480347 | 0.678802  | 2.096342  |
| C | 0.382034  | 6.74278   | -0.772109 | H | -7.288595 | -1.239613 | -2.111224 |
| C | 3.153845  | 5.96272   | -0.580835 | H | -8.236213 | -0.418139 | -0.871276 |
| C | 1.183321  | 6.472456  | -1.938782 | H | -7.396744 | 0.521429  | -2.105681 |
| C | 1.136291  | 6.892749  | 0.446301  | H | -4.717566 | 0.621088  | -2.013228 |
| C | 2.438956  | 6.529362  | 0.536698  | H | -3.912627 | -0.135617 | -0.643793 |
| C | 2.48307   | 6.104081  | -1.850063 | H | -4.603236 | -1.136081 | -1.917444 |
| H | 0.726292  | 6.506616  | -2.920518 | H | -6.955059 | -1.565289 | 2.371802  |
| H | 0.637173  | 7.240973  | 1.342999  | C | -4.854932 | -1.185241 | 2.678765  |
| H | 2.927564  | 6.595562  | 1.501929  | C | -4.991235 | -0.092573 | 3.731376  |
| H | 3.012156  | 5.857082  | -2.762984 | H | -6.021733 | 1.833036  | 3.550324  |
| H | 1.263779  | -6.352908 | -2.945303 | H | -4.665099 | 1.618625  | 2.452045  |
| C | 4.293387  | 5.178867  | -0.420652 | H | -4.727731 | -2.186155 | 3.095706  |
| C | 4.906177  | 4.924882  | 0.867997  | H | -3.98454  | -0.98782  | 2.041686  |
| C | 4.862313  | 4.40508   | -1.505252 | H | -5.729834 | -0.391913 | 4.485605  |
| C | 5.640876  | 3.817245  | 1.094906  | H | -4.056168 | 0.123593  | 4.25706   |

**2DR-8CPP (anti)**

opt freq BS-UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 2.2385$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.561700 Hartree

Sum of electronic and thermal Energies = -2624.506825 Hartree

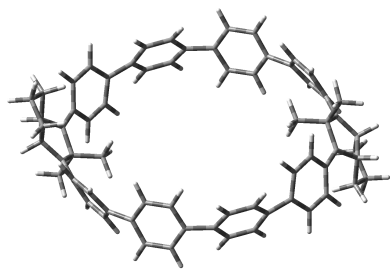
Sum of electronic and enthalpy Energies = -2624.505881 Hartree

Sum of electronic and thermal Free Energies = -2624.651621 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.428906             | 2.519621  | -0.496304 | H    | 5.573342              | -5.173162 | -1.488081 |
| C    | -4.671039             | 4.787752  | -0.431815 | C    | 5.683275              | -2.939329 | 1.63353   |
| C    | -6.45249              | 3.46611   | 0.566493  | H    | 4.202564              | -4.221229 | 2.458866  |
| C    | -5.683297             | 2.939316  | -1.633508 | C    | 6.428892              | -2.519612 | 0.49634   |
| C    | -4.830804             | 4.022072  | -1.596301 | H    | 7.095449              | -3.30663  | -1.418478 |
| C    | -5.581991             | 4.539072  | 0.60608   | H    | 5.736007              | -2.369627 | 2.550589  |
| H    | -7.095504             | 3.306709  | 1.418473  | C    | 6.985943              | -1.205534 | 0.381668  |
| H    | -5.736024             | 2.36959   | -2.550552 | C    | 8.062892              | -0.78282  | -0.586312 |
| H    | -4.202599             | 4.221207  | -2.458877 | C    | 6.500687              | -0.000004 | 1.205268  |
| H    | -5.573404             | 5.173251  | 1.488033  | H    | 9.032514              | -1.144847 | -0.211927 |
| C    | -6.428922             | -2.519598 | -0.496333 | C    | 8.062903              | 0.782787  | -0.586312 |
| C    | -4.671059             | -4.787733 | -0.431836 | C    | 7.969064              | -1.189858 | -2.080259 |
| C    | -6.452508             | -3.466084 | 0.566467  | C    | 6.985946              | 1.205511  | 0.381654  |
| C    | -5.683303             | -2.93929  | -1.633531 | C    | 7.176517              | 0.000017  | 2.601921  |
| C    | -4.830814             | -4.022049 | -1.596321 | C    | 4.956637              | 0.000007  | 1.303305  |
| C    | -5.582013             | -4.539048 | 0.606056  | C    | 7.969098              | 1.189831  | -2.080259 |
| H    | -7.095516             | -3.306673 | 1.418449  | H    | 9.032525              | 1.144801  | -0.211912 |
| H    | -5.736012             | -2.369554 | -2.550571 | C    | 8.567754              | -0.000022 | -2.823503 |
| H    | -4.202601             | -4.221182 | -2.458893 | H    | 8.49115               | -2.125009 | -2.298623 |
| H    | -5.57343              | -5.173226 | 1.48801   | H    | 6.921711              | -1.312644 | -2.375521 |
| C    | -3.44903              | 5.60156   | -0.254372 | C    | 6.428903              | 2.519589  | 0.496326  |
| C    | -0.74157              | 6.414254  | -0.022908 | H    | 6.902256              | -0.881419 | 3.186864  |
| C    | -2.782873             | 5.564229  | 0.975671  | H    | 8.265955              | 0.000005  | 2.497885  |
| C    | -2.781324             | 6.211347  | -1.324549 | H    | 6.902273              | 0.881486  | 3.186824  |
| C    | -1.458241             | 6.61121   | -1.210056 | H    | 4.575779              | 0.876451  | 1.82192   |
| C    | -1.459238             | 5.957783  | 1.087408  | H    | 4.524457              | 0         | 0.298635  |
| H    | -3.265985             | 5.100359  | 1.829391  | H    | 4.575768              | -0.876421 | 1.821938  |
| H    | -3.287923             | 6.325915  | -2.278249 | H    | 8.491216              | 2.124966  | -2.298618 |
| H    | -0.956997             | 7.03265   | -2.076259 | H    | 6.921751              | 1.312655  | -2.375526 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.942186 | 5.7896    | 2.026301  | H | 9.660216  | -0.000038 | -2.72111  |
| C | -3.449054 | -5.601546 | -0.254387 | H | 8.337332  | -0.000018 | -3.893292 |
| C | -0.741598 | -6.414251 | -0.022908 | C | 6.452477  | 3.466045  | -0.566499 |
| C | -2.782903 | -5.564218 | 0.97566   | C | 5.683275  | 2.939297  | 1.633511  |
| C | -2.781344 | -6.211336 | -1.32456  | C | 5.581971  | 4.539     | -0.606114 |
| C | -1.458264 | -6.611204 | -1.21006  | H | 7.095487  | 3.30662   | -1.418478 |
| C | -1.459269 | -5.957776 | 1.087404  | C | 4.830777  | 4.022047  | 1.596276  |
| H | -3.266016 | -5.100346 | 1.829377  | H | 5.735997  | 2.369585  | 2.550566  |
| H | -3.287938 | -6.325903 | -2.278263 | C | 4.671018  | 4.787704  | 0.431774  |
| H | -0.957016 | -7.032645 | -2.07626  | H | 5.573382  | 5.173157  | -1.488083 |
| H | -0.94222  | -5.789595 | 2.026299  | H | 4.202564  | 4.221197  | 2.458844  |
| C | 0.741565  | 6.414243  | 0.022828  | C | -6.985927 | 1.205538  | -0.381591 |
| C | 3.449014  | 5.601517  | 0.254312  | C | -6.50068  | 0.000016  | -1.205206 |
| C | 1.45824   | 6.611216  | 1.209972  | C | -8.06283  | 0.78281   | 0.586437  |
| C | 1.459226  | 5.957732  | -1.087476 | C | -6.985939 | -1.205514 | -0.381614 |
| C | 2.782857  | 5.564163  | -0.975729 | C | -7.176511 | 0.000035  | -2.601858 |
| C | 2.781316  | 6.211338  | 1.324475  | C | -4.956627 | -0.000002 | -1.303235 |
| H | 0.942171  | 5.789532  | -2.026364 | C | -8.062836 | -0.782798 | 0.586423  |
| H | 3.265961  | 5.100265  | -1.829438 | C | -7.968921 | 1.189837  | 2.080384  |
| H | 3.287917  | 6.325923  | 2.278172  | H | -9.032472 | 1.144836  | 0.212104  |
| C | 0.741537  | -6.414243 | 0.022834  | H | -6.902249 | -0.8814   | -3.186803 |
| C | 3.448988  | -5.601527 | 0.254325  | H | -8.265948 | 0.000018  | -2.497818 |
| C | 1.458207  | -6.61122  | 1.20998   | H | -6.902273 | 0.881505  | -3.186762 |
| C | 1.459202  | -5.957734 | -1.087468 | H | -4.575753 | 0.876365  | -1.821967 |
| C | 2.782833  | -5.56417  | -0.975718 | H | -4.524456 | 0.000123  | -0.298561 |
| C | 2.781285  | -6.211346 | 1.324486  | H | -4.575766 | -0.876509 | -1.821742 |
| H | 0.956964  | -7.032685 | 2.076171  | H | -9.032482 | -1.144813 | 0.212089  |
| H | 0.942149  | -5.789533 | -2.026358 | C | -7.968921 | -1.189856 | 2.08036   |
| H | 3.265941  | -5.100274 | -1.829425 | C | -8.567533 | -0.000018 | 2.82366   |
| H | 3.287883  | -6.325934 | 2.278184  | H | -8.491024 | 2.124969  | 2.29879   |
| H | 0.957001  | 7.032684  | 2.076164  | H | -6.921554 | 1.312658  | 2.37558   |
| C | 4.670997  | -4.787722 | 0.431788  | H | -8.491021 | -2.124994 | 2.298748  |
| C | 5.581942  | -4.539013 | -0.606107 | H | -6.921552 | -1.312681 | 2.37555   |
| C | 4.83077   | -4.022076 | 1.596293  | H | -9.660004 | -0.000017 | 2.721351  |
| C | 6.45245   | -3.46606  | -0.566492 | H | -8.337031 | -0.000029 | 3.893431  |



**2DR-8CPP (anti)**

opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 3,  $\langle S^2 \rangle = 3.2350$ 

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.561808 Hartree

Sum of electronic and thermal Energies = -2624.506943 Hartree

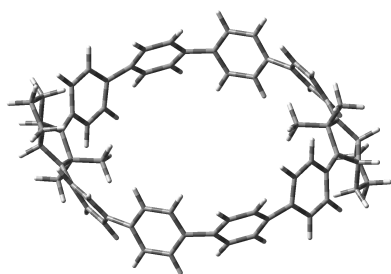
Sum of electronic and enthalpy Energies = -2624.505999 Hartree

Sum of electronic and thermal Free Energies = -2624.652618 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.417907             | 2.519823  | -0.498944 | H    | 5.590649              | -5.19752  | -1.464991 |
| C    | -4.661441             | 4.78836   | -0.426234 | C    | 5.676325              | -2.935913 | 1.637689  |
| C    | -6.44355              | 3.46336   | 0.565864  | H    | 4.196511              | -4.216256 | 2.46652   |
| C    | -5.669187             | 2.940952  | -1.633044 | C    | 6.425299              | -2.522385 | 0.499958  |
| C    | -4.81684              | 4.024029  | -1.591784 | H    | 7.107577              | -3.325995 | -1.402505 |
| C    | -5.573928             | 4.537248  | 0.609201  | H    | 5.721115              | -2.358336 | 2.550095  |
| H    | -7.08744              | 3.301388  | 1.416692  | C    | 6.974675              | -1.20691  | 0.373433  |
| H    | -5.719007             | 2.372365  | -2.550978 | C    | 8.036178              | -0.783773 | -0.612271 |
| H    | -4.185868             | 4.224677  | -2.452023 | C    | 6.488569              | 0.000505  | 1.189489  |
| H    | -5.567089             | 5.169491  | 1.492575  | H    | 9.012091              | -1.145892 | -0.255587 |
| C    | -6.417652             | -2.519191 | -0.501461 | C    | 8.036435              | 0.782759  | -0.612954 |
| C    | -4.660724             | -4.78787  | -0.430497 | C    | 7.912592              | -1.192    | -2.104105 |
| C    | -6.444001             | -3.464464 | 0.5622    | C    | 6.975117              | 1.206948  | 0.372361  |
| C    | -5.668297             | -2.939315 | -1.635906 | C    | 7.161891              | 0.001055  | 2.589429  |
| C    | -4.816412             | -4.022514 | -1.595685 | C    | 4.943203              | 0.000778  | 1.281857  |
| C    | -5.574015             | -4.53778  | 0.604923  | C    | 7.912742              | 1.189638  | -2.105169 |
| H    | -7.088873             | -3.303919 | 1.412566  | H    | 9.012567              | 1.14491   | -0.256842 |
| H    | -5.717947             | -2.369858 | -2.553309 | C    | 8.491894              | -0.001551 | -2.861301 |
| H    | -4.185675             | -4.22244  | -2.456239 | H    | 8.434399              | -2.124654 | -2.333278 |
| H    | -5.567859             | -5.171203 | 1.487442  | H    | 6.859798              | -1.320694 | -2.376781 |
| C    | -3.440341             | 5.604009  | -0.244971 | C    | 6.425378              | 2.523088  | 0.497473  |
| C    | -0.735477             | 6.422261  | -0.010565 | H    | 6.886258              | -0.879897 | 3.17393   |
| C    | -2.77252              | 5.557646  | 0.983549  | H    | 8.251368              | 0.000503  | 2.486801  |
| C    | -2.776116             | 6.223734  | -1.311142 | H    | 6.88706               | 0.882998  | 3.172819  |
| C    | -1.453674             | 6.626445  | -1.195166 | H    | 4.56069               | 0.877329  | 1.799078  |
| C    | -1.449747             | 5.954128  | 1.096761  | H    | 4.51568               | 0.001744  | 0.275189  |
| H    | -3.253449             | 5.085314  | 1.833877  | H    | 4.560324              | -0.876514 | 1.797528  |
| H    | -3.284329             | 6.343687  | -2.263349 | H    | 8.434558              | 2.12207   | -2.33523  |
| H    | -0.954181             | 7.055387  | -2.058735 | H    | 6.859932              | 1.318066  | -2.377906 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.9305   | 5.779811  | 2.033347  | H | 9.586547  | -0.001588 | -2.785779 |
| C | -3.44037  | -5.603468 | -0.249699 | H | 8.235504  | -0.002009 | -3.925174 |
| C | -0.735026 | -6.421716 | -0.012178 | C | 6.458455  | 3.478447  | -0.557196 |
| C | -2.77607  | -5.564871 | 0.981271  | C | 5.676901  | 2.93752   | 1.634819  |
| C | -2.772286 | -6.216901 | -1.317534 | C | 5.592051  | 4.555059  | -0.592904 |
| C | -1.450335 | -6.619659 | -1.200015 | H | 7.106218  | 3.323774  | -1.406465 |
| C | -1.45347  | -5.961023 | 1.095917  | C | 4.828398  | 4.023634  | 1.601488  |
| H | -3.259581 | -5.098158 | 1.833221  | H | 5.721786  | 2.360751  | 2.547731  |
| H | -3.277672 | -6.332289 | -2.271786 | C | 4.676829  | 4.798465  | 0.442177  |
| H | -0.948797 | -7.044276 | -2.064487 | H | 5.589919  | 5.196094  | -1.469969 |
| H | -0.937615 | -5.791749 | 2.035254  | H | 4.196312  | 4.218409  | 2.462252  |
| C | 0.74834   | 6.42552   | 0.034447  | C | -6.97699  | 1.205235  | -0.389056 |
| C | 3.455638  | 5.614692  | 0.264996  | C | -6.487525 | 0.000484  | -1.211213 |
| C | 1.46504   | 6.623852  | 1.220977  | C | -8.061089 | 0.782213  | 0.570548  |
| C | 1.46522   | 5.96889   | -1.075979 | C | -6.976433 | -1.205297 | -0.390131 |
| C | 2.789316  | 5.576117  | -0.964646 | C | -7.157369 | 0.000989  | -2.610808 |
| C | 2.788682  | 6.224757  | 1.335073  | C | -4.943211 | 0.000896  | -1.303171 |
| H | 0.947344  | 5.799909  | -2.014315 | C | -8.060769 | -0.783404 | 0.569853  |
| H | 3.272611  | 5.111787  | -1.818051 | C | -7.979149 | 1.188412  | 2.065379  |
| H | 3.295647  | 6.339054  | 2.288643  | H | -9.027957 | 1.144235  | 0.188985  |
| C | 0.748062  | -6.424935 | 0.036058  | H | -6.880109 | -0.880002 | -3.195036 |
| C | 3.455828  | -5.614038 | 0.269706  | H | -8.247253 | 0.000445  | -2.511547 |
| C | 1.461926  | -6.617047 | 1.22575   | H | -6.880903 | 0.882911  | -3.194009 |
| C | 1.469053  | -5.975668 | -1.075059 | H | -4.5607   | 0.877051  | -1.821112 |
| C | 2.792966  | -5.583191 | -0.962298 | H | -4.514858 | 0.001827  | -0.296864 |
| C | 2.78507   | -6.217867 | 1.34138   | H | -4.56015  | -0.875836 | -1.819699 |
| H | 0.958261  | -7.033546 | 2.092926  | H | -9.027372 | -1.145432 | 0.187682  |
| H | 0.954528  | -5.811713 | -2.01608  | C | -7.978955 | -1.190989 | 2.064295  |
| H | 3.278788  | -5.124431 | -1.817263 | C | -8.584515 | -0.001668 | 2.802877  |
| H | 3.289249  | -6.327634 | 2.296935  | H | -8.502147 | 2.123958  | 2.279894  |
| H | 0.963373  | 7.044618  | 2.0873    | H | -6.934151 | 1.309885  | 2.369379  |
| C | 4.676281  | -4.797875 | 0.446399  | H | -8.501936 | -2.126754 | 2.277886  |
| C | 5.592191  | -4.55539  | -0.58874  | H | -6.933978 | -1.312754 | 2.368255  |
| C | 4.828275  | -4.022137 | 1.605395  | H | -9.676019 | -0.001708 | 2.690775  |
| C | 6.458948  | -3.479346 | -0.553646 | H | -8.363533 | -0.002139 | 3.874654  |

### 2DR-8CPP (anti)



opt freq UCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 5,  $\langle S^2 \rangle = 6.2319$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.561912 Hartree

Sum of electronic and thermal Energies = -2624.507057 Hartree

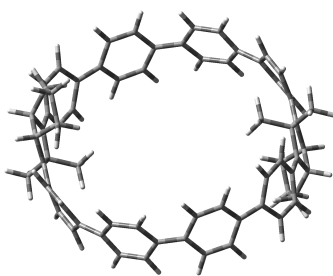
Sum of electronic and enthalpy Energies = -2624.506113 Hartree

Sum of electronic and thermal Free Energies = -2624.653060 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.413288             | -2.522835 | 0.503108  | H    | 5.585041              | 5.195395  | 1.467963  |
| C    | -4.666352             | -4.799327 | 0.44184   | C    | 5.660778              | 2.937465  | -1.637624 |
| C    | -6.449878             | -3.477603 | -0.551863 | H    | 4.178334              | 4.219642  | -2.460028 |
| C    | -5.660793             | -2.937417 | 1.637646  | C    | 6.413267              | 2.522817  | -0.503106 |
| C    | -4.81322              | -4.024225 | 1.601517  | H    | 7.099816              | 3.32197   | 1.399344  |
| C    | -5.584282             | -4.554813 | -0.590505 | H    | 5.702052              | 2.36057   | -2.550641 |
| H    | -7.099865             | -3.322105 | -1.399283 | C    | 6.964277              | 1.206738  | -0.380819 |
| H    | -5.702054             | -2.360461 | 2.550625  | C    | 8.032425              | 0.783242  | 0.597368  |
| H    | -4.178339             | -4.219543 | 2.46011   | C    | 6.473969              | -0.000007 | -1.195168 |
| H    | -5.585097             | -5.195538 | -1.467802 | H    | 9.005954              | 1.145378  | 0.234148  |
| C    | -6.413243             | 2.522867  | 0.503032  | C    | 8.032433              | -0.783315 | 0.597335  |
| C    | -4.666314             | 4.799366  | 0.44178   | C    | 7.919522              | 1.190641  | 2.090237  |
| C    | -6.449811             | 3.477624  | -0.551951 | C    | 6.964285              | -1.20678  | -0.380864 |
| C    | -5.660785             | 2.937473  | 1.637585  | C    | 7.141137              | 0.000026  | -2.598183 |
| C    | -4.813212             | 4.024281  | 1.601464  | C    | 4.928323              | -0.000014 | -1.281141 |
| C    | -5.584218             | 4.554838  | -0.590584 | C    | 7.919536              | -1.190779 | 2.090186  |
| H    | -7.09978              | 3.322119  | -1.399383 | H    | 9.005964              | -1.145428 | 0.234098  |
| H    | -5.702074             | 2.360534  | 2.550574  | C    | 8.504847              | -0.000082 | 2.842404  |
| H    | -4.178355             | 4.219614  | 2.460072  | H    | 8.442348              | 2.123564  | 2.316019  |
| H    | -5.585013             | 5.195552  | -1.467889 | H    | 6.868676              | 1.318311  | 2.370796  |
| C    | -3.446989             | -5.617572 | 0.261172  | C    | 6.413257              | -2.522849 | -0.503169 |
| C    | -0.741925             | -6.434176 | 0.024006  | H    | 6.863247              | 0.881491  | -3.180848 |
| C    | -2.782885             | -5.578106 | -0.969599 | H    | 8.231059              | 0.000032  | -2.500387 |
| C    | -2.779414             | -6.231091 | 1.328855  | H    | 6.86326               | -0.88142  | -3.180884 |
| C    | -1.456958             | -6.633224 | 1.211447  | H    | 4.543592              | -0.876937 | -1.796104 |
| C    | -1.45987              | -5.973563 | -1.08411  | H    | 4.504884              | -0.000058 | -0.272744 |
| H    | -3.266788             | -5.11112  | -1.821218 | H    | 4.54358               | 0.876946  | -1.796033 |
| H    | -3.284867             | -6.345933 | 2.283169  | H    | 8.442373              | -2.123706 | 2.315927  |
| H    | -0.95482              | -7.056998 | 2.07602   | H    | 6.868691              | -1.318474 | 2.370739  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.943472 | -5.80372  | -2.023084 | H | 9.598868  | -0.000073 | 2.758241  |
| C | -3.446947 | 5.61761   | 0.261131  | H | 8.256805  | -0.000106 | 3.908253  |
| C | -0.741877 | 6.434206  | 0.024009  | C | 6.449829  | -3.477565 | 0.55185   |
| C | -2.782822 | 5.57814   | -0.969629 | C | 5.660742  | -2.937469 | -1.63768  |
| C | -2.779388 | 6.23113   | 1.328825  | C | 5.584217  | -4.55476  | 0.590547  |
| C | -1.456929 | 6.633257  | 1.211438  | H | 7.099826  | -3.322038 | 1.399259  |
| C | -1.459805 | 5.973594  | -1.084118 | C | 4.813154  | -4.024263 | -1.601496 |
| H | -3.266713 | 5.111155  | -1.821255 | H | 5.702006  | -2.360559 | -2.550688 |
| H | -3.284857 | 6.345974  | 2.28313   | C | 4.666282  | -4.799311 | -0.441784 |
| H | -0.954804 | 7.05703   | 2.076018  | H | 5.585029  | -5.195446 | 1.467873  |
| H | -0.943392 | 5.803748  | -2.023083 | H | 4.178267  | -4.219614 | -2.460078 |
| C | 0.741861  | -6.434168 | -0.023875 | C | -6.964265 | -1.206749 | 0.380748  |
| C | 3.446919  | -5.617548 | -0.26108  | C | -6.473874 | 0.000026  | 1.195001  |
| C | 1.456896  | -6.633262 | -1.211308 | C | -8.032424 | -0.783272 | -0.597436 |
| C | 1.459803  | -5.973495 | 1.084218  | C | -6.964219 | 1.206778  | 0.380684  |
| C | 2.782814  | -5.578031 | 0.969689  | C | -7.140932 | 0.00008   | 2.598065  |
| C | 2.779348  | -6.231121 | -1.328735 | C | -4.928222 | -0.000014 | 1.280864  |
| H | 0.943403  | -5.803614 | 2.023185  | C | -8.032388 | 0.783285  | -0.597485 |
| H | 3.266714  | -5.111004 | 1.821287  | C | -7.919577 | -1.190753 | -2.090287 |
| H | 3.284803  | -6.346005 | -2.283043 | H | -9.005954 | -1.145359 | -0.234172 |
| C | 0.74191   | 6.434191  | -0.023849 | H | -6.862978 | 0.881547  | 3.1807    |
| C | 3.446964  | 5.61755   | -0.26102  | H | -8.230861 | 0.000103  | 2.500357  |
| C | 1.456964  | 6.63329   | -1.211268 | H | -6.86302  | -0.881367 | 3.18075   |
| C | 1.459831  | 5.973502  | 1.084252  | H | -4.543475 | -0.876941 | 1.795808  |
| C | 2.782841  | 5.578027  | 0.969739  | H | -4.504855 | -0.000078 | 0.272438  |
| C | 2.779415  | 6.23114   | -1.328679 | H | -4.543422 | 0.876939  | 1.795721  |
| H | 0.954846  | 7.057121  | -2.075824 | H | -9.005906 | 1.145434  | -0.234246 |
| H | 0.943416  | 5.803617  | 2.023209  | C | -7.91952  | 1.190665  | -2.090362 |
| H | 3.266724  | 5.110988  | 1.821339  | C | -8.504892 | -0.000054 | -2.842498 |
| H | 3.284885  | 6.346028  | -2.282979 | H | -8.442436 | -2.123673 | -2.316005 |
| H | 0.954761  | -7.057081 | -2.07586  | H | -6.868743 | -1.318467 | -2.37087  |
| C | 4.666321  | 4.799301  | -0.441716 | H | -8.442334 | 2.123595  | -2.316142 |
| C | 5.584235  | 4.554723  | 0.590627  | H | -6.86868  | 1.318308  | -2.370956 |
| C | 4.813203  | 4.02427   | -1.601438 | H | -9.59891  | -0.000025 | -2.7583   |
| C | 6.449834  | 3.477517  | 0.551928  | H | -8.256884 | -0.000094 | -3.908355 |

### 2DR-8CPP (anti)



opt freq RCAM-B3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -2624.467575 Hartree

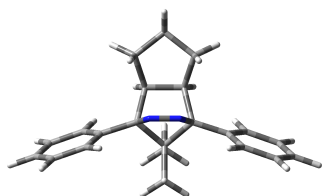
Sum of electronic and thermal Energies = -2624.412698 Hartree

Sum of electronic and enthalpy Energies = -2624.411754 Hartree

Sum of electronic and thermal Free Energies = -2624.554789 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -6.02897              | 2.447193  | -0.425586 | H    | 5.556697              | -5.39657  | -1.269178 |
| C    | -4.551255             | 4.912381  | -0.618061 | C    | 5.475521              | -2.924522 | 1.692729  |
| C    | -6.182383             | 3.521428  | 0.555882  | H    | 4.348742              | -4.355384 | 2.731954  |
| C    | -5.47546              | 2.924448  | -1.692753 | C    | 6.028944              | -2.447213 | 0.425545  |
| C    | -4.78875              | 4.082567  | -1.779481 | H    | 6.858422              | -3.379652 | -1.389528 |
| C    | -5.45708              | 4.655984  | 0.48232   | H    | 5.591833              | -2.326573 | 2.586372  |
| H    | -6.858428             | 3.379691  | 1.389464  | C    | 6.282338              | -1.144832 | 0.114263  |
| H    | -5.591748             | 2.32648   | -2.586387 | C    | 6.825984              | -0.692624 | -1.231728 |
| H    | -4.348583             | 4.355243  | -2.731965 | C    | 6.007972              | 0.068654  | 1.020839  |
| H    | -5.556615             | 5.396547  | 1.269118  | H    | 7.898852              | -0.92412  | -1.274848 |
| C    | -5.979552             | -2.590764 | -0.414246 | C    | 6.61751               | 0.844961  | -1.276941 |
| C    | -4.469715             | -5.034881 | -0.581389 | C    | 6.130825              | -1.248285 | -2.512008 |
| C    | -6.0718               | -3.628933 | 0.610801  | C    | 6.228989              | 1.285833  | 0.112745  |
| C    | -5.465156             | -3.094768 | -1.685947 | C    | 7.013487              | 0.086846  | 2.195811  |
| C    | -4.757119             | -4.24174  | -1.75797  | C    | 4.539298              | 0.058578  | 1.510727  |
| C    | -5.340763             | -4.75946  | 0.542816  | C    | 5.540957              | 1.062105  | -2.367341 |
| H    | -6.706417             | -3.458146 | 1.473082  | H    | 7.538455              | 1.343662  | -1.605157 |
| H    | -5.613035             | -2.520236 | -2.590811 | C    | 5.85557               | -0.024066 | -3.387551 |
| H    | -4.338814             | -4.534189 | -2.714501 | H    | 6.737426              | -1.997084 | -3.028014 |
| H    | -5.397773             | -5.473049 | 1.357778  | H    | 5.186971              | -1.730215 | -2.24198  |
| C    | -3.44903              | 5.759902  | -0.519595 | C    | 5.979565              | 2.59078   | 0.414306  |
| C    | -0.755416             | 6.708734  | -0.12733  | H    | 6.914319              | -0.801401 | 2.823833  |
| C    | -3.00221              | 6.284276  | 0.746249  | H    | 8.039684              | 0.11203   | 1.818034  |
| C    | -2.557899             | 6.015076  | -1.622899 | H    | 6.874953              | 0.962928  | 2.833338  |
| C    | -1.299285             | 6.483683  | -1.440184 | H    | 4.299555              | 0.961233  | 2.0724    |
| C    | -1.727314             | 6.706156  | 0.934142  | H    | 3.860929              | 0.014191  | 0.654131  |
| H    | -3.66927              | 6.251804  | 1.60007   | H    | 4.322931              | -0.800994 | 2.143755  |
| H    | -2.883806             | 5.788436  | -2.631561 | H    | 5.549457              | 2.072056  | -2.781492 |
| H    | -0.660979             | 6.594483  | -2.308732 | H    | 4.545681              | 0.897916  | -1.937321 |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -1.428064 | 7.020434  | 1.92771   | H | 6.752262  | 0.248845  | -3.958048 |
| C | -3.348938 | -5.858484 | -0.488207 | H | 5.048425  | -0.191504 | -4.107235 |
| C | -0.628264 | -6.731041 | -0.104949 | C | 6.071882  | 3.62897   | -0.610712 |
| C | -2.881642 | -6.366942 | 0.777044  | C | 5.46518   | 3.094781  | 1.686012  |
| C | -2.457775 | -6.098419 | -1.595403 | C | 5.340877  | 4.759517  | -0.54272  |
| C | -1.184435 | -6.528502 | -1.416408 | H | 6.706503  | 3.458175  | -1.472989 |
| C | -1.596177 | -6.755565 | 0.95991   | C | 4.757181  | 4.241774  | 1.758046  |
| H | -3.545532 | -6.355838 | 1.633575  | H | 5.613031  | 2.520228  | 2.590867  |
| H | -2.793387 | -5.883648 | -2.603494 | C | 4.469811  | 5.034937  | 0.581472  |
| H | -0.546355 | -6.623672 | -2.287152 | H | 5.397917  | 5.473118  | -1.357669 |
| H | -1.285969 | -7.063917 | 1.95185   | H | 4.338867  | 4.534217  | 2.714576  |
| C | 0.628351  | 6.731037  | 0.104969  | C | -6.282464 | 1.144833  | -0.114304 |
| C | 3.349036  | 5.858541  | 0.488274  | C | -6.008228 | -0.068669 | -1.020893 |
| C | 1.184506  | 6.528522  | 1.416438  | C | -6.826086 | 0.692656  | 1.231706  |
| C | 1.596282  | 6.75558   | -0.959874 | C | -6.229058 | -1.285825 | -0.112723 |
| C | 2.881752  | 6.366987  | -0.776985 | C | -7.013957 | -0.086928 | -2.195684 |
| C | 2.457851  | 6.098464  | 1.595455  | C | -4.539647 | -0.058553 | -1.511049 |
| H | 1.286082  | 7.063918  | -1.95182  | C | -6.617569 | -0.844924 | 1.276959  |
| H | 3.545654  | 6.355889  | -1.633506 | C | -6.130914 | 1.248373  | 2.511955  |
| H | 2.79345   | 5.883699  | 2.603553  | H | -7.898957 | 0.924135  | 1.274837  |
| C | 0.755507  | -6.708776 | 0.127328  | H | -6.87551  | -0.963021 | -2.833214 |
| C | 3.449141  | -5.759995 | 0.519569  | H | -8.040084 | -0.112139 | -1.817718 |
| C | 1.299402  | -6.483764 | 1.440178  | H | -6.91494  | 0.801303  | -2.823751 |
| C | 1.727389  | -6.706197 | -0.934159 | H | -4.323432 | 0.801024  | -2.144127 |
| C | 3.002293  | -6.284339 | -0.746278 | H | -3.86112  | -0.014132 | -0.654579 |
| C | 2.558027  | -6.015183 | 1.622883  | H | -4.299981 | -0.961207 | -2.072758 |
| H | 0.661107  | -6.594568 | 2.308734  | H | -7.538491 | -1.343646 | 1.605206  |
| H | 1.428116  | -7.020448 | -1.927729 | C | -5.540988 | -1.062005 | 2.367345  |
| H | 3.669338  | -6.251853 | -1.600112 | C | -5.855611 | 0.024188  | 3.38753   |
| H | 2.883951  | -5.788562 | 2.631543  | H | -6.737523 | 1.997173  | 3.02795   |
| H | 0.54641   | 6.623681  | 2.287171  | H | -5.187078 | 1.730318  | 2.241892  |
| C | 4.551359  | -4.912465 | 0.618029  | H | -5.549452 | -2.071943 | 2.781527  |
| C | 5.457153  | -4.656024 | -0.482366 | H | -4.545725 | -0.897802 | 1.937301  |
| C | 4.788865  | -4.082671 | 1.77946   | H | -6.752285 | -0.24873  | 3.95805   |
| C | 6.182405  | -3.521434 | -0.55593  | H | -5.048457 | 0.191668  | 4.107193  |

**AZ-2Ph**

opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$ 

Number of Imaginary Frequencies = 0

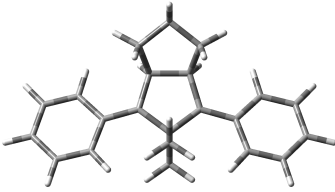
Sum of electronic and zero-point Energies = -961.840788 Hartree

Sum of electronic and thermal Energies = -961.821144 Hartree

Sum of electronic and enthalpy Energies = -961.820200 Hartree

Sum of electronic and thermal Free Energies = -961.888286 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 0.000001              | -0.815708 | 0.467939  | C    | -3.186367             | -0.886171 | -1.20283  |
| C    | 1.123827              | 0.115533  | -0.128115 | C    | -3.29772              | -0.263162 | 1.12165   |
| C    | 0.781565              | 1.507769  | 0.510293  | C    | -4.614126             | -0.720808 | 1.1837    |
| C    | -0.781567             | 1.507767  | 0.510293  | C    | -4.504468             | -1.339999 | -1.142322 |
| C    | -1.123826             | 0.11553   | -0.128116 | H    | -2.63583              | -0.937046 | -2.135941 |
| N    | -0.623121             | 0.256531  | -1.540017 | H    | -2.84524              | 0.161271  | 2.013245  |
| N    | 0.623123              | 0.256536  | -1.540017 | H    | -5.164209             | -0.649834 | 2.118538  |
| H    | -1.171232             | 1.533854  | 1.531472  | H    | -4.969836             | -1.754597 | -2.032892 |
| H    | 1.17123               | 1.533856  | 1.531472  | C    | 0.000001              | -2.231822 | -0.132683 |
| C    | 1.216454              | 2.787531  | -0.244131 | H    | 0.885699              | -2.781813 | 0.203354  |
| H    | 1.40107               | 2.569324  | -1.300609 | H    | 0.000001              | -2.232334 | -1.224949 |
| H    | 2.135722              | 3.218596  | 0.165349  | H    | -0.885698             | -2.781812 | 0.203354  |
| C    | 2.561399              | -0.340891 | -0.071147 | C    | 0.000003              | -0.961178 | 1.994782  |
| C    | 5.223821              | -1.261367 | 0.050643  | H    | 0.882345              | -1.527681 | 2.313074  |
| C    | 3.186357              | -0.886192 | -1.202825 | H    | -0.882328             | -1.527698 | 2.313076  |
| C    | 3.297731              | -0.263137 | 1.121642  | H    | -0.000006             | -0.013474 | 2.537957  |
| C    | 4.614137              | -0.720786 | 1.18369   | C    | -0.000003             | 3.726508  | -0.107725 |
| C    | 4.504458              | -1.340022 | -1.142319 | C    | -1.216459             | 2.787529  | -0.24413  |
| H    | 2.635813              | -0.937082 | -2.135931 | H    | 6.250491              | -1.61471  | 0.097016  |
| H    | 2.84526               | 0.161315  | 2.013232  | H    | -6.250492             | -1.614706 | 0.097022  |
| H    | 5.164227              | -0.649795 | 2.118522  | H    | -2.135727             | 3.218593  | 0.165351  |
| H    | 4.969817              | -1.754639 | -2.032885 | H    | -1.401075             | 2.569323  | -1.300608 |
| C    | -2.561398             | -0.340894 | -0.071147 | H    | -0.000004             | 4.533257  | -0.849233 |
| C    | -5.223821             | -1.261365 | 0.050647  | H    | -0.000003             | 4.198381  | 0.884528  |

| DR-2Ph                                                                            |                       |           |           |                                                                   |                       |           |           |
|-----------------------------------------------------------------------------------|-----------------------|-----------|-----------|-------------------------------------------------------------------|-----------------------|-----------|-----------|
|  |                       |           |           | opt freq UB3LYP/6-31G(d)                                          |                       |           |           |
|                                                                                   |                       |           |           | Charge = 0, Multiplicity = 3, <S <sup>2</sup> > = 2.0530          |                       |           |           |
|                                                                                   |                       |           |           | Number of Imaginary Frequencies = 0                               |                       |           |           |
|                                                                                   |                       |           |           | Sum of electronic and zero-point Energies = -852.325368 Hartree   |                       |           |           |
|                                                                                   |                       |           |           | Sum of electronic and thermal Energies = -852.306085 Hartree      |                       |           |           |
|                                                                                   |                       |           |           | Sum of electronic and enthalpy Energies = -852.305141 Hartree     |                       |           |           |
|                                                                                   |                       |           |           | Sum of electronic and thermal Free Energies = -852.373869 Hartree |                       |           |           |
| Atom                                                                              | Cartesian Coordinates |           |           | Atom                                                              | Cartesian Coordinates |           |           |
|                                                                                   | X                     | Y         | Z         |                                                                   | X                     | Y         | Z         |
| C                                                                                 | -0.000025             | -0.839735 | 0.190813  | C                                                                 | 4.433321              | -1.905867 | 0.260067  |
| C                                                                                 | -1.207268             | 0.056463  | -0.12304  | H                                                                 | 2.402844              | -2.319793 | 0.721937  |
| C                                                                                 | -0.778568             | 1.466107  | -0.444415 | H                                                                 | 3.261729              | 1.540706  | -1.033562 |
| C                                                                                 | 0.778463              | 1.466045  | -0.444754 | H                                                                 | 5.627829              | 0.964096  | -1.109315 |
| C                                                                                 | 1.207171              | 0.056397  | -0.12338  | H                                                                 | 4.761996              | -2.875002 | 0.627423  |
| H                                                                                 | 1.148293              | 1.768022  | -1.435184 | C                                                                 | 0.000257              | -1.248347 | 1.697957  |
| H                                                                                 | -1.148784             | 1.76826   | -1.434646 | H                                                                 | -0.882843             | -1.83805  | 1.957808  |
| C                                                                                 | -1.203411             | 2.560585  | 0.587893  | H                                                                 | 0.000429              | -0.35626  | 2.332878  |
| H                                                                                 | -1.361247             | 2.099871  | 1.571096  | H                                                                 | 0.883426              | -1.838124 | 1.957436  |
| H                                                                                 | -2.138141             | 3.060138  | 0.31505   | C                                                                 | -0.000214             | -2.090486 | -0.738018 |
| C                                                                                 | -2.586953             | -0.329696 | -0.150407 | H                                                                 | -0.882896             | -2.713661 | -0.579609 |
| C                                                                                 | -3.08289              | -1.58796  | 0.306941  | H                                                                 | 0.882481              | -2.713715 | -0.579906 |
| C                                                                                 | -3.571974             | 0.572804  | -0.657502 | H                                                                 | -0.000381             | -1.776854 | -1.787044 |
| C                                                                                 | -4.919158             | 0.244799  | -0.706349 | C                                                                 | 0.000209              | 3.51139   | 0.643061  |
| C                                                                                 | -4.433439             | -1.905985 | 0.259501  | C                                                                 | 1.203836              | 2.560622  | 0.587244  |
| H                                                                                 | -2.402971             | -2.320181 | 0.721134  | H                                                                 | 2.138376              | 3.060214  | 0.313827  |
| H                                                                                 | -3.26179              | 1.541157  | -1.032555 | H                                                                 | 1.362274              | 2.09998   | 1.570382  |
| H                                                                                 | -5.627896             | 0.964606  | -1.108602 | H                                                                 | 0.000441              | 4.152603  | 1.53206   |
| H                                                                                 | -4.762123             | -2.875299 | 0.626377  | H                                                                 | -0.000041             | 4.170845  | -0.235874 |
| C                                                                                 | 2.586854              | -0.329768 | -0.150634 | C                                                                 | -5.368986             | -0.998032 | -0.247565 |
| C                                                                                 | 3.082775              | -1.58781  | 0.307336  | H                                                                 | -6.424339             | -1.252761 | -0.28318  |
| C                                                                                 | 3.571893              | 0.572515  | -0.658081 | C                                                                 | 5.368883              | -0.998158 | -0.247414 |
| C                                                                                 | 4.919075              | 0.244473  | -0.706759 | H                                                                 | 6.424235              | -1.252909 | -0.282886 |



### Biphenyl

opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

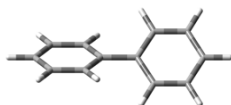
Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -463.124102 Hartree

Sum of electronic and thermal Energies = -463.115231 Hartree

Sum of electronic and enthalpy Energies = -463.114287 Hartree

Sum of electronic and thermal Free Energies = -463.158663 Hartree



| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | -2.859871             | 1.139436  | -0.393974 | C    | 0.7429                | -0.000028 | 0.000043  |
| C    | -1.465785             | 1.138697  | -0.394434 | C    | 1.465836              | -1.138729 | -0.394424 |
| C    | -0.7429               | -0.000051 | -0.000009 | C    | 1.465784              | 1.138689  | 0.394444  |
| C    | -1.465834             | -1.138721 | 0.394435  | C    | 2.859921              | -1.139409 | -0.393962 |
| C    | -2.859949             | -1.139394 | 0.393949  | H    | 0.928376              | -2.021812 | -0.728859 |
| C    | -3.56381              | 0.000026  | -0.000018 | C    | 2.859898              | 1.139421  | 0.39396   |
| H    | -3.397158             | 2.029568  | -0.710959 | H    | 0.928329              | 2.021797  | 0.728828  |
| H    | -0.928281             | 2.021781  | -0.728797 | C    | 3.56381               | 0.000032  | -0.000002 |
| H    | -0.928425             | -2.021828 | 0.728888  | H    | 3.397248              | -2.029502 | -0.710991 |
| H    | -3.397248             | -2.029506 | 0.710971  | H    | 3.397157              | 2.029572  | 0.710941  |
| H    | -4.650487             | 0.000077  | -0.000025 | H    | 4.650487              | 0.00004   | -0.000036 |

### Terphenyl

opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

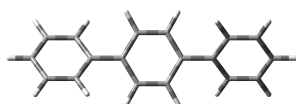
Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -694.100704 Hartree

Sum of electronic and thermal Energies = -694.087153 Hartree

Sum of electronic and enthalpy Energies = -694.086209 Hartree

Sum of electronic and thermal Free Energies = -694.142197 Hartree



| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 5.026248              | -0.95921  | -0.730253 | C    | -0.695673             | -1.201007 | -0.000315 |
| C    | 3.632272              | -0.958556 | -0.730605 | H    | 1.22686               | -2.148663 | 0.018636  |
| C    | 2.90897               | -0.000045 | 0.000002  | C    | -1.424577             | 0.000018  | -0.000077 |
| C    | 3.63218               | 0.958527  | 0.730597  | H    | -1.226774             | 2.148655  | 0.018502  |
| C    | 5.026165              | 0.959255  | 0.730316  | H    | -1.226866             | -2.148622 | -0.018457 |
| C    | 5.730199              | 0.000039  | 0.000059  | C    | -2.908962             | 0.000044  | -0.000009 |
| H    | 5.563469              | -1.705688 | -1.309588 | C    | -3.632214             | -0.958607 | 0.730464  |

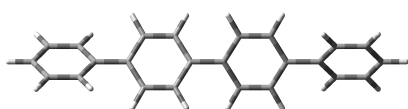
|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | 3.09547   | -1.692779 | -1.32492  | C | -3.632289 | 0.958633  | -0.73048  |
| H | 3.095316  | 1.692762  | 1.324845  | C | -5.026201 | -0.959318 | 0.73021   |
| H | 5.563306  | 1.705786  | 1.309656  | H | -3.095381 | -1.692922 | 1.324646  |
| H | 6.816845  | 0.000068  | 0.000075  | C | -5.026283 | 0.959297  | -0.73014  |
| C | 1.424623  | -0.000042 | -0.000069 | H | -3.095534 | 1.692929  | -1.324753 |
| C | 0.695732  | 1.201007  | -0.000402 | C | -5.730253 | -0.000015 | 0.000062  |
| C | 0.695684  | -1.20105  | 0.000349  | H | -5.563349 | -1.705906 | 1.30947   |
| C | -0.695614 | 1.201027  | 0.000254  | H | -5.563482 | 1.70585   | -1.309398 |
| H | 1.226971  | 2.14859   | -0.018586 | H | -6.816897 | -0.000044 | 0.000094  |

### Quaterphenyl

opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0



Sum of electronic and zero-point Energies = -925.077374 Hartree

Sum of electronic and thermal Energies = -925.059112 Hartree

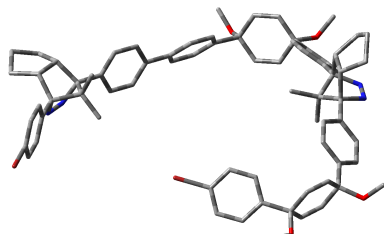
Sum of electronic and enthalpy Energies = -925.058168 Hartree

Sum of electronic and thermal Free Energies = -925.125729 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|-----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y         | Z         |
| C    | 2.861977              | 1.140621  | 0.376038  | C    | -5.798479             | 0.680297  | -0.994906 |
| C    | 1.470752              | 1.14062   | 0.376557  | C    | -5.798495             | -0.680313 | 0.994909  |
| C    | 0.741408              | 0.00002   | 0.000001  | C    | -7.19246              | 0.680968  | -0.994876 |
| C    | 1.470741              | -1.140587 | -0.376552 | H    | -5.261666             | 1.190492  | -1.789889 |
| C    | 2.861966              | -1.140599 | -0.376041 | C    | -7.192477             | -0.680982 | 0.994858  |
| C    | 3.591016              | 0.000009  | -0.000004 | H    | -5.261697             | -1.190497 | 1.789909  |
| H    | 3.393062              | 2.046213  | 0.655866  | C    | -7.896451             | -0.000009 | -0.000016 |
| H    | 0.94026               | 2.034988  | 0.691332  | H    | -7.729639             | 1.20745   | -1.779545 |
| H    | 0.940239              | -2.034955 | -0.691313 | H    | -7.729669             | -1.207456 | 1.779523  |
| H    | 3.393044              | -2.046195 | -0.655869 | H    | -8.983091             | -0.000006 | -0.000023 |
| C    | -0.741417             | 0.000018  | 0.000013  | C    | 5.075291              | 0         | -0.000009 |
| C    | -1.470748             | -1.140864 | 0.375747  | C    | 5.798497              | 0.678752  | 0.995956  |
| C    | -1.470767             | 1.140884  | -0.375737 | C    | 5.798488              | -0.67876  | -0.995961 |
| C    | -2.861967             | -1.14089  | 0.375221  | C    | 7.192479              | 0.679391  | 0.995938  |
| H    | -0.940236             | -2.035457 | 0.689855  | H    | 5.261721              | 1.187747  | 1.791735  |
| C    | -2.861986             | 1.140889  | -0.375207 | C    | 7.192479              | -0.679419 | -0.995941 |
| H    | -0.94027              | 2.035477  | -0.689867 | H    | 5.261719              | -1.187757 | -1.791744 |
| C    | -3.591033             | -0.000003 | 0.000016  | C    | 7.896457              | -0.000021 | -0.000004 |
| H    | -3.393071             | -2.046688 | 0.654356  | H    | 7.729667              | 1.204634  | 1.78143   |

|   |           |           |           |   |          |           |           |
|---|-----------|-----------|-----------|---|----------|-----------|-----------|
| H | -3.393106 | 2.046671  | -0.654363 | H | 7.729646 | -1.204679 | -1.781436 |
| C | -5.075273 | -0.000014 | 0.000003  | H | 8.983098 | -0.000024 | 0.000008  |

#### 4 (syn)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -8448.826947 Hartree

Sum of electronic and thermal Energies = -8448.753171 Hartree

Sum of electronic and enthalpy Energies = -8448.752227 Hartree

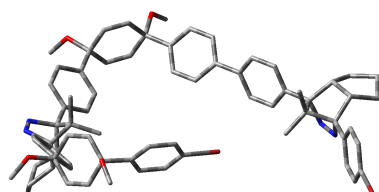
Sum of electronic and thermal Free Energies = -8448.949241 Hartree

| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |            |           |
|------|-----------------------|-----------|-----------|------|-----------------------|------------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y          | Z         |
| C    | 5.477283              | 0.657351  | 0.326556  | C    | 2.198138              | -6.463809  | 0.288348  |
| C    | 6.602525              | -0.287053 | 0.899161  | C    | 0.872787              | -6.083548  | 0.504041  |
| C    | 6.59316               | 0.065547  | 2.428239  | C    | 0.200036              | -7.399204  | -1.417019 |
| C    | 6.328817              | 1.606292  | 2.420802  | H    | 1.784439              | -8.429072  | -2.441022 |
| C    | 6.222965              | 1.927818  | 0.888456  | H    | 2.967841              | -6.088916  | 0.955892  |
| N    | 7.615416              | 1.656119  | 0.386601  | H    | 0.613266              | -5.428522  | 1.328771  |
| N    | 7.826022              | 0.427827  | 0.392173  | H    | -0.580555             | -7.761687  | -2.077307 |
| H    | 5.373192              | 1.840012  | 2.897846  | C    | 0.076126              | 7.218219   | -0.928223 |
| H    | 5.769363              | -0.468112 | 2.909837  | C    | -2.242422             | 5.608351   | -0.596079 |
| C    | 7.48307               | 2.249461  | 3.226964  | C    | -1.006855             | 7.321171   | -1.812436 |
| H    | 8.287693              | 2.572306  | 2.558968  | C    | -0.016449             | 6.296131   | 0.120164  |
| C    | 7.894979              | -0.148194 | 3.238102  | C    | -1.15293              | 5.50623    | 0.28344   |
| H    | 8.761985              | -0.189558 | 2.571427  | C    | -2.143118             | 6.534356   | -1.648572 |
| H    | 7.879137              | -1.080836 | 3.811371  | H    | -0.957515             | 8.039646   | -2.624625 |
| C    | 7.981945              | 1.105131  | 4.133454  | H    | 0.813014              | 6.182058   | 0.811831  |
| H    | 8.990867              | 1.280218  | 4.523399  | H    | -1.184061             | 4.780858   | 1.091912  |
| C    | 5.730526              | 3.299303  | 0.498519  | H    | -2.978646             | 6.65851    | -2.331977 |
| C    | 4.768616              | 5.8544    | -0.285157 | O    | 4.053995              | -8.999018  | -1.689525 |
| C    | 6.379108              | 4.042878  | -0.499026 | O    | 0.982439              | 9.317358   | -1.796188 |
| C    | 4.59265               | 3.860917  | 1.09497   | C    | 0.28309               | 10.254272  | -0.990572 |
| C    | 4.118505              | 5.11428   | 0.707772  | H    | -0.067217             | 11.034621  | -1.671518 |
| C    | 5.906839              | 5.296227  | -0.88204  | H    | 0.934644              | 10.716582  | -0.236611 |
| H    | 7.2717                | 3.637987  | -0.963739 | H    | -0.585798             | 9.807063   | -0.488179 |
| H    | 4.064449              | 3.32216   | 1.876476  | C    | 3.740881              | -10.115452 | -0.869229 |
| H    | 3.232354              | 5.517376  | 1.189123  | H    | 4.542749              | -10.336833 | -0.152187 |
| H    | 6.430864              | 5.860052  | -1.647661 | H    | 3.636392              | -10.966748 | -1.547064 |

|   |          |           |           |   |           |           |           |
|---|----------|-----------|-----------|---|-----------|-----------|-----------|
| C | 6.59424  | -1.746971 | 0.52019   | H | 2.797317  | -9.981127 | -0.3222   |
| C | 6.530748 | -4.4795   | -0.252667 | C | -3.454553 | 4.770309  | -0.421241 |
| C | 5.725302 | -2.656473 | 1.139405  | C | -5.783643 | 3.167175  | -0.087408 |
| C | 7.433878 | -2.233164 | -0.493478 | C | -3.94352  | 4.453833  | 0.856774  |
| C | 7.401984 | -3.573401 | -0.871883 | C | -4.158656 | 4.266858  | -1.527042 |
| C | 5.692602 | -3.997805 | 0.758069  | C | -5.297672 | 3.482245  | -1.364914 |
| H | 5.063455 | -2.323979 | 1.933939  | C | -5.083587 | 3.670995  | 1.019554  |
| H | 8.128542 | -1.553437 | -0.975299 | H | -3.44264  | 4.851311  | 1.735208  |
| H | 8.067501 | -3.930463 | -1.651794 | H | -3.796374 | 4.472832  | -2.530408 |
| H | 5.005663 | -4.673819 | 1.258672  | H | -5.82527  | 3.112393  | -2.237472 |
| C | 5.383851 | 0.635037  | -1.208627 | H | -5.434919 | 3.4649    | 2.026494  |
| H | 6.346764 | 0.797868  | -1.697885 | C | -6.995255 | 2.286301  | 0.085992  |
| H | 4.695915 | 1.414497  | -1.55344  | C | -8.075177 | 2.739008  | 1.131024  |
| H | 4.993973 | -0.331036 | -1.546945 | H | -7.581759 | 2.935208  | 2.086688  |
| C | 4.066954 | 0.416165  | 0.878968  | C | -9.001167 | 1.483653  | 1.23059   |
| H | 3.689322 | -0.548445 | 0.521519  | H | -8.969373 | 1.054312  | 2.235753  |
| H | 3.385726 | 1.18948   | 0.506754  | C | -8.325838 | 0.481988  | 0.22907   |
| H | 4.003682 | 0.414338  | 1.969488  | C | -6.790554 | 0.76476   | 0.447657  |
| C | 6.480131 | -5.94059  | -0.742287 | C | -6.256122 | 0.487539  | 1.858368  |
| C | 4.870225 | -7.648699 | 0.178408  | H | -5.220603 | 0.837353  | 1.93643   |
| C | 4.552229 | -6.771663 | -2.134868 | H | -6.248136 | -0.591767 | 2.047654  |
| C | 4.010457 | -7.695657 | -1.066077 | H | -6.827996 | 0.961775  | 2.659149  |
| C | 5.631875 | -6.00644  | -1.993513 | C | -5.885149 | 0.017682  | -0.546521 |
| C | 5.950964 | -6.879417 | 0.320028  | H | -6.188319 | 0.154144  | -1.586963 |
| H | 4.000002 | -6.781956 | -3.071251 | H | -5.896374 | -1.056309 | -0.331324 |
| H | 6.502575 | -6.906757 | 1.257407  | H | -4.852062 | 0.368355  | -0.449479 |
| H | 4.560786 | -8.287662 | 1.002877  | N | -7.787941 | 2.147024  | -1.185392 |
| H | 5.976858 | -5.380887 | -2.812846 | N | -8.525673 | 1.145889  | -1.106188 |
| C | 4.243497 | 7.217728  | -0.777646 | C | -8.99471  | 3.929682  | 0.766553  |
| C | 2.188686 | 7.376259  | -2.226077 | H | -8.663292 | 4.865942  | 1.226971  |
| C | 2.133265 | 8.28532   | 0.095411  | H | -9.012073 | 4.087158  | -0.31637  |
| C | 1.34274  | 8.061576  | -1.175403 | C | -8.819072 | -0.943608 | 0.236917  |
| C | 3.403391 | 7.916119  | 0.269342  | C | -9.672662 | -3.614043 | 0.243957  |
| C | 3.45554  | 7.007282  | -2.052083 | C | -8.971081 | -1.650804 | 1.439492  |
| H | 1.689971 | 7.210638  | -3.177739 | C | -9.110341 | -1.610681 | -0.962134 |
| H | 1.60604  | 8.77581   | 0.911036  | C | -9.536707 | -2.938741 | -0.965952 |
| H | 3.887982 | 8.113363  | 1.223322  | C | -9.392498 | -2.979995 | 1.451921  |
| H | 4.009729 | 6.535249  | -2.859434 | H | -8.763433 | -1.166024 | 2.388703  |
| O | 5.350949 | 8.04769   | -1.197975 | H | -9.015846 | -1.077224 | -1.901506 |
| O | 7.788309 | -6.353322 | -1.198509 | H | -9.762246 | -3.441612 | -1.900196 |

|   |          |           |           |    |            |           |           |
|---|----------|-----------|-----------|----|------------|-----------|-----------|
| C | 8.796798 | -6.500644 | -0.209349 | H  | -9.50577   | -3.514424 | 2.38892   |
| H | 8.872705 | -5.620577 | 0.443701  | C  | -10.435833 | 1.975581  | 0.920887  |
| H | 9.737467 | -6.61713  | -0.754027 | H  | -10.671914 | 1.836469  | -0.138726 |
| H | 8.639425 | -7.394695 | 0.409306  | C  | -10.388155 | 3.481276  | 1.2536    |
| C | 6.223331 | 8.517395  | -0.180866 | H  | -10.471236 | 3.628134  | 2.339226  |
| H | 7.084307 | 8.948752  | -0.698452 | Br | -1.938598  | -6.048046 | -0.062914 |
| H | 6.573796 | 7.708739  | 0.47509   | Br | -10.256727 | -5.436238 | 0.250125  |
| H | 5.757232 | 9.300633  | 0.432501  | H  | -11.203572 | 4.045557  | 0.787673  |
| C | 2.54625  | -7.310612 | -0.7698   | H  | -11.193608 | 1.435559  | 1.497624  |
| C | -0.11686 | -6.55734  | -0.352742 | H  | 7.155562   | 3.128269  | 3.791833  |
| C | 1.528916 | -7.768772 | -1.618493 | H  | 7.314553   | 0.994438  | 4.999232  |

#### 4 (anti)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -8448.827189 Hartree

Sum of electronic and thermal Energies = -8448.753375 Hartree

Sum of electronic and enthalpy Energies = -8448.752431 Hartree

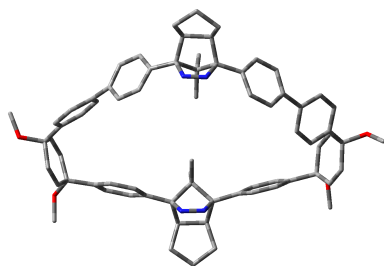
Sum of electronic and thermal Free Energies = -8448.950825 Hartree

| Atom | Cartesian Coordinates |           |          | Atom | Cartesian Coordinates |           |           |
|------|-----------------------|-----------|----------|------|-----------------------|-----------|-----------|
|      | X                     | Y         | Z        |      | X                     | Y         | Z         |
| C    | -5.409783             | -0.532381 | 0.45159  | C    | -1.88483              | 6.402462  | -0.164445 |
| C    | -6.454515             | 0.462748  | 1.08707  | C    | -0.562284             | 5.963105  | -0.09061  |
| C    | -6.336871             | 0.14251   | 2.618612 | C    | -0.045425             | 7.236063  | -2.087326 |
| C    | -6.126068             | -1.406478 | 2.627253 | H    | -1.684921             | 8.33103   | -2.943603 |
| C    | -6.151257             | -1.763825 | 1.099718 | H    | -2.593295             | 6.065935  | 0.586196  |
| N    | -7.568921             | -1.454074 | 0.700374 | H    | -0.243748             | 5.301307  | 0.707596  |
| N    | -7.737065             | -0.219173 | 0.693381 | H    | 0.673714              | 7.5589    | -2.832529 |
| H    | -5.144435             | -1.663015 | 3.034431 | C    | -0.300671             | -7.223042 | -0.891419 |
| H    | -5.460157             | 0.656969  | 3.021276 | C    | 2.050889              | -5.651273 | -0.612247 |
| C    | -7.234143             | -1.990873 | 3.536587 | C    | 0.741321              | -7.310441 | -1.825538 |
| H    | -8.100049             | -2.298067 | 2.941869 | C    | -0.148662             | -6.338047 | 0.181525  |
| C    | -7.562297             | 0.419386  | 3.522942 | C    | 1.004104              | -5.566956 | 0.319148  |
| H    | -8.47787              | 0.476935  | 2.926057 | C    | 1.89323               | -6.541626 | -1.688274 |
| H    | -7.469639             | 1.362934  | 4.070508 | H    | 0.648024              | -8.00359  | -2.655834 |
| C    | -7.619259             | -0.810631 | 4.452283 | H    | -0.943205             | -6.239102 | 0.915055  |
| H    | -8.598759             | -0.941259 | 4.925725 | H    | 1.081811              | -4.870535 | 1.149567  |
| C    | -5.734467             | -3.157922 | 0.702629 | H    | 2.696103              | -6.652968 | -2.411996 |

|   |           |           |           |   |           |            |           |
|---|-----------|-----------|-----------|---|-----------|------------|-----------|
| C | -4.909146 | -5.751207 | -0.11057  | O | -3.833022 | 9.000565   | -1.96879  |
| C | -4.58547  | -3.755801 | 1.239613  | O | -1.261675 | -9.301776  | -1.748615 |
| C | -6.464994 | -3.88567  | -0.248813 | C | -0.542739 | -10.248424 | -0.972097 |
| C | -6.059249 | -5.157518 | -0.647276 | H | -0.225423 | -11.029852 | -1.667823 |
| C | -4.178994 | -5.028141 | 0.838289  | H | -1.17066  | -10.707163 | -0.196204 |
| H | -3.994161 | -3.230536 | 1.984186  | H | 0.348233  | -9.811432  | -0.500468 |
| H | -7.368071 | -3.45275  | -0.665808 | C | -3.385365 | 10.115586  | -1.211282 |
| H | -6.643771 | -5.706659 | -1.378947 | H | -4.095449 | 10.385739  | -0.418271 |
| H | -3.28137  | -5.45891  | 1.27172   | H | -3.316937 | 10.949353  | -1.915026 |
| C | -6.426436 | 1.91346   | 0.674984  | H | -2.395445 | 9.947692   | -0.764653 |
| C | -6.328989 | 4.625996  | -0.162046 | C | 9.508397  | 3.391428   | 1.259519  |
| C | -5.470025 | 2.798711  | 1.191912  | C | 8.668194  | 0.784397   | 0.661524  |
| C | -7.337119 | 2.413958  | -0.267984 | C | 9.394884  | 2.995227   | -0.070816 |
| C | -7.288835 | 3.744532  | -0.67736  | C | 9.212621  | 2.509582   | 2.295499  |
| C | -5.420598 | 4.130087  | 0.778661  | C | 8.793836  | 1.2145     | 1.990741  |
| H | -4.751607 | 2.454846  | 1.930518  | C | 8.978977  | 1.695854   | -0.359315 |
| H | -8.098821 | 1.753709  | -0.668641 | H | 9.631684  | 3.689188   | -0.870026 |
| H | -8.010588 | 4.113828  | -1.399529 | H | 9.309542  | 2.828069   | 3.327854  |
| H | -4.665198 | 4.787077  | 1.199621  | H | 8.57537   | 0.520532   | 2.794862  |
| C | -5.436489 | -0.544356 | -1.086345 | H | 8.902138  | 1.397518   | -1.400543 |
| H | -6.439805 | -0.682756 | -1.495368 | C | 8.182344  | -0.607467  | 0.341512  |
| H | -4.804236 | -1.354132 | -1.466087 | C | 8.990837  | -1.425201  | -0.726384 |
| H | -5.042363 | 0.400595  | -1.47558  | H | 9.094569  | -0.81987   | -1.631097 |
| C | -3.952755 | -0.332888 | 0.886946  | C | 8.063785  | -2.657452  | -0.983063 |
| H | -3.567277 | 0.605636  | 0.472969  | H | 7.705583  | -2.667155  | -2.015929 |
| H | -3.332685 | -1.143069 | 0.487124  | C | 6.849984  | -2.377953  | -0.028253 |
| H | -3.804747 | -0.304433 | 1.968837  | C | 6.692056  | -0.811746  | -0.13031  |
| C | -6.269598 | 6.074104  | -0.687598 | C | 6.355084  | -0.269979  | -1.525061 |
| C | -4.492705 | 7.722017  | 0.008077  | H | 6.374447  | 0.825579   | -1.513086 |
| C | -4.476938 | 6.791326  | -2.30605  | H | 5.339353  | -0.571971  | -1.803795 |
| C | -3.777678 | 7.707349  | -1.3256   | H | 7.030008  | -0.606272  | -2.315325 |
| C | -5.565854 | 6.078861  | -2.027109 | C | 5.657022  | -0.237943  | 0.852454  |
| C | -5.58392  | 7.00688   | 0.286863  | H | 5.815151  | -0.568791  | 1.881382  |
| H | -4.034171 | 6.760082  | -3.298458 | H | 4.648069  | -0.541563  | 0.553279  |
| H | -6.025144 | 7.077324  | 1.278864  | H | 5.694913  | 0.856758   | 0.841412  |
| H | -4.06437  | 8.360531  | 0.777881  | N | 8.199873  | -1.506808  | 1.547378  |
| H | -6.028103 | 5.455576  | -2.788335 | N | 7.460899  | -2.488939  | 1.342261  |
| C | -4.453046 | -7.131856 | -0.624112 | C | 10.371525 | -1.996053  | -0.32175  |
| C | -2.448602 | -7.340247 | -2.135187 | H | 11.198954 | -1.377252  | -0.683569 |
| C | -2.34839  | -8.26456  | 0.178197  | H | 10.461038 | -2.055892  | 0.767428  |

|   |           |           |           |    |           |           |           |
|---|-----------|-----------|-----------|----|-----------|-----------|-----------|
| C | -1.58661  | -8.04481  | -1.110599 | C  | 5.628492  | -3.248736 | -0.183397 |
| C | -3.603097 | -7.864327 | 0.391292  | C  | 3.277818  | -4.829612 | -0.465306 |
| C | -3.700964 | -6.942746 | -1.923218 | C  | 5.081131  | -3.517805 | -1.447243 |
| H | -1.9754   | -7.185469 | -3.101589 | C  | 4.980208  | -3.789162 | 0.936923  |
| H | -1.812393 | -8.777985 | 0.973731  | C  | 3.831676  | -4.564329 | 0.797518  |
| H | -4.065954 | -8.05885  | 1.356506  | C  | 3.929503  | -4.288306 | -1.585474 |
| H | -4.268816 | -6.457498 | -2.713071 | H  | 5.551271  | -3.118271 | -2.341193 |
| O | -5.603178 | -7.91888  | -1.00899  | H  | 5.398347  | -3.617872 | 1.922981  |
| O | -7.60162  | 6.538326  | -1.00465  | H  | 3.371894  | -4.99371  | 1.683436  |
| C | -8.48815  | 6.740249  | 0.086426  | H  | 3.517142  | -4.454057 | -2.576878 |
| H | -8.530241 | 5.870709  | 0.756363  | C  | 8.928894  | -3.914064 | -0.721024 |
| H | -9.476862 | 6.893755  | -0.354    | H  | 8.799193  | -4.265022 | 0.307494  |
| H | -8.225151 | 7.632152  | 0.671428  | C  | 10.374274 | -3.415457 | -0.926354 |
| C | -6.453197 | -8.367747 | 0.036191  | H  | 10.604974 | -3.365468 | -1.999333 |
| H | -7.349314 | -8.759973 | -0.45207  | Br | 2.165922  | 5.796234  | -0.960108 |
| H | -6.747043 | -7.554542 | 0.713929  | H  | 11.120366 | -4.072276 | -0.465534 |
| H | -5.994038 | -9.17543  | 0.622409  | H  | 8.666193  | -4.744239 | -1.384482 |
| C | -2.308123 | 7.258796  | -1.186981 | H  | -6.893165 | -2.869056 | 4.094398  |
| C | 0.347845  | 6.386598  | -1.055178 | H  | -6.880699 | -0.705487 | 5.258911  |
| C | -1.370489 | 7.664934  | -2.146587 | Br | 10.083418 | 5.17054   | 1.666683  |

### 5 (syn)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -3305.411268 Hartree

Sum of electronic and thermal Energies = -3305.342684 Hartree

Sum of electronic and enthalpy Energies = -3305.341740 Hartree

Sum of electronic and thermal Free Energies = -3305.515236 Hartree

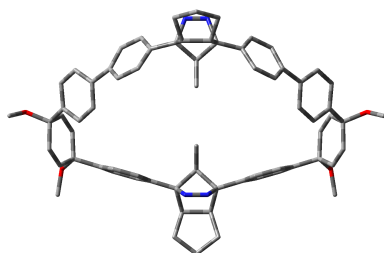
| Atom | Cartesian Coordinates |           |           | Atom | Cartesian Coordinates |          |           |
|------|-----------------------|-----------|-----------|------|-----------------------|----------|-----------|
|      | X                     | Y         | Z         |      | X                     | Y        | Z         |
| C    | 0.468365              | -6.157567 | 0.440329  | C    | -3.994088             | 4.470236 | 1.207256  |
| C    | -1.091281             | -6.107155 | 0.479238  | C    | -3.914315             | 4.724728 | -1.180983 |
| C    | -1.391138             | -4.571763 | 0.35213   | C    | -2.783983             | 5.528003 | -1.057293 |
| C    | -0.276445             | -4.093593 | -0.657832 | C    | -2.849021             | 5.256106 | 1.329618  |
| C    | 0.857737              | -4.643403 | 0.293259  | H    | -4.444293             | 4.040451 | 2.098375  |
| C    | -0.223601             | -2.567185 | -0.839572 | H    | -4.327312             | 4.526951 | -2.166666 |
| H    | -1.112991             | -2.220124 | -1.376928 | H    | -2.333204             | 5.940873 | -1.955078 |
| H    | 0.65618               | -2.292186 | -1.431688 | H    | -2.423865             | 5.443191 | 2.310175  |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| H | -0.170653 | -2.025945 | 0.107675  | C | 5.029855  | 3.790145  | 0.460681  |
| C | -0.357581 | -4.724661 | -2.053714 | C | 2.832612  | 5.591965  | 0.491777  |
| H | -1.277234 | -4.396288 | -2.552084 | C | 4.675038  | 4.522149  | -0.684229 |
| H | -0.352365 | -5.816874 | -2.055748 | C | 4.350331  | 4.088383  | 1.65414   |
| H | 0.48368   | -4.385782 | -2.668349 | C | 3.26999   | 4.966019  | 1.669022  |
| N | -0.835305 | -4.008351 | 1.634917  | C | 3.601796  | 5.41015   | -0.666991 |
| N | 0.409392  | -4.048424 | 1.599398  | H | 5.187761  | 4.331838  | -1.622867 |
| C | 2.315636  | -4.361481 | 0.010775  | H | 4.6439    | 3.594031  | 2.576136  |
| C | 5.089888  | -3.929033 | -0.448539 | H | 2.724303  | 5.136383  | 2.591528  |
| C | 2.982485  | -4.981637 | -1.058814 | H | 3.321799  | 5.908296  | -1.590709 |
| C | 3.06159   | -3.503838 | 0.83038   | C | 1.461301  | 6.209744  | 0.42198   |
| C | 4.42313   | -3.295299 | 0.604357  | C | -0.780671 | 6.276424  | 0.287602  |
| C | 4.340093  | -4.766871 | -1.286752 | C | 0.338267  | 5.244175  | -0.118252 |
| H | 2.444163  | -5.658782 | -1.716282 | C | 0.246436  | 3.914484  | 0.649941  |
| H | 2.575149  | -3.01966  | 1.669704  | H | -0.643562 | 3.361439  | 0.332824  |
| H | 4.971365  | -2.635947 | 1.270209  | H | 0.188337  | 4.048082  | 1.732449  |
| H | 4.837047  | -5.274435 | -2.107826 | H | 1.122484  | 3.294568  | 0.434499  |
| C | -2.835046 | -4.174752 | 0.148979  | C | 0.427049  | 4.902337  | -1.610994 |
| C | -5.586204 | -3.491092 | -0.082802 | H | -0.441543 | 4.304401  | -1.905811 |
| C | -3.71126  | -4.938118 | -0.63904  | H | 1.318403  | 4.293384  | -1.795915 |
| C | -3.356562 | -3.034723 | 0.77751   | H | 0.473865  | 5.769905  | -2.273101 |
| C | -4.705376 | -2.697185 | 0.660653  | C | -0.356207 | 7.558549  | -0.506688 |
| C | -5.059958 | -4.604265 | -0.751952 | H | -0.687134 | 7.460655  | -1.544495 |
| H | -3.351757 | -5.825731 | -1.151511 | C | 1.210987  | 7.511416  | -0.413031 |
| H | -2.707481 | -2.429582 | 1.401795  | H | 1.654316  | 7.390571  | -1.40531  |
| H | -5.077873 | -1.816205 | 1.174713  | H | 8.562176  | 1.417302  | -1.485704 |
| H | -5.725072 | -5.236245 | -1.332537 | N | 0.889884  | 6.550044  | 1.772345  |
| C | 6.592651  | -3.71587  | -0.731087 | N | -0.354565 | 6.587583  | 1.697122  |
| C | 8.106772  | -1.079652 | -0.482717 | H | 0.823572  | -6.695756 | -0.442883 |
| C | 6.760412  | -2.713838 | -1.851618 | H | -1.514431 | -6.624422 | -0.385646 |
| C | 7.349957  | -3.273083 | 0.50286   | C | 0.911969  | -6.908415 | 1.719114  |
| C | 8.019312  | -2.123481 | 0.609296  | H | 1.808851  | -7.514305 | 1.554801  |
| C | 7.433826  | -1.570348 | -1.747014 | H | 1.138278  | -6.20164  | 2.523602  |
| H | 6.302069  | -3.001162 | -2.794618 | C | -0.322717 | -7.752873 | 2.096402  |
| H | 7.327577  | -3.953587 | 1.351352  | H | -0.307565 | -8.086192 | 3.140133  |
| H | 8.526569  | -1.889977 | 1.543037  | H | -0.366652 | -8.653788 | 1.469039  |
| H | 7.530876  | -0.904986 | -2.601181 | C | -1.518531 | -6.831442 | 1.778942  |
| C | -7.098803 | -3.189064 | -0.169447 | H | -1.660959 | -6.112731 | 2.592016  |
| C | -8.227277 | -0.373528 | -0.42512  | H | -2.458779 | -7.37938  | 1.659461  |
| C | -7.427368 | -2.510981 | -1.482137 | C | 1.647776  | 8.868228  | 0.190079  |



|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| C | -7.566692 | -2.349589 | 0.997516  | H | 1.763977   | 8.788817  | 1.275486  |
| C | -8.067915 | -1.120797 | 0.881318  | H | 2.603859   | 9.214285  | -0.21614  |
| C | -7.935478 | -1.284124 | -1.597211 | C | -0.77965   | 8.940951  | 0.045482  |
| H | -7.240167 | -3.118277 | -2.36478  | H | -1.65856   | 9.341909  | -0.469841 |
| H | -7.461399 | -2.790382 | 1.985753  | H | -1.027943  | 8.872412  | 1.109233  |
| H | -8.393824 | -0.571671 | 1.761342  | C | 0.476911   | 9.815759  | -0.14323  |
| H | -8.136109 | -0.882976 | -2.587881 | H | 0.466626   | 10.7155   | 0.482025  |
| C | 7.431161  | 0.246571  | -0.079346 | H | 0.549095   | 10.147818 | -1.188071 |
| C | 5.956541  | 2.635641  | 0.359277  | O | 7.159143   | -4.929107 | -1.282099 |
| C | 7.75044   | 1.428439  | -0.764514 | O | 9.473591   | -0.84355  | -0.883828 |
| C | 6.409123  | 0.29619   | 0.873351  | O | -7.833407  | -4.433349 | -0.259103 |
| C | 5.68502   | 1.46548   | 1.08684   | O | -9.608335  | 0.051725  | -0.402747 |
| C | 7.030247  | 2.600445  | -0.547099 | C | 7.26418    | -6.044557 | -0.410335 |
| H | 6.142777  | -0.598808 | 1.426243  | H | 7.516693   | -6.899202 | -1.043931 |
| H | 4.852082  | 1.449956  | 1.783421  | H | 8.063272   | -5.914344 | 0.332122  |
| H | 7.297356  | 3.49819   | -1.098529 | H | 6.321137   | -6.257482 | 0.111562  |
| C | -7.296254 | 0.856187  | -0.404182 | C | 10.375937  | -0.43711  | 0.134026  |
| C | -5.530206 | 3.067217  | -0.203449 | H | 10.635744  | -1.263397 | 0.809849  |
| C | -6.141547 | 0.925215  | -1.189833 | H | 11.284583  | -0.11181  | -0.379706 |
| C | -7.571367 | 1.924051  | 0.464516  | H | 9.984487   | 0.400862  | 0.727079  |
| C | -6.698703 | 3.005054  | 0.572948  | C | -7.859934  | -5.251484 | 0.900796  |
| C | -5.280926 | 2.015834  | -1.098492 | H | -8.503229  | -4.832262 | 1.686018  |
| H | -5.894524 | 0.108043  | -1.859896 | H | -8.280282  | -6.209684 | 0.58333   |
| H | -8.481229 | 1.903777  | 1.057583  | H | -6.8564    | -5.4246   | 1.312824  |
| H | -6.9286   | 3.815482  | 1.260167  | C | -10.076384 | 0.763876  | -1.537494 |
| H | -4.373467 | 2.03157   | -1.695742 | H | -10.195198 | 0.113443  | -2.414947 |
| C | -4.507406 | 4.133205  | -0.05434  | H | -11.059229 | 1.15631   | -1.262963 |
| C | -2.188859 | 5.748845  | 0.194511  | H | -9.419701  | 1.60415   | -1.801019 |

### 5 (anti)



opt freq RB3LYP/6-31G(d)

Charge = 0, Multiplicity = 1,  $\langle S^2 \rangle = 0$

Number of Imaginary Frequencies = 0

Sum of electronic and zero-point Energies = -3305.412024 Hartree

Sum of electronic and thermal Energies = -3305.343544 Hartree

Sum of electronic and enthalpy Energies = -3305.342600 Hartree

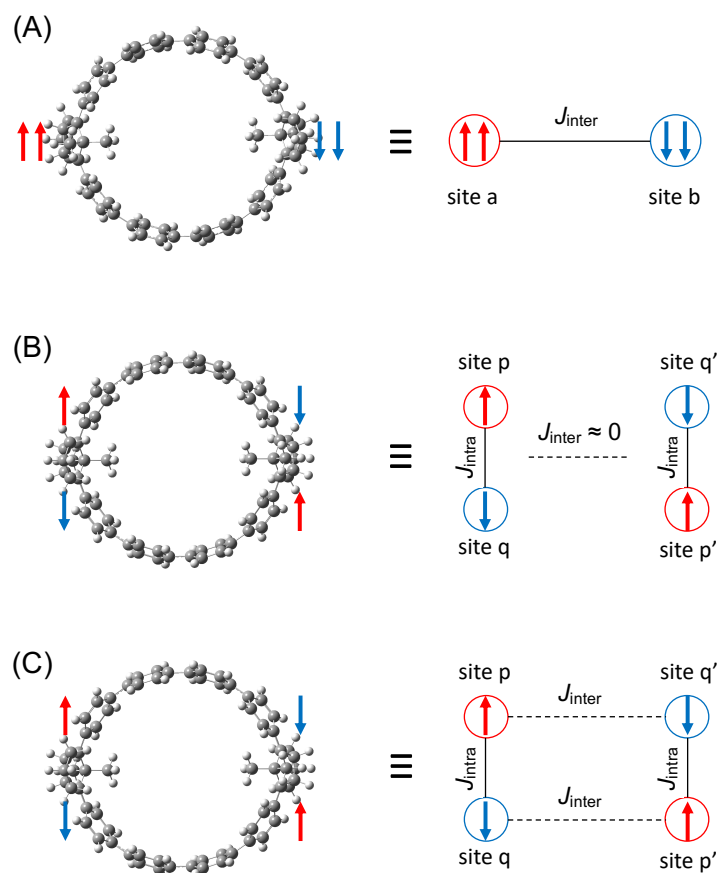
Sum of electronic and thermal Free Energies = -3305.515220 Hartree

| Cartesian Coordinates |   |   |   | Cartesian Coordinates |   |   |   |
|-----------------------|---|---|---|-----------------------|---|---|---|
| Atom                  | X | Y | Z | Atom                  | X | Y | Z |

|   |           |           |           |   |           |           |           |
|---|-----------|-----------|-----------|---|-----------|-----------|-----------|
| C | 1.123605  | -6.286919 | -0.157845 | H | 5.165606  | 5.404152  | 1.667522  |
| C | -0.00003  | -5.226179 | -0.469383 | H | 2.803955  | 5.879291  | 1.198969  |
| C | -1.123661 | -6.286915 | -0.157832 | H | 2.729258  | 2.681551  | -1.667948 |
| C | -0.000041 | -4.659377 | -1.89479  | C | -5.314795 | 3.777002  | 0.270043  |
| H | 0.882127  | -4.026597 | -2.039871 | C | -2.573578 | 4.309018  | -0.271668 |
| H | -0.882214 | -4.0266   | -2.039859 | C | -4.630983 | 4.800003  | 0.941306  |
| H | -0.000045 | -5.414852 | -2.683523 | C | -4.600727 | 3.015389  | -0.660707 |
| C | -0.000025 | -4.030496 | 0.498041  | C | -3.254727 | 3.273759  | -0.926697 |
| H | 0.884984  | -3.409564 | 0.326629  | C | -3.290049 | 5.063532  | 0.671499  |
| H | -0.000021 | -4.329489 | 1.548585  | H | -5.165609 | 5.404302  | 1.667393  |
| H | -0.885033 | -3.409561 | 0.326636  | H | -5.097141 | 2.207064  | -1.18926  |
| C | -2.51558  | -5.718762 | -0.070793 | H | -2.729192 | 2.681467  | -1.667836 |
| C | -4.78282  | -4.014699 | 0.076197  | H | -2.803956 | 5.87943   | 1.198832  |
| C | -3.036803 | -5.315847 | 1.167589  | C | -1.124583 | 4.613392  | -0.57243  |
| C | -3.227172 | -5.361989 | -1.226363 | C | 1.124619  | 4.613386  | -0.57242  |
| C | -4.336772 | -4.52287  | -1.154894 | C | 0.000011  | 4.220086  | 0.462324  |
| C | -4.150176 | -4.482809 | 1.239222  | C | 0.000007  | 2.723944  | 0.820758  |
| H | -2.52347  | -5.610002 | 2.077182  | H | 0.885682  | 2.481225  | 1.418029  |
| H | -2.886785 | -5.699064 | -2.201355 | H | 0.000007  | 2.074611  | -0.05743  |
| H | -4.845305 | -4.232147 | -2.070254 | H | -0.885671 | 2.481229  | 1.418027  |
| H | -4.478183 | -4.119279 | 2.209195  | C | 0.000001  | 5.01036   | 1.776989  |
| C | 2.515526  | -5.718768 | -0.070812 | H | 0.88149   | 4.743029  | 2.370268  |
| C | 4.782774  | -4.014717 | 0.076179  | H | -0.881523 | 4.743066  | 2.370232  |
| C | 3.036762  | -5.315871 | 1.16757   | H | 0.000027  | 6.095588  | 1.651298  |
| C | 3.227109  | -5.361984 | -1.226384 | C | 0.780468  | 6.112369  | -0.891354 |
| C | 4.336713  | -4.52287  | -1.154915 | H | 1.172733  | 6.740556  | -0.087093 |
| C | 4.150139  | -4.482839 | 1.239203  | C | -0.780405 | 6.112366  | -0.891397 |
| H | 2.523436  | -5.610037 | 2.077164  | H | -1.17271  | 6.740579  | -0.087176 |
| H | 2.886713  | -5.699046 | -2.201377 | H | -7.4537   | 0.858526  | 2.825009  |
| H | 4.845241  | -4.232138 | -2.070275 | N | -0.623281 | 3.882399  | -1.78781  |
| H | 4.478156  | -4.119323 | 2.209179  | N | 0.623318  | 3.882454  | -1.78784  |
| C | -5.765657 | -2.905755 | 0.150367  | N | -0.623497 | -6.825727 | 1.155636  |
| C | -7.355699 | -0.570264 | 0.438532  | N | 0.623456  | -6.825758 | 1.155618  |
| C | -6.698894 | -2.788272 | 1.194701  | C | -0.785787 | -7.435213 | -1.168405 |
| C | -5.692309 | -1.855171 | -0.777424 | H | -1.173108 | -7.168102 | -2.155658 |
| C | -6.471005 | -0.709343 | -0.635013 | C | 0.785723  | -7.435209 | -1.168421 |
| C | -7.479918 | -1.644224 | 1.333617  | H | 1.173026  | -7.168089 | -2.155678 |
| H | -6.808024 | -3.599764 | 1.90955   | H | 4.970479  | -1.908311 | -1.587443 |
| H | -4.970521 | -1.908312 | -1.587448 | C | -0.000028 | -9.737857 | -1.216994 |
| H | -6.357926 | 0.09876   | -1.35087  | H | -0.000021 | -10.72491 | -0.741278 |

|   |           |           |           |   |            |           |           |
|---|-----------|-----------|-----------|---|------------|-----------|-----------|
| H | -8.190758 | -1.570778 | 2.151511  | H | -0.000041  | -9.901533 | -2.303451 |
| C | 5.765621  | -2.905781 | 0.150355  | C | -1.216395  | -8.878904 | -0.813967 |
| C | 7.35569   | -0.570311 | 0.438531  | H | -1.398706  | -8.974533 | 0.260977  |
| C | 6.698866  | -2.788318 | 1.194683  | H | -2.136022  | -9.175731 | -1.328795 |
| C | 5.692275  | -1.855186 | -0.777424 | C | 1.216345   | -8.878899 | -0.813995 |
| C | 6.470985  | -0.709368 | -0.635008 | H | 2.13596    | -9.175722 | -1.328846 |
| C | 7.479903  | -1.644279 | 1.333605  | H | 1.398681   | -8.974528 | 0.260944  |
| H | 6.807992  | -3.599818 | 1.909524  | C | -1.216291  | 6.688558  | -2.260393 |
| H | 6.357908  | 0.098744  | -1.350855 | H | -1.401283  | 5.882757  | -2.977668 |
| H | 8.190748  | -1.570849 | 2.151495  | H | -2.135348  | 7.279093  | -2.188691 |
| C | -8.106721 | 0.745512  | 0.724014  | C | 1.216428   | 6.688595  | -2.260313 |
| C | -6.792718 | 3.495541  | 0.616314  | H | 2.135463   | 7.279157  | -2.18854  |
| C | -7.434402 | 1.425208  | 1.897667  | H | 1.401495   | 5.882808  | -2.977584 |
| C | -8.133365 | 1.661253  | -0.48029  | C | 0.000071   | 7.523084  | -2.712133 |
| C | -7.544714 | 2.857797  | -0.53226  | H | 0.000104   | 7.727299  | -3.788737 |
| C | -6.854401 | 2.62214   | 1.849345  | H | 0.00004    | 8.494082  | -2.198062 |
| H | -8.653571 | 1.288596  | -1.360272 | O | -7.436327  | 4.720899  | 1.039667  |
| H | -7.596757 | 3.435079  | -1.452843 | O | -9.43926   | 0.463512  | 1.207012  |
| H | -6.394827 | 3.050802  | 2.736401  | O | 7.436358   | 4.720849  | 1.039676  |
| C | 8.106732  | 0.745453  | 0.724019  | O | 9.439266   | 0.463429  | 1.207018  |
| C | 6.792746  | 3.49549   | 0.616322  | C | -7.664134  | 5.706761  | 0.043854  |
| C | 7.434422  | 1.425154  | 1.897675  | H | -8.482042  | 5.426932  | -0.633882 |
| C | 8.133392  | 1.661201  | -0.48028  | H | -7.954037  | 6.614836  | 0.579608  |
| C | 7.544747  | 2.857748  | -0.532248 | H | -6.763641  | 5.918315  | -0.549042 |
| C | 6.854428  | 2.622089  | 1.849354  | C | -10.331698 | -0.1557   | 0.293374  |
| H | 7.45372   | 0.85847   | 2.825015  | H | -11.202698 | -0.458689 | 0.880634  |
| H | 8.653607  | 1.288548  | -1.360258 | H | -10.666311 | 0.535877  | -0.492227 |
| H | 7.596806  | 3.435038  | -1.452825 | H | -9.893623  | -1.046511 | -0.177252 |
| H | 6.394858  | 3.050753  | 2.736411  | C | 7.664154   | 5.706716  | 0.043865  |
| C | 5.314825  | 3.776956  | 0.270052  | H | 7.954053   | 6.614791  | 0.579622  |
| C | 2.57361   | 4.30899   | -0.271656 | H | 8.482061   | 5.426896  | -0.633875 |
| C | 4.600777  | 3.015409  | -0.660766 | H | 6.763657   | 5.918266  | -0.549025 |
| C | 4.630995  | 4.799901  | 0.941384  | C | 10.331696  | -0.155799 | 0.293383  |
| C | 3.290063  | 5.063437  | 0.67158   | H | 10.666303  | 0.535764  | -0.492234 |
| C | 3.254778  | 3.273789  | -0.926757 | H | 11.202701  | -0.458776 | 0.880641  |
| H | 5.097205  | 2.207132  | -1.189378 | H | 9.893617   | -1.046618 | -0.177224 |

## 11. Computational details II [reference 24-27 in this paper]



**Figure S44:** Calculation model of **2DR-8CPP**.

### Appendix I

As shown in Figure S43 (A), if the BS-singlet 1 state is a model consisting of two spin sites, site a and b, the Heisenberg Hamiltonian is

$$H = -2J_{inter}S_aS_b.....(A1).$$

According to the approximate spin projection method,  $J_{inter}$ , excluding the spin contamination error, is denoted when the energies of the Quintet and BS-singlet 1 states are  $E_Q$  and  $E_{BS1}$ ,

$$J_{inter} = (E_{BS1} - E_Q) / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1}).....(A2).$$

Where  $\langle S^2 \rangle_X$  is the  $\langle S^2 \rangle$  value of the spin state X. Since this value does not include spin contamination errors, it is equivalent to the value calculated from the energy of the correct singlet state, considering static electron correlations, and excluding spin contamination errors.

$$J_{inter} = (E_S - E_Q) / (\langle S^2 \rangle_Q - \langle S^2 \rangle_S).....(A3)$$

Where  $E_S$  is the energy of the correct singlet state determined by the approximate spin projection method, and  $\langle S^2 \rangle_S$  is correct singlet  $\langle S^2 \rangle$  value ( $\langle S^2 \rangle_S = 0$ ).

$$\text{From (A2) and (A3), } E_S = E_{BS1} \langle S^2 \rangle_Q / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1}) - E_Q \langle S^2 \rangle_{BS1} / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1}) .....(A4)$$

Next, as shown in Figure S43(B), assuming the BS-singlet 2 state to be a dimer model consisting of two spin sites (p (p') and q(q')) and  $J_{inter} \approx 0$ , Heisenberg Hamiltonian is

$$H = -2J_{intra}S_pS_q - 2J_{intra}S_{p'}S_{q'} = -4J_{intra}S_pS_q.....(A5).$$

When the energies of the Quintet and BS-singlet 2 states are  $E_Q$  and  $E_{BS2}$ ,

$$E_Q = -4J_{intra}\langle S_pS_q \rangle_Q.....(A6i)$$

$$E_{BS2} = -4J_{intra}\langle S_pS_q \rangle_{BS2}.....(A6j).$$

Where  $\langle S_pS_q \rangle_X$  is the spin correlation function of state X. Total spin Hamiltonian is  $S = S_p + S_q$ .

$$\text{From } S^2 = S_p^2 + S_q^2 + 2S_pS_q.....(A7),$$

$$E_Q = -2J_{intra}\{\langle S^2 \rangle_Q - \langle S_p^2 \rangle_Q - \langle S_q^2 \rangle_Q\}.....(A8i)$$

$$E_{BS2} = -2J_{intra}\{\langle S^2 \rangle_{BS2} - \langle S_p^2 \rangle_{BS2} - \langle S_q^2 \rangle_{BS2}\}.....(A8j)$$

There is no change in the spin density distribution between quintet and BS-singlet 2 states,

$$\langle S_p^2 \rangle_Q = \langle S_p^2 \rangle_{BS2}.....(A9i)$$

$$\langle S_q^2 \rangle_Q = \langle S_q^2 \rangle_{BS2}.....(A9j)$$

$$\text{From } E_{BS2} - E_Q = 2J_{intra}(\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS2}).....(A10),$$

$$J_{intra} = (E_{BS2} - E_Q) / \{2(\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS2})\}.....(A11).$$

Since this  $J_{intra}$  is a value without spin contamination error, as in the discussion of (A2) and (A3),

$E_S$  is determined as follows from  $(E_S - E_Q) / \{2(\langle S^2 \rangle_Q - \langle S^2 \rangle_S)\} = (E_{BS2} - E_Q) / \{2(\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS2})\}... (A12).$

$$E_S = E_{BS2} \langle S^2 \rangle_Q / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1}) - E_Q \langle S^2 \rangle_{BS2} / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1}).....(A13).$$

## **Appendix II**

In the case of spin contamination error in the structure of BS-singlet 1 and BS-singlet 2 state, the nuclear coordinate derivatives of  $E_S$  in (A4) or (A13) should be obtained. nuclear coordinate derivatives of  $E_S$  is

$$G_s(R) = \{\alpha(R)G_{BS1}(R) - \beta(R)G_Q(R)\} + \partial\alpha / \partial R\{E_{BS1} - E_Q\}.....(A14i).$$

Also, for BS-singlet 2 state,

$$G_s(R) = \{\alpha(R)G_{BS2}(R) - \beta(R)G_Q(R)\} + \partial\alpha / \partial R\{E_{BS2} - E_Q\}.....(A14j).$$

Where  $G$  is nuclear coordinate derivative of energy,  $\alpha$  and  $\beta$  is put as follows,

$$\alpha = (\langle S^2 \rangle_Q - \langle S^2 \rangle_S) / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1 \text{ or } 2})$$

$$\beta = \alpha - 1$$

Also,

$$\partial\alpha / \partial R \approx \{(\langle S^2 \rangle_Q - \langle S^2 \rangle_S) / (\langle S^2 \rangle_Q - \langle S^2 \rangle_{BS1 \text{ or } 2})\} \partial\langle S^2 \rangle_{BS1 \text{ or } 2} / \partial R$$

In this open shell molecules, the second term in (A14) can be neglected because  $\langle S^2 \rangle$  value does not change with a small shift in coordinates, and the energy difference between the singlet and the quintet is very small.

Therefore, for BS-singlet 1 state,  $G_s(R) \approx \{\alpha(R)G_{BS1}(R) - \beta(R)G_Q(R)\}.....(A15i).$

Also, for BS-singlet 2 state,  $G_s(R) \approx \{\alpha(R)G_{BS2}(R) - \beta(R)G_Q(R)\}.....(A15j)$

Using this energy derivative equation, structural optimization was performed using the BS-singlet 1 and BS-singlet 2 structures as initial structures. The convergence condition conforms to the opt = loose condition of Gaussian 09.

### **Appendix III**

Finally, as shown in Figure S43 (C), all spins of p, q, p', and q' are regarded as spin sites. Considered a four-spin site model with  $J_{intra}$  and  $J_{inter}$  as the spin coupling parameters within and between biradical sites, respectively.

Spin Hamiltonian is

$$H = -2J_{intra}S_pS_q - 2J_{intra}S_{p'}S_{q'} - 2J_{inter}S_pS_{q'} - 2J_{inter}S_qS_{p'}..... (A16).$$

Since the energies of the Quintet state in the BS-singlet 1 and BS-singlet 2 structures and the energy in the Quintet structure are almost identical (maximum difference is  $0.03 \text{ kcal mol}^{-1}$ ), the structures can be regarded as almost energetically same. Assuming the most stable BS-singlet 1 structure,  $J_{intra}$  and  $J_{inter}$  are calculated from the energy difference between BS-singlet 1, BS-singlet 2 and Quintet states. The spin Hamiltonians corresponding to Quintet, BS-singlet1, and BS-singlet2 when assuming the Ising model are

$$\text{Quintet : } H^{Quintet} = -J_{intra} - J_{inter}.....(A17i)$$

$$\text{BS-singlet 1 : } H^{BS1} = -J_{intra} + J_{inter}.....(A17j)$$

$$\text{BS-singlet 2 : } H^{BS2} = J_{intra} + J_{inter}.....(A17k)$$

From these equations,  $J_{intra}$  and  $J_{inter}$  are calculated from

$$J_{inter} = -(E^{Quintet} - E^{BS1}) / 2.....(A18i)$$

$$J_{intra} = -(E^{BS1} - E^{BS2}) / 2.....(A18i)$$

**Table S7:** Optimized Cartesian coordinate in THF.

| Atom | Quintet |         |         | Triplet |         |         | BS-singlet 1 |         |         | BS-singlet 2 |         |         |
|------|---------|---------|---------|---------|---------|---------|--------------|---------|---------|--------------|---------|---------|
|      | x [Å]   | y [Å]   | z [Å]   | x [Å]   | y [Å]   | z [Å]   | x [Å]        | y [Å]   | z [Å]   | x [Å]        | y [Å]   | z [Å]   |
| C    | -6.4362 | -2.5313 | 0.5117  | -6.4632 | -2.541  | 0.2646  | -6.4158      | -2.5333 | 0.5267  | -6.3795      | 2.4813  | -0.2455 |
| C    | -4.6829 | -4.8252 | 0.4633  | -4.7145 | -4.8392 | 0.3035  | -4.6753      | -4.8378 | 0.4865  | -4.7219      | 4.8441  | -0.3334 |
| C    | -6.4673 | -3.494  | -0.5479 | -6.4567 | -3.5146 | -0.7869 | -6.4601      | -3.5065 | -0.5241 | -6.4332      | 3.4926  | 0.7705  |
| C    | -5.6942 | -2.9566 | 1.6602  | -5.7726 | -2.9627 | 1.4474  | -5.6714      | -2.9558 | 1.6758  | -5.6909      | 2.8945  | -1.4341 |
| C    | -4.8479 | -4.0503 | 1.6314  | -4.9288 | -4.0579 | 1.4604  | -4.8322      | -4.0544 | 1.6509  | -4.8901      | 4.0205  | -1.469  |
| C    | -5.5993 | -4.5746 | -0.581  | -5.5901 | -4.5956 | -0.7781 | -5.5979      | -4.5908 | -0.5543 | -5.6087      | 4.6047  | 0.7408  |
| H    | -7.1127 | -3.3405 | -1.4004 | -7.0713 | -3.3683 | -1.6631 | -7.113       | -3.3581 | -1.3717 | -7.066       | 3.3546  | 1.6353  |
| H    | -5.7486 | -2.387  | 2.5784  | -5.8647 | -2.3883 | 2.3595  | -5.7182      | -2.38   | 2.5904  | -5.7436      | 2.2888  | -2.3285 |
| H    | -4.2307 | -4.2513 | 2.5026  | -4.3502 | -4.2554 | 2.3584  | -4.213       | -4.2532 | 2.5211  | -4.3044      | 4.212   | -2.3638 |
| H    | -5.5996 | -5.2152 | -1.4594 | -5.5584 | -5.244  | -1.6502 | -5.6081      | -5.2393 | -1.4269 | -5.6188      | 5.2844  | 1.5893  |
| C    | -6.4362 | 2.5313  | 0.5117  | -6.4635 | 2.5249  | 0.2604  | -6.4158      | 2.5334  | 0.5269  | -6.3273      | -2.6052 | -0.2156 |
| C    | -4.6829 | 4.8252  | 0.4633  | -4.7165 | 4.8236  | 0.2965  | -4.6753      | 4.8379  | 0.4868  | -4.6224      | -4.9351 | -0.2756 |
| C    | -6.4673 | 3.4941  | -0.5479 | -6.4562 | 3.4957  | -0.7929 | -6.4603      | 3.5068  | -0.5237 | -6.3613      | -3.6057 | 0.8119  |
| C    | -5.6942 | 2.9566  | 1.6602  | -5.7738 | 2.9482  | 1.4425  | -5.6714      | 2.9558  | 1.676   | -5.6295      | -3.018  | -1.3989 |
| C    | -4.8479 | 4.0503  | 1.6315  | -4.9295 | 4.0436  | 1.4538  | -4.8321      | 4.0543  | 1.6512  | -4.8062      | -4.128  | -1.4206 |
| C    | -5.5993 | 4.5747  | -0.581  | -5.5906 | 4.5781  | -0.785  | -5.5981      | 4.5911  | -0.5539 | -5.5145      | -4.7012 | 0.7954  |
| H    | -7.1128 | 3.3406  | -1.4003 | -7.069  | 3.3469  | -1.6699 | -7.1133      | 3.3585  | -1.3713 | -6.9974      | -3.4705 | 1.6747  |
| H    | -5.7485 | 2.387   | 2.5784  | -5.8656 | 2.375   | 2.3553  | -5.7181      | 2.3799  | 2.5906  | -5.6937      | -2.4238 | -2.3003 |
| H    | -4.2307 | 4.2513  | 2.5026  | -4.3501 | 4.2422  | 2.3511  | -4.2128      | 4.253   | 2.5214  | -4.2162      | -4.318  | -2.3128 |
| H    | -5.5996 | 5.2153  | -1.4594 | -5.558  | 5.2246  | -1.6585 | -5.6083      | 5.2397  | -1.4263 | -5.5116      | -5.3711 | 1.6517  |
| C    | -3.4646 | -5.6447 | 0.2884  | -3.4899 | -5.6559 | 0.1808  | -3.4621      | -5.6616 | 0.3097  | -3.5238      | 5.6987  | -0.215  |
| C    | -0.7417 | -6.4685 | 0.0339  | -0.755  | -6.47   | 0.0285  | -0.7407      | -6.4906 | 0.0392  | -0.8071      | 6.5719  | -0.0518 |
| C    | -2.8175 | -5.6605 | -0.9625 | -2.8015 | -5.6894 | -1.0481 | -2.8275      | -5.6972 | -0.9477 | -2.8498      | 5.7734  | 1.0202  |
| C    | -2.7696 | -6.2212 | 1.3709  | -2.8276 | -6.2133 | 1.2943  | -2.756       | -6.225  | 1.3928  | -2.8627      | 6.2508  | -1.332  |
| C    | -1.4442 | -6.6262 | 1.2453  | -1.4977 | -6.6136 | 1.2183  | -1.4329      | -6.6329 | 1.2593  | -1.5428      | 6.6815  | -1.2502 |
| C    | -1.4905 | -6.0566 | -1.0846 | -1.4696 | -6.08   | -1.1205 | -1.502       | -6.0949 | -1.0775 | -1.5267      | 6.1915  | 1.0976  |
| H    | -3.3163 | -5.2365 | -1.8289 | -3.2731 | -5.2818 | -1.9373 | -3.3345      | -5.2857 | -1.8153 | -3.3224      | 5.3739  | 1.9126  |
| H    | -3.2555 | -6.307  | 2.3391  | -3.3446 | -6.2883 | 2.2472  | -3.2322      | -6.2986 | 2.3668  | -3.3713      | 6.2962  | -2.2913 |
| H    | -0.9328 | -7.0208 | 2.119   | -1.0149 | -6.9943 | 2.114   | -0.9149      | -7.0183 | 2.1331  | -1.0608      | 7.0566  | -2.1486 |
| H    | -0.9986 | -5.9267 | -2.0437 | -0.9481 | -5.9622 | -2.0653 | -1.0212      | -5.978  | -2.0437 | -1.014       | 6.1014  | 2.0501  |
| C    | -3.4646 | 5.6447  | 0.2884  | -3.4907 | 5.641   | 0.1732  | -3.462       | 5.6616  | 0.3101  | -3.4081      | -5.7651 | -0.1474 |
| C    | -0.7416 | 6.4685  | 0.0339  | -0.7571 | 6.4557  | 0.027   | -0.7406      | 6.4905  | 0.0393  | -0.6757      | -6.5857 | 0.0255  |
| C    | -2.8175 | 5.6605  | -0.9624 | -2.7961 | 5.6607  | -1.0519 | -2.8276      | 5.6973  | -0.9474 | -2.7329      | -5.8125 | 1.0885  |
| C    | -2.7696 | 6.2213  | 1.3709  | -2.8357 | 6.2097  | 1.2844  | -2.7558      | 6.2248  | 1.3932  | -2.7368      | -6.3177 | -1.258  |
| C    | -1.4441 | 6.6262  | 1.2453  | -1.5049 | 6.6102  | 1.2115  | -1.4328      | 6.6327  | 1.2596  | -1.4091      | -6.7229 | -1.1714 |
| C    | -1.4905 | 6.0566  | -1.0846 | -1.4639 | 6.0522  | -1.1214 | -1.5021      | 6.095   | -1.0773 | -1.4022      | -6.2049 | 1.1705  |
| H    | -3.3163 | 5.2366  | -1.8289 | -3.2628 | 5.2426  | -1.9388 | -3.3348      | 5.286   | -1.8151 | -3.2128      | -5.4114 | 1.9762  |

|   |         |         |         |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| H | -3.2555 | 6.307   | 2.3391  | -3.3578 | 6.2926  | 2.2338  | -3.232  | 6.2984  | 2.3672  | -3.2445 | -6.3837 | -2.2165 |
| H | -0.9328 | 7.0208  | 2.119   | -1.0257 | 6.9986  | 2.1059  | -0.9147 | 7.0179  | 2.1333  | -0.9203 | -7.0996 | -2.0654 |
| H | -0.9986 | 5.9267  | -2.0437 | -0.9361 | 5.925   | -2.0616 | -1.0214 | 5.9782  | -2.0435 | -0.8913 | -6.0939 | 2.1217  |
| C | 0.7416  | -6.4685 | -0.0339 | 0.7285  | -6.4653 | 0.0115  | 0.7408  | -6.4906 | -0.0407 | 0.6764  | 6.5858  | -0.0247 |
| C | 3.4646  | -5.6448 | -0.2885 | 3.4597  | -5.6395 | -0.1474 | 3.4623  | -5.6618 | -0.3107 | 3.4088  | 5.7654  | 0.1481  |
| C | 1.4442  | -6.6262 | -1.2454 | 1.4726  | -6.617  | -1.1764 | 1.4331  | -6.6323 | -1.2609 | 1.4098  | 6.7228  | 1.1722  |
| C | 1.4905  | -6.0567 | 1.0846  | 1.4401  | -6.0586 | 1.1566  | 1.5023  | -6.0957 | 1.0762  | 1.4029  | 6.2053  | -1.1698 |
| C | 2.8175  | -5.6606 | 0.9624  | 2.7701  | -5.6622 | 1.081   | 2.8277  | -5.6981 | 0.9467  | 2.7337  | 5.8131  | -1.0878 |
| C | 2.7696  | -6.2213 | -1.371  | 2.8007  | -6.211  | -1.2556 | 2.7562  | -6.2245 | -1.3941 | 2.7375  | 6.3177  | 1.2587  |
| H | 0.9986  | -5.9267 | 2.0437  | 0.9173  | -5.9333 | 2.0996  | 1.0214  | -5.9794 | 2.0425  | 0.8921  | 6.0945  | -2.121  |
| H | 3.3163  | -5.2366 | 1.8289  | 3.2389  | -5.2429 | 1.9662  | 3.3348  | -5.2872 | 1.8146  | 3.2137  | 5.4122  | -1.9756 |
| H | 3.2555  | -6.307  | -2.3392 | 3.3188  | -6.2936 | -2.2072 | 3.2325  | -6.2976 | -2.3681 | 3.2451  | 6.3835  | 2.2173  |
| C | 0.7417  | 6.4685  | -0.0339 | 0.7276  | 6.4512  | 0.0164  | 0.7409  | 6.4905  | -0.0406 | 0.8078  | -6.572  | 0.0526  |
| C | 3.4646  | 5.6447  | -0.2885 | 3.4579  | 5.6261  | -0.1367 | 3.4624  | 5.6617  | -0.3109 | 3.5245  | -5.6991 | 0.2156  |
| C | 1.4442  | 6.6262  | -1.2454 | 1.4769  | 6.6148  | -1.1659 | 1.433   | 6.6322  | -1.2609 | 1.5435  | -6.6815 | 1.2509  |
| C | 1.4905  | 6.0566  | 1.0846  | 1.4315  | 6.0306  | 1.1605  | 1.5024  | 6.0956  | 1.0762  | 1.5274  | -6.1919 | -1.097  |
| C | 2.8176  | 5.6605  | 0.9624  | 2.7619  | 5.6338  | 1.0876  | 2.8279  | 5.698   | 0.9465  | 2.8506  | -5.774  | -1.0196 |
| C | 2.7696  | 6.2212  | -1.371  | 2.806   | 6.2093  | -1.2422 | 2.7561  | 6.2243  | -1.3943 | 2.8634  | -6.2508 | 1.3327  |
| H | 0.9328  | 7.0208  | -2.119  | 1.0002  | 7.0153  | -2.0564 | 0.9149  | 7.017   | -2.1348 | 1.0615  | -7.0563 | 2.1494  |
| H | 0.9986  | 5.9267  | 2.0437  | 0.9023  | 5.895   | 2.0986  | 1.0217  | 5.9793  | 2.0425  | 1.0148  | -6.102  | -2.0495 |
| H | 3.3163  | 5.2366  | 1.8289  | 3.2258  | 5.2036  | 1.9702  | 3.3351  | 5.2871  | 1.8144  | 3.3232  | -5.3747 | -1.9121 |
| H | 3.2555  | 6.3069  | -2.3392 | 3.3294  | 6.3006  | -2.1902 | 3.2323  | 6.2975  | -2.3683 | 3.372   | -6.2961 | 2.292   |
| H | 0.9328  | -7.0208 | -2.119  | 0.9923  | -7.0093 | -2.0684 | 0.9151  | -7.0171 | -2.1349 | 0.9209  | 7.0993  | 2.0663  |
| C | 4.6829  | 4.8252  | -0.4633 | 4.6837  | 4.8095  | -0.2673 | 4.6757  | 4.8379  | -0.4873 | 4.7226  | -4.8444 | 0.3338  |
| C | 5.5994  | 4.5746  | 0.581   | 5.5558  | 4.5533  | 0.8131  | 5.5981  | 4.5911  | 0.5537  | 5.6094  | -4.6051 | -0.7404 |
| C | 4.8479  | 4.0502  | -1.6315 | 4.9007  | 4.0443  | -1.4333 | 4.8329  | 4.0543  | -1.6516 | 4.8908  | -4.0206 | 1.4693  |
| C | 6.4673  | 3.494   | 0.5479  | 6.4276  | 3.4755  | 0.8087  | 6.4603  | 3.5068  | 0.5239  | 6.4338  | -3.4929 | -0.7704 |
| H | 5.5996  | 5.2152  | 1.4594  | 5.5168  | 5.187   | 1.6956  | 5.6081  | 5.2398  | 1.4261  | 5.6196  | -5.2849 | -1.5888 |
| C | 5.6942  | 2.9566  | -1.6602 | 5.7518  | 2.9537  | -1.4346 | 5.6721  | 2.9557  | -1.676  | 5.6915  | -2.8945 | 1.4341  |
| H | 4.2308  | 4.2513  | -2.5027 | 4.3209  | 4.2507  | -2.3285 | 4.2139  | 4.2529  | -2.522  | 4.3053  | -4.2121 | 2.3642  |
| C | 6.4362  | 2.5313  | -0.5117 | 6.4457  | 2.5217  | -0.259  | 6.4161  | 2.5333  | -0.5266 | 6.3799  | -2.4814 | 0.2454  |
| H | 7.1127  | 3.3405  | 1.4003  | 7.0359  | 3.317   | 1.6871  | 7.113   | 3.3586  | 1.3717  | 7.0666  | -3.3549 | -1.6352 |
| H | 5.7486  | 2.387   | -2.5784 | 5.849   | 2.3922  | -2.3544 | 5.7192  | 2.3798  | -2.5906 | 5.7442  | -2.2887 | 2.3285  |
| C | 6.9814  | 1.2124  | -0.3798 | 6.9913  | 1.204   | -0.1153 | 6.9502  | 1.2132  | -0.3831 | 6.8477  | -1.1501 | 0.0285  |
| C | 8.0381  | 0.7867  | 0.6161  | 8.0177  | 0.7782  | 0.9112  | 7.9875  | 0.7867  | 0.6333  | 7.7436  | -0.7181 | -1.1137 |
| C | 6.5047  | 0       | -1.2074 | 6.5537  | -0.0061 | -0.9714 | 6.4779  | -0.0001 | -1.2116 | 6.4171  | 0.0663  | 0.8752  |
| H | 9.0179  | 1.1466  | 0.2631  | 9.0074  | 1.1384  | 0.5862  | 8.9742  | 1.1469  | 0.301   | 8.7684  | -1.0706 | -0.9181 |
| C | 8.0381  | -0.7867 | 0.6161  | 8.0177  | -0.7945 | 0.9094  | 7.9874  | -0.7866 | 0.6335  | 7.7284  | 0.8546  | -1.1222 |
| C | 7.9145  | 1.197   | 2.116   | 7.8539  | 1.1862  | 2.4075  | 7.8301  | 1.1979  | 2.1302  | 7.3775  | -1.1428 | -2.5719 |
| C | 6.9814  | -1.2124 | -0.3798 | 6.9911  | -1.2183 | -0.1179 | 6.9502  | -1.2133 | -0.3829 | 6.823   | 1.2814  | 0.0145  |



|   |         |         |         |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | 7.2023  | 0       | -2.6058 | 7.3171  | -0.0047 | -2.3327 | 7.1803  | -0.0002 | -2.6072 | 7.1994  | 0.0818  | 2.2251  |
| C | 4.954   | 0       | -1.3319 | 5.0117  | -0.0058 | -1.1716 | 4.9268  | -0.0002 | -1.3391 | 4.8732  | 0.0519  | 1.0841  |
| C | 7.9146  | -1.197  | 2.116   | 7.8545  | -1.2061 | 2.4048  | 7.8299  | -1.1974 | 2.1305  | 7.3561  | 1.2567  | -2.5851 |
| H | 9.0179  | -1.1466 | 0.2631  | 9.0071  | -1.154  | 0.583   | 8.9742  | -1.147  | 0.3014  | 8.746   | 1.2288  | -0.929  |
| C | 8.4927  | 0       | 2.8763  | 8.4152  | -0.0107 | 3.1806  | 8.3864  | 0.0003  | 2.9053  | 7.7895  | 0.056   | -3.429  |
| H | 8.441   | 2.1287  | 2.3438  | 8.371   | 2.1196  | 2.6498  | 8.3553  | 2.127   | 2.3707  | 7.8834  | -2.0605 | -2.8851 |
| H | 6.8613  | 1.3286  | 2.3903  | 6.7935  | 1.3132  | 2.6544  | 6.7712  | 1.3354  | 2.3785  | 6.2974  | -1.3105 | -2.658  |
| C | 6.4362  | -2.5313 | -0.5117 | 6.4464  | -2.5348 | -0.2647 | 6.4161  | -2.5334 | -0.5263 | 6.3278  | 2.6053  | 0.2157  |
| H | 6.9317  | 0.8823  | -3.1933 | 7.0751  | 0.8786  | -2.9318 | 6.9125  | 0.8822  | -3.1959 | 6.9783  | -0.7994 | 2.8341  |
| H | 8.2914  | 0       | -2.4888 | 8.3994  | -0.0049 | -2.1635 | 8.2689  | -0.0002 | -2.4852 | 8.2783  | 0.0918  | 2.0364  |
| H | 6.9317  | -0.8824 | -3.1933 | 7.075   | -0.8866 | -2.9337 | 6.9125  | -0.8826 | -3.1958 | 6.9603  | 0.9648  | 2.8247  |
| H | 4.5842  | -0.8776 | -1.86   | 4.6676  | -0.8823 | -1.7184 | 4.5577  | -0.878  | -1.8674 | 4.5227  | 0.9263  | 1.6305  |
| H | 4.5031  | 0       | -0.3337 | 4.5119  | -0.0065 | -0.197  | 4.4746  | -0.0001 | -0.3415 | 4.3711  | 0.0478  | 0.1106  |
| H | 4.5842  | 0.8775  | -1.86   | 4.6679  | 0.8716  | -1.7171 | 4.5576  | 0.8773  | -1.8677 | 4.539   | -0.8297 | 1.6292  |
| H | 8.441   | -2.1288 | 2.3438  | 8.3723  | -2.1397 | 2.6449  | 8.355   | -2.1266 | 2.3712  | 7.8462  | 2.1796  | -2.9081 |
| H | 6.8613  | -1.3286 | 2.3902  | 6.7942  | -1.3345 | 2.6516  | 6.771   | -1.3347 | 2.3787  | 6.2734  | 1.4047  | -2.6738 |
| H | 9.589   | 0       | 2.8073  | 9.5128  | -0.0103 | 3.137   | 9.4842  | 0.0002  | 2.8656  | 8.8795  | 0.0649  | -3.5649 |
| H | 8.2287  | 0       | 3.9399  | 8.1265  | -0.0119 | 4.2377  | 8.0944  | 0.0004  | 3.9615  | 7.334   | 0.0465  | -4.4257 |
| C | 6.4673  | -3.4941 | 0.5479  | 6.4299  | -3.4922 | 0.8006  | 6.4601  | -3.5067 | 0.5244  | 6.3619  | 3.606   | -0.8116 |
| C | 5.6942  | -2.9566 | -1.6602 | 5.7514  | -2.9646 | -1.4411 | 5.6723  | -2.956  | -1.6758 | 5.6301  | 3.018   | 1.3991  |
| C | 5.5994  | -4.5747 | 0.581   | 5.5576  | -4.569  | 0.8035  | 5.5979  | -4.5911 | 0.5542  | 5.5153  | 4.7016  | -0.7949 |
| H | 7.1128  | -3.3406 | 1.4003  | 7.0402  | -3.3366 | 1.6781  | 7.1126  | -3.3584 | 1.3724  | 6.9981  | 3.471   | -1.6743 |
| C | 4.8479  | -4.0503 | -1.6315 | 4.9014  | -4.0554 | -1.442  | 4.8331  | -4.0546 | -1.6513 | 4.8069  | 4.1281  | 1.421   |
| H | 5.7486  | -2.387  | -2.5784 | 5.8483  | -2.4012 | -2.3597 | 5.7196  | -2.3802 | -2.5904 | 5.6942  | 2.4236  | 2.3004  |
| C | 4.6829  | -4.8252 | -0.4633 | 4.6839  | -4.8229 | -0.2769 | 4.6757  | -4.838  | -0.487  | 4.6232  | 4.9354  | 0.2762  |
| H | 5.5996  | -5.2153 | 1.4594  | 5.5202  | -5.2052 | 1.6842  | 5.6077  | -5.2396 | 1.4267  | 5.5125  | 5.3718  | -1.651  |
| H | 4.2307  | -4.2513 | -2.5026 | 4.3222  | -4.2604 | -2.338  | 4.2143  | -4.2533 | -2.5219 | 4.217   | 4.318   | 2.3133  |
| C | -6.9814 | -1.2124 | 0.3798  | -6.9913 | -1.2218 | 0.0957  | -6.9501 | -1.2132 | 0.3837  | -6.8477 | 1.15    | -0.029  |
| C | -6.5048 | 0       | 1.2074  | -6.5618 | -0.0079 | 0.9462  | -6.4773 | 0       | 1.212   | -6.4168 | -0.0663 | -0.8756 |
| C | -8.0381 | -0.7867 | -0.6161 | -7.9791 | -0.7965 | -0.9696 | -7.9881 | -0.7866 | -0.632  | -7.7447 | 0.7181  | 1.1124  |
| C | -6.9814 | 1.2124  | 0.3798  | -6.992  | 1.2043  | 0.0938  | -6.9501 | 1.2133  | 0.3838  | -6.823  | -1.2814 | -0.0149 |
| C | -7.2023 | 0       | 2.6058  | -7.3359 | -0.007  | 2.3032  | -7.1788 | 0       | 2.6081  | -7.1988 | -0.0819 | -2.2257 |
| C | -4.954  | 0       | 1.3319  | -5.0194 | -0.0074 | 1.1538  | -4.9261 | -0.0001 | 1.3386  | -4.8729 | -0.0519 | -1.0841 |
| C | -8.0381 | 0.7867  | -0.6161 | -7.9795 | 0.7771  | -0.9709 | -7.9881 | 0.7867  | -0.6319 | -7.7295 | -0.8546 | 1.1209  |
| C | -7.9145 | -1.197  | -2.116  | -7.7498 | -1.2084 | -2.4569 | -7.8318 | -1.1976 | -2.129  | -7.3802 | 1.1429  | 2.5709  |
| H | -9.0179 | -1.1466 | -0.2632 | -8.9804 | -1.1571 | -0.6847 | -8.9746 | -1.1469 | -0.299  | -8.7694 | 1.0705  | 0.9158  |
| H | -6.9318 | 0.8824  | 3.1933  | -7.0994 | 0.8762  | 2.904   | -6.9107 | 0.8824  | 3.1965  | -6.9595 | -0.965  | -2.8251 |
| H | -8.2915 | 0       | 2.4888  | -8.4166 | -0.0076 | 2.1242  | -8.2675 | -0.0001 | 2.4868  | -8.2778 | -0.0919 | -2.0372 |
| H | -6.9317 | -0.8823 | 3.1933  | -7.0987 | -0.889  | 2.9055  | -6.9106 | -0.8823 | 3.1966  | -6.9776 | 0.7992  | -2.8347 |
| H | -4.5842 | -0.8775 | 1.8601  | -4.6776 | -0.8845 | 1.701   | -4.5567 | -0.8777 | 1.8668  | -4.5387 | 0.8296  | -1.6292 |

|   |         |         |         |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| H | -4.5031 | 0.0001  | 0.3337  | -4.5163 | -0.0079 | 0.1809  | -4.4745 | -0.0001 | 0.3408  | -4.371  | -0.0478 | -0.1106 |
| H | -4.5842 | 0.8776  | 1.8601  | -4.678  | 0.8703  | 1.7     | -4.5566 | 0.8776  | 1.8668  | -4.5223 | -0.9263 | -1.6305 |
| H | -9.0179 | 1.1466  | -0.2631 | -8.9812 | 1.1378  | -0.6871 | -8.9746 | 1.147   | -0.2989 | -8.7469 | -1.2288 | 0.9267  |
| C | -7.9145 | 1.197   | -2.116  | -7.7499 | 1.1866  | -2.4588 | -7.832  | 1.1977  | -2.129  | -7.3587 | -1.2566 | 2.5843  |
| C | -8.4927 | 0       | -2.8763 | -8.2692 | -0.0116 | -3.2582 | -8.389  | 0.0001  | -2.9034 | -7.7932 | -0.0559 | 3.4276  |
| H | -8.441  | -2.1287 | -2.3438 | -8.2623 | -2.138  | -2.7218 | -8.3571 | -2.1268 | -2.3692 | -7.8864 | 2.0606  | 2.8835  |
| H | -6.8613 | -1.3286 | -2.3902 | -6.6801 | -1.3453 | -2.6537 | -6.7731 | -1.3348 | -2.3781 | -6.3002 | 1.3104  | 2.6582  |
| H | -8.441  | 2.1288  | -2.3438 | -8.2624 | 2.1158  | -2.7253 | -8.3573 | 2.1269  | -2.3691 | -7.849  | -2.1796 | 2.9067  |
| H | -6.8613 | 1.3286  | -2.3902 | -6.6802 | 1.3231  | -2.6557 | -6.7732 | 1.3351  | -2.3781 | -6.276  | -1.4043 | 2.6742  |
| H | -9.589  | 0       | -2.8073 | -9.3676 | -0.0116 | -3.2697 | -9.4867 | 0       | -2.8628 | -8.8834 | -0.0649 | 3.5622  |
| H | -8.2287 | 0       | -3.9399 | -7.928  | -0.0124 | -4.2996 | -8.0979 | 0.0001  | -3.9599 | -7.3389 | -0.0464 | 4.4249  |

| Atom | Spin-projected for BS1 |         |         | Spin-projected for BS2 |         |         | Spin restricted |         |         |
|------|------------------------|---------|---------|------------------------|---------|---------|-----------------|---------|---------|
|      | x [Å]                  | y [Å]   | z [Å]   | x [Å]                  | y [Å]   | z [Å]   | x [Å]           | y [Å]   | z [Å]   |
| C    | -6.4159                | -2.533  | 0.5266  | -6.3796                | 2.481   | -0.2455 | -6.0826         | 2.4747  | -0.394  |
| C    | -4.675                 | -4.8381 | 0.4865  | -4.7215                | 4.8444  | -0.3334 | -4.5965         | 4.9531  | -0.5983 |
| C    | -6.4601                | -3.5065 | -0.5242 | -6.4332                | 3.4927  | 0.7707  | -6.2297         | 3.5512  | 0.5767  |
| C    | -5.6713                | -2.9559 | 1.676   | -5.6908                | 2.8946  | -1.4343 | -5.4852         | 2.9248  | -1.643  |
| C    | -4.8321                | -4.0544 | 1.6512  | -4.8899                | 4.0205  | -1.4693 | -4.7853         | 4.0931  | -1.7334 |
| C    | -5.5979                | -4.5908 | -0.5545 | -5.6086                | 4.6048  | 0.741   | -5.4954         | 4.7002  | 0.495   |
| H    | -7.113                 | -3.3581 | -1.3717 | -7.066                 | 3.3545  | 1.6353  | -6.8901         | 3.4159  | 1.4241  |
| H    | -5.7182                | -2.3801 | 2.5905  | -5.7436                | 2.2888  | -2.3285 | -5.5549         | 2.3063  | -2.5278 |
| H    | -4.2128                | -4.2533 | 2.5212  | -4.3043                | 4.2121  | -2.3639 | -4.297          | 4.3244  | -2.6744 |
| H    | -5.6081                | -5.2393 | -1.4269 | -5.6187                | 5.2844  | 1.5893  | -5.5842         | 5.4308  | 1.2939  |
| C    | -6.416                 | 2.5331  | 0.5268  | -6.3275                | -2.605  | -0.2156 | -6.0146         | -2.6017 | -0.381  |
| C    | -4.675                 | 4.8382  | 0.4868  | -4.6221                | -4.9355 | -0.2756 | -4.4932         | -5.0568 | -0.5608 |
| C    | -6.4603                | 3.5068  | -0.5238 | -6.3612                | -3.6058 | 0.8121  | -6.106          | -3.646  | 0.6289  |
| C    | -5.6713                | 2.9558  | 1.6762  | -5.6294                | -3.018  | -1.3991 | -5.4505         | -3.0732 | -1.6361 |
| C    | -4.832                 | 4.0544  | 1.6514  | -4.806                 | -4.1281 | -1.4209 | -4.7295         | -4.2304 | -1.7126 |
| C    | -5.598                 | 4.5911  | -0.5541 | -5.5143                | -4.7013 | 0.7957  | -5.3618         | -4.7883 | 0.554   |
| H    | -7.1133                | 3.3585  | -1.3713 | -6.9974                | -3.4705 | 1.6747  | -6.732          | -3.4862 | 1.4999  |
| H    | -5.7181                | 2.3799  | 2.5906  | -5.6937                | -2.4238 | -2.3003 | -5.5496         | -2.475  | -2.5325 |
| H    | -4.2127                | 4.2531  | 2.5214  | -4.2161                | -4.3181 | -2.3129 | -4.2622         | -4.4785 | -2.66   |
| H    | -5.6083                | 5.2397  | -1.4263 | -5.5114                | -5.3711 | 1.6518  | -5.4105         | -5.4942 | 1.3777  |
| C    | -3.4623                | -5.6614 | 0.3097  | -3.524                 | 5.6986  | -0.2151 | -3.4774         | 5.8386  | -0.5051 |
| C    | -0.7403                | -6.4906 | 0.0391  | -0.8067                | 6.572   | -0.0518 | -0.7842         | 6.805   | -0.1296 |
| C    | -2.8275                | -5.6971 | -0.948  | -2.8497                | 5.7734  | 1.0205  | -3.0264         | 6.3493  | 0.7535  |
| C    | -2.7559                | -6.2249 | 1.3932  | -2.8626                | 6.2507  | -1.3324 | -2.6002         | 6.0873  | -1.6082 |
| C    | -1.4329                | -6.6328 | 1.2597  | -1.5427                | 6.6814  | -1.2507 | -1.3221         | 6.5573  | -1.4297 |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | -1.502  | -6.0948 | -1.0778 | -1.5266 | 6.1914  | 1.098   | -1.739  | 6.7965  | 0.9334  |
| H | -3.3345 | -5.2856 | -1.8155 | -3.3224 | 5.3739  | 1.9128  | -3.6751 | 6.2892  | 1.6215  |
| H | -3.2322 | -6.2985 | 2.3669  | -3.3713 | 6.2961  | -2.2916 | -2.9198 | 5.844   | -2.6161 |
| H | -0.9149 | -7.0182 | 2.1333  | -1.0608 | 7.0565  | -2.1489 | -0.6841 | 6.6412  | -2.3031 |
| H | -1.0212 | -5.978  | -2.0439 | -1.014  | 6.1013  | 2.0503  | -1.436  | 7.0955  | 1.9317  |
| C | -3.4623 | 5.6615  | 0.3101  | -3.4083 | -5.7649 | -0.1474 | -3.3549 | -5.9181 | -0.4713 |
| C | -0.7403 | 6.4905  | 0.0393  | -0.6753 | -6.5857 | 0.0255  | -0.6383 | -6.8222 | -0.1024 |
| C | -2.8276 | 5.6973  | -0.9478 | -2.7328 | -5.8125 | 1.0889  | -2.8881 | -6.4188 | 0.786   |
| C | -2.7557 | 6.2247  | 1.3935  | -2.7367 | -6.3176 | -1.2584 | -2.4751 | -6.1483 | -1.5766 |
| C | -1.4327 | 6.6326  | 1.26    | -1.4091 | -6.7228 | -1.1719 | -1.185  | -6.5869 | -1.4011 |
| C | -1.5021 | 6.0949  | -1.0776 | -1.4021 | -6.2048 | 1.1709  | -1.5912 | -6.8375 | 0.9622  |
| H | -3.3348 | 5.2859  | -1.8152 | -3.2128 | -5.4114 | 1.9764  | -3.5356 | -6.3787 | 1.6557  |
| H | -3.232  | 6.2983  | 2.3674  | -3.2445 | -6.3836 | -2.2168 | -2.8005 | -5.9096 | -2.5837 |
| H | -0.9147 | 7.0178  | 2.1336  | -0.9203 | -7.0995 | -2.0657 | -0.5466 | -6.6533 | -2.2757 |
| H | -1.0213 | 5.9782  | -2.0437 | -0.8912 | -6.0938 | 2.122   | -1.2803 | -7.133  | 1.9592  |
| C | 0.7405  | -6.4907 | -0.0406 | 0.676   | 6.5858  | -0.0248 | 0.6383  | 6.8222  | 0.1024  |
| C | 3.4626  | -5.6616 | -0.3108 | 3.4091  | 5.7652  | 0.1481  | 3.3549  | 5.9181  | 0.4712  |
| C | 1.433   | -6.6323 | -1.2612 | 1.4097  | 6.7227  | 1.1727  | 1.185   | 6.5869  | 1.401   |
| C | 1.5022  | -6.0957 | 1.0766  | 1.4029  | 6.2053  | -1.1701 | 1.5911  | 6.8375  | -0.9623 |
| C | 2.8277  | -5.698  | 0.947   | 2.7336  | 5.813   | -1.0882 | 2.8881  | 6.4188  | -0.786  |
| C | 2.7561  | -6.2244 | -1.3945 | 2.7374  | 6.3177  | 1.2592  | 2.475   | 6.1483  | 1.5766  |
| H | 1.0214  | -5.9793 | 2.0428  | 0.892   | 6.0944  | -2.1213 | 1.2803  | 7.133   | -1.9592 |
| H | 3.3348  | -5.2871 | 1.8148  | 3.2137  | 5.4121  | -1.9758 | 3.5356  | 6.3787  | -1.6557 |
| H | 3.2325  | -6.2975 | -2.3684 | 3.2451  | 6.3834  | 2.2176  | 2.8004  | 5.9096  | 2.5837  |
| C | 0.7405  | 6.4905  | -0.0406 | 0.8074  | -6.5721 | 0.0525  | 0.7841  | -6.805  | 0.1296  |
| C | 3.4626  | 5.6615  | -0.311  | 3.5247  | -5.6989 | 0.2156  | 3.4774  | -5.8386 | 0.5051  |
| C | 1.4329  | 6.6322  | -1.2613 | 1.5434  | -6.6814 | 1.2514  | 1.3221  | -6.5573 | 1.4297  |
| C | 1.5024  | 6.0955  | 1.0766  | 1.5274  | -6.1918 | -1.0974 | 1.739   | -6.7965 | -0.9334 |
| C | 2.8279  | 5.6979  | 0.9469  | 2.8506  | -5.7739 | -1.02   | 3.0264  | -6.3493 | -0.7536 |
| C | 2.756   | 6.2243  | -1.3946 | 2.8633  | -6.2508 | 1.333   | 2.6002  | -6.0873 | 1.6082  |
| H | 0.9149  | 7.0169  | -2.1351 | 1.0614  | -7.0562 | 2.1497  | 0.6841  | -6.6412 | 2.3031  |
| H | 1.0217  | 5.9792  | 2.0427  | 1.0147  | -6.1019 | -2.0497 | 1.436   | -7.0955 | -1.9318 |
| H | 3.3351  | 5.287   | 1.8145  | 3.3232  | -5.3746 | -1.9124 | 3.6751  | -6.2892 | -1.6216 |
| H | 3.2323  | 6.2974  | -2.3685 | 3.372   | -6.2961 | 2.2922  | 2.9198  | -5.8441 | 2.6161  |
| H | 0.915   | -7.017  | -2.1351 | 0.9208  | 7.0992  | 2.0666  | 0.5466  | 6.6534  | 2.2757  |
| C | 4.6754  | 4.8381  | -0.4873 | 4.7223  | -4.8447 | 0.3338  | 4.5964  | -4.9531 | 0.5983  |
| C | 5.5981  | 4.5912  | 0.5539  | 5.6093  | -4.6052 | -0.7407 | 5.4954  | -4.7002 | -0.495  |
| C | 4.8328  | 4.0543  | -1.6518 | 4.8907  | -4.0207 | 1.4696  | 4.7853  | -4.0931 | 1.7334  |
| C | 6.4603  | 3.5067  | 0.5241  | 6.4337  | -3.493  | -0.7706 | 6.2297  | -3.5512 | -0.5767 |
| H | 5.6081  | 5.2398  | 1.4261  | 5.6195  | -5.285  | -1.5889 | 5.5842  | -5.4308 | -1.2939 |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | 5.672   | 2.9557  | -1.6762 | 5.6914  | -2.8946 | 1.4343  | 5.4852  | -2.9248 | 1.643   |
| H | 4.2138  | 4.253   | -2.5221 | 4.3051  | -4.2122 | 2.3642  | 4.297   | -4.3245 | 2.6744  |
| C | 6.4162  | 2.533   | -0.5266 | 6.38    | -2.4811 | 0.2453  | 6.0826  | -2.4747 | 0.394   |
| H | 7.113   | 3.3585  | 1.3718  | 7.0666  | -3.3549 | -1.6352 | 6.89    | -3.4159 | -1.4241 |
| H | 5.7192  | 2.3798  | -2.5906 | 5.7442  | -2.2887 | 2.3285  | 5.5549  | -2.3063 | 2.5278  |
| C | 6.9502  | 1.2133  | -0.3831 | 6.8476  | -1.1503 | 0.0285  | 6.3748  | -1.1556 | 0.0867  |
| C | 7.9874  | 0.7867  | 0.6334  | 7.7433  | -0.7181 | -1.1137 | 6.984   | -0.7019 | -1.2311 |
| C | 6.4779  | -0.0001 | -1.2116 | 6.417   | 0.0664  | 0.876   | 6.0527  | 0.0612  | 0.9814  |
| H | 8.9741  | 1.1469  | 0.3011  | 8.7681  | -1.0705 | -0.9181 | 8.0591  | -0.9367 | -1.2247 |
| C | 7.9874  | -0.7867 | 0.6336  | 7.7282  | 0.8546  | -1.1222 | 6.7853  | 0.8464  | -1.2853 |
| C | 7.8299  | 1.1979  | 2.1303  | 7.3772  | -1.1428 | -2.5719 | 6.3592  | -1.2553 | -2.5584 |
| C | 6.9502  | -1.2134 | -0.383  | 6.8229  | 1.2817  | 0.0145  | 6.3128  | 1.2829  | 0.0814  |
| C | 7.1803  | -0.0002 | -2.6072 | 7.1992  | 0.0818  | 2.2251  | 7.0042  | 0.0834  | 2.2151  |
| C | 4.9268  | -0.0002 | -1.3391 | 4.8732  | 0.0519  | 1.0841  | 4.5502  | 0.045   | 1.3999  |
| C | 7.8297  | -1.1974 | 2.1306  | 7.3559  | 1.2567  | -2.5851 | 5.7902  | 1.0748  | -2.4633 |
| H | 8.974   | -1.147  | 0.3014  | 8.7457  | 1.2287  | -0.9289 | 7.731   | 1.3406  | -1.5496 |
| C | 8.3862  | 0.0003  | 2.9054  | 7.7891  | 0.056   | -3.4293 | 6.1643  | -0.0258 | -3.4565 |
| H | 8.3552  | 2.127   | 2.3706  | 7.8831  | -2.0605 | -2.885  | 6.9875  | -2.0172 | -3.0296 |
| H | 6.7711  | 1.3354  | 2.3784  | 6.2973  | -1.3105 | -2.658  | 5.3882  | -1.7165 | -2.3501 |
| C | 6.4162  | -2.5331 | -0.5262 | 6.3279  | 2.605   | 0.2157  | 6.0146  | 2.6017  | 0.3811  |
| H | 6.9125  | 0.8822  | -3.1959 | 6.9783  | -0.7994 | 2.8342  | 6.8756  | -0.8059 | 2.8383  |
| H | 8.2688  | -0.0002 | -2.4851 | 8.2781  | 0.0918  | 2.0363  | 8.0486  | 0.1129  | 1.8872  |
| H | 6.9125  | -0.8826 | -3.1958 | 6.9602  | 0.9649  | 2.8247  | 6.8285  | 0.959   | 2.846   |
| H | 4.5577  | -0.878  | -1.8674 | 4.5227  | 0.9263  | 1.6305  | 4.2737  | 0.9557  | 1.932   |
| H | 4.4745  | 0       | -0.3415 | 4.3711  | 0.0478  | 0.1106  | 3.9174  | -0.0239 | 0.509   |
| H | 4.5576  | 0.8773  | -1.8677 | 4.539   | -0.8297 | 1.6292  | 4.3111  | -0.8035 | 2.041   |
| H | 8.3548  | -2.1266 | 2.3712  | 7.8459  | 2.1796  | -2.908  | 5.8521  | 2.0805  | -2.8857 |
| H | 6.7709  | -1.3347 | 2.3787  | 6.2733  | 1.4047  | -2.6738 | 4.7611  | 0.93    | -2.11   |
| H | 9.4841  | 0.0002  | 2.8655  | 8.8792  | 0.0649  | -3.5648 | 7.1043  | 0.2321  | -3.963  |
| H | 8.0941  | 0.0004  | 3.9617  | 7.3335  | 0.0465  | -4.4259 | 5.4061  | -0.185  | -4.2316 |
| C | 6.46    | -3.5067 | 0.5245  | 6.3619  | 3.6061  | -0.8118 | 6.106   | 3.646   | -0.6289 |
| C | 5.6722  | -2.9561 | -1.6759 | 5.63    | 3.0181  | 1.3993  | 5.4505  | 3.0733  | 1.6361  |
| C | 5.5978  | -4.5911 | 0.5544  | 5.5152  | 4.7016  | -0.7951 | 5.3618  | 4.7883  | -0.554  |
| H | 7.1126  | -3.3584 | 1.3724  | 6.9981  | 3.471   | -1.6743 | 6.732   | 3.4862  | -1.4999 |
| C | 4.833   | -4.0546 | -1.6516 | 4.8068  | 4.1281  | 1.4213  | 4.7295  | 4.2304  | 1.7126  |
| H | 5.7196  | -2.3802 | -2.5904 | 5.6942  | 2.4236  | 2.3004  | 5.5495  | 2.475   | 2.5326  |
| C | 4.6754  | -4.8383 | -0.487  | 4.6229  | 4.9358  | 0.2761  | 4.4932  | 5.0568  | 0.5608  |
| H | 5.6076  | -5.2396 | 1.4267  | 5.5124  | 5.3718  | -1.6511 | 5.4105  | 5.4942  | -1.3777 |
| H | 4.2142  | -4.2534 | -2.5219 | 4.2168  | 4.3181  | 2.3134  | 4.2622  | 4.4785  | 2.66    |
| C | -6.9501 | -1.2133 | 0.3837  | -6.8477 | 1.1502  | -0.029  | -6.3748 | 1.1556  | -0.0867 |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | -6.4772 | 0       | 1.212   | -6.4167 | -0.0664 | -0.8763 | -6.0528 | -0.0612 | -0.9813 |
| C | -7.988  | -0.7866 | -0.632  | -7.7445 | 0.7182  | 1.1124  | -6.984  | 0.7019  | 1.2311  |
| C | -6.95   | 1.2134  | 0.3838  | -6.8229 | -1.2816 | -0.0149 | -6.3128 | -1.2829 | -0.0813 |
| C | -7.1788 | 0       | 2.6081  | -7.1986 | -0.0819 | -2.2256 | -7.0042 | -0.0834 | -2.215  |
| C | -4.9261 | 0       | 1.3386  | -4.8729 | -0.0519 | -1.0842 | -4.5503 | -0.045  | -1.3999 |
| C | -7.988  | 0.7867  | -0.6319 | -7.7293 | -0.8546 | 1.1209  | -6.7853 | -0.8464 | 1.2853  |
| C | -7.8317 | -1.1976 | -2.129  | -7.3799 | 1.1429  | 2.5709  | -6.3592 | 1.2554  | 2.5584  |
| H | -8.9745 | -1.1469 | -0.299  | -8.7691 | 1.0705  | 0.9157  | -8.0591 | 0.9367  | 1.2247  |
| H | -6.9107 | 0.8824  | 3.1965  | -6.9595 | -0.965  | -2.8251 | -6.8286 | -0.959  | -2.8459 |
| H | -8.2674 | 0       | 2.4867  | -8.2775 | -0.0919 | -2.0372 | -8.0486 | -0.1129 | -1.8871 |
| H | -6.9105 | -0.8823 | 3.1966  | -6.9776 | 0.7992  | -2.8347 | -6.8757 | 0.8058  | -2.8383 |
| H | -4.5567 | -0.8777 | 1.8668  | -4.5387 | 0.8296  | -1.6292 | -4.3111 | 0.8034  | -2.0411 |
| H | -4.4745 | -0.0001 | 0.3407  | -4.371  | -0.0478 | -0.1106 | -3.9174 | 0.0239  | -0.5091 |
| H | -4.5566 | 0.8776  | 1.8668  | -4.5223 | -0.9263 | -1.6305 | -4.2738 | -0.9557 | -1.932  |
| H | -8.9745 | 1.147   | -0.2989 | -8.7466 | -1.2287 | 0.9267  | -7.731  | -1.3406 | 1.5497  |
| C | -7.8318 | 1.1978  | -2.1291 | -7.3585 | -1.2566 | 2.5843  | -5.7902 | -1.0747 | 2.4633  |
| C | -8.3888 | 0       | -2.9035 | -7.7928 | -0.0559 | 3.4279  | -6.1643 | 0.0258  | 3.4565  |
| H | -8.3569 | -2.1268 | -2.3692 | -7.8861 | 2.0606  | 2.8834  | -6.9875 | 2.0173  | 3.0296  |
| H | -6.773  | -1.3348 | -2.3781 | -6.3001 | 1.3104  | 2.6583  | -5.3882 | 1.7166  | 2.35    |
| H | -8.3572 | 2.1269  | -2.3691 | -7.8487 | -2.1795 | 2.9066  | -5.852  | -2.0805 | 2.8857  |
| H | -6.7732 | 1.3351  | -2.3781 | -6.2759 | -1.4043 | 2.6742  | -4.761  | -0.93   | 2.11    |
| H | -9.4866 | 0       | -2.8627 | -8.8831 | -0.0649 | 3.5621  | -7.1043 | -0.232  | 3.963   |
| H | -8.0976 | 0       | -3.96   | -7.3384 | -0.0464 | 4.4251  | -5.406  | 0.1851  | 4.2316  |

**Table S8:** Optimized Cartesian coordinate in the gas phase.

| Atom | Quintet |         |         | Triplet |         |         | BS-singlet 1 |         |         | BS-singlet 2 |         |         |
|------|---------|---------|---------|---------|---------|---------|--------------|---------|---------|--------------|---------|---------|
|      | x [Å]   | y [Å]   | z [Å]   | x [Å]   | y [Å]   | z [Å]   | x [Å]        | y [Å]   | z [Å]   | x [Å]        | y [Å]   | z [Å]   |
| C    | -6.4095 | -2.5354 | 0.523   | -6.4101 | -2.54   | 0.2795  | -6.4039      | -2.5358 | 0.5314  | -6.3558      | 2.4833  | -0.2574 |
| C    | -4.6729 | -4.8408 | 0.4758  | -4.6616 | -4.8375 | 0.2845  | -4.6706      | -4.8443 | 0.4885  | -4.7131      | 4.855   | -0.3432 |
| C    | -6.4611 | -3.5094 | -0.5249 | -6.4136 | -3.5053 | -0.778  | -6.4591      | -3.5139 | -0.5134 | -6.4267      | 3.5008  | 0.7508  |
| C    | -5.656  | -2.9543 | 1.6657  | -5.7014 | -2.9651 | 1.4486  | -5.6514      | -2.9545 | 1.6756  | -5.6585      | 2.8936  | -1.4411 |
| C    | -4.8179 | -4.0534 | 1.6372  | -4.8583 | -4.0605 | 1.4459  | -4.8158      | -4.055  | 1.6493  | -4.8646      | 4.0238  | -1.4745 |
| C    | -5.6011 | -4.5956 | -0.5582 | -5.5481 | -4.5868 | -0.7849 | -5.6005      | -4.6004 | -0.5452 | -5.6095      | 4.6177  | 0.7223  |
| H    | -7.1193 | -3.3613 | -1.3686 | -7.0369 | -3.3529 | -1.6469 | -7.1199      | -3.3681 | -1.3555 | -7.0702      | 3.3644  | 1.608   |
| H    | -5.693  | -2.3731 | 2.5773  | -5.7809 | -2.3924 | 2.3631  | -5.6878      | -2.372  | 2.5864  | -5.6968      | 2.28    | -2.3307 |
| H    | -4.1869 | -4.249  | 2.4997  | -4.2638 | -4.2617 | 2.3327  | -4.1859      | -4.2504 | 2.5127  | -4.2668      | 4.2129  | -2.3618 |
| H    | -5.6166 | -5.2479 | -1.428  | -5.5258 | -5.2323 | -1.6595 | -5.6188      | -5.2555 | -1.4129 | -5.6326      | 5.3053  | 1.5643  |
| C    | -6.4096 | 2.5355  | 0.5234  | -6.4107 | 2.5237  | 0.2756  | -6.4036      | 2.5356  | 0.531   | -6.3062      | -2.6095 | -0.2283 |
| C    | -4.6729 | 4.8408  | 0.476   | -4.6637 | 4.8217  | 0.2779  | -4.6701      | 4.844   | 0.488   | -4.6169      | -4.949  | -0.2867 |

|   |         |         |         |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | -6.4612 | 3.5095  | -0.5245 | -6.4135 | 3.4863  | -0.7836 | -6.4584 | 3.5134  | -0.514  | -6.3578 | -3.6169 | 0.791   |
| C | -5.6559 | 2.9543  | 1.666   | -5.7029 | 2.9504  | 1.4441  | -5.6513 | 2.9545  | 1.6753  | -5.6    | -3.0191 | -1.4069 |
| C | -4.8179 | 4.0534  | 1.6374  | -4.8594 | 4.046   | 1.4396  | -4.8156 | 4.055   | 1.649   | -4.784  | -4.1337 | -1.4274 |
| C | -5.6013 | 4.5956  | -0.5579 | -5.549  | 4.5692  | -0.7914 | -5.5997 | 4.5999  | -0.5458 | -5.5185 | -4.7175 | 0.7756  |
| H | -7.1195 | 3.3613  | -1.3681 | -7.0352 | 3.3316  | -1.6532 | -7.1188 | 3.3674  | -1.3563 | -7.0048 | -3.484  | 1.6462  |
| H | -5.6929 | 2.3731  | 2.5776  | -5.7822 | 2.3789  | 2.3593  | -5.688  | 2.3722  | 2.5862  | -5.6498 | -2.4165 | -2.3035 |
| H | -4.1867 | 4.249   | 2.4999  | -4.2642 | 4.2482  | 2.3258  | -4.186  | 4.2504  | 2.5126  | -4.1819 | -4.321  | -2.3122 |
| H | -5.6168 | 5.2479  | -1.4277 | -5.5259 | 5.2129  | -1.6674 | -5.6177 | 5.2547  | -1.4137 | -5.5286 | -5.396  | 1.6252  |
| C | -3.4601 | -5.6663 | 0.2947  | -3.4428 | -5.6601 | 0.1456  | -3.4597 | -5.6701 | 0.3072  | -3.5193 | 5.7143  | -0.2186 |
| C | -0.7413 | -6.4951 | 0.034   | -0.7169 | -6.4944 | -0.0244 | -0.7407 | -6.4998 | 0.0372  | -0.8064 | 6.5917  | -0.0511 |
| C | -2.812  | -5.6686 | -0.9547 | -2.7533 | -5.6666 | -1.082  | -2.8195 | -5.6863 | -0.9467 | -2.8428 | 5.7734  | 1.0149  |
| C | -2.7692 | -6.2575 | 1.3708  | -2.7874 | -6.2499 | 1.2451  | -2.7607 | -6.2505 | 1.3846  | -2.8633 | 6.2821  | -1.3296 |
| C | -1.4455 | -6.6655 | 1.242   | -1.4616 | -6.6606 | 1.1603  | -1.4382 | -6.6591 | 1.2512  | -1.5448 | 6.7153  | -1.2455 |
| C | -1.4866 | -6.0671 | -1.0797 | -1.4255 | -6.0673 | -1.1627 | -1.4948 | -6.0846 | -1.0762 | -1.521  | 6.1936  | 1.0943  |
| H | -3.309  | -5.2276 | -1.8138 | -3.2206 | -5.2272 | -1.9583 | -3.322  | -5.2552 | -1.8075 | -3.3125 | 5.3557  | 1.9006  |
| H | -3.2598 | -6.3537 | 2.3359  | -3.3093 | -6.3446 | 2.1938  | -3.2449 | -6.3386 | 2.3537  | -3.3774 | 6.3391  | -2.2855 |
| H | -0.9344 | -7.0745 | 2.1094  | -0.9811 | -7.0701 | 2.0448  | -0.9227 | -7.0607 | 2.1194  | -1.0638 | 7.1056  | -2.1382 |
| H | -0.9891 | -5.9229 | -2.034  | -0.8979 | -5.9258 | -2.101  | -1.0048 | -5.9497 | -2.0356 | -1.0016 | 6.0878  | 2.0417  |
| C | -3.4601 | 5.6663  | 0.2948  | -3.4439 | 5.6448  | 0.138   | -3.4592 | 5.6698  | 0.3069  | -3.4066 | -5.7833 | -0.1521 |
| C | -0.7414 | 6.495   | 0.034   | -0.7192 | 6.4797  | -0.0268 | -0.7403 | 6.4997  | 0.0373  | -0.6773 | -6.6061 | 0.0255  |
| C | -2.8121 | 5.6685  | -0.9547 | -2.7485 | 5.6381  | -1.0857 | -2.8188 | 5.6862  | -0.9469 | -2.7292 | -5.8148 | 1.082   |
| C | -2.7691 | 6.2575  | 1.3709  | -2.7951 | 6.2454  | 1.235   | -2.7603 | 6.2502  | 1.3845  | -2.7395 | -6.3514 | -1.2564 |
| C | -1.4455 | 6.6655  | 1.242   | -1.4686 | 6.6563  | 1.1527  | -1.4379 | 6.6588  | 1.2514  | -1.413  | -6.7581 | -1.1673 |
| C | -1.4867 | 6.067   | -1.0798 | -1.4205 | 6.0397  | -1.1641 | -1.4942 | 6.0847  | -1.0761 | -1.3996 | -6.2084 | 1.1662  |
| H | -3.3091 | 5.2275  | -1.8137 | -3.2114 | 5.1888  | -1.9594 | -3.3213 | 5.2552  | -1.8078 | -3.207  | -5.3959 | 1.9627  |
| H | -3.2597 | 6.3537  | 2.336   | -3.3218 | 6.3476  | 2.1802  | -3.2446 | 6.3382  | 2.3535  | -3.2524 | -6.4294 | -2.2115 |
| H | -0.9344 | 7.0745  | 2.1094  | -0.9915 | 7.0733  | 2.0356  | -0.9225 | 7.0604  | 2.1196  | -0.9244 | -7.1496 | -2.0554 |
| H | -0.9893 | 5.9228  | -2.0341 | -0.8872 | 5.8894  | -2.0977 | -1.0041 | 5.9499  | -2.0355 | -0.8824 | -6.0813 | 2.1121  |
| C | 0.7413  | -6.495  | -0.0329 | 0.766   | -6.5016 | -0.0422 | 0.7405  | -6.4997 | -0.0375 | 0.6765  | 6.6061  | -0.026  |
| C | 3.46    | -5.6662 | -0.2939 | 3.4975  | -5.6842 | -0.2011 | 3.4594  | -5.6699 | -0.3074 | 3.4057  | 5.783   | 0.1516  |
| C | 1.4455  | -6.6658 | -1.2409 | 1.5085  | -6.6538 | -1.2303 | 1.438   | -6.6588 | -1.2516 | 1.4123  | 6.7581  | 1.1667  |
| C | 1.4865  | -6.0666 | 1.0807  | 1.4793  | -6.1014 | 1.1029  | 1.4945  | -6.0848 | 1.0759  | 1.3987  | 6.208   | -1.1667 |
| C | 2.812   | -5.6681 | 0.9555  | 2.81    | -5.7091 | 1.0274  | 2.8192  | -5.6864 | 0.9465  | 2.7283  | 5.8143  | -1.0825 |
| C | 2.7691  | -6.2578 | -1.3698 | 2.837   | -6.2514 | -1.3097 | 2.7604  | -6.2502 | -1.3849 | 2.7388  | 6.3513  | 1.2558  |
| H | 0.9891  | -5.9221 | 2.0349  | 0.954   | -5.9729 | 2.0443  | 1.0045  | -5.9501 | 2.0353  | 0.8815  | 6.0808  | -2.1126 |
| H | 3.3089  | -5.2268 | 1.8144  | 3.2815  | -5.2886 | 1.9107  | 3.3217  | -5.2554 | 1.8074  | 3.206   | 5.3952  | -1.9632 |
| H | 3.2597  | -6.3543 | -2.3349 | 3.357   | -6.3331 | -2.2607 | 3.2446  | -6.3381 | -2.354  | 3.2518  | 6.4294  | 2.2109  |
| C | 0.7413  | 6.495   | -0.033  | 0.7649  | 6.4874  | -0.0391 | 0.7409  | 6.4998  | -0.0372 | 0.8057  | -6.5916 | 0.0506  |
| C | 3.46    | 5.6661  | -0.2941 | 3.4955  | 5.6707  | -0.1931 | 3.4599  | 5.6702  | -0.3071 | 3.5185  | -5.714  | 0.2183  |
| C | 1.4454  | 6.6657  | -1.241  | 1.5118  | 6.6504  | -1.2222 | 1.4385  | 6.6593  | -1.2512 | 1.5441  | -6.7152 | 1.245   |

|   |        |         |         |        |         |         |        |         |         |        |         |         |
|---|--------|---------|---------|--------|---------|---------|--------|---------|---------|--------|---------|---------|
| C | 1.4865 | 6.0666  | 1.0806  | 1.4713 | 6.0746  | 1.1052  | 1.4949 | 6.0846  | 1.0761  | 1.5203 | -6.1932 | -1.0947 |
| C | 2.8119 | 5.668   | 0.9553  | 2.8024 | 5.6821  | 1.032   | 2.8196 | 5.6863  | 0.9468  | 2.842  | -5.7729 | -1.0153 |
| C | 2.769  | 6.2577  | -1.37   | 2.8413 | 6.2484  | -1.2992 | 2.761  | 6.2507  | -1.3845 | 2.8625 | -6.2819 | 1.3292  |
| H | 0.9343 | 7.0751  | -2.1083 | 1.0304 | 7.0483  | -2.1116 | 0.9231 | 7.061   | -2.1194 | 1.0631 | -7.1056 | 2.1377  |
| H | 0.9891 | 5.9221  | 2.0348  | 0.9404 | 5.9369  | 2.0422  | 1.0048 | 5.9495  | 2.0355  | 1.0008 | -6.0874 | -2.0421 |
| H | 3.3089 | 5.2267  | 1.8143  | 3.2699 | 5.2519  | 1.9129  | 3.3221 | 5.2551  | 1.8076  | 3.3116 | -5.3551 | -1.9009 |
| H | 3.2596 | 6.3541  | -2.3351 | 3.3658 | 6.3379  | -2.2469 | 3.2453 | 6.3389  | -2.3535 | 3.3766 | -6.339  | 2.2851  |
| H | 0.9344 | -7.0752 | -2.1081 | 1.0241 | -7.0446 | -2.1211 | 0.9224 | -7.0603 | -2.1198 | 0.9239 | 7.1498  | 2.0547  |
| C | 4.6727 | 4.8406  | -0.4755 | 4.7157 | 4.8468  | -0.3217 | 4.6708 | 4.8444  | -0.4883 | 4.7124 | -4.8547 | 0.3429  |
| C | 5.6011 | 4.5951  | 0.5583  | 5.6032 | 4.6097  | 0.7499  | 5.6007 | 4.6005  | 0.5454  | 5.6088 | -4.6173 | -0.7226 |
| C | 4.8177 | 4.0536  | -1.6372 | 4.9068 | 4.05    | -1.4704 | 4.816  | 4.0552  | -1.6492 | 4.8639 | -4.0236 | 1.4743  |
| C | 6.4611 | 3.509   | 0.5246  | 6.4576 | 3.5193  | 0.7613  | 6.4593 | 3.5139  | 0.5136  | 6.426  | -3.5005 | -0.7509 |
| H | 5.6166 | 5.2471  | 1.4283  | 5.5894 | 5.2722  | 1.6119  | 5.6189 | 5.2555  | 1.4131  | 5.6316 | -5.3048 | -1.5646 |
| C | 5.6557 | 2.9545  | -1.6661 | 5.7397 | 2.947   | -1.4553 | 5.6516 | 2.9547  | -1.6755 | 5.658  | -2.8935 | 1.441   |
| H | 4.1865 | 4.2494  | -2.4996 | 4.3135 | 4.2415  | -2.3602 | 4.1863 | 4.2506  | -2.5126 | 4.266  | -4.2127 | 2.3616  |
| C | 6.4095 | 2.5353  | -0.5236 | 6.4388 | 2.5319  | -0.2762 | 6.4041 | 2.5359  | -0.5313 | 6.3554 | -2.4832 | 0.2574  |
| H | 7.1194 | 3.3606  | 1.3681  | 7.0859 | 3.3784  | 1.6287  | 7.12   | 3.368   | 1.3556  | 7.0696 | -3.3642 | -1.6081 |
| H | 5.6927 | 2.3736  | -2.5779 | 5.8122 | 2.3588  | -2.3601 | 5.6881 | 2.3722  | -2.5863 | 5.6964 | -2.2801 | 2.3307  |
| C | 6.9447 | 1.2138  | -0.3844 | 6.949  | 1.2076  | -0.0934 | 6.9341 | 1.2143  | -0.3862 | 6.8146 | -1.1502 | 0.0361  |
| C | 7.9909 | 0.787   | 0.6229  | 7.9061 | 0.7793  | 0.9993  | 7.9693 | 0.787   | 0.6326  | 7.6971 | -0.7181 | -1.117  |
| C | 6.4607 | -0.0001 | -1.2048 | 6.5167 | -0.0065 | -0.9409 | 6.4545 | 0.0001  | -1.2088 | 6.3791 | 0.0672  | 0.8784  |
| H | 8.976  | 1.148   | 0.2848  | 8.9166 | 1.1418  | 0.7506  | 8.9581 | 1.1479  | 0.3056  | 8.7246 | -1.073  | -0.9383 |
| C | 7.991  | -0.787  | 0.623   | 7.9061 | -0.7949 | 0.9982  | 7.9691 | -0.787  | 0.6326  | 7.6839 | 0.855   | -1.1246 |
| C | 7.8437 | 1.1975  | 2.1212  | 7.6219 | 1.1894  | 2.4786  | 7.8041 | 1.1977  | 2.1291  | 7.3056 | -1.1417 | -2.5693 |
| C | 6.9448 | -1.2139 | -0.3842 | 6.949  | -1.2219 | -0.0952 | 6.9339 | -1.2141 | -0.3862 | 6.7913 | 1.2831  | 0.0223  |
| C | 7.1421 | -0.0002 | -2.611  | 7.2818 | -0.0053 | -2.3027 | 7.1423 | 0.0001  | -2.6116 | 7.1499 | 0.0821  | 2.2346  |
| C | 4.9074 | -0.0002 | -1.3065 | 4.9711 | -0.0066 | -1.1333 | 4.9014 | 0.0002  | -1.317  | 4.8326 | 0.0534  | 1.0717  |
| C | 7.8437 | -1.1973 | 2.1214  | 7.622  | -1.2071 | 2.4769  | 7.8037 | -1.1975 | 2.1291  | 7.289  | 1.2584  | -2.5817 |
| H | 8.976  | -1.1479 | 0.285   | 8.9165 | -1.157  | 0.7488  | 8.9578 | -1.1481 | 0.3058  | 8.7048 | 1.2287  | -0.9463 |
| C | 8.4043 | 0.0002  | 2.8935  | 8.1025 | -0.0094 | 3.3001  | 8.3535 | 0       | 2.9086  | 7.7008 | 0.0567  | -3.4347 |
| H | 8.37   | 2.1271  | 2.3589  | 8.1298 | 2.1149  | 2.7665  | 8.3294 | 2.1261  | 2.3736  | 7.8057 | -2.059  | -2.8931 |
| H | 6.7863 | 1.3365  | 2.375   | 6.5464 | 1.3371  | 2.6313  | 6.744  | 1.3393  | 2.3694  | 6.2246 | -1.313  | -2.6343 |
| C | 6.4095 | -2.5355 | -0.5235 | 6.4399 | -2.5453 | -0.2801 | 6.4037 | -2.5356 | -0.5313 | 6.3058 | 2.6094  | 0.2282  |
| H | 6.8675 | 0.8824  | -3.1962 | 7.0443 | 0.8782  | -2.9024 | 6.8707 | 0.883   | -3.1979 | 6.9258 | -0.7997 | 2.8417  |
| H | 8.2326 | -0.0002 | -2.5059 | 8.3636 | -0.0058 | -2.1301 | 8.2322 | -0.0001 | -2.5012 | 8.2305 | 0.0921  | 2.0551  |
| H | 6.8675 | -0.8829 | -3.1961 | 7.0438 | -0.8875 | -2.9042 | 6.8704 | -0.8825 | -3.1981 | 6.908  | 0.9654  | 2.8328  |
| H | 4.5286 | -0.879  | -1.8261 | 4.6225 | -0.8853 | -1.6736 | 4.5245 | -0.8785 | -1.8381 | 4.4754 | 0.9298  | 1.6105  |
| H | 4.473  | -0.0001 | -0.3014 | 4.479  | -0.0066 | -0.1552 | 4.4628 | 0.0003  | -0.3138 | 4.3411 | 0.0478  | 0.0933  |
| H | 4.5284 | 0.8783  | -1.8263 | 4.6223 | 0.8719  | -1.6738 | 4.5245 | 0.8788  | -1.8382 | 4.4914 | -0.8282 | 1.6124  |
| H | 8.37   | -2.1268 | 2.3593  | 8.1301 | -2.1329 | 2.7635  | 8.3286 | -2.126  | 2.3738  | 7.7775 | 2.179   | -2.9141 |

|   |         |         |         |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| H | 6.7863  | -1.3362 | 2.3752  | 6.5465  | -1.3553 | 2.6294  | 6.7435  | -1.3387 | 2.3694  | 6.206   | 1.4149  | -2.6499 |
| H | 9.5021  | 0.0002  | 2.8484  | 9.1995  | -0.0094 | 3.3623  | 9.4518  | -0.0002 | 2.8783  | 8.7878  | 0.0633  | -3.5948 |
| H | 8.1197  | 0.0003  | 3.9519  | 7.7161  | -0.0101 | 4.3257  | 8.0547  | 0.0001  | 3.9631  | 7.2255  | 0.0484  | -4.4223 |
| C | 6.4611  | -3.5091 | 0.5247  | 6.4592  | -3.5349 | 0.7559  | 6.4586  | -3.5136 | 0.5136  | 6.357   | 3.6165  | -0.7914 |
| C | 5.6558  | -2.9545 | -1.6659 | 5.7409  | -2.9594 | -1.4601 | 5.6513  | -2.9544 | -1.6756 | 5.5996  | 3.0192  | 1.4067  |
| C | 5.6011  | -4.5953 | 0.5585  | 5.6044  | -4.6243 | 0.7434  | 5.6     | -4.6001 | 0.5453  | 5.5176  | 4.717   | -0.7762 |
| H | 7.1195  | -3.3608 | 1.3683  | 7.0885  | -3.3957 | 1.6229  | 7.1192  | -3.3677 | 1.3558  | 7.0039  | 3.4835  | -1.6467 |
| C | 4.8177  | -4.0536 | -1.637  | 4.9089  | -4.0626 | -1.477  | 4.8157  | -4.0549 | -1.6493 | 4.7834  | 4.1336  | 1.427   |
| H | 5.6927  | -2.3737 | -2.5777 | 5.8137  | -2.3702 | -2.3643 | 5.6879  | -2.3719 | -2.5864 | 5.6496  | 2.4167  | 2.3034  |
| C | 4.6727  | -4.8407 | -0.4754 | 4.7163  | -4.8604 | -0.3286 | 4.6703  | -4.8441 | -0.4885 | 4.6161  | 4.9487  | 0.2862  |
| H | 5.6167  | -5.2473 | 1.4285  | 5.5912  | -5.2884 | 1.6042  | 5.6181  | -5.255  | 1.413   | 5.5274  | 5.3953  | -1.626  |
| H | 4.1865  | -4.2495 | -2.4994 | 4.3168  | -4.2534 | -2.3678 | 4.186   | -4.2503 | -2.5129 | 4.1814  | 4.321   | 2.3118  |
| C | -6.9446 | -1.2138 | 0.3834  | -6.948  | -1.2211 | 0.1308  | -6.934  | -1.2143 | 0.3864  | -6.8145 | 1.1502  | -0.0358 |
| C | -6.4605 | -0.0001 | 1.2039  | -6.4998 | -0.008  | 0.9766  | -6.4544 | 0       | 1.2088  | -6.3793 | -0.0672 | -0.8781 |
| C | -7.9907 | -0.7869 | -0.6239 | -7.9731 | -0.7963 | -0.8983 | -7.9695 | -0.7871 | -0.6322 | -7.6961 | 0.7181  | 1.118   |
| C | -6.9447 | 1.2139  | 0.3837  | -6.9489 | 1.2035  | 0.129   | -6.9339 | 1.2141  | 0.3862  | -6.7913 | -1.2831 | -0.0221 |
| C | -7.1417 | -0.0003 | 2.6103  | -7.2429 | -0.0071 | 2.3491  | -7.1422 | 0       | 2.6117  | -7.1503 | -0.082  | -2.2343 |
| C | -4.9072 | 0       | 1.3055  | -4.9541 | -0.0074 | 1.1507  | -4.9013 | 0       | 1.317   | -4.8327 | -0.0534 | -1.0716 |
| C | -7.9909 | 0.7871  | -0.6236 | -7.9736 | 0.7769  | -0.8994 | -7.9694 | 0.7869  | -0.6324 | -7.683  | -0.855  | 1.1255  |
| C | -7.8435 | -1.197  | -2.1223 | -7.8012 | -1.2071 | -2.3934 | -7.8048 | -1.1979 | -2.1287 | -7.3031 | 1.1416  | 2.57    |
| H | -8.9758 | -1.1479 | -0.2859 | -8.9644 | -1.1571 | -0.5777 | -8.9582 | -1.148  | -0.305  | -8.7238 | 1.0731  | 0.9404  |
| H | -6.867  | 0.8822  | 3.1955  | -6.9947 | 0.8764  | 2.9453  | -6.8702 | 0.8827  | 3.1981  | -6.9087 | -0.9654 | -2.8323 |
| H | -8.2322 | -0.0004 | 2.5054  | -8.3277 | -0.0078 | 2.1962  | -8.2321 | 0.0001  | 2.5014  | -8.2308 | -0.0916 | -2.0546 |
| H | -6.8669 | -0.8831 | 3.1952  | -6.9938 | -0.8894 | 2.9468  | -6.8705 | -0.8828 | 3.198   | -6.9259 | 0.7996  | -2.8415 |
| H | -4.5282 | -0.8787 | 1.825   | -4.5993 | -0.8852 | 1.6885  | -4.5244 | -0.8786 | 1.8382  | -4.4916 | 0.8286  | -1.6117 |
| H | -4.4729 | 0.0002  | 0.3004  | -4.4719 | -0.0075 | 0.1676  | -4.4628 | -0.0001 | 0.3137  | -4.3412 | -0.0486 | -0.0932 |
| H | -4.5283 | 0.8786  | 1.8253  | -4.5998 | 0.8708  | 1.6881  | -4.5245 | 0.8788  | 1.838   | -4.4757 | -0.9295 | -1.6111 |
| H | -8.9759 | 1.1479  | -0.2855 | -8.9652 | 1.1375  | -0.5797 | -8.958  | 1.148   | -0.3054 | -8.7041 | -1.2287 | 0.9482  |
| C | -7.8436 | 1.1978  | -2.1219 | -7.8016 | 1.1855  | -2.3951 | -7.8044 | 1.1973  | -2.129  | -7.2867 | -1.2586 | 2.5823  |
| C | -8.4041 | 0.0005  | -2.8943 | -8.3539 | -0.0114 | -3.1741 | -8.3544 | -0.0003 | -2.9082 | -7.6974 | -0.0568 | 3.4357  |
| H | -8.3697 | -2.1265 | -2.3603 | -8.3196 | -2.1393 | -2.6383 | -8.3301 | -2.1263 | -2.3731 | -7.803  | 2.0589  | 2.8944  |
| H | -6.7861 | -1.3358 | -2.3762 | -6.7395 | -1.3405 | -2.6313 | -6.7447 | -1.3394 | -2.3693 | -6.2221 | 1.313   | 2.6338  |
| H | -8.37   | 2.1273  | -2.3596 | -8.3202 | 2.1173  | -2.6414 | -8.3293 | 2.1258  | -2.3736 | -7.7751 | -2.179  | 2.9151  |
| H | -6.7863 | 1.3368  | -2.3757 | -6.7399 | 1.3187  | -2.6331 | -6.7442 | 1.3384  | -2.3696 | -6.2037 | -1.4153 | 2.6493  |
| H | -9.502  | 0.0004  | -2.8492 | -9.4522 | -0.0116 | -3.1415 | -9.4528 | -0.0001 | -2.8775 | -8.7843 | -0.0633 | 3.5971  |
| H | -8.1195 | 0.0007  | -3.9527 | -8.0569 | -0.0121 | -4.229  | -8.056  | -0.0005 | -3.9628 | -7.221  | -0.0486 | 4.4228  |



| Atom | Spin-projected for BS1 |         |         | Spin-projected for BS2 |         |         | Spin restricted |         |         |
|------|------------------------|---------|---------|------------------------|---------|---------|-----------------|---------|---------|
|      | x [Å]                  | y [Å]   | z [Å]   | x [Å]                  | y [Å]   | z [Å]   | x [Å]           | y [Å]   | z [Å]   |
| C    | -6.404                 | -2.5351 | 0.5314  | -6.3559                | 2.4828  | -0.2574 | -6.082          | 2.4706  | -0.3951 |
| C    | -4.6704                | -4.8445 | 0.4885  | -4.7129                | 4.8552  | -0.3433 | -4.5917         | 4.9463  | -0.5992 |
| C    | -6.4592                | -3.5135 | -0.5143 | -6.4267                | 3.5005  | 0.7517  | -6.2302         | 3.5491  | 0.5729  |
| C    | -5.6513                | -2.9542 | 1.6766  | -5.6584                | 2.8933  | -1.4421 | -5.4826         | 2.9194  | -1.6432 |
| C    | -4.8151                | -4.0553 | 1.6504  | -4.8641                | 4.0239  | -1.4757 | -4.7804         | 4.0854  | -1.7327 |
| C    | -5.6003                | -4.6005 | -0.5464 | -5.6093                | 4.6178  | 0.7237  | -5.4937         | 4.6959  | 0.4913  |
| H    | -7.1199                | -3.3677 | -1.3562 | -7.0701                | 3.3639  | 1.6089  | -6.8952         | 3.4163  | 1.4172  |
| H    | -5.6871                | -2.3725 | 2.5876  | -5.6961                | 2.2805  | -2.332  | -5.5527         | 2.2994  | -2.5269 |
| H    | -4.1855                | -4.25   | 2.514   | -4.2668                | 4.2121  | -2.3634 | -4.2886         | 4.3154  | -2.6722 |
| H    | -5.6189                | -5.2548 | -1.4145 | -5.6324                | 5.3044  | 1.5665  | -5.585          | 5.4302  | 1.2868  |
| C    | -6.4037                | 2.535   | 0.5311  | -6.3063                | -2.6089 | -0.2284 | -6.0217         | -2.6054 | -0.3822 |
| C    | -4.6699                | 4.8442  | 0.4881  | -4.6166                | -4.9492 | -0.2867 | -4.4995         | -5.0595 | -0.5627 |
| C    | -6.4585                | 3.513   | -0.5149 | -6.3579                | -3.6165 | 0.792   | -6.1123         | -3.6501 | 0.6271  |
| C    | -5.6512                | 2.9542  | 1.6763  | -5.5999                | -3.0189 | -1.4079 | -5.4606         | -3.0784 | -1.6377 |
| C    | -4.815                 | 4.0552  | 1.6501  | -4.7834                | -4.134  | -1.4285 | -4.7383         | -4.2338 | -1.7136 |
| C    | -5.5995                | 4.5999  | -0.5471 | -5.5183                | -4.7175 | 0.7771  | -5.368          | -4.7914 | 0.5514  |
| H    | -7.1188                | 3.367   | -1.3571 | -7.0047                | -3.4835 | 1.647   | -6.7391         | -3.4899 | 1.4976  |
| H    | -5.6872                | 2.3727  | 2.5875  | -5.649                 | -2.4171 | -2.3048 | -5.5624         | -2.4797 | -2.5335 |
| H    | -4.1856                | 4.2501  | 2.5139  | -4.1818                | -4.3203 | -2.3137 | -4.2703         | -4.4823 | -2.6605 |
| H    | -5.6177                | 5.254   | -1.4154 | -5.5284                | -5.3949 | 1.6275  | -5.417          | -5.4994 | 1.3735  |
| C    | -3.4597                | -5.6701 | 0.3072  | -3.5195                | 5.7142  | -0.2186 | -3.4719         | 5.8295  | -0.5055 |
| C    | -0.7406                | -6.4998 | 0.0372  | -0.8062                | 6.5917  | -0.0511 | -0.7795         | 6.7962  | -0.1285 |
| C    | -2.8198                | -5.6859 | -0.948  | -2.843                 | 5.7731  | 1.0163  | -3.0218         | 6.3416  | 0.7521  |
| C    | -2.7606                | -6.2502 | 1.3861  | -2.8633                | 6.2818  | -1.3311 | -2.593          | 6.0755  | -1.6069 |
| C    | -1.4375                | -6.659  | 1.2528  | -1.5442                | 6.7151  | -1.2471 | -1.3156         | 6.5449  | -1.4278 |
| C    | -1.4944                | -6.0845 | -1.0776 | -1.5206                | 6.1934  | 1.0958  | -1.7355         | 6.7893  | 0.9324  |
| H    | -3.3218                | -5.2548 | -1.8089 | -3.312                 | 5.3554  | 1.9021  | -3.6716         | 6.2803  | 1.6191  |
| H    | -3.2441                | -6.3383 | 2.3554  | -3.3766                | 6.3386  | -2.2873 | -2.9122         | 5.8297  | -2.6144 |
| H    | -0.9228                | -7.06   | 2.1215  | -1.0639                | 7.1047  | -2.1404 | -0.6746         | 6.6238  | -2.2993 |
| H    | -1.0048                | -5.9492 | -2.037  | -1.0016                | 6.0873  | 2.0432  | -1.4326         | 7.0921  | 1.9297  |
| C    | -3.4592                | 5.6698  | 0.307   | -3.4066                | -5.7833 | -0.1521 | -3.36           | -5.918  | -0.4737 |
| C    | -0.7401                | 6.4997  | 0.0373  | -0.6771                | -6.6062 | 0.0255  | -0.6421         | -6.8159 | -0.1046 |
| C    | -2.8191                | 5.6859  | -0.9482 | -2.7294                | -5.8144 | 1.0834  | -2.8925         | -6.4197 | 0.7821  |
| C    | -2.7603                | 6.2499  | 1.3861  | -2.7395                | -6.3511 | -1.2579 | -2.4789         | -6.1446 | -1.578  |
| C    | -1.4372                | 6.6587  | 1.2528  | -1.4124                | -6.7579 | -1.1689 | -1.1885         | -6.5792 | -1.4022 |
| C    | -1.4938                | 6.0845  | -1.0776 | -1.3991                | -6.2082 | 1.1677  | -1.5957         | -6.8358 | 0.9583  |
| H    | -3.321                 | 5.2548  | -1.8093 | -3.2066                | -5.3955 | 1.9643  | -3.5411         | -6.3806 | 1.6511  |
| H    | -3.2438                | 6.3378  | 2.3553  | -3.2516                | -6.429  | -2.2133 | -2.8052         | -5.9049 | -2.5845 |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| H | -0.9226 | 7.0597  | 2.1217  | -0.9245 | -7.1489 | -2.0575 | -0.5474 | -6.6397 | -2.2754 |
| H | -1.0042 | 5.9494  | -2.037  | -0.8824 | -6.0808 | 2.1137  | -1.2839 | -7.1348 | 1.9539  |
| C | 0.7403  | -6.4997 | -0.0375 | 0.6764  | 6.6061  | -0.026  | 0.642   | 6.8159  | 0.1046  |
| C | 3.4594  | -5.6698 | -0.3074 | 3.4059  | 5.7829  | 0.1515  | 3.36    | 5.918   | 0.4737  |
| C | 1.4373  | -6.6587 | -1.2531 | 1.4117  | 6.758   | 1.1683  | 1.1884  | 6.5792  | 1.4022  |
| C | 1.4941  | -6.0846 | 1.0773  | 1.3983  | 6.2078  | -1.1682 | 1.5957  | 6.8358  | -0.9583 |
| C | 2.8194  | -5.686  | 0.9478  | 2.7285  | 5.814   | -1.0839 | 2.8925  | 6.4196  | -0.7822 |
| C | 2.7604  | -6.2499 | -1.3864 | 2.7389  | 6.351   | 1.2573  | 2.4789  | 6.1446  | 1.5779  |
| H | 1.0046  | -5.9495 | 2.0368  | 0.8815  | 6.0802  | -2.1142 | 1.2838  | 7.1348  | -1.954  |
| H | 3.3215  | -5.255  | 1.8088  | 3.2057  | 5.3948  | -1.9647 | 3.5411  | 6.3806  | -1.6512 |
| H | 3.2438  | -6.3377 | -2.3557 | 3.251   | 6.429   | 2.2127  | 2.8052  | 5.9049  | 2.5845  |
| C | 0.7408  | 6.4998  | -0.0372 | 0.8055  | -6.5916 | 0.0506  | 0.7795  | -6.7963 | 0.1285  |
| C | 3.46    | 5.6701  | -0.3071 | 3.5186  | -5.7139 | 0.2183  | 3.4718  | -5.8296 | 0.5055  |
| C | 1.4378  | 6.6591  | -1.2528 | 1.5434  | -6.715  | 1.2466  | 1.3156  | -6.5449 | 1.4277  |
| C | 1.4945  | 6.0844  | 1.0775  | 1.5198  | -6.1931 | -1.0962 | 1.7355  | -6.7893 | -0.9324 |
| C | 2.8198  | 5.686   | 0.948   | 2.8421  | -5.7726 | -1.0166 | 3.0218  | -6.3416 | -0.7521 |
| C | 2.761   | 6.2504  | -1.386  | 2.8625  | -6.2816 | 1.3307  | 2.593   | -6.0755 | 1.6069  |
| H | 0.9231  | 7.0602  | -2.1215 | 1.0631  | -7.1048 | 2.1399  | 0.6745  | -6.6238 | 2.2993  |
| H | 1.0049  | 5.949   | 2.0369  | 1.0008  | -6.0869 | -2.0436 | 1.4326  | -7.0921 | -1.9297 |
| H | 3.3219  | 5.2548  | 1.809   | 3.3112  | -5.3548 | -1.9024 | 3.6716  | -6.2803 | -1.6191 |
| H | 3.2444  | 6.3386  | -2.3552 | 3.3758  | -6.3385 | 2.2869  | 2.9122  | -5.8298 | 2.6143  |
| H | 0.9225  | -7.0595 | -2.1219 | 0.9239  | 7.149   | 2.0568  | 0.5474  | 6.6397  | 2.2754  |
| C | 4.6706  | 4.8446  | -0.4883 | 4.7121  | -4.8549 | 0.3429  | 4.5917  | -4.9463 | 0.5992  |
| C | 5.6005  | 4.6005  | 0.5466  | 5.6084  | -4.6175 | -0.724  | 5.4937  | -4.6959 | -0.4913 |
| C | 4.8154  | 4.0554  | -1.6503 | 4.8633  | -4.0238 | 1.4755  | 4.7803  | -4.0854 | 1.7327  |
| C | 6.4594  | 3.5135  | 0.5144  | 6.4261  | -3.5003 | -0.7519 | 6.2302  | -3.5491 | -0.5729 |
| H | 5.619   | 5.2549  | 1.4147  | 5.6315  | -5.304  | -1.5668 | 5.585   | -5.4302 | -1.2868 |
| C | 5.6515  | 2.9544  | -1.6765 | 5.6579  | -2.8933 | 1.442   | 5.4826  | -2.9194 | 1.6432  |
| H | 4.1858  | 4.2503  | -2.514  | 4.2659  | -4.212  | 2.363   | 4.2886  | -4.3154 | 2.6722  |
| C | 6.4042  | 2.5352  | -0.5313 | 6.3555  | -2.4827 | 0.2574  | 6.0819  | -2.4706 | 0.3951  |
| H | 7.12    | 3.3677  | 1.3564  | 7.0695  | -3.3638 | -1.6089 | 6.8952  | -3.4163 | -1.4172 |
| H | 5.6873  | 2.3728  | -2.5876 | 5.6956  | -2.2807 | 2.332   | 5.5527  | -2.2994 | 2.5268  |
| C | 6.9342  | 1.2138  | -0.3861 | 6.8146  | -1.1499 | 0.0361  | 6.3751  | -1.1525 | 0.0883  |
| C | 7.9693  | 0.7869  | 0.6325  | 7.697   | -0.7181 | -1.117  | 6.9824  | -0.6994 | -1.2314 |
| C | 6.4544  | 0.0001  | -1.2092 | 6.379   | 0.0672  | 0.8792  | 6.0554  | 0.0651  | 0.983   |
| H | 8.958   | 1.1475  | 0.3055  | 8.724   | -1.0728 | -0.9381 | 8.0585  | -0.9319 | -1.2282 |
| C | 7.9691  | -0.7869 | 0.6326  | 7.6838  | 0.855   | -1.1246 | 6.7799  | 0.8487  | -1.2863 |
| C | 7.8039  | 1.198   | 2.1293  | 7.305   | -1.1419 | -2.5694 | 6.3541  | -1.2549 | -2.5563 |
| C | 6.934   | -1.2136 | -0.3861 | 6.7913  | 1.2829  | 0.0223  | 6.3171  | 1.2867  | 0.0834  |
| C | 7.1422  | 0.0001  | -2.6118 | 7.1499  | 0.0821  | 2.2346  | 7.0063  | 0.0851  | 2.2169  |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| C | 4.9014  | 0.0002  | -1.3168 | 4.8327  | 0.0533  | 1.0713  | 4.5526  | 0.052   | 1.4008  |
| C | 7.8035  | -1.1978 | 2.1293  | 7.2885  | 1.2586  | -2.5818 | 5.7737  | 1.0717  | -2.4553 |
| H | 8.9577  | -1.1477 | 0.3057  | 8.7042  | 1.2285  | -0.9461 | 7.7225  | 1.3446  | -1.561  |
| C | 8.3536  | 0       | 2.9083  | 7.7007  | 0.0567  | -3.4343 | 6.1445  | -0.026  | -3.4527 |
| H | 8.3293  | 2.126   | 2.3735  | 7.8054  | -2.0588 | -2.8929 | 6.9844  | -2.0122 | -3.0333 |
| H | 6.7438  | 1.3395  | 2.3696  | 6.2242  | -1.313  | -2.6346 | 5.3889  | -1.7251 | -2.3423 |
| C | 6.4038  | -2.5349 | -0.5313 | 6.3059  | 2.6089  | 0.2282  | 6.0217  | 2.6054  | 0.3823  |
| H | 6.8705  | 0.883   | -3.1979 | 6.9255  | -0.7996 | 2.8416  | 6.8794  | -0.806  | 2.8382  |
| H | 8.2323  | 0       | -2.5004 | 8.2302  | 0.092   | 2.0536  | 8.0508  | 0.1151  | 1.8894  |
| H | 6.8702  | -0.8825 | -3.1982 | 6.9079  | 0.9654  | 2.8327  | 6.8326  | 0.9602  | 2.8493  |
| H | 4.5244  | -0.8786 | -1.8383 | 4.4754  | 0.9298  | 1.6105  | 4.2753  | 0.9652  | 1.9282  |
| H | 4.4629  | 0.0003  | -0.313  | 4.3411  | 0.0473  | 0.0924  | 3.9206  | -0.0186 | 0.5097  |
| H | 4.5245  | 0.8789  | -1.8384 | 4.4914  | -0.8283 | 1.6125  | 4.3097  | -0.7956 | 2.0417  |
| H | 8.3285  | -2.1259 | 2.3736  | 7.7771  | 2.1788  | -2.9138 | 5.823   | 2.0789  | -2.8762 |
| H | 6.7433  | -1.3388 | 2.3697  | 6.2056  | 1.4149  | -2.6503 | 4.749   | 0.9218  | -2.0917 |
| H | 9.4519  | -0.0002 | 2.8773  | 8.7877  | 0.0632  | -3.5934 | 7.0784  | 0.237   | -3.9684 |
| H | 8.0547  | 0.0001  | 3.9628  | 7.2251  | 0.0484  | -4.4217 | 5.3813  | -0.189  | -4.2222 |
| C | 6.4588  | -3.5131 | 0.5145  | 6.3571  | 3.6162  | -0.7924 | 6.1123  | 3.6501  | -0.627  |
| C | 5.6512  | -2.954  | -1.6766 | 5.5995  | 3.0189  | 1.4077  | 5.4605  | 3.0784  | 1.6377  |
| C | 5.5998  | -4.6002 | 0.5466  | 5.5173  | 4.7171  | -0.7776 | 5.368   | 4.7914  | -0.5514 |
| H | 7.1192  | -3.3674 | 1.3566  | 7.0038  | 3.4831  | -1.6475 | 6.7391  | 3.4899  | -1.4976 |
| C | 4.8151  | -4.0551 | -1.6505 | 4.7828  | 4.1339  | 1.4282  | 4.7383  | 4.2338  | 1.7136  |
| H | 5.6872  | -2.3724 | -2.5876 | 5.6488  | 2.4173  | 2.3047  | 5.5624  | 2.4797  | 2.5336  |
| C | 4.6701  | -4.8443 | -0.4885 | 4.6158  | 4.9489  | 0.2862  | 4.4995  | 5.0595  | 0.5627  |
| H | 5.6181  | -5.2544 | 1.4147  | 5.5272  | 5.3943  | -1.6282 | 5.417   | 5.4994  | -1.3735 |
| H | 4.1857  | -4.2499 | -2.5142 | 4.1813  | 4.3203  | 2.3133  | 4.2703  | 4.4823  | 2.6605  |
| C | -6.9341 | -1.2137 | 0.3862  | -6.8146 | 1.15    | -0.0358 | -6.3752 | 1.1525  | -0.0883 |
| C | -6.4543 | 0       | 1.2092  | -6.3792 | -0.0672 | -0.879  | -6.0555 | -0.0651 | -0.983  |
| C | -7.9695 | -0.787  | -0.6322 | -7.6961 | 0.7181  | 1.1181  | -6.9824 | 0.6994  | 1.2314  |
| C | -6.934  | 1.2136  | 0.386   | -6.7914 | -1.2829 | -0.0221 | -6.3171 | -1.2867 | -0.0834 |
| C | -7.1421 | 0       | 2.6118  | -7.1502 | -0.082  | -2.2343 | -7.0064 | -0.0851 | -2.2168 |
| C | -4.9013 | 0       | 1.3168  | -4.8328 | -0.0533 | -1.0712 | -4.5527 | -0.052  | -1.4008 |
| C | -7.9694 | 0.7868  | -0.6324 | -7.6829 | -0.855  | 1.1255  | -6.7799 | -0.8487 | 1.2863  |
| C | -7.8045 | -1.1982 | -2.1289 | -7.3026 | 1.1419  | 2.5701  | -6.3541 | 1.255   | 2.5563  |
| H | -8.9581 | -1.1476 | -0.3049 | -8.7232 | 1.0729  | 0.9402  | -8.0586 | 0.9319  | 1.2282  |
| H | -6.8701 | 0.8827  | 3.1981  | -6.9084 | -0.9654 | -2.8322 | -6.8327 | -0.9603 | -2.8492 |
| H | -8.2322 | 0.0001  | 2.5005  | -8.2305 | -0.0915 | -2.0531 | -8.0509 | -0.1151 | -1.8894 |
| H | -6.8703 | -0.8827 | 3.1981  | -6.9256 | 0.7996  | -2.8414 | -6.8796 | 0.8059  | -2.8382 |
| H | -4.5243 | -0.8786 | 1.8384  | -4.4915 | 0.8287  | -1.6117 | -4.3098 | 0.7956  | -2.0417 |
| H | -4.4629 | -0.0001 | 0.313   | -4.3412 | -0.0481 | -0.0923 | -3.9206 | 0.0186  | -0.5097 |

|   |         |         |         |         |         |         |         |         |         |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| H | -4.5244 | 0.8788  | 1.8381  | -4.4757 | -0.9295 | -1.6111 | -4.2754 | -0.9652 | -1.9283 |
| H | -8.9579 | 1.1476  | -0.3052 | -8.7035 | -1.2284 | 0.948   | -7.7225 | -1.3446 | 1.561   |
| C | -7.8042 | 1.1976  | -2.1292 | -7.2862 | -1.2588 | 2.5824  | -5.7736 | -1.0717 | 2.4553  |
| C | -8.3545 | -0.0003 | -2.9079 | -7.6972 | -0.0568 | 3.4354  | -6.1445 | 0.0261  | 3.4527  |
| H | -8.33   | -2.1262 | -2.3729 | -7.8026 | 2.0587  | 2.8941  | -6.9844 | 2.0122  | 3.0332  |
| H | -6.7445 | -1.3396 | -2.3696 | -6.2217 | 1.313   | 2.6342  | -5.3889 | 1.7252  | 2.3423  |
| H | -8.3293 | 2.1257  | -2.3734 | -7.7747 | -2.1789 | 2.9149  | -5.8229 | -2.0789 | 2.8762  |
| H | -6.744  | 1.3386  | -2.3699 | -6.2033 | -1.4153 | 2.6498  | -4.749  | -0.9217 | 2.0917  |
| H | -9.4529 | -0.0001 | -2.8764 | -8.7841 | -0.0632 | 3.5957  | -7.0783 | -0.237  | 3.9685  |
| H | -8.056  | -0.0005 | -3.9624 | -7.2206 | -0.0486 | 4.4223  | -5.3813 | 0.1891  | 4.2222  |

**Table S9.** Calculated total energies and  $\langle S^2 \rangle$  values of optimized structures in THF.

|                                                         | Singlet        |                |                    |                    |
|---------------------------------------------------------|----------------|----------------|--------------------|--------------------|
|                                                         | BS-singlet 1   | BS-singlet 2   | Spin-projected BS1 | Spin-projected BS2 |
| Total energy [a.u.]                                     | -2627.21466602 | -2627.21427358 | -2627.21466072     | -2627.21426700     |
| $\langle S^2 \rangle$                                   | 2.1252         | 2.1273         | 2.1255             | 2.1277             |
| Quintet state energy on the singlet structure           |                |                |                    |                    |
| Total energy [a.u.]                                     | -2627.21400633 | -2627.21403090 | -2627.21398820     | -2627.21400289     |
| $\langle S^2 \rangle$                                   | 6.1230         | 6.1282         | 6.1230             | 6.1286             |
| BS-singlet 2 state energy on the BS-singlet 1 structure |                |                |                    |                    |
| Total energy [a.u.]                                     | -2627.21416072 |                |                    |                    |
| $\langle S^2 \rangle$                                   | 2.1204         |                |                    |                    |
| Spin-projected energy [a.u.]                            | -2627.21501671 | -2627.21440261 | -2627.21501830     | -2627.21440746     |

|                       | Quintet        | Triplet        | Singlet<br>Spin restricted |
|-----------------------|----------------|----------------|----------------------------|
| Total energy [a.u.]   | -2627.21404832 | -2627.21418431 | -2627.17432752             |
| $\langle S^2 \rangle$ | 6.1206         | 3.1190         | 0.0000                     |

**Table S10:** Calculated total energies and  $\langle S^2 \rangle$  values of optimized structures in the gas phase.

|                                               | Singlet        |                |                 |
|-----------------------------------------------|----------------|----------------|-----------------|
|                                               | BS-singlet 1   | BS-singlet 2   | Spin restricted |
| Total energy [a.u.]                           | -2627.20205681 | -2627.20170505 | -2627.16048075  |
| $\langle S^2 \rangle$                         | 2.1268         | 2.1295         | 0.00000         |
| Quintet state energy on the singlet structure |                |                |                 |
| Total energy [a.u.]                           | -2627.20147181 | -2627.20152844 |                 |
| $\langle S^2 \rangle$                         | 6.1243         | 6.1302         |                 |
| AP energy [a.u.]                              | -2627.20236805 | -2627.20179906 |                 |

|                       | Quintet        | Triplet        |
|-----------------------|----------------|----------------|
| Total energy [a.u.]   | -2627.20150186 | -2627.20161387 |
| $\langle S^2 \rangle$ | 6.1228         | 3.1214         |